

Constitutive structure of the vacuum and electromagnetic propagation

Derivation of Catalan's constant, permeability, and permittivity

Oumar Haidara Fall · Rubén Ramos Balsa

Coauthorship without precedence of contribution

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Abstract

A positive, invertible constitutive response of the electromagnetic vacuum is constructed: two oriented channels jointly determine the dimensionless coordinates of permittivity, permeability, impedance and propagation speed. Their exact relations and the unique recovery of the angular pair producing them are proved. The same response reconstructs the shape of the action ellipse and expresses the Barbero–Immirzi functional through its two labelled eigenvalues. The result thus relates the determination of values to the structural information preserved by their evaluation.

The construction starts from a formal mathematical core called Triadic Modular Holography, henceforth HMT. *All Possible Paths* (APP) is the arithmetic path structure on the toroidal support $(\mathbb{Z}/9\mathbb{Z})^2$; its cardinal displacements form the additive Cayley graph, and its sheets preserve the sum and product evaluations with residue and quotient. TRIT is the oriented signature $\{-1, 0, +1\}$ distinguishing local regimes. The *Topological Phase Kernel* (TPK) composes selection, transport, memory update and return on this enriched state. Nonadic prolongation and dodecaphasic closure generate the correlated coordinates and records from which the action, angular pair and incidences used by the response arise.

The sectors 90 and 120 of the transport T determine $\mathcal{V} = (I - T^{90})(I - T^{120})^{-1}$. Its scalar function $f(s) = (1 + s + s^2)/(1 + s + s^2 + s^3)$ is a decreasing bijection from $(0, 1)$ onto $(3/4, 1)$: this property establishes cubic inversion and its domain. Sheet interchange permutes permittivity and permeability, inverts impedance and preserves speed. The identity between constitutive speed and the clock's kinematic speed is also proved in the same dimensional chart.

Realization on the twelve-phase cycle supplies the trace defect $-1/12$, a representation of f as a quotient of determinants and a quarter-turn whose square is $-I$. Double logarithmic integration of its resolvent produces the Catalan moment. At the Pell section $2 - \sqrt{3}$, an exact identity with Catalan's constant and $\log(2 + \sqrt{3})$ is proved; the two constitutive arguments retain their own angular determination. Inversion of f subsequently composes the vacuum response with the oriented Barbero–Immirzi functional. Integral reconstruction also proves that the constitutive function determines the complete oriented moment under the condition $\mathcal{C}_2(s) \rightarrow 0$ at the origin; its limit at $s = 1$ recovers Catalan's constant. A functional correspondence, with its orientation and arguments, is thereby preserved alongside its evaluations.

Discrete polarization is obtained from the bounded vacancy balance, and the electronic prefactor $\sqrt{3}/4$ from the commutator norm of projections associated with the sum and product fibres of the three-dimensional arithmetic support. The positive charge section, constructed from α , action and impedance, determines the von Klitzing, conductance, magnetic flux and Josephson relations, including their cross-identities and covariance under action return. Dimensional realizations preserve the action, charge and speed sections and their changes of units. The numerical evaluation concerns the internal coordinates; physical comparison explicitly retains the representation and dimensional chart used.

Positive separation of the common and oriented modes specifies the information preserved by unimodular normalization. Fourier resolution of the dodecaphasic incidence identifies four modes and their exact weights, distinguishing the rank defect, signed multiplicities and singular coefficient of return.

The inversion also induces an associative composition law for responses, with an additive angular coordinate. In the tree-level electroweak realisation, the gauge and Higgs readers jointly recover the channels and impedance within an explicit domain.

Keywords: structural determination of constants; electromagnetic vacuum; constitutive response; spectral inversion; Triadic Modular Holography; sheet orientation; Catalan's constant; Pell unit; Barbero–Immirzi; polarization; electrical quantization.

Contents

Abstract	1
1 Introduction	6
1.1 Generating kernel and provenance of the response	6
1.2 Constitutive determination and recovery of shape	7
1.3 Cycle, constants and oriented functionals	7
1.4 Polarization, charge and dimensional realization	8
2 Formal core of Triadic Modular Holography	9
2.1 APP: positional support and arithmetic evaluations	9
2.2 TRIT: orientation and quadratic regimes	12
2.3 TPK: topological phase kernel	18
2.4 Accumulated memory and transport resolvents	24
3 Topological Phase Kernel and generation of the seed catalogue	28
3.1 Phase selection, transport and enriched state	28
3.2 Initial conditions and reversible enumeration	29
3.3 Sheet evaluation and arithmetic record	30
3.4 Six windows and decimal block emission	31
3.5 Complete calculation of an emission beginning with 501	32
3.6 Factorisation of the finite census	33
3.7 Oriented ternary reading and preservation of fibres	35
3.8 Dihedral action and the forty-three orbits	36
3.9 Self-contained enumeration and verification algorithm	37
3.10 Preservation of the record and the function of the seeds	37
4 Development of the formal kernel: initial conditions, geometry, and refinement	38
4.1 Initial states and the emission recurrence	38
4.2 Calendar closure and memory advancement	39
4.3 Observable compression and orthogonal conservation	41
4.4 Local development: commutation, Lorentzian signature, and discrete curvature	41
4.5 Cubic completion and two notions of boundary	42
4.6 Cubic incidence and Lorentzian realization	44
4.7 Compatibility between modular and positional reductions	45
4.8 Number as a compatible section and value as a readout	46
4.9 Composition of histories and memory multiplier	46
4.10 Double variance of refinement and preservation of the unit	47
4.11 Phase return, Weyl representation, and two readouts	48
5 Coinductive generation of π, φ, e to arbitrary depth	49
5.1 Orbits, regions, and characters of closure, propagation, and self-scaling	49
5.2 Finite transport and the prolongation boundary	50
5.3 Five closure regions and angular compensation	50
5.4 Propagation readout and normalization of the unit	51
5.5 Modal incidence and generation of self-scaling	52
5.6 Exponential realizations of the generated coordinates	54
5.7 Compatible cylinders and arbitrary depth	56
5.8 Specular involution and synchronization of the regions	57
5.9 Sturmian helical suspension	57
5.10 Nonadic connection, holonomy and monodromic representation	60
5.11 Block clock and capacity clock	62
5.12 Symbolic suspension and aperiodic time crystal	63
5.13 Mirror symmetry of the regions and dual realization	63
5.14 Temporal transport and tritic exponential composition	65
5.15 Multiplicative irreducibility and return periods	66
6 Regional generation and refinement	69
6.1 Positional emission through rational separation of cells	69
6.2 Regional blocks of six trits at arbitrary depth	70
6.3 Transition registers and reconstruction of linear lifts	71
6.4 Simultaneous refinement in ternary and decimal bases	72
6.5 Selection of the next block by the regional reader	73
6.6 Ternary signature, carry, Witt incidence, and reading clock	74
6.7 Composition theorem for the block, cylinder, and signature	75

6.8	Calendar census and simultaneous compatibility of the regional biographies	76
7	Regional determination of the dodecaphasic descriptor and selection of the coupling register	77
7.1	Regional bands, encoding, and integer signatures	77
7.2	Conversion clock and first zero advance	78
7.3	Oriented construction of the twenty-four-trit descriptor	79
7.4	Functional panel and terminal repertoire	80
7.5	Comparative readers, affine reader, and temporal incidence	84
7.6	Enumeration of cylinders and decimal candidates	85
7.7	Exceptional incidence and heterotypy condition	87
7.8	Regional composition and continuity of the realizations	89
7.9	Executable certification and traceability of the operations	90
7.10	Complete census of cylinders and selection of the terminal boundary	90
8	Dodecaphase record and the rational phase constant	95
8.1	Cyclic extension and memory grading	95
8.2	The oriented record and the integral $1 + 5 + 6$ chart	96
8.3	Evaluation of incidences and determination of the transverse coordinates	97
8.4	Integral image and inversion of the transverse coordinates	99
8.5	Conjugation, covariance, and memory of the incidence register	100
8.6	Two readouts of the terminal state	102
8.7	From the channel record to the signed vector	102
8.8	Orbitwise Hadamard transformation and integer reconstruction	103
8.9	Cyclic differences and the uniform component	105
8.10	Two rational readouts: κ_{36} and κ_{per}	105
8.11	Reversibility of the rational constant and phase change	106
8.12	Periodicity and the cokernel of the transport operator	107
9	Derivation of the fine-structure constant	108
9.1	First route: positional determination through integer closure	108
9.2	Second route: analytic determination through vacancies	114
9.3	Identification criterion for compatible readouts	116
9.4	Compatibility between the dodecaphasic and analytic representations of alpha	117
10	Exceptional prolongation: from dodecaphasic incidence to Moonshine	123
10.1	Compatibility of the Archimedean and incidence realizations	123
10.2	The finite realization and the ternary code	124
10.3	Dodecaphasic flag associated with K and α	125
10.4	Isometric intertwining of the incidence representations	128
10.5	The Witt star and the generation of M_{12}	129
10.6	The binary residue and the five regions of π	130
10.7	From the ternary code to the Niemeier lattice	132
10.8	Incidence selection of the rootless unimodular neighbor	133
10.9	The vertex operator algebra and the Monster	134
10.10	Graded Moonshine traces and the Monster representation	135
11	Dodecaphasic update and expressed memory	137
11.1	Quadratic balance	137
11.2	Transport resolvents and preservation of memory	138
12	Dodecaphasic seal and reversible rational readout	139
13	Cyclic decomposition and determinantal representations	140
13.1	Oriented information and the complex sector	140
14	Planck constants and the determinantal structure of action	141
14.1	Determinantal action scale and decimal valuation	141
14.2	Action sections and the dimensionless return quotient	142
14.3	Angular coordinates, winding normalization, and action units	144
14.4	Covariance of the channels and the incidence functional	145
15	Dodecaphase fundamental chain and hexad–octad functional	146
16	Hexadic–octadic incidence and the Barbero–Immirzi functional	148
16.1	The weights precede the pole coefficient	148
16.2	Parity and structural sensitivity	149
16.3	Structural dependence of the functional and areal normalization	149

17 Action ellipse and modular reciprocity	150
17.1 Semiaxes, foci, and interior radii	150
17.2 Parametric phase, polar angle, and action area	151
17.3 Shape reciprocity and interior distance	152
17.4 Bohr radius: a consequence of the action scale	152
17.5 Inductive–capacitive realization and constitutive continuity	153
17.6 Comparison of the reduced Planck constant	153
18 Discrete polarization and the tritic mode of arithmetic projections	155
18.1 Refinement discrepancy and endpoint convention	155
18.2 Bounded polarization and exact temporal balance	155
18.3 Realization on the charge and time lines	156
18.4 Pairing the additive and multiplicative fibres	157
18.5 Commutator norm and electronic prefactor	158
18.6 Quadratic defect and response of the 90/120 sectors	159
19 Sheet return, areal–volumetric incidence and constitutive orientation	160
19.1 Projected return and restoration of the oriented cover	160
19.2 Descent of incidence from the tritic calendar	160
19.3 Boundary marking and areal–volumetric ratio	161
19.4 Transport of orientation to the constitutive channels	161
20 Constitutive operator and reconstruction of the vacuum channels	162
20.1 Positive response on two sheets	162
20.2 Four coordinates of a common response	162
20.3 Cubic inversion and angular recovery	163
20.4 Transported composition of constitutive responses	164
20.5 Logarithmic coordinates and refinement	166
20.6 Dimensional types of the four evaluations	166
20.7 Reconstruction of the action shape from the constitutive response	166
20.8 Constitutive speed and kinematic realization of the clock	168
21 Joint reconstruction of gauge couplings, Higgs and the constitutive response	170
21.1 Angular provenance and route signatures	170
21.2 Rationalized tree-level realization	171
21.3 Joint inverse and exact domain	171
21.4 Higgs–impedance reconstruction	172
21.5 Retained information and transport of orientation	173
22 Cyclic decomposition, quarter-turn, and oriented Catalan moment	174
22.1 Two subspaces of the dodecaphase cycle	174
22.2 Faithful representation of the quarter-turn	174
22.3 Resolvent and unlimited depth	175
22.4 Composition with the constitutive response	175
22.5 Return traces and moments of the constitutive quotient	176
22.6 Residual composition and products of moments	178
22.7 Functional reconstruction of the oriented moment	179
22.8 Compatibility of reconstruction with channels and speed	180
23 Evaluation at the Pell scale and the constitutive Barbero–Immirzi functional	181
23.1 Oriented moment and evaluation at the Pell unit	181
23.2 Composition of phase and weighted depth	182
23.3 Dodecaphasic incidence and return functional	183
23.4 Factorization through the constitutive operator	184
23.5 Reconstruction from vacuum coordinates	185
24 Constitutive normalization and spectral incidences of return	186
24.1 Exact separation of the common and oriented modes	186
24.2 Operational role of the Catalan moment	186
24.3 Three distinct readings of the 90/120 incidences	187
24.4 Fourier resolution of the dodecaphasic pattern	188
25 Differential reciprocity and recovery of constitutive channels	189
25.1 Spectral potential and constitutive reciprocity	189
25.2 Global inversion through speed and Barbero–Immirzi	190
25.3 Angular recovery, action, and periodic record	191
25.4 Stability bounds in the contractive domain	191

26	Charge section and electrical quantization relations	192
26.1	Positive determination of the charge section	192
26.2	Resistance, conductance, flux, and frequency	192
26.3	Action return and transport of the sections	193
27	Constitutive evaluation and dimensional realization	194
27.1	Evaluation of the internal chain	194
27.2	Unit charts and covariance	195
27.3	Identity checks and scope of comparison	195
28	Conclusions	196
A	Dependencies, procedures and reproduction	199
A.1	Order of reproduction	199
A.2	Three checks with different functions	199
B	Exhaustive enumeration and finite determination of transport operators	200
B.1	Complete domain of initial conditions	200
B.2	Census of signatures and emissions	201
B.3	Transition register and reconstruction of the lifts	202
B.4	Exhaustiveness of the calendar and stability of reconstruction	203
B.5	Reproduction certificate and location of the data	203
C	Finite transport realization with phase memory	204
C.1	Finite transfer on positions, modes and phase	204
C.2	Spectral decomposition and parametric sensitivity	205
C.3	Reciprocal readouts and sensitivity tests	206
	References	208

1 Introduction

A constitutive relation organizes a system's response, but its functional form and its coefficients raise different questions. Once the coefficients are known, the equation permits responses to be calculated; determining why those coefficients jointly have their values requires describing the structure that fixes them. This work addresses the second question through a mathematical construction of two oriented channels. Its thesis is that the preceding structure determines the equation and that evaluation preserves a recoverable part of that structure. The electromagnetic vacuum provides a precise realization of this thesis: permittivity, permeability, impedance and speed are obtained as four related readouts of a single response.

The principal result comprises both direct construction and inversion. On the contractive domain, the constitutive operator is positive, its eigenvalues lie in a specified interval and its four coordinates satisfy exact identities. Strict monotonicity of the response function provides a unique inverse; with the sheets labelled, that inverse recovers the channels and angular pair. This recovery composes three descriptions: the vacuum response, the shape of the action ellipse and the Barbero–Immirzi functional. The comparison therefore extends beyond numerical values: it identifies the operations transmitting orientation and spectral information between these descriptions.

1.1 Generating kernel and provenance of the response

The model is constructed from a formal mathematical core called Triadic Modular Holography, henceforth HMT. Its first object, *All Possible Paths* (APP), combines paths and arithmetic evaluations on the 81 positions of the support $X_9 = (\mathbb{Z}/9\mathbb{Z})^2$. Translations with generators $\{(1, 0), (-1, 0), (0, 1), (0, -1)\}$ define its additive Cayley graph, a two-dimensional cyclic lattice with toroidal realization. Sum and product evaluations act on this support; each preserves its residue and quotient. The path structure and the evaluations thus perform different functions within APP: displacements specify routes, while the arithmetic sheets determine the data transported along them.

The second object, TRIT, is the oriented ternary signature $\{-1, 0, +1\}$, obtained through the balanced representation of classes modulo 3. Its quadratic realization distinguishes the hyperbolic, parabolic and elliptic types, respectively. The third object is the *Topological Phase Kernel*, henceforth TPK: it composes phase selection, cardinal transport and memory update. The dynamics acts on positions, sheets, quotients, orientation, prefixes and boundary data. Phase return preserves the accumulated history, so a winding may close one coordinate without restoring the complete state.

Nonadic prolongation transmits this structure through compatible prefixes and boundaries. The coordinates π_{HMT} , φ_{HMT} , e_{HMT} and α_{HMT} are generated within its joint development. The dodecaphasic record K and its periodic evaluation enter the determination of α through dodecaphasic closure; the complete analytic representation expresses that same coordinate in the constructed tail class. The action sections and angular pair are subsequently obtained. Their use as antecedents preserves their proofs within the article. This sequence fixes the genealogy of the channels before the constitutive response is evaluated.

Sheet orientation and areal–volumetric incidence perform complementary roles. The former distinguishes route return from restoration of its orientation; the latter arises from the mode calendar and its incidence matrix. Chronological frequency, the measure of the symbolic system and the regional boundary mark retain their own definitions. These distinctions, developed in Section 19, specify the information reaching the angular transport and prevent its replacement by an isolated scalar ratio.

1.2 Constitutive determination and recovery of shape

With $T = q_+P_+ + q_-P_-$ and $0 < q_{\pm} < 1$, the temporal incidences 90 and 120 determine

$$\mathcal{V} = (I - T^{90})(I - T^{120})^{-1}.$$

Writing $S = T^{30}$, the equation $\mathcal{V}(I + S + S^2 + S^3) = I + S + S^2$ exhibits the rule fixing the response for this transport. Section 20 proves existence and uniqueness and establishes the scalar function $f : (0, 1) \rightarrow (3/4, 1)$. Its values $r_{\pm} = f(q_{\pm}^{30})$ produce

$$\hat{\varepsilon} = r_+^2, \quad \hat{\mu} = r_-^2, \quad \hat{Z} = r_-/r_+, \quad \hat{c} = (r_+r_-)^{-1}.$$

Positivity and the product and quotient readouts determine the two squares once the constitutive relations are fixed. The asserted uniqueness belongs to this explicit system of rules and its domain.

Cubic inversion recovers $q_{\pm}$ and then the coordinates $x = -\frac{1}{2} \log(q_+q_-)$, $y = \frac{1}{2} \log(q_-/q_+)$. With sheet labels preserved, impedance and speed suffice for this reconstruction. Proposition 20.7 subsequently obtains the anisotropy of the action ellipse and the relative inductive–capacitive impedance. Action, frequency and impedance references complete its dimensional realization. Proposition 20.8 proves $c_{\text{int}} = c_{\text{vac}}$ in the same chart and relates path length to elementary duration. Inversion thus reconstructs specified spectral antecedents, while route and memory remain in the enriched state from which they arise.

This inverse also induces a composition law between responses: multiplication of transports is expressed by the transported operation and admits an additive coordinate. Section 21 then composes the same channels with the angular gauge and Higgs signatures. The joint inverse, its exact domain, and impedance reconstruction retain the scale, refinement, and normalization conditions of that realization.

1.3 Cycle, constants and oriented functionals

The function f admits a realization on the twelve-phase cycle. Its subspaces of dimensions 3 and 4 produce the determinant quotient; their normalized trace difference gives $-1/12$. On an oriented plane of the same cycle, a quarter-turn J satisfies $J^2 = -I$ and its resolvent determines $s/(1 + s^2)$. Two logarithmic integrations, with their conditions at the origin, generate the moment $\mathcal{C}_2(s)$ and its endpoint value, Catalan’s constant. Theorem 23.2 relates evaluation at $2 - \sqrt{3}$ exactly to that endpoint value and $\log(2 + \sqrt{3})$. The algebraic Pell section and the angular arguments $q_{\pm}^{30}$ are distinct evaluations of the same collection.

The traces of the transport powers also determine a periodic table that reconstructs the logarithm of the constitutive quotient. Its moment series, Mellin representation, and remainder bounds are proved in Section 22.5. Diagonal and tensorial composition make explicit the depth retained by each operation; spectral return remains distinct from return of the state.

Theorem 22.8 completes the relation through integral reconstruction: the constitutive function determines the entire oriented moment, and the zero-limit condition at the origin fixes its normalization. The proof distinguishes recovery of that collection from inversion of its two channel evaluations. Both operations preserve the speed readout $\hat{c} = (\det \mathcal{V})^{-1}$ and its normalized trace, linking constitutive transport to the duration and length of the same state.

Composition with Barbero–Immirzi preserves additional information: the relative sign of the channels. Hexad–octad incidence and the calendar determine the terms of the angular functional; the inverse of f expresses it on $\mathcal{V}$. Theorem 23.4 proves this factorization. Channel interchange relative to a fixed orientation mark reverses the functional’s sign; simultaneous conjugation of operator and mark preserves it. The constitutive response thereby transmits the information required by its angular readout.

Section 25 assembles the spectral potential of constitutive response, global channel recovery through normalized speed and Barbero–Immirzi, and stability bounds. On the generated image, retaining labels and bases, the composition recovers the angular pair, action section, and periodic reading of the dodecaphasic record.

Section 24 makes the normalizations of this composition explicit. Positive factorization separates the common response factor from its oriented unimodular component; together these data restore all four constitutive coordinates. The Fourier representation then distinguishes the modes of the 90/120 incidences, retaining the chain’s signed weights and a declared transform normalization. This organization compares antecedent realizations with the operator used in the article, preserving each realization’s function and domain.

Appendix C also develops the finite realization on positions, modes, and phase: it proves equivariance and decomposition into twenty-seven components, retains the phases of their eigenvalues, and evaluates

parametric sensitivity. The antecedent angular readouts are reproduced with their declared operations and data. Comparison with the constitutive readouts in the main text distinguishes the weights, normalizations, and domains of each realization.

1.4 Polarization, charge and dimensional realization

The electrical development also relies on two results of the arithmetic sheets. The mismatch between nonadic and decimal capacities produces a uniformly bounded accumulated discrepancy; its temporal difference defines the polarization increment. In addition, the pairing of additive and multiplicative fibres in D_3^3 determines the tritic mode and the commutator norm $\sqrt{3}/4$ of their projections. Section 18 preserves the proofs and domains of both constructions. Their relation to the limit $3/4$ of the temporal response specifies the datum they share without identifying their operator spaces.

Section 26 composes impedance, α and Planck's constant through the unique positive section satisfying $Z_{\text{vac}} e_a^2 = 2\alpha h_a$. It yields the von Klitzing resistance, conductance quantum, magnetic flux and Josephson constant, with exact cross-identities. Action return leaves the first two invariant and transforms the latter two with opposite powers of the action ratio. This electrical content follows from the preceding sections rather than constituting a list of independent constants.

The final evaluation retains the action, charge and speed lines, as well as the transformations of their coordinates under changes of units. The table in Section 27 presents the internal coordinates obtained by evaluation. Physical comparison is subsequently formulated with the dimensional chart and coupling regime stated. The argument is organized by one common question: which structure determines the response, which equations it imposes and which information can be recovered from the values thus obtained.

2 Formal core of Triadic Modular Holography

Triadic Modular Holography, abbreviated HMT, begins with a finite structure of positions and operations. The order of construction matters: positions receive arithmetic evaluations, transitions acquire a ternary signature, and the dynamics preserves the information required to prolong them. Only subsequently are numerical realizations formed. A constant recognized at the end of this process does not replace either the rule that produced its region or the history that determines its evaluation.

The designation *All Possible Paths*, APP, expresses the role of paths on the support. It is adopted as the name of the construction, with a conceptual reference to the path formulation of quantum mechanics; it does not imply that its arithmetic law is the Feynman path integral. The *Topological Phase Kernel*, TPK,¹ denotes the composition of selection, transport, memory updating, and prolongation. The acronyms are retained only after the objects they represent have been defined.

2.1 APP: positional support and arithmetic evaluations

Consider the nine interior marks of a line graduated between zero and ten,

$$D_9 = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

Their horizontal and vertical arrangement produces the eighty-one positions of D_9^2 . The interior marks must not be confused with the ten unit intervals of the reference line. This arrangement fixes an alphabet and two directions; it does not yet use a real expansion to determine a trajectory.

Cyclic identification of the positions yields the support

$$X_9 = (\mathbb{Z}/9\mathbb{Z})^2.$$

The chart of positive representatives retains the symbol nine for the zero class. For every positive integer, define

$$\rho_9(n) = 1 + ((n - 1) \bmod 9), \quad q_9(n) = \frac{n - \rho_9(n)}{9}, \quad n = \rho_9(n) + 9q_9(n). \quad (2.1)$$

The positive digital root is therefore a choice of representative of the residue class. By itself, it does not retain the integer prior to reduction; the coordinate q_9 supplies precisely the missing information.

At each position (i, j) , accumulating i a total of j times produces ij . The two integer evaluations and their projections are

$$S(i, j) = i + j, \quad P(i, j) = ij, \quad \Sigma(i, j) = \rho_9(i + j), \quad \Pi(i, j) = \rho_9(ij).$$

The symmetry $(i, j) \mapsto (j, i)$ interchanges the directions of accumulation and preserves both evaluations. This is a property of a single support, not a subsequent identification between two independent matrices.

Definition 2.1 (Lifted arithmetic evaluation). The APP evaluation that preserves the two sheets is

$$\mathcal{A}(i, j) = ((R_{ij}^+, q_{ij}^+), (R_{ij}^\times, q_{ij}^\times)) = ((\rho_9(i + j), q_9(i + j)), (\rho_9(ij), q_9(ij))). \quad (2.2)$$

The term *sheet* here denotes an evaluation on the same support: additive or multiplicative. The two quotient coordinates belong to the arithmetic state and are not discarded when the residues are reduced.

Proposition 2.2 (Reconstruction of the evaluations). *The two pairs in (2.2) determine $i + j$ and ij exactly. The residues alone determine neither these integer evaluations nor their quotients.*

Proof. The first assertion is (2.1) applied to each sheet. For the second, $10 = 1 + 9$ and $19 = 1 + 2 \cdot 9$ have the same residue and different quotients. On the two-dimensional support, the positions $(5, 5)$ and $(8, 2)$ give the same sum and the same product residue, but products $25 = 7 + 18$ and $16 = 7 + 9$. The residual projection identifies data that the lifted evaluation distinguishes. $\square$

¹The authors also dedicate the initials TPK to Tan Pengkiang. This dedication is independent of the mathematical definition.

	1	2	3	4	5	6	7	8	9	
1	1	2	3	4	5	6	7	8	9	One support: 81 positions.
2	2	4	6	8	1	3	5	7	9	Two evaluations: $S(i, j) = i + j, P(i, j) = ij.$
3	3	6	9	3	6	9	3	6	9	Each row is generated by accumulation: $P(i, j + 1) - P(i, j) = i.$
4	4	8	3	7	2	6	1	5	9	
5	5	1	6	2	7	3	8	4	9	
6	6	3	9	6	3	9	6	3	9	The chart represents 0 (mod 9) by 9.
7	7	5	3	1	8	6	4	2	9	The marked block is
8	8	7	6	5	4	3	2	1	9	$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix},$
9	9	9	9	9	9	9	9	9	9	with eigenvalues 9 and 5.

Figure 1: APP multiplication table in positive representatives. Nonunit rows and columns are distinguished by grey shading; the frame identifies the adjacent diagonal block with equal diagonals. The displayed table is an observable of the arithmetic torus; its representational edges do not constitute an intrinsic topological boundary of the torus.

2.1.1 Cayley graph, toroidal topology, and winding number

The cardinal translations are given, in row and column coordinates, by

$$\nu_N = (-1, 0), \quad \nu_E = (0, 1), \quad \nu_S = (1, 0), \quad \nu_O = (0, -1), \quad T_d(x) = x + \nu_d.$$

The first coordinate increases southward and the second eastward. The graph

$$\mathcal{G}_9 = \text{Cay}(X_9, \{\nu_N, \nu_E, \nu_S, \nu_O\})$$

is the two-dimensional cyclic lattice: it has eighty-one vertices and four outgoing oriented edges per vertex. It is a Cayley graph for the *additive* structure of the support. Multiplication of residues has zero divisors and does not turn all APP rows into a multiplicative group.

A cardinal word $w = d_1 \cdots d_r$ and an origin x_0 determine

$$x_k = x_0 + \sum_{j=1}^k \nu_{d_j} \pmod{9}.$$

There are $81 \cdot 4^r$ oriented paths of length r , including returns and backtracking. When the path closes, its integer displacement prior to reduction defines

$$\text{wind}(w) = \frac{1}{9}(\#S(w) - \#N(w), \#E(w) - \#O(w)) \in \mathbb{Z}^2. \quad (2.3)$$

Path reversal changes the sign of this vector. Return to the same residual position therefore does not require the winding to vanish. This is the first instance of a distinction that will recur in phase return: one coordinate may close while another records a nontrivial evolution.

2.1.2 Transport quotient and cocycle identity

Let $s_9 : \mathbb{Z}/9\mathbb{Z} \rightarrow D_9$ be the section of positive representatives. The transport quotient associated with additive composition, also called the *carry* in positional arithmetic, is

$$c_9(a, b) = \frac{s_9(a) + s_9(b) - s_9(a + b)}{9}.$$

The compositional character of this memory is expressed by an exact identity.

Lemma 2.3 (Cocycle identity). *For $a, b, c \in \mathbb{Z}/9\mathbb{Z}$,*

$$c_9(a, b) + c_9(a + b, c) = c_9(b, c) + c_9(a, b + c).$$

Proof. Both sums equal

$$\frac{s_9(a) + s_9(b) + s_9(c) - s_9(a + b + c)}{9}.$$

Associativity of the sum of lifted representatives requires precisely this additional term when composing in the quotient. $\square$

With positive representatives, this cocycle is not normalized at the identity: $c_9(0, a) = 1$. If s_0 takes representatives in $\{0, \dots, 8\}$, its normalized carry c_0 satisfies

$$c_9(a, b) = c_0(a, b) + \mathbf{1}_{a=0} + \mathbf{1}_{b=0} - \mathbf{1}_{a+b=0}.$$

The difference is the change of section; it does not alter the cocycle identity. The subsequent decomposition of the calendar into phase and window will use representatives from zero to eight.

The totals of the two evaluations distinguish the visible contribution from that retained in the quotient:

$$\begin{aligned} \sum S = 810, \quad \sum P = 2025, \quad \sum \Sigma = 405, \quad \sum \Pi = 459, \\ \sum q^+ = 45, \quad \sum q^\times = 174. \end{aligned}$$

It follows that

$$2025 - 810 = (459 - 405) + 9(174 - 45) = 54 + 9 \cdot 129. \quad (2.4)$$

The sum-product difference is not exhausted by the visible defect 54. The identity also preserves the quotient defect 129. Suppressing it would change the arithmetic being transported, even if the residual table remained identical.

2.1.3 Ternary reduction and multiplicative radical

The ring $R = \mathbb{Z}/9\mathbb{Z}$ has unit group $\{1, 2, 4, 5, 7, 8\}$ and maximal ideal

$$\mathfrak{m} = (3) = \{0, 3, 6\}, \quad \mathfrak{m}^2 = 0.$$

Every additive row is a permutation of the nine representatives. A multiplicative row is such a permutation exactly when its index is a unit. Rows three and six contain three repeated values; row nine contains only the representative of the zero class. Thus multiplicative folding already distinguishes the units from the nilpotent sector before any continuous geometry is introduced.

The map $x \mapsto 3x$ has kernel and image both equal to $\mathfrak{m}$. It induces an isomorphism of vector spaces over $\mathbb{F}_3$,

$$R/\mathfrak{m} \longrightarrow \mathfrak{m}, \quad [x] \longmapsto 3x,$$

but not a ring isomorphism. In particular, the visible sequence 3, 6, 9 admits two different readouts: its direct reduction modulo three is 0, 0, 0, whereas extraction of the factor three followed by reduction of its internal coordinate gives 1, 2, 0. This second operation preserves a coordinate of the ideal; it is not the same projection as direct reduction of the original number.

2.1.4 Latin realization of the additive evaluation

The distinction between the two evaluations can also be expressed through operators. Let $(e_r)_{r \in \mathbb{Z}/9\mathbb{Z}}$ be the orthonormal basis of $\mathbb{C}^9$. The assignment $v_{ab} = e_{a+b}$ produces an orthonormal basis in each row and column. The projectors $P_{ab} = v_{ab}v_{ab}^*$ satisfy

$$\sum_b P_{ab} = I_9 = \sum_a P_{ab}, \quad P_{ab}^2 = P_{ab}.$$

Indeed, each translation $b \mapsto a + b$ permutes the nine residues. This yields a rank-one-projector realization of a Latin square, in the terminology of quantum Latin squares [4, 6]. This construction starts from the additive sheet already defined and makes explicit the map to its operator representation.

The multiplicative evaluation preserves different information: for $a = 3$, the vectors e_{3b} repeat the residues 0, 3, 6, and

$$\sum_{b \in \mathbb{Z}/9\mathbb{Z}} |e_{3b}\rangle\langle e_{3b}| = 3(|e_0\rangle\langle e_0| + |e_3\rangle\langle e_3| + |e_6\rangle\langle e_6|).$$

The multiplicity is recorded in the operator and characterizes the multiplicative radical. Accordingly, the toroidal support, the Latin property, and the operator resolution of the identity are properties requiring different definitions. The quantum magic squares studied by De las Cuevas, Drescher, and Netzer require positive entries and row and column sums equal to the identity; their theory also distinguishes dilations with commuting entries[5]. This reference provides precise terminology for comparing APP realizations.

2.2 TRIT: orientation and quadratic regimes

Definition 2.4 (Ternary signature and local lift). The TRIT signature uses the balanced identification

$$\mathbb{F}_3 \longrightarrow \{-1, 0, +1\}, \quad 0 \mapsto 0, \quad 1 \mapsto +1, \quad 2 \mapsto -1.$$

For $n \in \mathbb{Z}$, its lift preserves $n = r + 3c$, where $r \in \{-1, 0, +1\}$ and $c \in \mathbb{Z}$. In a transition of bounded local amplitude, the three states $(0, -1)$, $(0, 0)$, $(0, +1)$ may exhibit the same zero residue and different carries.

The signature does not replace the trajectory. A residual zero does not mean the absence of information about orientation, phase, or quotient. Nor does it by itself introduce a physical quantity. Its function is to distinguish algebraically the transport classes to be realized next.

2.2.1 Three quadratic regimes and a common exponential

Once the ternary signature has been obtained, its quadratic realization is defined by

$$\mathbb{A}_\tau = \mathbb{R}[X]/(X^2 + \tau), \quad \tau \in \{-1, 0, +1\}. \quad (2.5)$$

If J_τ is the class of X , then $J_\tau^2 = -\tau I$. The structure distinguishes the elliptic type $\tau = +1$, the parabolic type $\tau = 0$, and the hyperbolic type $\tau = -1$. The real realization does not generate the signature: it expresses the types already produced by oriented reduction.

Proposition 2.5 (Exponentiation of the three types). *In each algebra (2.5),*

$$\exp(tJ_\tau) = C_\tau(t)I + S_\tau(t)J_\tau, \quad (2.6)$$

where

$$C_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n}}{(2n)!}, \quad S_\tau(t) = \sum_{n \geq 0} \frac{(-\tau)^n t^{2n+1}}{(2n+1)!}.$$

The three realizations are

τ	J_τ^2	$\exp(tJ_\tau)$
+1	$-I$	$\cos t I + \sin t J_\tau$
0	0	$I + t J_\tau$
-1	I	$\cosh t I + \sinh t J_\tau$

Proof. Separating the even and odd terms of the exponential series and substituting $J_\tau^{2n} = (-\tau)^n I$ and $J_\tau^{2n+1} = (-\tau)^n J_\tau$ gives the formulas. For $\tau = 0$, every term of degree at least two vanishes. In the other two cases, the trigonometric or hyperbolic series are recovered. $\square$

The exponential is common to the three regimes: it is not assigned exclusively to the parabolic branch. The invariants

$$C_\tau(t)^2 + \tau S_\tau(t)^2 = 1$$

follow from $\exp(tJ_\tau)\exp(-tJ_\tau) = I$. The same algebraic principle preserves unity across realizations that differ in their quadratic type.

In the hyperbolic regime, $P_\pm = (I \pm J_{-1})/2$ are complementary projectors, and

$$\exp(tJ_{-1}) = e^t P_+ + e^{-t} P_-.$$

Growth and contraction occur jointly in two eigendirections. In the elliptic regime, the parameter describes a rotation; in the parabolic regime, a nilpotent translation. The temporal interpretation adopted by HMT associates +1 with return with memory, 0 with the threshold, and -1 with bidirectional opening. This interpretation requires an ordering of the parameterization; it does not add a fourth algebraic type.

2.2.2 Balanced division and composition of lifts

Ternary reduction makes it possible to express an integer APP evaluation in a visible coordinate and a prolongation coordinate. Its function is not merely to abbreviate the writing of an integer: it provides a composition rule determining how the quotient changes when evaluations are added or multiplied. The balanced choice also makes explicit the compatibility of that rule with sign reversal.

Definition 2.6 (Balanced ternary coordinates). For $n \in \mathbb{Z}$, let

$$q_3(n) = \left\lfloor \frac{n+1}{3} \right\rfloor, \quad r_3(n) = n - 3q_3(n).$$

The map

$$\mathcal{B} : \mathbb{Z} \longrightarrow \{-1, 0, +1\} \times \mathbb{Z}, \quad n \longmapsto (r_3(n), q_3(n))$$

is called the balanced ternary decomposition. Its inverse is $\mathcal{L}(r, c) = r + 3c$.

Indeed, $r_3(n) \in \{-1, 0, +1\}$. If two pairs represent the same integer, $r + 3c = s + 3d$, then $r - s$ is a multiple of 3 lying in the interval $[-2, 2]$; necessarily $r = s$ and $c = d$. The representation is therefore unique. Division can be repeated on $q_3(n)$, so that each level preserves the datum needed to reconstruct the preceding one. The quotient is not a perturbation of the residue: it constitutes its complementary coordinate.

The signature obtained from an evaluation F , with $F = S$ or $F = P$ in APP, is $r_3 \circ F$. When reduction acts on the phase cursor, the same balanced section produces the classes $\{1, 4, 7\}$, $\{2, 5, 8\}$, and $\{3, 6, 9\}$, with values +1, -1, 0. The TPK selector defined next uses these classes to determine the operational mode. Reading an evaluation and selecting a mode thus have the same ternary reduction, but different domains and functions.

Proposition 2.7 (Arithmetic of balanced pairs). Let $n = r + 3c$ and $m = s + 3d$, with $r, s \in \{-1, 0, +1\}$. Define

$$\chi(r, s) = \frac{r + s - r_3(r + s)}{3}.$$

Integer operations are transported to pairs by

$$(r, c) \boxplus (s, d) = (r_3(r + s), c + d + \chi(r, s)), \quad (2.7)$$

$$(r, c) \boxtimes (s, d) = (rs, rd + sc + 3cd). \quad (2.8)$$

The map $\mathcal{L}$ preserves both operations.

Proof. The equality $r + s = r_3(r + s) + 3\chi(r, s)$ gives

$$n + m = r_3(r + s) + 3(c + d + \chi(r, s)).$$

On the other hand,

$$nm = rs + 3(rd + sc + 3cd).$$

Since the product of two balanced representatives again belongs to $\{-1, 0, +1\}$, it requires no additional residue reduction. Uniqueness of the decomposition proves both formulas. $\square$

For example, $2 = (-1) + 3$ and $4 = 1 + 3$. Their sum and product are reconstructed as

$$(-1, 1) \boxplus (1, 1) = (0, 2), \quad (-1, 1) \boxtimes (1, 1) = (-1, 3).$$

The resulting evaluations are 6 and 8. Adding quotients suffices in this additive example; the product involves the cross terms rd , sc , and $3cd$. The discrepancy between the two operations also resides in the prolongation coordinate, not only in the residue.

The term χ is a cocycle of the balanced section. If $r \oplus s = r_3(r + s)$, it satisfies

$$\chi(r, s) + \chi(r \oplus s, t) = \chi(s, t) + \chi(r, s \oplus t).$$

Both sides represent the quotient of the same sum of three integers. This identity guarantees that regrouping the operations does not change the reconstructed evaluation. At the same time, projection onto the first component removes that accounting: $1 + 1 = -1 + 3$, whereas the residue readout retains only -1 . The discarded information is determined, in this case, by $\chi(1, 1) = 1$.

The relation to modulus 9 preserves a second discrete scale. The map

$$\beta : \{-1, 0, +1\} \times \mathbb{F}_3 \longrightarrow \mathbb{Z}/9\mathbb{Z}, \quad (r, \bar{c}) \longmapsto r + 3c \pmod{9}$$

is well-defined and bijective. Changing the representative of $\bar{c}$ adds a multiple of 9; conversely, equality of images first fixes r by reduction modulo 3, and then $\bar{c}$. The addition transported by this bijection contains the term $\chi(r, s)$ from (2.7), now reduced modulo 3. Thus the two ternary levels are not composed by independent coordinate addition: normalizing the first level modifies the second.

In particular, the classes 3, 6, and 0 in $\mathbb{Z}/9\mathbb{Z}$ have first component 0, but second components 1, 2, and 0, respectively. The distinction already observed between direct reduction and extraction of the ideal coordinate appears as a property of β . Preserving that coordinate prevents the zero-residue sector from being artificially turned into a single position. The modulus 1000, used later to organize decimal blocks, serves another positional-representation function; its quotient is preserved through the corresponding division. A change of base preserves the integer when it retains residue and quotient, whereas agreement of residues in different bases does not by itself identify their transport states.

2.2.3 Reversal, residue class, and nonvisible coordinates

Additive inversion lifts unambiguously:

$$\mathcal{B}(-n) = (-r_3(n), -q_3(n)).$$

In particular, 3, 0, and -3 have coordinates $(0, 1)$, $(0, 0)$, and $(0, -1)$. They share a zero signature but represent different evaluations. In a transition restricted to a quotient of maximum magnitude 1, they are the three possibilities in the visible fiber over zero; in a general integer lift, that fiber contains all pairs $(0, c)$, with $c \in \mathbb{Z}$.

The quotient coordinate recovers the evaluated integer, but does not by itself recover the trajectory that produced it. Two APP paths may share evaluation, residue, and quotient while differing in orientation, mode sequence, or winding. Thus the decomposition $\mathcal{B}$ is integrated into the transport state instead of replacing it. The arithmetic memory of a division and the memory of a trajectory are related coordinates, not synonyms.

For a signature sequence of length n , oriented reversal is

$$C_{\text{rev}}(\tau_1, \dots, \tau_n) = (-\tau_n, \dots, -\tau_1).$$

Applying it twice restores the sequence. The operation reverses the order of positions and the sign of each signature; it preserves the length and the zero positions in reverse order. In the adopted parametrization, it interchanges return with memory, associated with $+1$, and bidirectional opening, associated with -1 , while maintaining the threshold 0. Transport of the remaining trajectory data is specified through the corresponding TPK involution.

This signature reversal must be distinguished from conjugation within a quadratic algebra. The former may change the type τ ; the latter, constructed next, acts within the type already fixed. The distinction prevents attributing to sign reversal an algebraic equivalence between different regimes.

2.2.4 Quadratic product, conjugation, and algebraic norm

An element of $\mathbb{A}_\tau$ has a unique expression $aI + bJ_\tau$. The relation $J_\tau^2 = -\tau I$ determines its product:

$$(aI + bJ_\tau)(cI + dJ_\tau) = (ac - \tau bd)I + (ad + bc)J_\tau. \quad (2.9)$$

This is a real realization of the type already selected. The coordinates a, b are not APP residues, and the product (2.9) does not replace (2.8); it expresses another operation in the declared quadratic codomain.

Definition 2.8 (Conjugation and algebraic norm). Conjugation of type τ and its algebraic norm are

$$\overline{aI + bJ_\tau} = aI - bJ_\tau, \quad N_\tau(aI + bJ_\tau) = a^2 + \tau b^2.$$

Proposition 2.9 (Multiplicativity and invertibility). For $z, w \in \mathbb{A}_\tau$,

$$z\bar{z} = N_\tau(z)I, \quad N_\tau(zw) = N_\tau(z)N_\tau(w).$$

Moreover, z is invertible if and only if $N_\tau(z) \neq 0$, in which case

$$z^{-1} = \frac{\bar{z}}{N_\tau(z)}.$$

Proof. The product of $aI + bJ_\tau$ and $aI - bJ_\tau$ is $(a^2 + \tau b^2)I$. Conjugation is multiplicative and the algebra is commutative, so $(zw)\bar{z}\bar{w} = (z\bar{z})(w\bar{w})$. If the norm is nonzero, the stated formula provides the inverse. Conversely, if $zw = I$, multiplicativity implies $N_\tau(z)N_\tau(w) = 1$. $\square$

The quadratic form N_τ is positive definite only in the elliptic type. For $\tau = 0$, it equals a^2 , and the element J_0 is nonzero although its square and norm are zero. For $\tau = -1$, it equals $a^2 - b^2$; the nonzero elements $I + J_{-1}$ and $I - J_{-1}$ have zero norm and their product vanishes. The algebraic norm here is a multiplicative quadratic form: it does not presuppose a positive vector-space norm.

All three types can be faithfully realized by

$$aI + bJ_\tau \mapsto \begin{pmatrix} a & -\tau b \\ b & a \end{pmatrix}.$$

The determinant of this matrix is $N_\tau(aI + bJ_\tau)$. Thus the complex structure, the dual-number structure, and the split-real structure are represented by the same matrix expression. Each preserves its law and zero divisors. The form of signature $(1, 1)$ in the last case has a Lorentzian interpretation in dimension $1 + 1$; identifying a physical metric additionally requires a specific space, field, and transport rule.

2.2.5 Multiplicative conservation and composition of evolutions

The exponential already obtained belongs to the set of norm-one elements:

$$N_\tau(\exp(tJ_\tau)) = 1.$$

This conservation persists under composition of parameters. Indeed, all factors are functions of the same generator, and

$$\exp(tJ_\tau)\exp(sJ_\tau) = \exp((t+s)J_\tau).$$

Comparing the two components yields the addition laws

$$C_\tau(t+s) = C_\tau(t)C_\tau(s) - \tau S_\tau(t)S_\tau(s), \quad (2.10)$$

$$S_\tau(t+s) = S_\tau(t)C_\tau(s) + C_\tau(t)S_\tau(s). \quad (2.11)$$

Conjugation changes t to $-t$ and coincides with inversion within this subgroup. It does not change τ : it reverses an evolution in the chosen regime.

In the elliptic type, the preserved form is $a^2 + b^2$, and the exponential trajectory traverses the norm-one circle. In the hyperbolic type, the eigen-coordinates are $u = a + b$, $v = a - b$, with $uv = N_{-1}(z)$. The exponential gives

$$(u, v) = (e^t, e^{-t}), \quad uv = 1.$$

Growth of one coordinate is accompanied by contraction of the other. Conservation expresses a relation between the two coordinates; it does not assert that each remains constant.

In the parabolic type,

$$(I + tJ_0)(I + sJ_0) = I + (t + s)J_0, \quad (I + tJ_0)^{-1} = I - tJ_0.$$

Evolution does not stop because $J_0^2 = 0$. Nilpotence eliminates terms of order at least two, but preserves the linear term transporting the parameter. The visible symbol 0, the zero element of the algebra, and the nonzero generator J_0 are distinct. This distinction is necessary to describe the threshold without identifying it with the absence of a state or dynamics.

Proposition 2.10 (Composition of affine coordinates). *In the chart where the scalar component is nonzero, represent a direction by $I + uJ_\tau$. If $1 - \tau uv \neq 0$, the direction of its product with $I + vJ_\tau$ has coordinate*

$$u \star_\tau v = \frac{u + v}{1 - \tau uv}.$$

Proof. We have

$$(I + uJ_\tau)(I + vJ_\tau) = (1 - \tau uv)I + (u + v)J_\tau.$$

Dividing by the scalar component gives the stated expression. $\square$

The affine chart preserves the ratio between components, not the scalar factor divided out. When $1 - \tau uv = 0$, the product must be examined before changing charts: it may have a defined direction with zero scalar component, or be the zero element in the presence of zero divisors. The composition law still belongs to the complete algebra; a coordinate singularity does not automatically amount to a singularity of the object. This formula will reappear in the elliptic regional readout. Its use requires retaining the denominator and, where applicable, the normalization factor.

2.2.6 Transport of type and subsequent realizations

The algebras $\mathbb{A}_{+1}$, $\mathbb{A}_0$, and $\mathbb{A}_{-1}$ are not isomorphic as unital real algebras. The first is a field; the second contains a nonzero nilpotent; the third has nontrivial idempotents and is isomorphic to $\mathbb{R} \times \mathbb{R}$. Thus the signature change organizing a route cannot be interpreted as an undifferentiated isomorphism between the three realizations. The TPK preserves the regime index, domains, and effective operator acting at each transition.

In the electron chapter, the pair of projections of the APP sheets produces a normalized commutator whose square is $-I$. The same block contains an involution squaring to I and two nilpotent directions. The joint realization will be proved on that concrete pair of projections; the preceding classification explains why the three quadratic relations can appear in the same matrix algebra without being identified with one another.

In the exceptional prolongation, the Paley matrix reduced modulo 3 satisfies $A_W^2 = -I_6$. The relation equips the ternary space with a structure of dimension 3 over $\mathbb{F}_9$. Its link to a real operator root is expressed through the integral presentation

$$\mathbb{Z}[u]/(u^2 + 1).$$

Base changes to $\mathbb{F}_3$ and $\mathbb{R}$ produce representations of the same relation. Each preserves its characteristic, dimension, and incidence information. The development of the respective operators establishes the link, not equality of their cardinalities.

Preservation of norm one therefore describes the quadratic realizations of evolutions. Preservation of quotients, orientation, and trajectory belongs to the transported arithmetic state. Both properties enter the construction, with different functions: one determines the identities of the representations; the other makes it possible to compose, reconstruct, and prolong the operations giving rise to them.

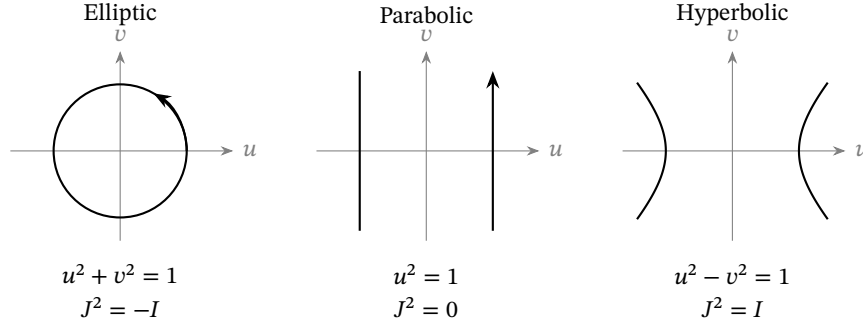


Figure 2: Quadratic realizations of TRIT. For $J_\tau = \begin{pmatrix} 0 & -\tau \\ 1 & 0 \end{pmatrix}$, the exponential preserves $u^2 + \tau v^2$. The parabolic degeneration preserves u and transports v linearly. The three diagrams represent orbits of quadratic forms; their geometric type corresponds to the algebraic relation of the generator.

2.3 TPK: topological phase kernel

The dynamics uses the phase selector

$$M_{\text{ph}}(r) = \begin{cases} +1, & r \in \{1, 4, 7\}, \\ -1, & r \in \{2, 5, 8\}, \\ 0, & r \in \{3, 6, 9\}. \end{cases}$$

Its value determines which evaluation acts. Cardinal transport displaces the corresponding cursor; the record preserves the previous value, the new quotient, the orientation, and the transition. Selection and transport are therefore distinct functions: a ternary sign is not a displacement on the torus.

We denote by $\hat{X}$ the space of arithmetic states with trajectory memory. An element contains the positions and directions of the two cursors, the active mode, the TRIT signature, the phase, the residues and quotients of both evaluations, the carries, the constructed prefix, and the boundary data. When a refinement cylinder is introduced, it forms part of the state. This notation avoids replacing the object with a reduced version that retains only the visible symbol.

Definition 2.11 (TPK update). Let Sel_t be mode selection by M_{ph} , Tra_t the lifted cardinal transport, and Mem_t the record update. The transition is

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (2.12)$$

Each application preserves the fields it does not modify and updates the remaining fields through Euclidean division and trajectory concatenation.

The notation (2.12) expresses the order of composition; it does not replace the effective rules. The next section fully specifies its finite two-cursor realization. Prolongation will then be obtained not by erasing its memory and repeating the first emission, but by transporting the prefixes and their compatible boundaries. Thus APP supplies the evaluations, TRIT distinguishes their regimes, and TPK composes their evolution.

2.3.1 Dynamic state, observables, and dependence of the operations

The Topological Phase Kernel organizes the evolution of the two APP sheets with the orientation determined by TRIT. Its definition comprises the state on which it acts, the transition rules, and the relations that make prolongations compatible. Equation (2.12) acquires operational content by specifying which coordinates each of its factors receives and modifies. Position belongs to the toroidal support; phase belongs to the calendar; orientation determines transport; quotients record the arithmetic evaluation before its reduction. These coordinates are related through the transition and preserve different mathematical functions.

The observable projection of a state is

$$q = (x_+, d_+, x_-, d_-, \theta) \in \mathcal{Q}_{\text{obs}} := (X_9 \times \mathcal{D})^2 \times \Theta_{108}, \quad \mathcal{D} = \{N, E, S, O\}, \quad \Theta_{108} = \mathbb{Z}/108\mathbb{Z}. \quad (2.13)$$

The subscripts distinguish the additive and multiplicative cursors. The fibers of complete states over q preserve the ordered trajectory, the integer evaluations of both sheets, their residues and quotients, and the orientation and boundary registers. A precision prolongation adds the prefix and the compatible cylinder. Thus two states with the same projection q may preserve different registers.

The dependence of the operations has three levels. At the local level, the selector activates a sheet, transport modifies its position, and the update calculates the new arithmetic coordinates. At the trajectory level, these transitions are composed and preserve the registers of earlier steps. At the refinement level, extension relations determine which continuation preserves the prefix, signature, and boundary of its predecessor. The finite calendar describes the phase of these operations; holonomy describes their composition on the history.

2.3.2 Trit selection and lifted cardinal transport

Let $\bar{\theta} \in \{0, \dots, 107\}$ be the calendar representative. The operational phase and dodecaphasic sector are

$$r(\theta) = 1 + (\bar{\theta} \bmod 9), \quad \beta(\theta) = \left[\left[\frac{\bar{\theta}}{9} \right] \right]_{12}. \quad (2.14)$$

The selector already defined can be written $\varepsilon_9 = M_{\text{ph}}$. Its value vector, in the ordered chart of D_9 , is

$$m_{\text{ph}} = (1, -1, 0, 1, -1, 0, 1, -1, 0).$$

The function $M_{\text{ph}} : D_9 \rightarrow \{-1, 0, +1\}$ and this vector are two presentations of the same selector, with their respective domains.

Define the activation indicators

$$a_+(\theta) = \mathbf{1}_{\{1,4,7\}}(r(\theta)), \quad a_\times(\theta) = \mathbf{1}_{\{2,5,8\}}(r(\theta)).$$

Cursor transport is then

$$x'_+ = x_+ + a_+(\theta)v_{d_+}, \quad x'_\times = x_\times + a_\times(\theta)v_{d_\times} \quad \text{in } X_9. \quad (2.15)$$

In phases 3, 6, 9, both positions remain unchanged; the neutral event is part of the chronology and its register. Cardinal directions determine the vectors v_d , while the phase signature determines which of those vectors is applied.

The integer lift preserves boundary crossing. If $s : X_9 \rightarrow D_9^2$ chooses positive representatives and $x' = x + av_d$, with $a \in \{0, 1\}$, define

$$b_d(x, a) = \frac{s(x) + av_d - s(x')}{9} \in \mathbb{Z}^2. \quad (2.16)$$

Divisibility by 9 follows from equality in X_9 . If $z \in \mathbb{Z}^2$ records earlier crossings, the update is $z' = z + b_d(x, a)$. Thus

$$s(x') + 9z' = s(x) + 9z + av_d.$$

The integer transport datum remains reconstructible after reducing the position on the torus.

Proposition 2.12 (Preservation of lifted displacement). *For a cardinal trajectory of n steps, whose activation indicators are a_t , we have*

$$s(x_n) + 9z_n = s(x_0) + 9z_0 + \sum_{t=0}^{n-1} a_t v_{d_t}.$$

The same equality is preserved when two compatible trajectories are concatenated.

Proof. The one-step identity is (2.16). Summing it over the n steps cancels the intermediate coordinates. Dividing the summation interval into two segments produces exactly the concatenation law. For a closed trajectory, the difference $z_n - z_0$ is the winding number defined earlier. $\square$

The same principle governs APP evaluations. If a sheet takes the integer value N_t , its register simultaneously preserves $N_t = R_t + 9q_t$. The difference between two steps satisfies

$$N_{t+1} - N_t = (R_{t+1} - R_t) + 9(q_{t+1} - q_t).$$

Spatial-crossing memory z_t and the arithmetic quotient q_t have different definitions; their joint preservation maintains the geometric and arithmetic provenance of each transition.

2.3.3 Two-cursor chronology and composite returns

The cardinal involution j interchanges N with S and E with O . After (2.15) is applied, both directions are reversed whenever a block of 27 steps ends. Writing

$$h(\theta) = \mathbf{1}_{\{0,27,54,81\}}((\bar{\theta} + 1) \bmod 108),$$

the complete rule on the observable projection is

$$F_{\text{obs}}(q) = (x'_+, j^{h(\theta)} d_+, x'_\times, j^{h(\theta)} d_\times, \theta + 1). \quad (2.17)$$

The readout forming the emission uses the positions before transport, according to the explicit recurrence in Section 4. The observable update and emission are therefore composed in a fixed order.

Proposition 2.13 (Reversibility and period of the observable calendar). *The map F_{obs} is a permutation of $\mathcal{Q}_{\text{obs}}$ and satisfies $F_{\text{obs}}^{108} = I$. In a block of 54 steps, positions and directions are restored; the coordinate in Θ_{108} advances by 54.*

Proof. Given the final state, one first recovers the previous phase by subtracting one in Θ_{108} . That phase determines whether the directions were reversed and which cursor moved. Applying the corresponding involution and subtracting the cardinal vector uniquely recovers the initial state.

Each interval of 27 steps contains nine activations of each cursor. The next block preserves the activations and reverses the directions, so the integer displacements cancel in 54 steps. The rule has period 54 in its position and orientation increments; the same cancellation holds for any calendar origin. After 108 steps, θ is also restored. $\square$

We reserve

$$\mathcal{K}_{27} = F_{\text{obs}}^{27} : \mathcal{Q}_{\text{obs}} \longrightarrow \mathcal{Q}_{\text{obs}}$$

for the iterate of 27 transitions. Its fourth iterate is the identity on the observable state defined in (2.13). Projecting history onto the calendar makes this finite check possible. The trajectory register, its evaluations, and new prefixes are preserved in the dynamic state on which $\mathcal{U}_t$ acts; the observable equality does not reset them.

2.3.4 Affine memory transport and state composition

The observable projection admits fibers of arithmetic and geometric data. Over a configuration q , write a state in the form

$$x = (q, o, h, c, \sigma, C, b, \lambda).$$

Here o preserves orientation and sheet; h , residues and quotients; c , additive transport memory; σ , the boundary signature; C , the precision cylinder; b , its boundary label; and λ , the recorded trajectory. The signature is determined by a map

$$\text{Sig}_q : H_q \times C_q \times \text{Hist}_q \longrightarrow S_q, \quad \sigma = \text{Sig}_q(h, c, \lambda).$$

The symbols H_q , C_q , Hist_q and S_q denote, respectively, the sets of residue–quotient data, the abelian group of transport memory, the recorded trajectories, and the boundary signatures. Realizations of this signature are specified with their components in the regional and terminal constructions. This dependence prevents treating the signature as a free parameter once the data producing it are fixed.

The elementary residue-and-quotient chart is explicit. For $n \in \mathbb{Z}$, let

$$r = 1 + ((n - 1) \bmod 9), \quad u = \frac{n - r}{9}.$$

Then $n = r + 9u$, with $r \in D_9$ and $u \in \mathbb{Z}$. Uniqueness follows because two representatives in D_9 congruent modulo 9 are equal. This chart provides local integer reconstruction. The precision towers additionally incorporate truncation maps and the cylinders developed in 4 and 5.

Let $\gamma : q \rightarrow q'$ be an observable trajectory. Fiber transport is expressed by maps

$$H(\gamma) : H_q \rightarrow H_{q'}, \quad C(\gamma) : C_q \rightarrow C_{q'}, \quad \text{Hist}(\gamma) : \text{Hist}_q \rightarrow \text{Hist}_{q'}.$$

The maps respect identities and composition, and $C(\gamma)$ is an abelian-group homomorphism. The increment $k_{\text{tr}}(\gamma) \in C_{q'}$ satisfies

$$\begin{aligned} k_{\text{tr}}(\text{id}_q) &= 0, \\ k_{\text{tr}}(\gamma_2 \circ \gamma_1) &= k_{\text{tr}}(\gamma_2) + C(\gamma_2)k_{\text{tr}}(\gamma_1). \end{aligned} \tag{2.18}$$

The symbol k_{tr} here denotes the transport cocycle; the rational coordinate κ of the register K retains its notation in the chapter on dodecaphasic closure.

The transportable coordinates are updated by

$$\begin{aligned} h' &= H(\gamma)h, & c' &= C(\gamma)c + k_{\text{tr}}(\gamma), \\ \lambda' &= \gamma \circ \lambda, & \sigma' &= \text{Sig}_{q'}(h', c', \lambda'). \end{aligned} \tag{2.19}$$

The second equation is an affine action: the linear term transports the previous memory, and the cocycle adds the increment produced by the trajectory. In the spatial-crossing realization for each cursor separately, $C(\gamma) = I$, and $k_{\text{tr}}(\gamma)$ is the sum of the vectors in (2.16). This concrete instance links the abstract law to the cardinal operations already defined.

Proposition 2.14 (Compatibility of the update with concatenation). *Updating (2.19) by γ_1 , followed by updating by γ_2 , coincides with updating by $\gamma_2 \circ \gamma_1$.*

Proof. For the affine coordinate, two updates give

$$c'' = C(\gamma_2)C(\gamma_1)c + C(\gamma_2)k_{\text{tr}}(\gamma_1) + k_{\text{tr}}(\gamma_2).$$

Functoriality of C and (2.18) turn this expression into $C(\gamma_2 \circ \gamma_1)c + k_{\text{tr}}(\gamma_2 \circ \gamma_1)$. The assertion for h follows from composition of H , and that for λ from associativity of trajectories. The final signature agrees because it is evaluated on the same three reconstructed coordinates. $\square$

The cylinder and its boundary complete this law through the relations Ext_γ , defined in 2.3.7. Their elements determine which of the preceding updates preserve an admissible prolongation. States, together with the triples (γ, x, x') admitted by these relations, form a category over observable trajectories:

$$(\gamma_2, x', x'') \circ (\gamma_1, x, x') = (\gamma_2 \circ \gamma_1, x, x'').$$

The identity of x is (id_q, x, x) . Composition preserves both arithmetic transport and boundary prolongability. This is the structure in which returns with memory and successive refinements take place.

2.3.5 Dodecaphasic register, phase, and accumulated memory

The 108 transitions are distributed over the ordered windows

$$E_m = \{9m - 8, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The register preserves the phase origin, orientation, and data produced during each window. Its map is

$$\mathcal{R}_{12}(x) = \left((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m) \right)_{m=1}^{12},$$

where τ_m is the window subroute, σ_m its orientation, κ_m the integer boundary increment, and Λ_m the local incidence register. We adopt the chart to be developed in 8.2; its window indices correspond to $m = \beta + 1$. The domain comprises compatible TPK histories, and the codomain their oriented temporal registers. The composition of segments and their increments is governed by (2.19). The group C_{12} , by

contrast, describes translation of the window index; Θ_{108} preserves the step calendar. This distinction makes it possible to specify what is transformed and what is observed when a cycle is completed.

In coordinates $\theta = 9b + r$, with $0 \leq r < 9$, advancement by 9 steps preserves r and increases b by one. Phase addition introduces the quotient

$$c_0(r, r') = \left\lfloor \frac{r + r'}{9} \right\rfloor,$$

whose cocycle identity ensures associativity of composition. The finite calendar retains $[b]_{12}$; a winding register retains $b \in \mathbb{Z}$, or $b \in \mathbb{N}_0$ for a history with fixed origin. Section 4 proves the nonsplit exact sequence (4.5). Its role is to describe the relation between these coordinates before they are used in constructing the signed state.

The signed register receives phase and memory data through the composition developed in Section 8:

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12}. \quad (2.20)$$

Its domain is the terminal state with memory; its output belongs to $\mathbb{Z}^{12}$. The reversible transformation between that signed state and K , as well as the rational readout of K , are proved in the corresponding chapter. The registration operations precede those transformations. Reconstructing a register from its cyclic differences verifies recoverability; its dynamic provenance must be preserved through the events that produced it.

For the encoded trace (d_t) preserved by the register, the event readout counts the codes $\{2, 4, 6, 8\}$ in each window with orientation $(+, +, +, -, -, -, +, +, +, -, -, -)$. These codes and the cardinal symbols of $\mathcal{D}$ are different coordinates of the register. Chapter 8 develops this readout, the integral decomposition $1 + 5 + 6$, the congruences characterizing its image, and its exact inverse. The inversion proof establishes which data that chart preserves; the definition of $\mathcal{R}_{12}$ specifies the oriented history on which it acts.

2.3.6 Phase incidence and internal construction of the channels

The selector determines three subsets of D_9 :

$$H_+ = \{1, 4, 7\}, \quad H_- = \{2, 5, 8\}, \quad H_0 = \{3, 6, 9\}.$$

Let $H_{\text{act}} = H_+ \sqcup H_-$ be the set of active phases, and let $O_{\text{ph}} = D_9 \setminus \{9\}$ be the chart preceding the marked return. Denote by $P_2(H_{\text{act}})$ the set of unordered pairs of active phases. Internal incidence is given by

$$\mathcal{F}_6^{\text{ph}} = H_{\text{act}} \times P_2(H_{\text{act}}), \quad \mathcal{F}_8^{\text{ph}} = O_{\text{ph}} \times P_2(H_{\text{act}}), \quad \Theta_{\text{ph}} = D_9 \times P_2(H_{\text{act}}). \quad (2.21)$$

The indices describe the number of positions in the first component. The construction uses the phases and calendar origin already fixed; subsequent exceptional realizations will transport these incidences to their own domains.

Proposition 2.15 (Cardinalities and oriented phase incidence). *The sets in (2.21) have cardinalities 90, 120, 135, respectively. If*

$$\partial_{\text{ph}} = \Theta_{\text{ph}} \times \{-1, +1\}, \quad E_{\text{ph}} = \partial_{\text{ph}} \sqcup \{\infty\}, \quad L_{\text{ph}} = H_0 \times P_2(H_{\text{act}}),$$

then

$$|\partial_{\text{ph}}| = 270, \quad |E_{\text{ph}}| = 271, \quad |L_{\text{ph}}| = 45, \quad |\partial_{\text{ph}} \sqcup L_{\text{ph}}| = 315.$$

Simultaneously translating the phase origin and marked return preserves all these cardinalities and their inclusions.

Proof. There are six active phases, so $|P_2(H_{\text{act}})| = \binom{6}{2} = 15$. Products by 6, 8, 9 give 90, 120, 135. Orientation doubles 135; the return mark adds one element; the three neutral phases give $3 \cdot 15 = 45$. A calendar translation transports each factor by a bijection and carries the marked point to its image. Products and disjoint unions thus preserve incidence. $\square$

The cardinality 271 here has a realization through phase incidence. The cubic completion in Section 4 has its own construction of the shell $1000 - 729$. The correspondence between both realizations must preserve the maps and markings that identify them, in addition to cardinality. Likewise, the selector ε_9 , the incidence $(90, 120)$, and the extractor $\varepsilon_{108}^{90,120}$ preserve the domains and operations just distinguished.

2.3.7 Trajectory reversal and conjugate prolongations

Bidirectionality begins with an operation on trajectories, before scalar values are assigned to them. For $\gamma = (x_0; d_1, \dots, d_n)$, with endpoint x_n , define

$$\text{rev } \gamma = (x_n; jd_n, \dots, jd_1).$$

The origin of the reverse trajectory is the endpoint of the original. Composition satisfies

$$\text{rev}^2 \gamma = \gamma, \quad \text{rev}(\gamma_2 \circ \gamma_1) = \text{rev } \gamma_1 \circ \text{rev } \gamma_2,$$

and the integer displacement changes sign. These identities are obtained by reversing the order of steps and replacing each cardinal vector by its opposite. For a loop it follows that $\text{wind}(\text{rev } \gamma) = -\text{wind}(\gamma)$.

On registers, reversal also requires transporting the order of their components and the orientation of their events. Route reversal and signature conjugation are distinguishable operations: their composition is specified when applied to a concrete register. In particular, the regional involution interchanging the closure and propagation components preserves the self-scaling axis; its formula is presented in Section 5.13. This regional operation acts on a different domain from reversal of a cardinal trajectory.

Extensions over a route $r : q \rightarrow q'$ are expressed through relations between fibers of compatible states. If $\mathcal{F}_q$ denotes the fiber of complete states over q , one requires

$$\text{Ext}_r \subseteq \mathcal{F}_q \times \mathcal{F}_{q'}, \quad \Delta_{\mathcal{F}_q} \subseteq \text{Ext}_{\text{id}_q},$$

$$\text{Ext}_{r_2} \circ \text{Ext}_{r_1} \subseteq \text{Ext}_{r_2 \circ r_1}.$$

Their composition preserves the evaluations of both sheets, orientation, quotients, prefix, and boundary. Route reversal does not by itself imply that every extension relation is a bijection. Conjugate prolongation is defined on the domain where transport of these data satisfies the corresponding compatibilities.

2.3.8 Nonadic holonomy and continuity of refinement

One phase winding composes transitions on the state with memory:

$$\Gamma_{9,k} = \mathcal{U}_{k+8} \circ \dots \circ \mathcal{U}_k. \quad (2.22)$$

On an individualized history, this functional composition realizes the holonomy relation; on the complete domain one retains $\Gamma_{9,k} \subseteq \widehat{X}_k \times \widehat{X}_{k+9}$, with the admissible prolongations developed below. The cyclic base has nine phases, whereas the fiber preserves the trajectory, quotients, and boundary. Monodromy is the action of one winding of that base on the fibers; holonomy $\Gamma_{9,k}$ expresses its realization by the effective TPK transitions.

If g observes the phase and M the number of retained windings,

$$g(\Gamma_{9,k}x) = g(x), \quad M(\Gamma_{9,k}x) = M(x) + 1.$$

The first identity expresses phase closure, and the second identifies the new temporal depth. Coinductive prolongation additionally preserves its cylinders and prefixes. Restrictions between depths compose an inverse system; a compatible section gathers its predecessors without replacing them by the last observation. The development of this arithmetic of histories and its unit occupies 4.8–4.10; the construction of regions and their readouts is carried out in Section 5.

Phase transfer on emissions already constructed has a different domain. Let $\mathcal{A}_+$ and $\mathcal{A}_\times$ be the alphabets of additive and multiplicative signatures, with cardinalities 18 and 26. Their product identifies the catalogue $\mathcal{U}_6$. For $f : \mathcal{A}_+ \times \mathcal{A}_\times \rightarrow \mathbb{R}$, the sheet averages are

$$(E_+f)(s, e) = \frac{1}{18} \sum_{s' \in \mathcal{A}_+} f(s', e), \quad (E_\times f)(s, e) = \frac{1}{26} \sum_{e' \in \mathcal{A}_\times} f(s, e').$$

These emissions and their cardinalities are constructed through the recurrence in Section 4. Once they have been obtained, the transfer operator is defined on $\mathcal{U}_6 \times C_9$ by

$$P_+ = E_+, \quad P_- = E_\times, \quad P_0 = I, \quad \mathcal{K}_{\text{ph}}((u, r), (v, r+1)) = P_{M_{\text{ph}}(r)}(u, v),$$

with zero weight for the remaining phase transitions. The index r uses the positive representative of the cyclic class. The phase order $(+1, -1, 0)^3$ then produces the renewal

$$\mathcal{K}_{\text{ph}}^9|_r = E_+ E_\times.$$

Indeed, both averages are idempotent, commute, and average different factors; their composition assigns weight $1/(18 \cdot 26) = 1/468$ to each emission. This equality characterizes a subsequent transfer observable. Transport of histories remains given by (2.22), with its memory and boundary.

The resulting hierarchy establishes the role of the following chapters. Finite emission constructs the domain and regions; extension relations preserve their prefixes; the nonadic connection prolongs them; the dodecaphasic register determines the data entering K and the closure of α . The numerical and incidence realizations receive that prior structure. This order makes it possible to present each consequence in its chapter while preserving, from the formal kernel, the objects that make it possible.

2.4 Accumulated memory and transport resolvents

Calendar return does not eliminate the increments produced during the traversal. To express this difference in operator terms, an accumulative register of trajectory events is constructed first, followed by its generating series. Memory variables will be reduced on that transport, while preserving the source and output readout as well.

2.4.1 Oriented events and register update

Let (d_t) be the sequence of integer direction codes preserved by a TPK trajectory. The code d_t and the cursor's cardinal direction are different coordinates; no identification between their alphabets is assumed. Number the windows from zero:

$$I_m = \{9m + 1, \dots, 9m + 9\}, \quad m \geq 0.$$

In the first cycle, I_m is window E_{m+1} of the dodecaphasic register. Its orientation is $\sigma = (1, 1, 1, -1, -1, -1, 1, 1, 1, -1, -1, -1)$. For a prolonged history, $\sigma_m \in \{-1, 1\}$ is the sign preserved in the corresponding window. The signed event is

$$a_m = \sigma_m \sum_{t \in I_m} \mathbf{1}_{\{2,4,6,8\}}(d_t), \quad |a_m| \leq 9. \quad (2.23)$$

Each summand is incorporated when its transition occurs. This definition counts the events in the trace; it does not choose directions through a value to be obtained later.

Let $\mathbf{e}_0, \dots, \mathbf{e}_{11}$ be the basis of $\mathbb{Z}^{12}$, and $\mathbf{S}\mathbf{e}_j = \mathbf{e}_{j+1 \bmod 12}$. Write $R = S^{-1}$. The accumulated register in fixed coordinates satisfies

$$z_{m+1} = z_m + a_m \mathbf{e}_{m \bmod 12}.$$

A prefix with no accumulated history starts at $z_0 = 0$; at another origin, z_0 preserves the previous register. The frame accompanying the calendar is $y_m = S^{-m} z_m$.

Proposition 2.16 (Update and return with memory). *The register in the moving frame satisfies*

$$y_{m+1} = Ry_m + \eta_m, \quad \eta_m = a_m R \mathbf{e}_0. \quad (2.24)$$

For every $n \geq 1$,

$$y_n = R^n y_0 + \sum_{j=0}^{n-1} R^{n-1-j} \eta_j. \quad (2.25)$$

In particular,

$$y_{12} = y_0 + \sum_{j=0}^{11} a_j \mathbf{e}_j. \quad (2.26)$$

Proof. Multiplying the update of z_m by $S^{-(m+1)}$ gives $y_{m+1} = S^{-1}y_m + a_m S^{-1} \mathbf{e}_0$, because $S^{-m} \mathbf{e}_{m \bmod 12} = \mathbf{e}_0$. Formula (2.25) follows by induction. For $n = 12$, $R^{12} = I$ and $R^{11-j} \eta_j = a_j R^{12-j} \mathbf{e}_0 = a_j \mathbf{e}_j$, proving (2.26). $\square$

Thus twelve windows restore the frame and preserve the event vector. The count alone does not recover the order of all steps within each window; that order remains in the recorded trajectory.

To specify the balance, define

$$D_3 = I - S^3, \quad D_4 = I - S^4, \quad B = D_3^\top D_3 + D_4^\top D_4,$$

$$\mathcal{M}(y) = \frac{1}{2} y^\top B y, \quad q(y) = \sum_{j=0}^{11} y_j.$$

The first is a quadratic form of the register; the second records the sum of its coordinates. They are not identified here with physical energy or charge.

Proposition 2.17 (Balance produced by the event). *The update (2.24) satisfies*

$$\begin{aligned} \mathcal{M}(y_{m+1}) - \mathcal{M}(y_m) &= a_m (B y_m)_0 + 2a_m^2, \\ q(y_{m+1}) - q(y_m) &= a_m. \end{aligned} \quad (2.27)$$

Proof. Expanding the products gives $B = 4I - S^3 - S^{-3} - S^4 - S^{-4}$, hence $R^\top B R = B$ and $B_{00} = 4$. Since $y_{m+1} = R(y_m + a_m \mathbf{e}_0)$, the quadratic difference is $a_m \mathbf{e}_0^\top B y_m + \frac{1}{2} a_m^2 B_{00}$. The sum of coordinates is invariant under the permutation R , giving the second equality. $\square$

2.4.2 Generating series and uniform control of memory

The series preserves all increments, not only their sum at the end of a cycle. With a formal variable u , let

$$Y(u) = \sum_{m \geq 0} y_m u^m, \quad H_\eta(u) = \sum_{m \geq 0} \eta_m u^m.$$

Proposition 2.18 (Resolvent and tail bound). *The formal identity*

$$Y(u) = (I - uR)^{-1}(y_0 + uH_\eta(u)). \quad (2.28)$$

holds. Both series converge for $|u| < 1$. If $\ell : \mathbb{R}^{12} \rightarrow \mathbb{R}$ is linear, $L = \|\ell\|_{1 \rightarrow \mathbb{R}}$, $M_0 = \|y_0\|_1$, and $0 < r < 1$, then, for every integer $N \geq 0$ and $|u| \leq r$,

$$\left| \sum_{m > N} \ell(y_m) u^m \right| \leq L r^{N+1} \left(\frac{M_0 + 9(N+1)}{1-r} + \frac{9r}{(1-r)^2} \right). \quad (2.29)$$

The derivative series converges uniformly on each disk $|u| \leq r < 1$.

Proof. Multiplying (2.24) by u^{m+1} and summing gives $Y - y_0 = uRY + uH_\eta$. The constant term of $I - uR$ is invertible, so its formal inverse is $\sum_{k \geq 0} u^k R^k$. Moreover, R preserves the ℓ^1 norm and $\|\eta_m\|_1 \leq 9$, whence $\|y_m\|_1 \leq M_0 + 9m$. For the tail, sum the majorants $L(M_0 + 9m)r^m$, using

$$\sum_{m > N} r^m = \frac{r^{N+1}}{1-r}, \quad \sum_{m > N} m r^m = r^{N+1} \left(\frac{N+1}{1-r} + \frac{r}{(1-r)^2} \right).$$

Finally, $\sum_{m \geq 1} Lm(M_0 + 9m)r^{m-1} < \infty$ majorizes the derivative series and proves uniform convergence. $\square$

The variable u counts windows of nine transitions. Identifying it with the variable of another readout requires preserving that clock and the map relating the two readouts.

2.4.3 Elimination of memory: operator, source, and readout

On a domain where the update is fixed as $U : X \rightarrow X$, the free vector space $V = \mathbb{Q}^{(X)}$ on the states admits the operator $T\delta_x = \delta_{U(x)}$. The clock is included in x if the transition depends on it. This linear extension preserves distinct states even when they share an observable phase.

Consider a decomposition $V = V_0 \oplus V_1$, where V_1 is the auxiliary sector to be eliminated, and write

$$T = \begin{pmatrix} A & B_1 \\ C & D \end{pmatrix}, \quad v = \begin{pmatrix} v_0 \\ v_1 \end{pmatrix}, \quad \ell = (\ell_0, \ell_1).$$

Here $A : V_0 \rightarrow V_0$, $B_1 : V_1 \rightarrow V_0$, $C : V_0 \rightarrow V_1$, and $D : V_1 \rightarrow V_1$. The following identities are formal; they also hold wherever the resolvents converge.

Proposition 2.19 (Schur reduction with source and readout). *Let*

$$D_u = (I - uD)^{-1}, \quad \mathcal{F}(u) = I - uA - u^2 B_1 D_u C.$$

The retained component of $(I - uT)^{-1}v$ is

$$w_0 = \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1), \quad w_1 = D_u(v_1 + uCw_0). \quad (2.30)$$

Its complete readout satisfies

$$\begin{aligned} \ell(I - uT)^{-1}v &= (\ell_0 + u\ell_1 D_u C) \mathcal{F}(u)^{-1}(v_0 + uB_1 D_u v_1) \\ &\quad + \ell_1 D_u v_1. \end{aligned} \quad (2.31)$$

Proof. The two equations of $(I - uT)w = v$ are $w_0 = v_0 + uAw_0 + uB_1w_1$ and $w_1 = v_1 + uCw_0 + uDw_1$. Solving the second and substituting into the first gives (2.30); applying ℓ_0 and ℓ_1 to both components gives (2.31). All formal inverses exist because their constant terms are identities. $\square$

The term $u^2B_1D_uC = \sum_{k \geq 0} u^{k+2}B_1D^kC$ describes an exit into memory, k steps within it, and a return. By contrast, $u\ell_1D_uC$ reads memory before that return. Thus reducing only the operator does not necessarily preserve the observation: source and readout must also be transformed.

The difference is visible in the cycle already constructed. If P retains $\mathbf{e}_0$, then $PRP = 0$, but

$$\langle \mathbf{e}_0, (I - uR)^{-1}\mathbf{e}_0 \rangle = \frac{1}{1 - u^{12}}, \quad \langle \mathbf{e}_{11}, (I - uR)^{-1}\mathbf{e}_0 \rangle = \frac{u}{1 - u^{12}}. \quad (2.32)$$

Indeed, $R^n\mathbf{e}_0 = \mathbf{e}_{-n \bmod 12}$: the two series collect, respectively, $n \equiv 0$ and $n \equiv 1 \pmod{12}$. The resolvent of the compression is I , whereas compression of the resolvent preserves the omitted returns. This example corresponds to the twelve-window register, not to the entire TPK state.

2.4.4 Composition of segments and coherence of reduction

Let there be three composable affine transports $F_i(y) = R_iy + b_i$, with compatible domains and codomains. The memory produced by the complete traversal is

$$F_3F_2F_1(y) = R_3R_2R_1y + b_3 + R_3b_2 + R_3R_2b_1. \quad (2.33)$$

The proof is successive substitution of the three affine formulas. Grouping either the first two segments or the last two first produces the same expression; the factors transport each increment from the point where it originated. In (2.24), $R_i = R$ and b_i is the event in its window.

Eliminating several layers preserves a similar coherence. To make it explicit, now decompose $V = V_0 \oplus V_1 \oplus V_2$ and write

$$T = \begin{pmatrix} A & B_1 & B_2 \\ C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{pmatrix}, \quad Q_2 = (I - uD_{22})^{-1}.$$

Eliminating the third component leaves the four blocks

$$\begin{aligned} A' &= A + uB_2Q_2C_2, & B'_1 &= B_1 + uB_2Q_2D_{21}, \\ C'_1 &= C_1 + uD_{12}Q_2C_2, & D'_{11} &= D_{11} + uD_{12}Q_2D_{21}. \end{aligned} \quad (2.34)$$

Proposition 2.20 (Transitivity of elimination). *If $D_* = (D_{ij})_{i,j=1}^2$, the complement obtained in two stages coincides with that obtained by eliminating both memories together:*

$$\begin{aligned} I - uA' - u^2B'_1(I - uD'_{11})^{-1}C'_1 &= I - uA \\ &\quad - u^2(B_1 \ B_2)(I - uD_*)^{-1} \begin{pmatrix} C_1 \\ C_2 \end{pmatrix}. \end{aligned} \quad (2.35)$$

Proof. Solving the third equation of $(I - uT)w = v$ and substituting into the first two produces (2.34). Subsequently eliminating the second component gives the left-hand side of (2.35). Solving both memory equations simultaneously gives the right-hand side, by Proposition 2.19. The formal solution is unique because $I - uT$ has constant term I . In both procedures, source and readout are also transformed according to (2.30)–(2.31). $\square$

The balance (2.27) describes register increments. The preceding identities allow them to be transported when memory is reused in dodecaphasic closure; they do not by themselves determine additional analytic coefficients.

3 Topological Phase Kernel and generation of the seed catalogue

The two APP evaluations acquire generative efficacy once their activation, evaluation position, positional transport and the information preserved by each transition have been specified. The *Topological Phase Kernel*, abbreviated TPK, combines these operations into a dynamics on enriched states. Its definition comprises selection, transport, arithmetic updating and preservation of the record. The finite construction developed in this section provides the seeds on which the regional lifts and coinductive prolongation subsequently act.

The principal result of this stage is a generated and classified catalogue: the initial conditions of the two cursors produce integer emissions; their ternary reading determines oriented words; and the dihedral action organises these words into orbits. Each step preserves its preimages and the relations that permit recovery of the preceding description. The chain of cardinalities

$$104\,976 \longrightarrow 468 \longrightarrow 243 \subset 729, \quad 243 \longrightarrow 43,$$

abbreviates distinct objects. Each object is constructed and each count justified below, before any of them is used as data for regional selection.

3.1 Phase selection, transport and enriched state

3.1.1 Tritic selector of operative phase

The nonadic phase takes its values in $D_9 = \{1, \dots, 9\}$. The tritic selector is the function

$$M_{\text{ph}} : D_9 \longrightarrow \{-1, 0, +1\}, \quad M_{\text{ph}}(g) = \begin{cases} +1, & g \in \{1, 4, 7\}, \\ -1, & g \in \{2, 5, 8\}, \\ 0, & g \in \{3, 6, 9\}. \end{cases} \quad (3.1)$$

In the emitter realisation, the value $+1$ activates the additive evaluation, -1 activates the multiplicative evaluation, and 0 records the neutral event. The cardinal orientation of the cursor is another state coordinate. This distinction specifies both the operation being read and the direction in which its evaluation point is transported.

For transitions indexed by $t \geq 1$, the phase preceding the reading is

$$g_t = 1 + [(t - 1)]_9, \quad \epsilon_t = M_{\text{ph}}(g_t), \quad (3.2)$$

where $[n]_b$ denotes the representative of n modulo b in $\{0, \dots, b - 1\}$. A window of nine transitions contains the sequence $(+1, -1, 0, +1, -1, 0, +1, -1, 0)$.

3.1.2 Oriented cursors and cardinal transport

The indices $I_9 = \{0, \dots, 8\}$ are used for positions, and the positive labels $(a, b) = (i + 1, j + 1)$ for APP evaluations. A finite cursor is

$$c = (i, j; d) \in \mathcal{S} := I_9^2 \times \mathcal{D}, \quad \mathcal{D} = \{N, E, S, O\}.$$

The four directions act by

$$v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

The cardinal map corresponding to d is

$$T_d(i, j) = ([i + (v_d)_1]_9, [j + (v_d)_2]_9).$$

Thus M_{ph} and T_d have distinct domains, codomains and actions. Tritic activation selects one of the two cursors; cardinal transport changes its position in the direction already retained by that cursor.

The integer lift of the cursor is $\tilde{c} = (x, y; d) \in \mathbb{Z}^2 \times \mathcal{D}$, with finite position $([x]_9, [y]_9)$. Its winding counts are recovered from

$$x = 9[x/9] + [x]_9, \quad y = 9[y/9] + [y]_9. \quad (3.3)$$

These two divisions concern spatial transport. The division of an additive or multiplicative evaluation constitutes a distinct arithmetic record, which is preserved together with them.

3.1.3 Composition of the transition

The state of the finite realisation includes the two lifted cursors, their directions, the phase, the accumulators of the active window, the ordered list of blocks already emitted and the transition record. Each local record contains the preceding positions, the integer evaluations of both sheets, their residues and quotients, the active regime and the orientations. Prefix and boundary fields arising in a prolongation are incorporated into the same state with their compatibility rules.

The transition is ordered as

$$\mathcal{U}_t = \text{Mem}_t \circ \text{Tra}_t \circ \text{Sel}_t. \quad (3.4)$$

Selection determines the regime and preserves the sample taken before any displacement. Transport applies the active movement and the cardinal reversal prescribed by the schedule. Updating incorporates that sample into the accumulators and the record; at the end of a window, it emits the corresponding block. The composition therefore uses a sample taken before transport, although its definitive storage occurs at the end of the transition.

3.2 Initial conditions and reversible enumeration

The additive and multiplicative sheets have independent cursors. The domain of initial conditions is the ordered product

$$\Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times, \quad |\mathcal{S}| = 9^2 \cdot 4 = 324.$$

Consequently,

$$|\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (3.5)$$

The number 81 counts positions; 324 counts the oriented positions of one cursor; 104 976 counts ordered pairs of cursors. Choosing a position in APP retains only part of an initial condition of the emitter.

Fix the numbering

$$\text{ind}(N) = 0, \quad \text{ind}(E) = 1, \quad \text{ind}(S) = 2, \quad \text{ind}(O) = 3,$$

and define

$$\nu(i, j; d) = 4(9i + j) + \text{ind}(d). \quad (3.6)$$

Proposition 3.1 (Exact enumeration of the initial conditions). *The map ν is a bijection from $\mathcal{S}$ onto $\{0, \dots, 323\}$. The pair of indices is reversibly encoded by $324\nu_+ + \nu_\times \in \{0, \dots, 104\,975\}$.*

Proof. Division by four recovers

$$\text{ind}(d) = [\nu]_4, \quad j = [\lfloor \nu/4 \rfloor]_9, \quad i = \lfloor \nu/36 \rfloor.$$

The bounds on the domain ensure that these three quantities belong to their respective index sets and reconstruct ν through (3.6). For the pair, Euclidean division by 324 recovers the quotient ν_+ and the residue $\nu_\times$. $\square$

The enumeration traverses the complete domain before any regional grouping. The indices locate the preimages of an emission while keeping explicit the condition that produced each trajectory.

3.3 Sheet evaluation and arithmetic record

At a positive position (a, b) , the integer evaluations are $A_+(a, b) = a + b$ and $A_\times(a, b) = ab$. For $n > 0$, retain

$$\rho_9(n) = 1 + [n - 1]_9, \quad q_9(n) = \lfloor (n - 1)/9 \rfloor, \quad n = \rho_9(n) + 9q_9(n). \quad (3.7)$$

Additive activation incorporates $\rho_9(A_+)$ into the corresponding accumulator. Multiplicative activation uses an observable of the residue $\rho_9(A_\times)$, defined next.

3.3.1 Discrete exponent in the units modulo nine

The powers of 2 modulo 9 traverse

$$2^0, 2^1, \dots, 2^5 \equiv 1, 2, 4, 8, 7, 5 \pmod{9}, \quad 2^6 \equiv 1 \pmod{9}.$$

The discrete logarithm in this group of six units is represented by

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5. \quad (3.8)$$

The emission law extends the observable by $\ell_9(3) = \ell_9(6) = \ell_9(9) = 0$. To preserve its provenance, the recorded sample includes the original residue and the indicator of membership in the group of units. An observed value of zero can therefore originate from the unit 1 or from one of the three non-invertible residues, with its origin remaining recoverable.

3.3.2 Order of reading and movement

Evaluations are performed at the positions preceding the transition. If $\epsilon_t = +1$, the additive reading is accumulated and its cursor is displaced. If $\epsilon_t = -1$, the multiplicative discrete exponent is accumulated and the second cursor is displaced. If $\epsilon_t = 0$, the neutral counter is incremented and both positions are preserved.

At the end of steps 27 and 54, both directions are replaced by their opposites. The windows are concatenated with positions and orientations preserved. At the start of each window, only its three local accumulators are reset to zero; the preceding record remains available.

Proposition 3.2 (Inversion of the lifted transport). *Let $a, b \in \{0, 1\}$ be the activation and reversal indicators of a cursor. If opp interchanges $N \leftrightarrow S$ and $E \leftrightarrow O$, the transformation*

$$F_{a,b}(z, d) = (z + av_d, \text{opp}^b(d))$$

has inverse

$$F_{a,b}^{-1}(z', d') = (z' - av_{\text{opp}^b(d')}, \text{opp}^b(d')).$$

Spatial reduction modulo nine intertwines this transformation with the finite transport.

Proof. Applying opposition twice recovers the initial direction. Once that direction has been recovered, subtracting av_d reverses the displacement. For the spatial projection, it suffices to use $[z + av_d]_9 = [[z]_9 + av_d]_9$ componentwise. Directional reversal preserves this equality. $\square$

The phase determines the indicators used by the inverse. The spatial and arithmetic records can therefore be preserved jointly, including when the boundary of the representative I_9^2 is crossed.

3.4 Six windows and decimal block emission

3.4.1 Accumulation within a nonadic window

Window m , with $1 \leq m \leq 6$, comprises steps $9m - 8, \dots, 9m$. Let S_m be the sum of its active positive additive residues, E_m the sum of its active discrete exponents, and Z_m its number of neutral events.

Lemma 3.3 (Activation multiplicities). *Each window contains three additive activations, three multiplicative activations and three neutral activations. In particular, $Z_m = 3$.*

Proof. The phases of a window traverse exactly $1, \dots, 9$. Each of the three sets in (3.1) has cardinality three. $\square$

The six windows comprise 54 transitions and produce six blocks. This length belongs to the initial emitter defined here. Continuation in depth subsequently acts on the words and their compatible records.

3.4.2 Decimal encoding rule

The emission law specifies the digits

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \quad (3.9)$$

The weights 100, 10, 1 are the hundreds, tens and units positions. The subscript 10 denotes the decimal residue, and each block satisfies $0 \leq U_m \leq 999$. The emitted word is the ordered sequence $\mathbf{U} = (U_1, \dots, U_6)$, with a fixed width of three digits per block.

The coefficients 3, 5, 7 constitute the explicit weights of this emission law. Once the rule has been fixed, each digit is calculated from the accumulators, and the results of this section follow from that rule. The number 1 appearing in its reduced form has a further specific provenance: $7Z_m = 21 \equiv 1 \pmod{10}$.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, one obtains

$$G(s_m, e_m) = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10} = U_m. \quad (3.10)$$

Here the letter e_m denotes a component of the discrete-exponent signature. Its definition belongs to the finite emitter.

Proposition 3.4 (Recovery of the two signatures). *The map $G : \{0, \dots, 9\}^2 \rightarrow \{0, \dots, 999\}$ is injective. For a block $U = 100a + 10b + c$ in its image, the inverse is*

$$s = a, \quad e = [b - a]_{10}, \quad c = [3s + 5e + 1]_{10}.$$

Consequently, the word $\mathbf{U}$ determines the two six-component signatures $\mathbf{s}$ and $\mathbf{e}$.

Proof. The last two summands of (3.10) belong to $\{0, \dots, 99\}$, so the hundreds digit recovers s . The tens digit is $[s + e]_{10}$; subtracting s and reducing modulo ten recovers e . The units digit verifies the third congruence. The reconstruction applies independently to the six positions of the word. $\square$

The integer totals S_m, E_m are recovered from the recorded readings; the hundreds and tens digits recover their residues. For example, a total $S_m = 15$ produces $s_m = 5$. This distinction identifies precisely which information each level of encoding preserves.

3.4.3 Positional encoding of the six-block word

The six blocks admit a fixed-width integer encoding,

$$\mathcal{N}(\mathbf{U}) = \sum_{m=1}^6 U_m 1000^{6-m}, \quad 0 \leq \mathcal{N}(\mathbf{U}) < 1000^6. \quad (3.11)$$

The first block occupies the position of greatest weight, and the sixth the base-thousand units position. A block 058 occupies a complete position with integer value 58; its three-digit width belongs to the decimal presentation of that position.

Proposition 3.5 (Positional recovery of the emissions). *The encoding (3.11) determines the complete word by*

$$U_m = \left\lfloor \left\lfloor \frac{\mathcal{N}(\mathbf{U})}{1000^{6-m}} \right\rfloor \right\rfloor_{1000}, \quad 1 \leq m \leq 6.$$

Proof. On division by 1000^{6-m} , the preceding blocks yield integer multiples of 1000, block m yields U_m , and the contribution of the subsequent blocks belongs to $[0, 1)$. Taking the integer part eliminates the latter contribution, and the residue modulo 1000 eliminates the former. $\square$

This encoding preserves the six emissions. Its domain has ordered base-thousand positions, whereas the APP domain has spatial positions modulo nine. The composition retains both indices: m identifies an emission window, and (i, j) identifies the cell visited during a transition in that window.

3.5 Complete calculation of an emission beginning with 501

Fix the initial conditions

$$c_{+,0} = (0, 4; N), \quad c_{\times,0} = (2, 3; N), \quad \nu_+ = 16, \quad \nu_{\times} = 84.$$

The initial positive positions are $(1, 5)$ and $(3, 4)$. The preceding chronological index is $t = 0$, the first phase read is $g_1 = 1$, and the record is empty. These coordinates are the inputs to the calculation; the blocks are obtained by executing the recurrence.

Table 1: First window: preceding readings and active accumulation.

t	ϵ_t	position +	position $\times$	ΔS	ΔE	(S, E, Z)
1	+1	(1, 5)	(3, 4)	6	0	(6, 0, 0)
2	-1	(9, 5)	(3, 4)	0	0	(6, 0, 0)
3	0	(9, 5)	(2, 4)	0	0	(6, 0, 1)
4	+1	(9, 5)	(2, 4)	5	0	(11, 0, 1)
5	-1	(8, 5)	(2, 4)	0	3	(11, 3, 1)
6	0	(8, 5)	(1, 4)	0	0	(11, 3, 2)
7	+1	(8, 5)	(1, 4)	4	0	(15, 3, 2)
8	-1	(7, 5)	(1, 4)	0	2	(15, 5, 2)
9	0	(7, 5)	(9, 4)	0	0	(15, 5, 3)

Note. Positions are positive labels of the cells preceding movement. Neutral events increment only Z .

The three active additive evaluations are

$$1 + 5 = 6, \quad 9 + 5 = 14 = 5 + 9, \quad 8 + 5 = 13 = 4 + 9,$$

so $S_1 = 6 + 5 + 4 = 15$. The three active multiplicative evaluations are $3 \cdot 4 = 12$, $2 \cdot 4 = 8$ and $1 \cdot 4 = 4$. Their positive residues are 3, 8, 4, with observables 0, 3, 2; hence $E_1 = 5$. Finally, $Z_1 = 3$. The decimal rule gives

$$d_{1,1} = [15]_{10} = 5, \quad d_{1,2} = [15 + 5]_{10} = 0, \quad d_{1,3} = [45 + 25 + 21]_{10} = 1,$$

and one obtains $U_1 = 501$.

Table 2: The six windows of initial condition (16, 84).

m	steps	S_m	E_m	Z_m	s_m	U_m	$[-U_m]_3$
1	1–9	15	5	3	5	501	0
2	10–18	6	5	3	6	614	1
3	19–27	24	5	3	4	498	0
4	28–36	21	5	3	1	169	2
5	37–45	12	5	3	2	272	1
6	46–54	12	5	3	2	272	1

The second additive reading above has arithmetic quotient 1, whereas its lifted cursor is at $(-1, 4; N)$, with vertical winding count -1 . These are distinct data in the same record. Both participate in the complete recovery of the evaluation and its position.

The resulting word and its signatures are

$$\mathbf{U} = (501, 614, 498, 169, 272, 272),$$

$$\mathbf{s} = (5, 6, 4, 1, 2, 2),$$

$$\mathbf{e} = (5, 5, 5, 5, 5, 5).$$

Cardinal reversal after step 27 changes the trajectories of the final three windows. The repeated block 272 corresponds to two successive windows with their own positions and chronological record.

As a second case, the minimal initial condition $(\nu_+, \nu_-) = (0, 0)$ produces

$$\mathbf{U} = (252, 109, 252, 903, 850, 850), \quad \mathbf{s} = (2, 1, 2, 9, 8, 8), \quad \mathbf{e} = (3, 9, 3, 1, 7, 7).$$

In this case, both cursors return to their initial positions and orientations after 54 transitions. The nonadic phase completes six cycles, the chronological counter advances by 54 units, and the record contains 54 samples. The positional reading of the return and the recorded history describe distinct properties of the execution.

3.6 Factorisation of the finite census

The signatures $f_+(\nu) = \mathbf{s}$ and $f_-(\nu) = \mathbf{e}$ are calculated by executing the corresponding cursor through the six windows. The independence of positions and directions between the two cursors gives

$$T_{\text{em}}(\nu_+, \nu_-) = G^{(6)}(f_+(\nu_+), f_-(\nu_-)), \quad (3.12)$$

where $G^{(6)}$ applies (3.10) at each position. This factorisation first permits enumeration of $324 + 324$ trajectories and then combination of the distinct signatures, preserving the multiplicities of their preimages.

3.6.1 Images of the two signatures

The following tables include every signature. Its multiplicity is the number of initial conditions producing it; the minimal index locates one such condition. The complete preimage is defined by

$$f_{\sigma}^{-1}(a) = \{\nu \in \{0, \dots, 323\} : f_{\sigma}(\nu) = a\}.$$

The recurrence of the preceding sections determines every element of this set by a finite integer calculation.

Table 3: The eighteen additive signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
122564	18	17	564122	18	16
122898	18	24	645212	18	4
212645	18	5	654889	18	33
212988	18	0	889221	18	13
221456	18	29	889654	18	32
221889	18	12	898122	18	25
456221	18	28	898465	18	20
465898	18	21	988212	18	1
546988	18	9	988546	18	8

Table 4: The twenty-six multiplicative signatures.

signature	multiplicity	$\nu_{\min}$	signature	multiplicity	$\nu_{\min}$
000000	108	8	555555	24	84
177393	8	1	555717	8	14
177555	8	54	555771	8	18
177771	8	11	555933	8	24
339555	8	6	717555	8	12
339771	8	15	717717	8	21
339933	8	59	717933	8	19
393177	8	0	771177	8	9
393393	8	45	771339	8	13
393555	8	40	771555	8	16
555177	8	52	933339	8	57
555339	8	4	933555	8	26
555393	8	41	933717	8	17

Proposition 3.6 (Image and multiplicities of the emitter). *The image $\mathcal{U}_6 = \text{im } T_{\text{em}}$ contains 468 distinct words. Their preimages are distributed into 432 fibres of cardinality 144, 18 of cardinality 432 and 18 of cardinality 1944.*

Proof. Exhaustive evaluation of the algorithm in Section 3.9 on the indices $0, \dots, 323$ gives the two tables above. The first partition has 18 classes of cardinality 18; the second has 24 classes of cardinality 8, one of cardinality 24 and one of cardinality 108. The sums $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$ verify the total mass of both partitions. Proposition 3.4 makes the coupling of their images injective; there are therefore $18 \cdot 26 = 468$ emissions. By (3.12), the preimage of a pair of signatures is the product of their two preimages. This gives $18 \cdot 24 = 432$ products of cardinality $18 \cdot 8 = 144$, 18 of cardinality $18 \cdot 24 = 432$, and 18 of cardinality $18 \cdot 108 = 1944$. Finally,

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

The enumerative part is an exact finite check of the recurrence; the factorisation explains why this count covers every pair. $\square$

The emission in Example 3.5 has a fibre of cardinality $18 \cdot 24 = 432$. The example with indices $(0, 0)$ has a fibre of cardinality $18 \cdot 8 = 144$. In both cases, the emitted word recovers the signatures, whereas identification of a particular initial condition uses its index or its trajectory record.

3.7 Oriented ternary reading and preservation of fibres

Oriented reduction acts separately on each block:

$$\Psi : \mathcal{U}_6 \longrightarrow \mathbb{F}_3^6, \quad \Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (3.13)$$

Its sign fixes an orientation of the catalogue. The balanced identification $0 \mapsto 0, 1 \mapsto +1, 2 \mapsto -1$ is then applied componentwise.

Modulo nine organises APP evaluations and positions; residues modulo ten form each digit; base thousand combines three decimal positions; and modulo three determines the ternary signature of each block. These four uses have defined objects and functions. In particular, Ψ reads six blocks and produces six trits; a positional conversion of the integer formed by eighteen digits would be a different operation.

For the two conditions already calculated, one obtains

$$\begin{aligned} \Psi(501, 614, 498, 169, 272, 272) &= (0, 1, 0, 2, 1, 1), \\ \Psi(252, 109, 252, 903, 850, 850) &= (0, 2, 0, 0, 2, 2). \end{aligned}$$

The decimal emitter is defined before this reduction. The function of Ψ is to produce the ternary object on which orbital symmetry and regional selectors act. This function explains its position in the genealogy: it connects an already calculated emission with a structural classification.

The image $\mathcal{W}_6 := \Psi(\mathcal{U}_6)$ contains exactly 243 words within the ambient set of $3^6 = 729$ words. Enumeration by the final algorithm gives the histogram

$ \Psi^{-1}(w) $	1	2	3	4	5	6
number of words w	132	66	18	3	6	18.

(3.14)

The sums for the two readings are

$$132 + 66 + 18 + 3 + 6 + 18 = 243, \quad 132 + 2 \cdot 66 + 3 \cdot 18 + 4 \cdot 3 + 5 \cdot 6 + 6 \cdot 18 = 468.$$

The first counts ternary words; the second recovers the number of emissions distributed among their fibres.

Example 3.7 (Two emissions with the same ternary signature). The emissions

$$(498, 501, 614, 252, 252, 109), \quad (870, 810, 923, 252, 252, 109)$$

are distinct and both have image $(0, 0, 1, 0, 0, 2)$ under Ψ . Each has 144 initial conditions. The ternary word preserves a reading class; the emissions and their preimages distinguish the data that converge in that class.

Preservation of fibres is expressed concretely by

$$\mathcal{F}_w = \{\mathbf{U} \in \mathcal{U}_6 : \Psi(\mathbf{U}) = w\}, \quad \Omega_{\mathbf{U}} = T_{\text{em}}^{-1}(\mathbf{U}).$$

The complete preimage of w in the initial domain is therefore the disjoint union of the sets $\Omega_{\mathbf{U}}$, with $\mathbf{U} \in \mathcal{F}_w$. The enriched record can retain the emission, its signatures and the initial condition together with its ternary reading. The exact inverse then concerns the recorded datum with its type; the isolated six-trit word retains the precise scope of (3.13).

3.8 Dihedral action and the forty-three orbits

Define the following permutations on six coordinates:

$$r(u_1, u_2, u_3, u_4, u_5, u_6) = (u_3, u_1, u_2, u_5, u_6, u_4), \quad (3.15)$$

$$s(u_1, u_2, u_3, u_4, u_5, u_6) = (u_4, u_5, u_6, u_1, u_2, u_3). \quad (3.16)$$

The first rotates the two groups of three coordinates in conjugate directions; the second interchanges the groups. Their composition satisfies $r^3 = s^2 = 1$ and $srs = r^{-1}$. They therefore generate an action of the six-element dihedral group D_3 .

Proposition 3.8 (Equivariance and stability of the catalogue). *Reduction by Ψ commutes with the action of D_3 . The sets $\mathcal{U}_6$ and $\mathcal{W}_6$ are stable under its generators.*

Proof. The equality $\Psi(g\mathbf{U}) = g\Psi(\mathbf{U})$ holds at every coordinate because g permutes positions and Ψ applies the same reduction at each position. Stability of the catalogue is a further check: generating its 468 elements by (3.12), one verifies that $r\mathbf{U}$ and $s\mathbf{U}$ again belong to that set. The algorithm in Section 3.9 performs these membership checks. Equivariance then transfers stability to the ternary image. $\square$

Proposition 3.9 (Complete orbital decomposition). *The set $\mathcal{W}_6$ decomposes into 43 orbits: 38 of cardinality six and 5 of cardinality three.*

Proof. For each $w \in \mathcal{W}_6$, form

$$\mathcal{O}(w) = \{w, rw, r^2w, sw, srw, sr^2w\}.$$

Choose the lexicographically least word in this set as representative. Two words have the same representative exactly when they belong to the same orbit. The algorithm generates the list of representatives in Table 5; their cardinalities sum to $38 \cdot 6 + 5 \cdot 3 = 243$. Since every word in the catalogue participates in this operation and the catalogue is stable, the list covers the entire image. $\square$

Table 5: Minimal representatives and cardinalities of the forty-three orbits.

representative	cardinality	representative	cardinality	representative	cardinality
001002	6	002112	6	012222	6
001011	6	002211	6	021022	6
001012	6	002220	6	021121	6
001021	6	011021	6	021202	6
001022	6	011112	6	022022	3
001100	3	011120	6	022112	6
001101	6	011201	6	022121	6
001110	6	011211	6	022202	3
001112	6	011220	6	022211	6
001202	6	011222	6	022222	6
001210	6	012112	6	112112	3
001211	6	012121	6	112211	3
001220	6	012202	6	112222	6
001222	6	012210	6		
002012	6	012220	6		

The orbits inherit integer invariants from their fibres. For each word, $n(w) = |\mathcal{F}_w|$ counts emissions, whereas

$$M(w) = \sum_{\mathbf{U} \in \mathcal{F}_w} |\Omega_{\mathbf{U}}|$$

counts initial conditions. The symbol census $c(w) = (n_0, n_1, n_2)$ records the number of occurrences of each ternary class. Together with the orientation and the support of the trajectories, these data provide the predicates for subsequent regional selection. The classification thus preserves more information than the form of a representative word.

3.9 Self-contained enumeration and verification algorithm

The enumeration uses integers, finite lists and the operations already defined. The following procedure specifies its execution; σ takes the values $+$ and $\times$.

1. For each $v = 0, \dots, 323$, recover $(i, j; d)$ using the inverse of (3.6). Initialise an empty signature list.
2. For each window $m = 1, \dots, 6$, initialise the accumulator $A = 0$. Traverse $t = 9m - 8, \dots, 9m$, calculating ϵ_t by (3.2).
3. If $\sigma = +$ and $\epsilon_t = +1$, add $\rho_9(i + j + 2)$ to A and update $(i, j) \leftarrow T_d(i, j)$. If $\sigma = \times$ and $\epsilon_t = -1$, add $\ell_9(\rho_9((i + 1)(j + 1)))$ and perform the same transport.
4. At the end of $t = 27$ or $t = 54$, replace d by its opposite. At the end of the window, append $[A]_{10}$ to the signature list. Preserve the position and direction for the next window.
5. Group the 324 indices according to equality of their six-component lists. These two groupings give f_+ , $f_\times$ and all their preimages.
6. For each pair of distinct signatures, apply $G^{(6)}$. Assign to the emission the product multiplicity of the two preimages. Verify signature recovery in each block and the sum of multiplicities 104 976.
7. Apply Ψ to each emission and group by equality of words. Count the 243 values and the multiplicities in (3.14).
8. Apply r, s to each emission and verify membership in the catalogue. For each ternary word, form its six dihedral images, remove repetitions and record its lexicographic minimum. Count the 43 orbits and the two sizes in Table 5.

The execution terminates: the first stage evaluates 648 trajectories of 54 transitions; the second combines 468 pairs of signatures; the third groups a set of 243 words. An additional direct check can execute the 104 976 pairs and compare the result with the factorisation (3.12); its scope is the same finite domain.

3.10 Preservation of the record and the function of the seeds

Let $\text{Sample}(q_t)$ be the sample preceding the transition, and $\mathcal{M}_t$ the accumulated record. The memory update is

$$\mathcal{M}_{t+1} = \mathcal{M}_t \parallel \text{Sample}(q_t), \quad (3.17)$$

where $\parallel$ denotes ordered concatenation. Induction gives

$$\mathcal{M}_n = \mathcal{M}_0 \parallel (\text{Sample}(q_0), \dots, \text{Sample}(q_{n-1})).$$

The accumulators are reconstructed by summing the active contributions of each segment of nine records. Their quotients, the winding counts of the cursors and the orientations remain associated with the same transitions. Execution determines that record through the recurrence; storage preserves the specific provenance of the readings.

At the end of this stage, the complete domain of initial conditions, decimal emission, its two recoverable signatures, the ternary image, its fibres and its orbital classification have been constructed. The counts 104 976, 468, 243, 729 and 43 now have a domain and a mechanism of derivation. Regional selections receive these structures with their invariants, and the lifts act on oriented words with preserved provenance. This continuity makes the finite catalogue a constructive antecedent of the coinductive development.

4 Development of the formal kernel: initial conditions, geometry, and refinement

The eighty-one-position support does not coincide with the domain of initial conditions of the dynamics. An initial condition must also specify an orientation; and the additive and multiplicative sheets have independent cursors. This distinction makes it possible to reconstruct the emission exhaustively without selecting a numerical region in advance.

4.1 Initial states and the emission recurrence

For enumeration we use indices $I_9 = \{0, \dots, 8\}$, related to the positive labels by $i \mapsto i + 1$. Let

$$\mathcal{S} = I_9^2 \times \{N, E, S, O\}, \quad \Omega_{\text{seed}} = \mathcal{S}_+ \times \mathcal{S}_\times.$$

Each initial condition $\omega = (s_+, s_\times)$ determines two positions and two directions. The domain comprises all ordered pairs, so

$$|\mathcal{S}| = 324, \quad |\Omega_{\text{seed}}| = 324^2 = 104\,976. \quad (4.1)$$

The independent annex enumerates all these conditions; the main text presents the law that transforms them. The rows of an emission table are fibers of this domain of initial conditions.

The finite realization uses six windows of nine transitions. At each time t , the phase signature is $M_{\text{ph}}(1 + ((t-1) \bmod 9))$. If it equals $+1$, the positive residue $\Sigma(i+1, j+1) = \rho_9(i+j+2)$ of the first cursor is read, and that cursor is then moved; if it equals -1 , the positive residue $\Pi(i+1, j+1) = \rho_9((i+1)(j+1))$ of the second cursor is read, and the second cursor is then moved. The value zero increments the neutral counter. At the end of transitions twenty-seven and fifty-four, the cardinal orientations are reversed. Integer evaluations and their quotients are retained in the register, but the emitter uses the residue readouts specified here.

In the unit sector, the multiplicative readout uses the base-two discrete logarithm in $(\mathbb{Z}/9\mathbb{Z})^\times$:

$$\ell_9(1) = 0, \quad \ell_9(2) = 1, \quad \ell_9(4) = 2, \quad \ell_9(8) = 3, \quad \ell_9(7) = 4, \quad \ell_9(5) = 5.$$

For the finite emission it is extended by zero to the nonunit sector. This extension is an observable, not an identification of the elements 3, 6, 9 with the multiplicative identity. The sheet and its membership in the radical sector remain in the trajectory register.

Let S_m be the sum of the three additive positive residues Σ in window m , E_m the sum of the three discrete exponents of the multiplicative residues Π , and $Z_m = 3$ the number of neutral events. Thus S_m does not sum the integer evaluations $S = i + j$, whose quotients remain in the register. With $[a]_{10} \in \{0, \dots, 9\}$, the decimal emission rule is

$$\begin{aligned} d_{m,1} &= [S_m]_{10}, \\ d_{m,2} &= [S_m + E_m]_{10}, \\ d_{m,3} &= [3S_m + 5E_m + 7Z_m]_{10}, \\ U_m &= 100d_{m,1} + 10d_{m,2} + d_{m,3}. \end{aligned} \quad (4.2)$$

The integer coefficients and calendar of this recurrence are an explicit part of the declared emission law. No values of π, φ, e or α are received. The distinction between an emission rule and a constant recognized from its output prevents the genealogy from being hidden in a table of decimals.

If $s_m = [S_m]_{10}$ and $e_m = [E_m]_{10}$, the same rule reads

$$U_m = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10}. \quad (4.3)$$

The final one is the residue of $7Z_m = 21$ modulo ten. The oriented ternary reduction of the catalogue adopts the convention

$$\Psi(U_1, \dots, U_6) = ([-U_1]_3, \dots, [-U_6]_3). \quad (4.4)$$

The sign of this chart is explicit: changing it modifies the words and requires transporting subsequent orientations. The balanced identification of each component of $\mathbb{F}_3$ with the TRIT is made after (4.4).

Proposition 4.1 (Injective factorization of the emission). *The word $U = (U_1, \dots, U_6)$ determines the two signatures $s = (s_1, \dots, s_6)$ and $e = (e_1, \dots, e_6)$. The emission image has $18 \cdot 26 = 468$ elements. The ternary projection has 243 distinct values in the ambient space of 729 words.*

Proof. The hundreds digits of U_m recover s_m . If b_m is its tens digit, then $e_m = [b_m - s_m]_{10}$. The units digit checks the third congruence in (4.3). Thus the coupling is injective on the product of signatures.

Enumerating the 324 conditions of a cursor using the preceding recurrence produces eighteen additive signatures, each with eighteen preimages; and twenty-six multiplicative signatures, of which twenty-four have eight preimages, one has twenty-four, and one has one hundred and eight. The annex preserves these signatures and their initial conditions, allowing the finite enumeration to be repeated without a reference numerical catalogue. Injectivity of the coupling gives $18 \cdot 26$ emissions and multiplicities

$$\underbrace{432}_{18 \cdot 24} \text{ fibers of } 144, \quad 18 \text{ fibers of } 432, \quad 18 \text{ fibers of } 1944.$$

Their sum is $432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976$. Applying (4.4) to the 468 emissions produces exactly 243 words. This last number is the cardinality of the image, not of the space $\mathbb{F}_3^6$. $\square$

This factorization explains the difference between finite exhaustiveness and unlimited depth. The domain (4.1) is complete for the declared support and emitter; it does not already contain every prefix of every prolongation. Depth belongs to the system of compatible extensions constructed over those conditions.

4.2 Calendar closure and memory advancement

The calendar of one hundred and eight transitions consists of twelve nonadic windows. Its group structure is more precise than an informal product of two periods:

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0, \quad \iota([b]) = [9b], \quad p([t]) = [t]_9. \quad (4.5)$$

If $t = 9b + r$ with $0 \leq r < 9$, adding two phases produces the carry

$$(b, r) + (b', r') = \left(b + b' + \left\lfloor \frac{r + r'}{9} \right\rfloor, [r + r']_9 \right),$$

where the first coordinate is reduced modulo twelve. Composition preserves the term linking the two coordinates.

Proposition 4.2 (Nonsplitting of the calendar). *The sequence (4.5) is exact and admits no section that is a group homomorphism.*

Proof. The kernel of reduction modulo nine is the subgroup of multiples of nine, the image of ι . If the extension split, commutativity would give $C_{108} \cong C_{12} \times C_9$. The first group has exponent 108, whereas the second has exponent $\text{mcm}(12, 9) = 36$, a contradiction. $\square$

Nine transitions restore the nonadic phase and advance one window; one hundred and eight restore the finite calendar. Neither fact requires erasing the trajectory or restoring the prefix of a prolongation. The nonadic holonomy Γ_9 denotes the transport of one winding on states with memory, so that

$$g(\Gamma_9 x) = g(x), \quad M(\Gamma_9 x) = M(x) + 1. \quad (4.6)$$

Equality of phase and increment of memory are compatible and constitute two observations of the same transport.

4.2.1 Temporal parametrization of compatibility and tritic orientation

The temporal interpretation of the TRIT is established on transitions of the complete state. Let $x \xrightarrow{\text{adm}} y$ be the admissibility relation determined by the TPK operations on $\hat{X}$. A trajectory parametrized by a discretely ordered set $\mathcal{T}$ is a map

$$\gamma : \mathcal{T} \longrightarrow \hat{X}, \quad \gamma(t) \xrightarrow{\text{adm}} \gamma(t^+),$$

where t^+ is the successor of t . The compatibility relation precedes the choice of parameter: the order expresses its transitions as an evolution. When the term section is used, the projection $p : E \rightarrow \mathcal{T}$ and condition $p \circ \gamma = \text{id}_{\mathcal{T}}$ are also declared. This specifies the state space, temporal order and information transported.

The signature τ selects a quadratic regime and admits a temporal readout relative to that parametrization. Type +1, with $J_{+1}^2 = -I$, represents return and memory; type 0, with $J_0^2 = 0$, represents the transition threshold; type -1, with $J_{-1}^2 = I$, represents opening in two eigendirections. In the model's temporal terminology, these functions correspond, respectively, to past, present and future. The correspondence concerns the operational position of a transition within a history: preserved antecedent, update and admissible continuation.

This interpretation acquires content through three distinct operations. Restricting a trajectory recovers the prefix already constructed. The update $\mathcal{U}_t$ composes selection, transport and registration in the observed transition. Prolongation considers the continuations that preserve the boundary conditions of that prefix. If $\mathcal{P}_n(x)$ denotes the set of admissible continuations of length n from x , deletion of the last step defines

$$r_n : \mathcal{P}_{n+1}(x) \longrightarrow \mathcal{P}_n(x).$$

An unbounded continuation is a collection $(\gamma_n)_{n \geq 1}$ satisfying $r_n(\gamma_{n+1}) = \gamma_n$. The inverse system thus preserves the restrictions that each depth imposes on the preceding ones. Existence of a compatible collection is established in the corresponding prolongation domain; merely setting up the sets $\mathcal{P}_n(x)$ does not replace that proof.

Bidirectionality therefore combines reconstruction of antecedents with compatibility of continuations. On the signature sequence, the involution C_{rev} reverses the order and changes the sign of each component. The trajectory is also acted on by the transformations of position, sheet and boundary specified by transport. The reversed signature is an observation of the conjugate trajectory, not a replacement for its remaining coordinates. The threshold remains distinguished: the visible value 0 retains its transition function even when its quotient or historical register is nonzero.

The relationship with exponential geometries is equally precise. In each fixed regime, $\exp(tJ_\tau)$ composes parameters and preserves the algebraic norm. Elliptic closure belongs to this local representation. The phase return of Γ_9 , by contrast, acts on the trajectory with memory, according to (4.6). A trajectory may recover its phase while retaining a new antecedent: its historical data have changed even though the phase observation is the same. The parabolic threshold retains a linear displacement; hyperbolic opening separates two eigendirections of expansion and contraction. These three realizations provide composition laws, while the TPK determines their selection and effective concatenation.

The aperiodic suspension of Section 5 will make this distinction explicit through a finite phase factor, an irrational rotation and a memory degree. The connection between TRIT and temporality is thus expressed through operations on trajectories and their quadratic representations. Its content goes beyond a classification of signs: it preserves the regime, direction of transport and relationship between antecedent, transition and continuation.

The monodromic representation of this return is developed in Section 5.10. There, the holonomy of the state is distinguished from its representation on phase and counter. The symbolic suspension in Section 5.12 defines the aperiodic time-crystal model, and Section 5.14 composes this temporality with the three Euler realizations.

4.3 Observable compression and orthogonal conservation

An isometric realization makes it possible to quantify what a reduced observation loses. Let $U : \mathcal{H} \rightarrow \mathcal{H}$ be an isometric realization of the transport of one winding, $U^*U = I$, and $J : \mathcal{H}_{\text{vis}} \rightarrow \mathcal{H}$ an isometric inclusion. Define

$$T = J^*UJ, \quad D = (I - JJ^*)UJ.$$

Then

$$UJ = JT + D, \quad J^*D = 0, \quad I - T^*T = D^*D. \quad (4.7)$$

The last equality follows by taking the adjoint product of the first and using orthogonality. Contraction of the visible component does not by itself constitute destruction of information in the total state.

Proposition 4.3 (Finite-horizon conservation). *If $I - T^*T = D^*D$, then, for every $n \geq 1$,*

$$I = \sum_{r=0}^{n-1} T^{*r}D^*DT^r + T^{*n}T^n. \quad (4.8)$$

Proof. Multiplying $D^*D = I - T^*T$ by T^{*r} and T^r and summing produces a telescoping sum. The interior terms cancel, leaving $I - T^{*n}T^n$. $\square$

The last summand preserves the terminal state. Removing it before justifying a limit would alter the equality. This distinction is the operational content of preservation of the unit under compression: the norm is distributed among the visible component, memory, and terminal, without being replaced by a single residue.

4.4 Local development: commutation, Lorentzian signature, and discrete curvature

The adjacent diagonal block with equal diagonal entries modulo nine is unique. Indeed,

$$k^2 \equiv (k+1)^2 \pmod{9} \iff 2k+1 \equiv 0 \pmod{9}$$

gives $k = 4$. Its visible matrix is

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix} = 7I + 2R, \quad R = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad R^2 = I.$$

With $P_{\pm} = (I \pm R)/2$, one obtains $B_c = 9P_+ + 5P_-$. The unit of stochastic normalization and preservation of an indefinite form are two realizations of this block:

$$\frac{B_c}{9} = P_+ + \frac{5}{9}P_-, \quad \frac{B_c}{\sqrt{45}} = \exp(\eta R), \quad \eta = \frac{1}{2} \log \frac{9}{5}.$$

For $Q = \text{diag}(1, -1)$, the second matrix satisfies $B_c^T Q B_c = 45Q$. This gives a local realization in $SO^+(1, 1)$; no prior introduction of a four-dimensional spacetime is needed to prove it.

Uniform averaging over the classes $C_1 = \{1, 4, 7\}$, $C_2 = \{2, 5, 8\}$, $C_0 = \{3, 6, 9\}$ gives

$$M_{\Pi} = \begin{pmatrix} 4 & 5 & 6 \\ 5 & 4 & 6 \\ 6 & 6 & 9 \end{pmatrix}, \quad M_{\Sigma} = \begin{pmatrix} 5 & 6 & 4 \\ 6 & 4 & 5 \\ 4 & 5 & 6 \end{pmatrix}.$$

The characteristic polynomial of M_{Π} is

$$(\lambda + 1)((\lambda - 9)^2 - 72),$$

so its quadratic form has signature $(2, 1)$. The local spectral gap $9 - 5 = 4$, the coefficient 2, the aggregate difference $51 - 45 = 6$, and the integrated defect $9 \cdot 6 = 54$ describe different levels of the same arithmetic comparison. They are not interchangeable indices of the same operator.

Discrete curvature can be described directly before this spectral realization. In residues, let $S(x, y) = x + y$, $P(x, y) = xy$, and Δ_x, Δ_y be the forward differences. The link variables are potential differences, $A_x^T = \Delta_x T$, $A_y^T = \Delta_y T$, and we set

$$F(T_x, T_y) = \Delta_x \Delta_y T_y - \Delta_y \Delta_x T_x.$$

Since $\Delta_x S = \Delta_y S = 1$, $\Delta_x P = y$, and $\Delta_y P = x$,

$$F(S, S) = F(P, P) = 0, \quad F(S, P) = +1, \quad F(P, S) = -1 \pmod{9}. \quad (4.9)$$

Sum and product, taken separately, induce flat connections; their ordered mixing changes the sign of plaquette curvature. Summing over a rectangle with sides a, b , the interior edges cancel and the oriented boundary sum equals $\pm ab$ modulo nine. An exact relation thus appears between composition order and boundary orientation.

4.5 Cubic completion and two notions of boundary

Enlarging the cubic support of side nine to that of side ten admits an exact positional stratification:

$$10^3 = 9^3 + 3 \cdot 9^2 + 3 \cdot 9 + 1 = 729 + 243 + 27 + 1. \quad (4.10)$$

The first term corresponds to three interior coordinates; the second to one new coordinate; the third to two; and the last to the vertex with three new coordinates. The enlargement shell contains 271 positions. By contrast, the internal boundary of the discrete cube of side nine contains $9^3 - 7^3 = 386$. The difference between these counts is a difference of domain, not a conflict of values.

Dividing (4.10) by 1000 gives a normalized partition of unity. Cube incidence distinguishes six faces, twelve edges, and eight vertices. The subsequent link with Steiner hexads and octads requires the corresponding combinatorial maps; the cardinalities of the cube do not replace them. This completion here provides the relation between the bases 729 and 1000, while the exceptional prolongation will be developed after numerical generation and the electron.

4.5.1 Stratification of the completion and restoration of the apex

The completion admits a set-theoretic description that makes its operation explicit. Let E_9 be the set of the nine residue classes, represented positively by $1, \dots, 9$, and let $*$ be an additional element. The symbol $*$ differs from the zero class, already represented by 9. Consequently, enlarging E_9 to $\hat{E} = E_9 \sqcup \{*\}$ increases the cardinality from nine to ten without modifying the internal modular identification. The cubic completion is $\hat{C} = \hat{E}^3$, and the initial cube is $C = E_9^3$.

Definition 4.4 (Strata of combinatorial codimension). For $0 \leq r \leq 3$, define

$$S_r = \{(x_1, x_2, x_3) \in \hat{E}^3 : \#\{j : x_j = *\} = r\}.$$

The apex of the completion is $a_* = (*, *, *)$, and the punctured completion is $\hat{C}^\circ = \hat{C} \setminus \{a_*\}$.

Proposition 4.5 (Exact decomposition of the completion). *The strata are disjoint and satisfy*

$$\hat{C} = S_0 \sqcup S_1 \sqcup S_2 \sqcup S_3, \quad |S_r| = \binom{3}{r} 9^{3-r}, \quad (4.11)$$

$$|\hat{C}^\circ| = 729 + 243 + 27 = 999, \quad |\hat{C}| = 999 + 1 = 1000. \quad (4.12)$$

The punctured enlargement contains 270 elements in addition to the initial cube; the full enlargement contains 271.

Proof. An element of S_r is obtained by choosing the r coordinates containing $*$, in $\binom{3}{r}$ ways, and assigning to each remaining coordinate one of the nine classes of E_9 . Classification by the number of additional coordinates is exhaustive, and its classes are disjoint. The only element of S_3 is a_* . Removing it subtracts one from the total cardinality; restoring it recovers the complete Cartesian product. $\square$

The expression *holographic extreme and mean ratio*, abbreviated EMRH in the corpus, here denotes this articulation between the initial body, the enlargement strata, and unit restoration. Its immediate content is the exact partition of a finite product and its normalization. The number 999 represents a definite object, the punctured completion; 1000 represents its closure by restoration of the apex. The added unit therefore has an explicit incidence location.

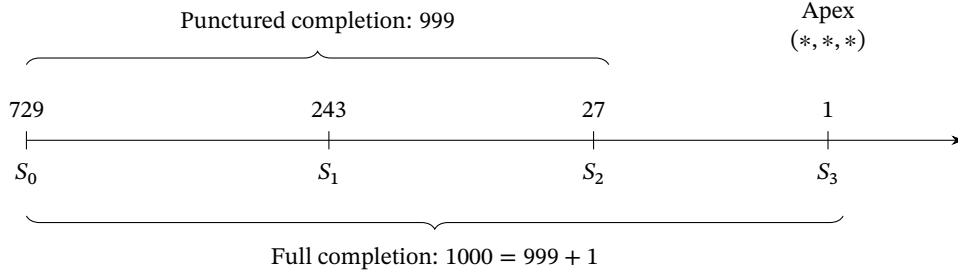


Figure 3: Strata of the cubic completion. The layout represents the set-theoretic decomposition, not a metric scale. The 243 elements of S_1 form three coordinate strata of 81 elements; the 27 elements of S_2 form three strata of 9. The apex completes the partition.

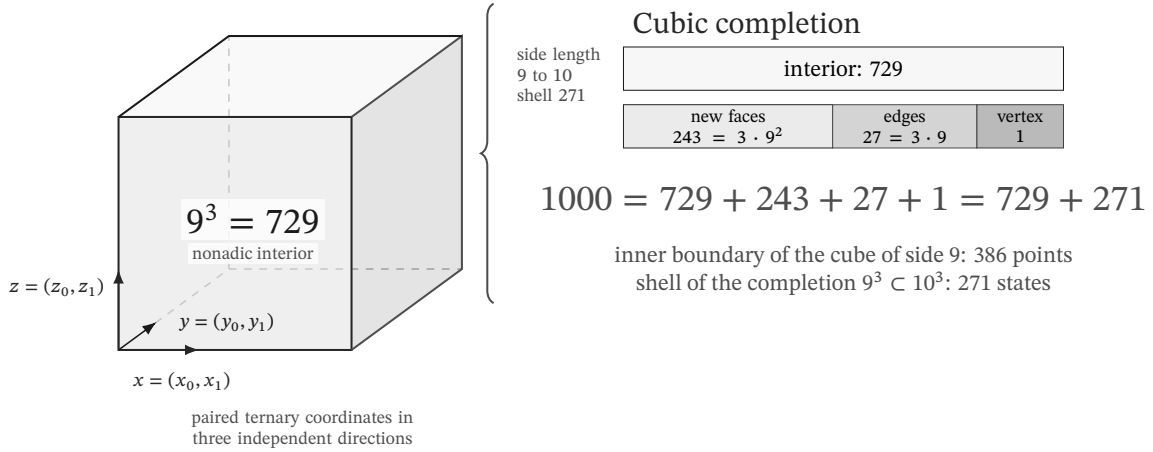


Figure 4: Cubic representation of the expansion from nine to ten classes per coordinate. Strata S_1, S_2, S_3 locate the added incidences on faces, edges, and the apex; their cardinalities form the 271-element shell. The 386-element boundary belongs to the initial cube. This geometric perspective complements the partition in Figure 3; graphical spacing is schematic.

4.5.2 Preservation of unity and dimensional compatibility

Normalization is not limited to dividing an isolated identity. In dimension d , the product $\hat{E}^d$ carries the uniform measure $\mu_d(\{x\}) = 10^{-d}$. The coordinate projection $p_d : \hat{E}^{d+1} \rightarrow \hat{E}^d$ removes the last coordinate.

Proposition 4.6 (Projective compatibility of the normalized measures). *For every $d \geq 1$,*

$$(p_d)_* \mu_{d+1} = \mu_d, \quad \mu_d(\hat{E}^d) = 1.$$

The mass of the stratum with r additional coordinates is $\binom{d}{r} 9^{d-r} / 10^d$, and the sum of the masses of all strata is one.

Proof. Each point of $\hat{E}^d$ has ten preimages. Their total mass is $10 \cdot 10^{-(d+1)} = 10^{-d}$. Equality for any subset follows by summing over its points. The stratified formula follows from the same count as (4.11); the binomial $(9 + 1)^d$ gives the total normalization. $\square$

In dimension three, removing the apex leaves mass 999/1000, and restoring it adds 1/1000. Renormalizing before restoration would define another measure: the distinction is operationally relevant. In particular, preservation of probability mass, preservation of transport registers, and thermodynamic dissipation are statements about different objects. The proposition proves the first and provides the compatible support for studying the second; the third requires a specified dynamical and energetic realization.

The internal geometric boundary also remains distinct. On the Cartesian representative $\{1, \dots, 9\}^3$, removing $\{2, \dots, 8\}^3$ leaves 386 points. This boundary belongs to the initial cube, whereas $\hat{C} \setminus C$ belongs to its enlargement. The cubic completion, the positional base-thousand expansion, and the exceptional incidence readout share support data; their respective maps determine which information from those data each representation uses.

4.6 Cubic incidence and Lorentzian realization

Cubic completion provides a relation between interior and boundary. Its geometry can be developed through the incidence of the six faces with the eight vertices. Label the faces by (i, ε) , $i \in \{1, 2, 3\}$, $\varepsilon \in \{+1, -1\}$, and the vertices by $\nu \in \{+1, -1\}^3$. The incidence matrix is

$$M_{(i,\varepsilon),\nu} = \mathbf{1}_{\{\nu_i=\varepsilon\}}.$$

Let $u = (1, \dots, 1)^\top \in \mathbb{R}^6$, let R_F be face opposition, and let $P_u = uu^\top/6$, $P_- = (I - R_F)/2$.

Proposition 4.7 (Four-dimensional incidence quotient). *The incidence Gram matrix satisfies*

$$MM^\top = 12P_u + 4P_-.$$

Its eigenvalues 12, 4, 0 have multiplicities 1, 3, 2. Consequently,

$$Q_\square = \mathbb{R}^6 / \ker(MM^\top) \simeq \mathbb{R}u \oplus E_-, \quad E_- = \text{im } P_-.$$

Proof. A face contains four vertices; two opposite faces have empty intersection, and two distinct nonopposite faces share two vertices. The matrix $12P_u + 4P_-$ has precisely these entries. The uniform space is one-dimensional; differences between opposite faces generate a three-dimensional space orthogonal to the uniform space. Its complement is two-dimensional and belongs to the kernel. $\square$

The decomposition $1 + 3$ comes from incidence. The temporal choice is declared through the bilinear form

$$B_\square([x], [y]) = \langle x, (P_- - P_u)y \rangle.$$

It is well defined on the quotient because both projectors annihilate the kernel. In the basis

$$t = [u]/\sqrt{6}, \quad d_i = [e_{i,+} - e_{i,-}]/\sqrt{2},$$

its matrix is $\text{diag}(-1, 1, 1, 1)$. This makes explicit the domain, decomposition and sign convention producing the Minkowski realization. The positive Gram matrix determines the quotient; the temporal assignment determines the indefinite form on it.

The operator

$$W_\square = \sqrt{6}P_u + \sqrt{2}P_-, \quad W_\square e_{i,\varepsilon} = t + \varepsilon d_i$$

sends the six face labels to null directions, since $B_\square(t + \varepsilon d_i, t + \varepsilon d_i) = -1 + 1 = 0$. Reversal of a direction and face opposition thus have a concrete geometric realization.

4.6.1 Soldering, reversal and subdivision

The cellular realization uses an invertible transport $U_m(a)$, preserving the Lorentzian form, between the fibres at the endpoints of an oriented dual edge a , a face label $\lambda_m(a)$ and an orientation $\sigma_m(a)$. With $\ell_m = \ell_0/9^m$, the soldering map is

$$\beta_m(a) = \ell_m \sigma_m(a) W_{\square, s(a)} e_{\lambda_m(a)}.$$

Compatible reversal satisfies

$$\beta_m(\bar{a}) = -U_m(a) \beta_m(a).$$

Identify the source fibres at two consecutive levels through the declared refinement isometry. If the nine subsegments $a_1, \dots, a_9$, at level $m+1$, preserve the label and orientation when transported to that common fibre, then

$$\beta_m(a) = \sum_{j=1}^9 U_{m+1}(a_1 \cdots a_{j-1})^{-1} \beta_{m+1}(a_j). \quad (4.13)$$

Each transported term equals $\ell_m/9$ times the same null direction, proving the identity. The compatibility hypothesis determines which subdivisions realize this refinement. The equation brings together direction, scale and transport, instead of limiting the geometric correspondence to a dimension count.

With orientation and the Lorentzian form also fixed, Hodge duality on $\Lambda^2 Q_{\square}^*$ satisfies $\star^2 = -I$. In dimension four the sign is $(-1)^{2(4-2)+1} = -1$. This yields a real complex structure on two-forms and, after complexification, eigenspaces with eigenvalues $+i$ and $-i$. The electromagnetic realization will use this domain after constructing the electronic section and declaring its coupling.

4.7 Compatibility between modular and positional reductions

The blockwise decimal base enters through a Euclidean division that simultaneously preserves remainder and quotient:

$$n = 1000q + r, \quad 0 \leq r < 1000.$$

Since $1000 \equiv 1 \pmod{9}$ and $1000 \equiv 1 \pmod{3}$,

$$n \equiv q + r \pmod{9}, \quad n \equiv q + r \pmod{3}.$$

Thus, reducing a block requires the quotient's contribution to recover the integer's class. The numbers 0 and 1000 have the same remainder modulo 1000, but different classes modulo 9 and modulo 3. This is an operational reason to preserve memory during a change of resolution.

Ternary refinement and determination of a decimal prefix can be coordinated through a single integer residual. If $N/3^L$ is a cylinder endpoint and $D/1000^r$ a decimal endpoint, define

$$A = 1000^r N - D 3^L.$$

On adjoining six ternary symbols, $N' = 729N + \nu$, $L' = L + 6$, with $0 \leq \nu < 729$. If the decimal prefix remains fixed, one obtains

$$A' = 729A + \nu 1000^r.$$

If a decimal block δ is also incorporated, then $D' = 1000D + \delta$, $r' = r + 1$, and

$$A' = 729000A + \nu 1000^{r+1} - 729\delta 3^L. \quad (4.14)$$

Both identities follow by substituting the definitions. Endpoint comparison is thus reduced to integer operations that retain each prolongation's contribution. This compatibility allows a region to be refined before its real coordinate is evaluated.

4.8 Number as a compatible section and value as a readout

The distinction between state and observation makes it possible to specify which object is prolonged as resolution increases. At depth n , let X_n be the set of admissible states produced by APP-TRIT-TPK. Their data include position, sum and product sheets, residues and quotients, regime, orientation, carry, phase, route, boundary, memory, and survival. They are not independent coordinates: arithmetic evaluations and previous transports impose their relations. A prolongation preserves these relations in every predecessor, even as it adds new steps and new memory data. This is the meaning of the arithmetic of histories developed in (see 4.9; 4.10).

Let $r_{m,n} : X_m \rightarrow X_n$, for $m \geq n$, be the depth restrictions, with $r_{n,n} = \text{id}$ and $r_{\ell,n} = r_{m,n} \circ r_{\ell,m}$. Each restriction preserves the prefix and the information already determined within it.

Definition 4.8 (Compatible numerical section). An HMT number is a compatible history of the state system:

$$x = (x_n)_{n \geq 0} \in X_\infty := \varprojlim_n X_n, \quad r_{m,n}(x_m) = x_n. \quad (4.15)$$

A readout is an additional operation on a specified domain of these histories. A digit is a finite observation; a scalar value is an evaluation of the readout. The pair consisting of the section and its readout constitutes a read presentation, not a redefinition of the section's identity.

On the Archimedean domain X_∞^{Arch} , the readout associates a closed cylinder with each prefix, preserves their nesting, and makes their diameters tend to zero. It thus determines

$$\text{ev}_\mathbb{R} : X_\infty^{\text{Arch}} \longrightarrow \mathbb{R}, \quad \{\text{ev}_\mathbb{R}(x)\} = \bigcap_n I_n(x). \quad (4.16)$$

The concrete construction of these cylinders and the decision of their boundaries are developed in Section 5. Definition (4.15) does not by itself assert that the image of (4.16) is the whole line: a coverage assertion requires its own theorem. Nor does equality of values establish, without an injectivity result, equality of their histories. This separation allows precision to be understood as depth while preserving the difference between the arithmetic predecessor and its scalar realization.

4.9 Composition of histories and memory multiplier

Fix a sector of states at depth n and a category $\mathcal{C}$ whose arrows are their admissible histories. Composition is written $vu = v \circ u$: first traverse u , then v . Elementary steps arise from TPK selection, transport, and update. A local trit fiber admits the notation $\mathbb{R}[J_x]/(J_x^2 + \tau(x)1_x)$, where $\tau(x) \in \{+1, 0, -1\}$. Regime and orientation belong to the transported state; a transition between regimes does not amount to identifying their quadratic algebras.

The following formulation specifies an algebraic sector of that composition. For each object x , a unital associative real algebra A_x is given, possibly noncommutative. For each $u : x \rightarrow y$, a unital homomorphism $\rho_u : A_x \rightarrow A_y$ is given. Assume a strict action:

$$\rho_{\text{id}_x} = \text{id}_{A_x}, \quad \rho_v \rho_u = \rho_{vu}. \quad (4.17)$$

For each composable pair $u : x \rightarrow y, v : y \rightarrow z$, fix an invertible central multiplier $\Omega(v, u) \in Z(A_z)^\times$. Require the normalization $\Omega(\text{id}_y, u) = \Omega(u, \text{id}_x) = 1_y$ and, for each composable triple, the identity

$$\rho_w(\Omega(v, u))\Omega(w, vu) = \Omega(w, v)\Omega(wv, u). \quad (4.18)$$

These are explicit hypotheses on the transports and their coefficients, not a consequence of calling the dynamics holonomic. When this presentation is applied to a TPK sector, its ρ_u and $\Omega(v, u)$ must be identified. Transitions expressed through bimodules require preserving that structure and are not replaced by (4.17).

The space of finite sums

$$\mathcal{H}(\mathcal{C}, A, \rho, \Omega) = \bigoplus_{u \in \text{Mor}(\mathcal{C})} A_{I(u)} T_u$$

receives the bilinear product

$$(aT_v)(bT_u) = a\rho_v(b)\Omega(v, u)T_{vu}, \quad (4.19)$$

if the arrows are composable, and zero otherwise. Here $t(u)$ denotes the terminal state. The sum gathers histories with coefficients; the product concatenates histories while preserving the register multiplier.

Proposition 4.9 (Sectorial associativity with a central multiplier). *Under (4.17) and (4.18), the product (4.19) is associative. If the set of objects is finite, its unit is $\sum_x 1_x T_{\text{id}_x}$.*

Proof. For $u : x \rightarrow y, v : y \rightarrow z, w : z \rightarrow t$, let $a \in A_y, b \in A_z, c \in A_t$. The two coefficients of T_{wvu} are

$$\begin{aligned} ((cT_w)(bT_v))(aT_u) &: c\rho_w(b)\Omega(w, v)\rho_{wv}(a)\Omega(wv, u), \\ (cT_w)((bT_v)(aT_u)) &: c\rho_w(b)\rho_w\rho_v(a)\rho_w(\Omega(v, u))\Omega(w, vu). \end{aligned}$$

The strict action identifies $\rho_w\rho_v(a)$ with $\rho_{wv}(a)$. Centrality of $\Omega(w, v)$ allows it to be moved to the right of that factor; (4.18) then equates the remaining products. The coefficients a, b, c are not permuted. If the triple is not composable, both parenthesizations are zero by their initial and terminal states. Bilinearity completes the proof. Local identities and the normalization of Ω verify the stated unit. $\square$

Calendar carry provides an effective example. In the category with one object and arrows C_9 , take coefficients $A = \mathbb{R}[z, z^{-1}]$, the identity action, and

$$\Omega(b, a) = z^{c_0(a, b)}, \quad c_0(a, b) = \left\lfloor \frac{a+b}{9} \right\rfloor, \quad 0 \leq a, b < 9.$$

Both carry sums of a triple equal $\lfloor (a+b+c)/9 \rfloor$; hence (4.18) holds. The formal variable z records the winding without assigning it a numerical value. Subsequently imposing $z^{12} = 1$ recovers the dodecaphasic register of the calendar (4.5); retaining z without that relation distinguishes successive windings. The example realizes the memory of this calendar without identifying it with all TPK memory.

4.10 Double variance of refinement and preservation of the unit

Restricting states prolongs observables in the opposite direction. Consider the function algebras, with pointwise product, $D_n = \text{Fun}(X_n, \mathbb{R})$. Precomposition defines

$$r_{m,n}^* : D_n \longrightarrow D_m, \quad r_{m,n}^* f = f \circ r_{m,n}. \quad (4.20)$$

These are unital morphisms; they are also injective when the corresponding restrictions are surjective. The direct limit $D_{\text{cyl}} = \varinjlim_n (D_n, r_{n+1,n}^*)$ gathers finite-depth observations. This algebra is not identified here with the entire route algebra: prolonging transport operators also requires their compatibilities with refinement.

Lemma 4.10 (Unit and naturality of evaluation). *If e_x is the indicator of $x \in X_n$, then*

$$r_{m,n}^* e_x = \sum_{y: r_{m,n}(y)=x} e_y, \quad r_{m,n}^* 1_n = 1_m, \quad (4.21)$$

and, for every $f \in D_n, y \in X_m$,

$$\langle r_{m,n}^* f, y \rangle = \langle f, r_{m,n} y \rangle.$$

Each section $x \in X_\infty$ determines a well-defined evaluation $D_{\text{cyl}} \rightarrow \mathbb{R}$, given by $[f] \mapsto f(x_n)$ for $f \in D_n$.

Proof. The fibers of $r_{m,n}$ partition X_m . Their indicators are orthogonal and sum to 1_m , proving (4.21). Naturality is the definition of precomposition. Finally, $(r_{m,n}^* f)(x_m) = f(x_n)$, so two compatible representatives give the same evaluation in the direct limit. $\square$

The global unit is the constant-one function on all possibilities; the individual section selects a compatible history. Preserving the former does not mean immobilizing the latter. Refinement can increase the number of extensions and the depth of each history while maintaining the unit and all evaluations of its predecessors.

4.11 Phase return, Weyl representation, and two readouts

The law (4.6) acquires an operator realization in the bilateral suspension to be constructed in Section 5. There, T is one-step advancement, V_9 observes the nonadic phase, W_δ the rotation phase, and D_M the memory degree. Writing $\hat{\Gamma}_9 = T^9$ for the winding in this representation, the Weyl relations (5.17) imply, on finitely supported vectors,

$$\begin{aligned} V_9 \hat{\Gamma}_9 &= \hat{\Gamma}_9 V_9, & W_\delta \hat{\Gamma}_9 &= e^{2\pi i 9\delta} \hat{\Gamma}_9 W_\delta, \\ [D_M, \hat{\Gamma}_9] &= \hat{\Gamma}_9. \end{aligned} \tag{4.22}$$

The first equality follows from $\zeta_9^9 = 1$; the second from nine applications of the covariance of W_δ ; the third expresses the unit memory increment. The finite phase returns, while irrational rotation and memory distinguish the next state. The characters and parameter δ are subsequently realized from the numerical construction specified there: they do not retrospectively select the seeds. This representation describes observables and translations of the suspension; it does not replace all coordinates or all operators of the TPK.

The double readout prolongs the same distinction between history and observable. On its Archimedean domain, (4.16) is applied; on the calibrated incidence domain, the maps Q_n in Section 10.1 preserve words and boundary frames. The equality proved there, $r_{n,m} Q_n = Q_m p_{n,m}$, determines their prolongation Q_∞ . On the common domain of both readouts one may jointly consider $\text{ev}_\mathbb{R}(x)$ and $Q_\infty(x)$: the former retains the value; the latter, the transported incidence that its codomain preserves. The section precedes both. This articulation permits passing from the arithmetic of histories to constants and the exceptional readout without turning their realizations into new initial data.

5 Coinductive generation of π, φ, e to arbitrary depth

The six-block emission is a finite stage of the construction, not a complete decimal expansion. Its role is to produce regions with multiplicity, orientation, and transport rules. Prolongation must preserve these data and determine a compatible collection of prefixes of arbitrary length. The term *generation* denotes this operation; *genealogy* denotes the reconstruction of its dependencies. Both notions are involved, but they serve different purposes.

5.1 Orbits, regions, and characters of closure, propagation, and self-scaling

For a word $w = (w_1, \dots, w_6)$, define

$$r(w) = (w_3, w_1, w_2, w_5, w_6, w_4), \quad s(w) = (w_4, w_5, w_6, w_1, w_2, w_3).$$

We have $r^3 = s^2 = I$ and $srs = r^{-1}$; these permutations therefore realize the dihedral group D_3 . The action preserves the catalogue and the multiplicities and commutes with Ψ . The quotient of the 243 visible words has 43 orbits: thirty-eight of cardinality six and five of cardinality three.

The closure class is recognized by a fibre predicate: its six orientations have five emissions per word, total multiplicity 1008, and distribution $432 + 4 \cdot 144$. There is exactly one orbit with these properties in the catalogue. To specify the other two selections, write $f(w) = \#\Psi^{-1}(w)$, $m(w) = \sum_{U \in \Psi^{-1}(w)} \text{mult}(U)$, and $\nu(w) = (n_0, n_1, n_2)$. The additive diagonal class of an emission is $d(U) = [i + j]_9$ in indices from zero to eight; its additive signature retains this class and the orientation *ES* or *NO*. Let $\mathcal{D}(\mathcal{O})$ be the set of these classes for all emissions over an orbit.

Among the orbits of size six, the joint predicate

$$f(w) = 1, \quad m(w) = 144 \quad (w \in \mathcal{O}), \quad \mathcal{D}(\mathcal{O}) = \{0, 3, 6\}$$

leaves exactly six orbits. In this sector, the census $\nu = (2, 3, 1)$, the same as for closure, uniquely selects propagation; the census $\nu = (2, 2, 2)$ uniquely selects self-scaling. The diagonal support cannot be omitted: without it there are four balanced single-preimage orbits of multiplicity 144, not one. Finally, the existence of an emission with $d = 6$ and orientation *ES* selects one representative of each orbit and yields

$$w^{\text{cl}} = 010211, \quad w^{\text{pr}} = 201101, \quad w^{\text{au}} = 121200. \quad (5.1)$$

These symbols first designate regions of the catalogue. Scalar identification will be established through their readouts; they are not chosen because they coincide with digits of a known constant.

The closure word 010211 has the following five preimages:

Six-block emission	Multiplicative signature	Role	Multiplicity
501, 614, 498, 169, 272, 272	555555	transverse	432
810, 923, 870, 169, 272, 272	339555	positive orientation	144
810, 983, 810, 169, 272, 272	393555	positive orientation	144
870, 923, 810, 169, 272, 272	933555	positive orientation	144
870, 923, 810, 418, 674, 521	933717	oriented return	144

The multiplicity 432 and the constant signature 555555 characterize the transverse region. They do not identify the central position (5, 5) of APP. Confusing a six-window signature with a position in the support would erase the distinction between numerical region and electronic structure.

The microfibre also has a precise bipartite morphology. Each emission is a product of an additive cursor class and a multiplicative class. The three positive regions share the same additive class of eighteen conditions; together they contain three multiplicative classes of eight conditions each. The transverse region has a product 18×24 ; the three positive regions taken together, another 18×24 ; and the return, a product 18×8 . The decomposition by emissions has five parts, whereas the decomposition into connected components of the bipartite graph has three. Both preserve the same 1008 seeds without reducing form to a cardinality.

5.2 Finite transport and the prolongation boundary

The endomorphisms L_0, L_1 of the ternary word space act before the Archimedean realization. Their determination begins with the transitions of the arithmetic state, not with the prescription of two matrices. If U_t is the composite step of selection, transport, and update, the observed block satisfies

$$b_j^\chi = \Pi_6(x_j^\chi), \quad x_{j+1}^\chi = U_{9j+9} \cdots U_{9j+1}(x_j^\chi), \quad \chi \in \{\text{cl}, \text{pr}, \text{au}\}.$$

Six transitions are formed for each regime: two consecutive transitions in each of the three regions. The pairs of source and target matrices obtained in the record are

X_0	Y_0	X_1	Y_1
010211	012222	010211	002111
012222	010211	002111	110221
201101	121221	102011	012222
121221	102011	012222	102011
121200	112202	121020	010210
112202	121020	010210	010200

Each string of six digits represents a row in $\mathbb{F}_3^6$. Since $\det X_0 = 1$ and $\det X_1 = -1$ in $\mathbb{F}_3$, the equations $X_a L_a = Y_a$ uniquely determine $L_a = X_a^{-1} Y_a$. In this row-vector chart, they yield

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix} \pmod{3}.$$

The schedule (L_0, L_0, L_1, L_1) produces four new blocks of six components. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate accumulated length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace the production of those transitions by U_t . The provenance of their record and the examination of the sixteen binary schedules are documented in the technical supplement. The indicated schedule is the only one of these sixteen that jointly reproduces the three sequences of five blocks.

5.3 Five closure regions and angular compensation

The closure region contains one transverse component and four accompanying components. The symmetric readout acts on the space of its five positions through

$$P_5 = \frac{1}{5} \mathbf{1} \mathbf{1}^\top, \quad \mathbf{1} = (1, 1, 1, 1, 1)^\top.$$

Invariance $P_5 \lambda = \lambda$ and conservation $\mathbf{1}^\top \lambda = 1$ force $\lambda = \mathbf{1}/5$. This normalization distributes the unit among positions of the readout; it does not replace the seed multiplicities 432, 144, 144, 144, 144 with uniform frequencies. The elliptic chart of the readout realizes each coordinate through $1 + \lambda_j J$; its slope is λ_j , so the primitive slope is $1/5$. This makes explicit the rule taking a normalized coordinate to a slope, rather than identifying the two notions without a map. The readout composes the four accompanying positions by the law

$$a \oplus b = \frac{a + b}{1 - ab}. \quad (5.2)$$

This law is obtained by multiplying $(1 + aJ)(1 + bJ)$ in the regime $J^2 = -I$ and dividing by its scalar component. The composition of slopes is therefore a realization of the quadratic structure of TRIT, not an ordinary sum of four fractions.

Proposition 5.1 (Closure compensator). *The composition of four slopes $1/5$ is $120/119$. The condition of return to slope one determines a unique positive compensator of the form $1/q$, with $q > 1$, and yields $q = 239$.*

Proof. The first doubling gives $(1/5) \oplus (1/5) = 5/12$; the second, $(5/12) \oplus (5/12) = 120/119$. The condition

$$\frac{120/119 - 1/q}{1 + (120/119)/q} = 1$$

is equivalent to $(120 - 119)q = 120 + 119$, hence $q = 239$. $\square$

The resulting closure character has the realization

$$\Pi_{\text{cl}} = 4 \left(4 \arctan \frac{1}{5} - \arctan \frac{1}{239} \right). \quad (5.3)$$

Machin's angular identity recognizes $\Pi_{\text{cl}} = \pi$. The direction of the construction is precise: the slope and the number of compositions are specified by the pentafibre readout; the compensation equation forces 239. Once this character has been produced, its evaluation by alternating series does not select the seed orbit anew.

For $0 < x < 1$, the alternating sums

$$A_n(x) = \sum_{j=0}^n \frac{(-1)^j x^{2j+1}}{2j+1}$$

enclose $\arctan x$ between two consecutive truncations, with difference $x^{2n+3}/(2n+3)$. Applied to the two slopes in (5.3), they provide rational endpoints with explicit error bounds. The numerical value is thus calculated as a consequence of the exact character, without introducing a target prefix.

5.4 Propagation readout and normalization of the unit

The propagation region is realized by a formal collection $P_t \in \mathcal{A}[[t]]$, where $\mathcal{A}$ is a unital commutative algebra over $\mathbb{Q}$. Its composition satisfies $P_{t+s} = P_t P_s$, with unit $P_0 = 1$, and the oriented elementary transport fixes the scalar generator $P'_0 = +1$. The condition fixes its sign, not merely its magnitude. Writing

$$P_t = \sum_{n \geq 0} a_n t^n,$$

the normalization determines $a_0 = a_1 = 1$. Comparing the coefficient of $s^n t$ in $P_{s+t} = P_s P_t$ gives $(n+1)a_{n+1} = a_n a_1$; equivalently, $P'_t = P_t$. Thus,

$$(n+1)a_{n+1} = a_n, \quad a_n = \frac{1}{n!}. \quad (5.4)$$

The character at one unit of the parameter is $\Pi_{\text{pr}} = P_1$, subsequently recognized as e .

Normalization of the generator matters. Without this datum, the semigroup law would admit $P_t = e^{ct}$ for different c ; the condition fixing the unit of the parameter in the readout is therefore not omitted. This condition precedes decimal comparison and belongs to the exposition of the operational rule, not to the choice of an experimental value.

Proposition 5.2 (Rational propagation bounds). *For $n \geq 1$, let $E_n = \sum_{j=0}^n 1/j!$. Then*

$$E_n < \Pi_{\text{pr}} < E_n + \frac{n+2}{(n+1)(n+1)!}.$$

Proof. The first omitted term is $1/(n+1)!$. Each subsequent ratio is at most $1/(n+2)$. The geometric upper bound for the tail is

$$\frac{1}{(n+1)!} \frac{1}{1 - 1/(n+2)} = \frac{n+2}{(n+1)(n+1)!}.$$

Strict inequality follows from the decrease of the successive ratios. $\square$

5.5 Modal incidence and generation of self-scaling

The tritic selector produces the cycle $(+1, -1, 0)$. On the modes Σ, Π , the rule is $+1 : m \mapsto \Sigma$, $-1 : m \mapsto \Pi$, $0 : m \mapsto m$. With cyclic closure, the observed mode is (Σ, Π, Π) . Its edges are $\Sigma \rightarrow \Pi$, $\Pi \rightarrow \Pi$, and $\Pi \rightarrow \Sigma$. Its incidence, in this ordering of the modes, is therefore

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (5.5)$$

The four entries are obtained from the schedule, not from an equation that has already received the value of φ . Repeating the schedule gives counts $3F_{\text{av}}, 9F_{\text{av}}, 36F_{\text{av}}$ at lengths nine, twenty-seven, and one hundred and eight. These counts are not powers of the matrix.

Theorem 5.3 (Positive self-scaling character). *The incidence (5.5) has a unique strictly positive eigenray. When normalized as $(1, x)^T$, its coordinate x satisfies $x^2 - x - 1 = 0$, $x > 0$, and equals $\varphi = (1 + \sqrt{5})/2$.*

Proof. The eigenvalue equation gives $x = \lambda$ and $1 + x = \lambda x$. Eliminating λ yields the polynomial. Only one of its two roots is positive. Since all entries of F_{av}^2 are positive, the positive ray is unique and simple. The radical expression recognizes the coordinate obtained; it does not participate in the production of F_{av} . $\square$

With $F_0 = 0, F_1 = 1, F_{n+1} = F_n + F_{n-1}$, the incidence induces, for $n \geq 1$,

$$F_{\text{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}.$$

The consecutive ratios F_{n+1}/F_n alternate around φ . The determinant identity

$$F_{n+1}^2 - F_n F_{n+2} = (-1)^n$$

gives an exact width $1/(F_n F_{n+1})$ between two consecutive ratios. Their growth makes this rational bracket arbitrarily narrow.

The measure on all compatible histories does not coincide with the mode frequency of the three-event schedule. The chronological frequency is $(1/3, 2/3)$. The measure of maximal entropy of the symbolic envelope defined by F_{av} is constructed from its eigenvectors and has weight ratio φ^{-2} . This latter measure enters the areal–volumetric readout; it is not obtained by simply replacing one counting frequency with another.

5.5.1 Trajectory space and invariant measure between sheets

The toroidal topology of APP fixes the support of the cardinal translations and their winding classes. The correspondence between sheets arises from another operation on the same support: temporal selection of the additive or multiplicative evaluation. Its geometric typing is

$$\eta_{\text{av}}(\Sigma) = \mathcal{H}_{\text{ar}}, \quad \eta_{\text{av}}(\Pi) = \mathcal{H}_{\text{vol}}.$$

The first sheet corresponds to the additive boundary readout; the second, to the multiplicative interior readout. This map transports the mode labels and preserves the incidence (5.5). The square of the self-scaling ratio is determined through the trajectories of that incidence.

Definition 5.4 (One-step-memory symbolic envelope). Let

$$X_{\text{av}} = \{(s_k)_{k \in \mathbb{Z}} \in \{\mathcal{H}_{\text{ar}}, \mathcal{H}_{\text{vol}}\}^{\mathbb{Z}} : (F_{\text{av}})_{s_k, s_{k+1}} = 1 \text{ for all } k\}.$$

The index shift acts on this space. It is the smallest one-step-memory symbolic system containing the modal sequences and preserving their consecutive pairs: it admits both cross-transitions and volumetric self-retention. Phase, cardinal orientation, and trajectory records remain in the TPK state from which this envelope is obtained.

Proposition 5.5 (Weighting of trajectories between sheets). *In areal–volumetric order, the invariant measure of maximal entropy on X_{av} has transition matrix and stationary distribution*

$$P_{\text{av}} = \begin{pmatrix} 0 & 1 \\ \varphi^{-2} & \varphi^{-1} \end{pmatrix}, \quad (\mu_{\text{ar}}, \mu_{\text{vol}}) = \frac{(1, \varphi^2)}{1 + \varphi^2}. \quad (5.6)$$

In particular, $\mu_{\text{ar}}/\mu_{\text{vol}} = \varphi^{-2}$.

Proof. The already constructed incidence has positive eigenvector $r = (1, \varphi)^\top$ and eigenvalue φ . The normalization

$$(P_{\text{av}})_{ij} = \frac{(F_{\text{av}})_{ij} r_j}{\varphi r_i}$$

gives the stated matrix. Its rows sum to one because $\varphi^{-2} + \varphi^{-1} = 1$. Symmetry of F_{av} implies that weights proportional to r_i^2 satisfy detailed balance; hence $\mu P_{\text{av}} = \mu$. Irreducibility and volumetric self-retention determine a unique stationary distribution.

To verify maximality, every allowed transition satisfies $\log(P_{\text{av}})_{ij} = \log r_j - \log r_i - \log \varphi$. Under any invariant measure, the endpoint terms cancel on average. If ν_i are its vertex weights and $q_i(j) = \nu(s_1 = j \mid s_0 = i)$, for $\nu_i > 0$, then

$$h_\nu \leq H_\nu(s_1 \mid s_0) = \log \varphi - \sum_{i: \nu_i > 0} \nu_i D(q_i \parallel R_i) \leq \log \varphi,$$

where $D(q \parallel p) = \sum_j q_j \log(q_j/p_j)$ is the relative entropy on the allowed pairs and the convention $0 \log 0 = 0$ is adopted. The Markov measure of (5.6) attains the bound. Equality in the first inequality requires the Markov property; equality in the second requires $q_i = R_i$. Stationarity and irreducibility then fix μ , establishing uniqueness. This is the Parry measure of the defined envelope. $\square$

The frequency $(1/3, 2/3)$ counts the instants of the primitive calendar; μ weights the trajectories of the envelope. Its quadratic ratio expresses the product of incoming and outgoing incidence amplitudes. TPK memory and phase restrictions retain their own domain; the entropy of the envelope pertains to the symbolic system specified in this definition.

5.5.2 Closure operator and limit of marked returns

The closure coordinate π , generated by the regional readout, enters after construction of the modal incidence. Let $e_{\text{ar}} = (1, 0)^\top$, $e_{\text{vol}} = (0, 1)^\top$, and $R_i = e_i e_i^\top$ be the sheet projectors. The readout conditions are preservation of both sheets, unit volumetric normalization, and an areal mark equal to the closure coordinate. These conditions determine the positive operator

$$D_\pi = \pi P_{\text{ar}} + P_{\text{vol}} = \begin{pmatrix} \pi & 0 \\ 0 & 1 \end{pmatrix}. \quad (5.7)$$

Diagonality expresses sheet preservation, and the two entries express their normalizations; this makes explicit the uniqueness of the operator for the fixed readout.

Proposition 5.6 (Boundary-return ratio). *For $n \geq 1$, the marked evaluation*

$$\Theta_n = \frac{\langle e_{\text{ar}}, D_\pi F_{\text{av}}^n e_{\text{ar}} \rangle}{\langle e_{\text{vol}}, F_{\text{av}}^n e_{\text{vol}} \rangle} = \pi \frac{F_{n-1}}{F_{n+1}} \quad (5.8)$$

satisfies

$$\lim_{n \rightarrow \infty} \Theta_n = \pi \frac{\mu_{\text{ar}}}{\mu_{\text{vol}}} = \frac{\pi}{\varphi^2}. \quad (5.9)$$

Proof. The diagonal entries of F_{av}^n count paths of length n returning to their initial sheet. The power formula already proved gives F_{n-1} and F_{n+1} . The operator D_π multiplies the first count by π . Finally, F_{n-1}/F_{n+1} is the product of two consecutive quotients converging to φ^{-1} ; its limit is φ^{-2} . The identity with the weight ratio follows from (5.6). $\square$

The closure operator appears exactly once, in the terminal evaluation of the return. The transfer F_{av} remains determined by the calendar. Information from two readouts of the same state is thus composed: modal incidence determines the quadratic weighting, and the closure region contributes its coordinate to the boundary.

The composition also preserves refinement. For positive cylinders $I = [a, b]$ and $J = [c, d]$, the image of the readout $\mathcal{R}(x, y) = x/y^2$ is the interval

$$\mathcal{Q}(I, J) = \left[\frac{a}{d^2}, \frac{b}{c^2} \right].$$

Monotonicity in each argument proves that nested closure and self-scaling cylinders produce nested intervals of the ratio; the positivity of their limits and contraction of their diameters determine π/φ^2 . The scalar expression thus retains a chain of operations compatible with depth, applied next in its two orientations.

Once the characters have been generated, the two orientations of the readout

$$R(x, y) = \frac{x}{y^2}$$

provide

$$\rho_A = \frac{\pi}{\varphi^2}, \quad \rho_E = \frac{\varphi}{\pi^2}, \quad \varphi^3 \rho_A^2 \rho_E = 1. \quad (5.10)$$

The second expression will appear in the electronic exponent. The first is also obtained as the limit of the marked readout $\pi F_{n-1}/F_{n+1}$. Interchanging the arguments relates the two, but does not turn a reparametrization of the electronic equation into a second independent derivation of all its coefficients.

5.6 Exponential realizations of the generated coordinates

The regional coordinates already determined allow recognition of the concrete realizations of the TRIT quadratic collection. The order is substantive: construction of the regions precedes evaluation of an exponential at the parameters π or $2 \log \varphi$. These expressions identify the role of the coordinates in operators constructed from APP and TPK; they do not enter the selection of the prefixes that produce them.

5.6.1 Ternary cycle, nilpotent transport, and sheet incidence

Multiplication $a(r) = 4r$ in $\mathbb{Z}/9\mathbb{Z}$ has order 3. On the units it determines the orbits $\{1, 4, 7\}$ and $\{2, 8, 5\}$. In the augmentation module of either orbit, formed by the integer combinations whose coefficients sum to zero, the basis $(e_r - e_{4r}, e_{4r} - e_{7r})$ gives

$$C = \begin{pmatrix} 0 & -1 \\ 1 & -1 \end{pmatrix}, \quad C^2 + C + I = 0.$$

The restricted form has Gram matrix $\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$. In residual evaluations modulo 9, the action distinguishes the sheets: $S(ax, ay) = aS(x, y)$, whereas $P(ax, ay) = a^{-1}P(x, y)$. The cyclic realization therefore preserves a contragredient orientation between sum and product. The quotients of the integer lift are preserved in the corresponding arithmetic register.

Elementary parabolic transport is represented by

$$N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}.$$

The areal–volumetric incidence already constructed in (5.5) provides

$$F_{av} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}, \quad A = F_{av}^2 = \begin{pmatrix} 1 & 1 \\ 1 & 2 \end{pmatrix}.$$

The three operators act on declared real spaces: the augmentation module tensored with $\mathbb{R}$, the nilpotent transport plane, and the modal incidence plane. Sharing a matrix dimension does not identify these spaces or their coordinates.

Proposition 5.7 (Concrete generators of the three types). *The operators*

$$J_e = \frac{2C + I}{\sqrt{3}}, \quad J_p = N, \quad J_h = \frac{2A - 3I}{\sqrt{5}}$$

satisfy, respectively,

$$J_e^2 = -I, \quad J_p^2 = 0, \quad J_h^2 = I.$$

Proof. The relation for C implies $(2C + I)^2 = -3I$. Direct multiplication gives $N^2 = 0$. Finally, A has trace 3 and determinant 1, so $A^2 - 3A + I = 0$ and $(2A - 3I)^2 = 5I$. $\square$

Proposition 5.8 (Three evaluations of the exponential identity). *In the preceding realizations,*

$$\exp(\pi J_e) + I = 0, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{av}^2. \quad (5.11)$$

Proof. The elliptic specialization of the tritic exponential gives $\cos \pi I + \sin \pi J_e = -I$. Nilpotence removes exactly the terms of degree at least 2 in the second expression. For the third, $\varphi^2 = \varphi + 1$ implies

$$\cosh(2 \log \varphi) = \frac{3}{2}, \quad \sinh(2 \log \varphi) = \frac{\sqrt{5}}{2}.$$

The exponential therefore equals $\frac{3}{2}I + \frac{\sqrt{5}}{2}J_h = A$. $\square$

The first equality represents the elliptic half-turn; the complete turn satisfies $\exp(2\pi J_e) = I$. The second retains a nonzero linear term: the parabolic regime expresses transport, not absence of evolution. The third realizes a unimodular hyperbolic advance. Its eigenvalues are φ^2 and φ^{-2} ; one direction expands and the other contracts while preserving the determinant. The three evaluations use different parameters of a common exponential collection.

In that collection, e enters through the scalar evaluation $\exp(I) = eI$ and the law $\exp(sI)\exp(tI) = \exp((s+t)I)$. Its role is not restricted to the parabolic type. The coordinate e , obtained previously from its region, is recognized here as the unit-parameter evaluation of the exponential; this identity does not replace its coinductive genealogy.

5.6.2 Polynomial resolution of the regimes

The three realizations admit a compact presentation on the direct sum of their spaces. Define

$$\mathbb{J} = J_e \oplus J_p \oplus J_h, \quad Q = \mathbb{J}^2.$$

The preceding relations imply

$$\mathbb{J}^6 = \mathbb{J}^2, \quad Q^3 = Q.$$

The use of Q keeps this operator square separate from the dodecaphasic register K .

Proposition 5.9 (Polynomial regime projectors). *The maps*

$$p_e = \frac{Q^2 - Q}{2}, \quad p_p = I - Q^2, \quad p_h = \frac{Q^2 + Q}{2}$$

are mutually orthogonal idempotents, sum to I , and recover the three summands of the representation.

Proof. The three polynomials take values $(1, 0, 0)$, $(0, 1, 0)$, and $(0, 0, 1)$ at the points $-1, 0, +1$, respectively. The differences between their squares and themselves, as well as their cross-products, are multiples of $X^3 - X$. Substituting Q , which annihilates that polynomial, proves the identities. On the three blocks, Q acts as $-I, 0, I$. $\square$

This yields

$$\begin{aligned} \exp(t\mathbb{J}) &= p_e(\cos tI + \sin t\mathbb{J}) + p_p(I + t\mathbb{J}) \\ &\quad + p_h(\cosh tI + \sinh t\mathbb{J}). \end{aligned} \tag{5.12}$$

Each summand preserves its quadratic relation, and conjugation reverses its evolution. The total representation brings together the three regimes, but does not by itself define a transition operator between them. Such a transition requires the TPK transport maps with their domains, orientation, and memory. The exponential articulation allows their types to be recognized without replacing that dynamics by an identification of the fibres.

5.7 Compatible cylinders and arbitrary depth

The realization of a ternary block $b = (b_1, \dots, b_6)$ is the integer

$$\nu(b) = \sum_{j=1}^6 b_j 3^{6-j} \in \{0, \dots, 728\}.$$

A sequence of blocks defines integer prefixes by $N_{n+1} = 729N_n + \nu(b_{n+1})$, starting from $N_0 = m$, where m is the integer part determined by the readout. The preceding brackets place the closure, propagation, and self-scaling characters in $(3, 4)$, $(2, 3)$, and $(1, 2)$, respectively; their initial values are therefore $m = 3, 2, 1$. Subsequent blocks refine the fractional part. The corresponding cylinder is

$$I_n = \left[\frac{N_n}{729^n}, \frac{N_n + 1}{729^n} \right], \quad |I_n| = 729^{-n}. \tag{5.13}$$

The half-open convention is used to select a representative when a value lies on a boundary. Passage to the limit uses the closures of these intervals and identifies the two equivalent boundary expansions. This qualification prevents the assumption that every intersection of nested half-open intervals is automatically nonempty.

Theorem 5.10 (Determination of the scalar realization). *A compatible block history determines a unique limit of the left endpoints of (5.13). If an exact character has rational brackets of arbitrarily small width and is separated from the boundary of every decimal cylinder examined, each decimal prefix of finite length is determined after a finite computation. The prefixes thus obtained are mutually compatible.*

Proof. Extending a block preserves the previous prefix by Euclidean division. The closed cylinders are nested and their diameter tends to zero; their left endpoints form a Cauchy sequence and have a unique limit. For a fixed decimal length, a value separated from the boundaries of the partition has positive distance from them. A sufficiently narrow bracket lies within a single cell and determines its prefix. Uniqueness of the cell and nesting of the partitions prove compatibility under truncation. At points with a terminating expansion, the exact boundary decision and its representation convention must be added; this decision does not follow from numerical width alone. $\square$

Corollary 5.11 (Finite determination and compatibility of regional prefixes). *For each of the realizations π, φ, e and every finite decimal length, the constructed brackets determine the corresponding prefix by a finite computation. The prefixes obtained at different depths are compatible under truncation.*

Proof. The closure, propagation, and self-scaling characters have the preceding brackets. After their recognition, the irrationality of π, e, φ ensures their separation from the rational boundaries of every finite decimal partition. The preceding theorem applies to each character. $\square$

Arbitrary depth expresses the quantifier in the corollary: for every finite horizon there is a computation determining its prefix. The mathematical object is the complete compatible collection; each execution obtains a finite projection of it.

5.8 Specular involution and synchronization of the regions

The joint nonadic connection transports the three characters together. In the ninth-phase chart, the self-scaling axis and the ternary sum of closure and propagation remain fixed under specular interchange. Helical inversion changes the direction of transport; the mirror of the two channels changes their relative orientation. These are different involutions, although both act on the same trajectory system.

The matching census values illustrate this coordination:

Phase	N_π	N_e	N_φ	Equality of counts
4	207	207	122	$N_\pi = N_e$
5	39	151	151	$N_e = N_\varphi$
6	110	110	69	$N_\pi = N_e$
7	81	29	81	$N_\varphi = N_\pi$
8	59	59	59	triple synchronization
9	43	43	43	triple synchronization

These integers count compatible coordinates of the regions. They do not assert equality between π, e , and φ as real numbers. If

$$\partial(a, b, c) = (a - b, b - c, c - a),$$

its image belongs to the lattice $A_2 = \{(u, v, w) \in \mathbb{Z}^3 : u + v + w = 0\}$. In phases eight and nine the relative component vanishes, but the diagonal component changes from (59, 59, 59) to (43, 43, 43). Relative synchronization coexists with a shift of -16 per channel. The closure that will lead to α must preserve both pieces of information: relative coincidence and memory evolution.

5.9 Sturmian helical suspension

The relation between bases nine and ten of the completion defines the dimensionless parameter

$$\delta = \frac{\log(10/9)}{\log 10}. \quad (5.14)$$

It is not obtained from a target physical constant. It is the relative logarithmic scale of two bases already present in the construction. Its use as a symbolic readout of refinement has an exact consequence.

Lemma 5.12 (Irrationality of the slope). *The number δ is irrational.*

Proof. If $\delta = p/q$, $q > 0$, then $(10/9)^q = 10^p$ and $9^q = 10^{q-p}$. The factorization of the left-hand side contains the prime three, whereas that of the right-hand side contains only the primes two and five. Equality is impossible. $\square$

The lower mechanical word, with intercept θ , is

$$z_n(\theta) = \lfloor (n+1)\delta + \theta \rfloor - \lfloor n\delta + \theta \rfloor \in \{0, 1\}.$$

It is obtained by observing the rotation $\theta \mapsto \theta + \delta \pmod{1}$ through the interval $[1 - \delta, 1)$. The event sequence has density δ , and its sum over any window of length L equals $\lfloor L\delta \rfloor$ or $\lceil L\delta \rceil$. Since $21 < 1/\delta < 22$, the gaps between events are twenty-one or twenty-two. The word indicating which intervals are short is another mechanical word, of slope $\sigma = 22 - 1/\delta$; its events are separated by six or seven primary intervals. The second scale is induced by the first and is not a second free parameter.

5.9.1 Capacity index, vacancies, and induction of the tritic regime

The vacancy sequence is linked to refinement through an increment identity. To exhibit it, fix the phase origin $\theta = 0$. Let

$$\rho = \log_{10} 9 = 1 - \delta, \quad \mathcal{C}_t = \lfloor (t+1)\rho \rfloor \quad (t \geq 0).$$

The index $\mathcal{C}_t$ expresses the decimal capacity of refinement in the normalization comparing a ternary block of 6 positions, with 729 possibilities, and a decimal block of 3 positions, with 1000 possibilities. Indeed, $\log_{1000} 729 = \log_{10} 9 = \rho$. The letter K is reserved for the dodecaphase record to be constructed later.

Proposition 5.13 (Complementarity of capacity and vacancy). *For $t \geq 1$,*

$$\mathcal{C}_t - \mathcal{C}_{t-1} = 1 - z_t(0). \quad (5.15)$$

Consequently, if $0 \leq a < b$,

$$\mathcal{C}_b - \mathcal{C}_a + \sum_{t=a+1}^b z_t(0) = b - a.$$

Proof. The irrationality of δ gives

$$\mathcal{C}_t = t + 1 - \lfloor (t+1)\delta \rfloor = t - \lfloor (t+1)\delta \rfloor.$$

Subtracting two consecutive terms yields (5.15). The sum over the interval telescopes. $\square$

Each transition increases the block counter by one. If $z_t = 0$, capacity also increases; if $z_t = 1$, capacity remains unchanged while a transition is added to the history. A vacancy is here an event of capacity retention, with a determined position and antecedent. The preceding balance preserves the length exactly, partitioned into acquired capacity and recorded vacancies. The observed phase $g_t^{\text{cap}} = \lfloor \mathcal{C}_t \rfloor_9$ satisfies

$$g_t^{\text{cap}} - g_{t-1}^{\text{cap}} \equiv 1 - z_t \pmod{9}.$$

This capacity phase and the chronological phase $[t]_9$ are two different observations. Local capacity retention and a complete holonomy turn have distinct mechanisms, although both allow the phase observation to remain unchanged while the state changes.

The relationship with TRIT can be followed at the vacancy positions themselves. For $j \geq 1$, define

$$p_j = \left\lfloor \frac{j}{\delta} \right\rfloor, \quad v_j = \mathcal{C}_{p_j}, \quad h_j = p_{j+1} - p_j.$$

The inequality $p_j\delta < j < (p_j + 1)\delta$ identifies p_j as the position of event j . Since $0 < \delta < 1$, it also gives $\lfloor (p_j + 1)\delta \rfloor = j$, and hence

$$v_j = p_j - j, \quad v_{j+1} - v_j = h_j - 1.$$

If $s_j = 22 - h_j$, then $s_j \in \{0, 1\}$, and

$$[v_{j+1}]_3 = [v_j - s_j]_3. \quad (5.16)$$

An interval of length 22 preserves the class; one of length 21 shifts it by one unit in the negative direction. The canonical selector is applied to the positive phase $r_j = 1 + [v_j]_9$:

$$\tau_j = M_{\text{ph}}(r_j).$$

A tritic regime is thus determined by the capacity index retained at each event, in addition to the binary vacancy coding.

The positions of the short intervals form the induced mechanical coding of slope $22 - 1/\delta$. Their spacings 6 and 7 determine the plateau lengths of the class $[v_j]_3$, once the origin and endpoint convention are fixed. The first classes are 2 for 6 events, 1 for 7, and 0 for 7; the first segment of 6 is the initial restriction of a plateau. Their signatures traverse 0, -1 , $+1$. Equations (5.15) and (5.16) make explicit the link between refinement, vacancies, and regime selection. The quadratic TRIT readout subsequently realizes these types through their respective generators; the temporal event and its geometric representation therefore retain an identifiable dependency.

Inverting the monotone index provides another useful observation:

$$\mathcal{T}(k) = \min\{t : \mathcal{C}_t = k\} = \left\lceil \frac{k}{\rho} \right\rceil - 1 \quad (k \geq 1).$$

If $\sigma = \rho^{-1} - 1$, the number of blocks required to gain m units of capacity is

$$\mathcal{T}(k+m) - \mathcal{T}(k) = m + [(k+m)\sigma] - [k\sigma] \in \{m + \lfloor m\sigma \rfloor, m + \lceil m\sigma \rceil\}.$$

In particular, 9 units of capacity require 9 or 10 blocks. The period of a finite phase remains exact, while the return time measured by the other index admits two values. This distinction belongs to the dynamics of refinement and is not an a posteriori correction of its numerical values.

Theorem 5.14 (Complexity of the phase readouts). *For $m \geq 1$, the mechanical word has factor complexity $p(m) = m + 1$. When the phase $n \bmod 9$ is retained, the complexity is $p_9(m) = 9(m + 1)$; when only its ternary class is retained, it is $p_3(m) = 3(m + 1)$. All three topological entropies vanish.*

Proof. For factors of length m , the predecessors of the two endpoints of the coding interval form $m + 1$ distinct points on the circle: coincidence between shifted endpoints removes duplicates, and irrationality prevents further coincidences. These points determine $m + 1$ distinct intervals on each of which the itinerary is constant. Every irrational orbit visits each interval, hence $p(m) = m + 1$.

For a fixed phase modulo nine, the positions of that phase follow a rotation with step 9δ , which is also irrational. Density implies that every factor occurs in every phase, and the first coordinate distinguishes its nine copies. The same argument with step 3δ gives three copies after ternary reduction. Finally, $m^{-1} \log(c(m + 1)) \rightarrow 0$ for $c = 1, 3, 9$. $\square$

The phase product $C_9 \times \mathbb{T}$, translated by $(1, \delta)$, has frequency characters

$$\frac{a}{9} + k\delta \pmod{1}, \quad a \in \mathbb{Z}/9\mathbb{Z}, k \in \mathbb{Z}.$$

The spectral module is not $\log 10$: it is the additive module generated by $1/9$ and δ , modulo the integers. The factor $\log 10$ belongs to the normalization in (5.14).

An operator presentation makes memory explicit. On $\ell^2(\mathbb{Z})$, let $T|n\rangle = |n + 1\rangle$, $V_9|n\rangle = \zeta_9^n|n\rangle$, $W_\delta|n\rangle = e^{2\pi i n \delta}|n\rangle$, and $D_M|n\rangle = \lfloor n/9 \rfloor |n\rangle$. On finitely supported vectors,

$$V_9 T = \zeta_9 T V_9, \quad W_\delta T = e^{2\pi i \delta} T W_\delta, \quad [D_M, T^9] = T^9. \quad (5.17)$$

The first two relations are Weyl covariance relations; the third expresses that a nonadic turn increases the memory degree. The base rotation is abelian, whereas the algebra of observables and translations

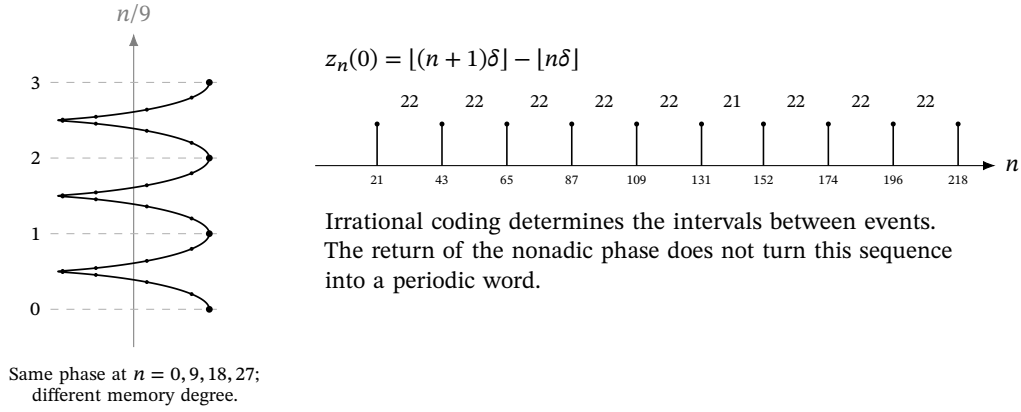


Figure 5: Complementary representations of transport. Left: projection of the helix $(\cos(2\pi n/9), \sin(2\pi n/9), n/9)$: advancing nine steps recovers the phase and increases the height. Right: the first ten events of the lower word for $\delta = \log_{10}(10/9)$, with exact gaps 21 or 22. The helical representation introduces no additional physical metric.

is not. The inversion $|n\rangle \mapsto |-n\rangle$ conjugates forward and backward motion. This realization provides an aperiodic helical suspension with finite phase return, not a periodic word of length nine.

For this count we fix the intercept $\theta = 0$ and the cumulative endpoints $(N_0, \dots, N_4) = (0, 729, 972, 999, 1000)$, corresponding to the cubic ordering $(729, 243, 27, 1)$. The lower counts are $[N_j\delta] - [N_{j-1}\delta]$, giving $(33, 11, 1, 0)$. The upper counts are $[N_j\delta] - [N_{j-1}\delta]$, with the same initial boundary zero, giving $(34, 11, 1, 0)$. These values depend on the intercept and the ordering of the strata; they are not asserted for an arbitrary initial phase. Their totals are forty-five and forty-six. The increment corresponds to a single boundary unit, located in the first stratum of this ordered partition. The numbers thirty-four and eleven thus appear in a single count preceding any chart of units; identifying them with physical exponents additionally requires the corresponding dimensional derivation.

The designation *aperiodic crystal* is used here for this symbolic order, with its complexity, spectrum, and return specified. Vanishing topological entropy does not by itself imply vanishing physical dissipation: a thermodynamic realization requires a defined energy dynamics and erasure operation. Likewise, this Sturmian readout neither replaces all TPK operators nor proves by itself periodicity or aperiodicity of the complete output of α . Its role is to make explicit an order structure that a mere return of the schedule does not describe.

5.10 Nonadic connection, holonomy and monodromic representation

The regional connection already constructed is now studied through three objects: transport of the complete state, representation of its windings and suspension of its itineraries. The APP-TRIT-TPK operations and the initial catalogue retain the preceding definitions. The necessary distinction here concerns what each representation retains of the same history(see 4.8).

At prolongation index k , a state chart preserves

$$\hat{x}_k = (B_k, C_k, p_k, c_k, \Sigma_k, W_k, g_k, m_k).$$

Here B_k collects the three regional blocks, C_k is the cylinder, p_k the prefix, c_k the carry register, Σ_k the boundary signature and W_k the incidence information. The phase is $g_k = 1 + [(k-1)]_9$; m_k counts complete windings. The reduction $\beta_k = [m_k]_{12}$ preserves the dodecaphasic position, while the integer m_k distinguishes its successive windings.

Definition 5.15 (Transport of a nonadic winding). Let $\mathcal{R}_k \subseteq \hat{X}_k \times \hat{X}_{k+1}$ be the TPK admissible-extension relations. The realization of holonomy at level k is the relational composition

$$\Gamma_{9,k} = \mathcal{R}_{k+8} \circ \cdots \circ \mathcal{R}_k.$$

For $s \geq 1$, its prolongation to s windings is

$$\Gamma_{9s,k} = \Gamma_{9,k+9(s-1)} \circ \cdots \circ \Gamma_{9,k}.$$

The relational formulation preserves admissible continuations. On an individually specified history, each arrow leads to the state actually prolonged; the isolated visible block may still admit more than one continuation. This difference is precisely the role of memory and boundary conditions in determining the trajectory.

Proposition 5.16 (Iteration of return and accumulated memory). *In every admissible composition of s windings,*

$$g_{k+9s} = g_k, \quad m_{k+9s} = m_k + s, \quad \beta_{k+9s} = \beta_k + s \pmod{12}.$$

Proof. The one-winding rule preserves phase and increments the counter by one. Applying it to the already transported state repeats exactly these two operations. Induction gives the first two equalities; reducing the second modulo 12 gives the third. $\square$

The counter thus provides an integer witness of the distinction between windings. After 12 windings the finite phase pair is restored, but the counter increases by 12. The nonsplit extension (4.5) expresses composition of the calendar through its cocycle; the trajectory also preserves the integer lift of this datum. Preservation here means transport of previous contributions, not immobility of every coordinate.

5.10.1 Bilateral representation of the return index

The shift T of (5.17) admits an expression in phase and winding coordinates. Number phases from 0 to 8, by shifting the origin relative to g_k , and write $k = 9m + r$, with $0 \leq r < 9$, on the integer line covering the phase cycle. On $\mathcal{H}_{\text{ind}} = \ell^2(\mathbb{Z})$, the basis $|k\rangle = |r, m\rangle$ expresses that same unitary shift:

$$T|r, m\rangle = \begin{cases} |r+1, m\rangle, & r < 8, \\ |0, m+1\rangle, & r = 8. \end{cases}$$

The inverse decreases r by one when $r > 0$, and takes $|0, m\rangle$ to $|8, m-1\rangle$. Passage through the phase origin thus incorporates the exact quotient unit.

Proposition 5.17 (Representation of cycle monodromy). *Let $S = T^9$. Then*

$$S|r, m\rangle = |r, m+1\rangle, \quad \mathbb{Z} \longrightarrow \mathcal{U}(\mathcal{H}_{\text{ind}}), \quad s \longmapsto S^s$$

is a faithful unitary representation of the cycle's winding group.

Proof. Nine shifts preserve the remainder upon division by 9 and increment the quotient. Powers satisfy $S^{s+t} = S^s S^t$, and $S^s|r, m\rangle = |r, m+s\rangle$ distinguishes every integer s . A permutation of an orthonormal basis is unitary. $\square$

This representation corresponds to the cover of the cycle, whose winding group is $\mathbb{Z}$, and represents phase together with its counter. The TPK state additionally contains the trajectory coordinates already described. The bilateral extension admits negative indices as coordinates of the cover; a trajectory beginning at an initial condition uses its admissible half-line. This provides an operator inverse of the index without turning every extension relation into a bijection of the complete state.

With $D_M|k\rangle = \lfloor k/9 \rfloor |k\rangle$, the identity $[D_M, S] = S$ expresses memory increment on the dense domain of finitely supported vectors. If $\mathcal{J}|k\rangle = |-k\rangle$ and P_0 projects onto indices divisible by 9, one also obtains

$$\mathcal{J}T\mathcal{J} = T^{-1}, \quad \mathcal{J}D_M\mathcal{J} = -D_M - (I - P_0). \quad (5.18)$$

Indeed, for $k = 9m + r$, $\lfloor -k/9 \rfloor = -m$ if $r = 0$, and equals $-m - 1$ if $r > 0$. Inversion thus preserves the boundary term of Euclidean division.

5.11 Block clock and capacity clock

The index of a refinement map and the number of available decimal positions obey related but different laws. To express them, reserve n for the ternary-block index and define

$$\rho = \log_{1000} 729 = \log_{10} 9, \quad \delta = 1 - \rho, \quad \mathcal{C}_n = \lfloor (n+1)\rho \rfloor, \quad n \geq 0.$$

The symbol $\mathcal{C}_n$ denotes capacity here, not the dodecaphasic register K . At the canonical phase origin, the preceding mechanical word gives

$$Z_n = \sum_{j=0}^n z_j(0) = \lfloor (n+1)\delta \rfloor.$$

Proposition 5.18 (Exact balance between capacity and vacancies). *For $n \geq 0$ and, in the second identity, $n \geq 1$,*

$$\mathcal{C}_n = n - Z_n, \quad \mathcal{C}_n - \mathcal{C}_{n-1} = 1 - z_n(0).$$

Proof. Irrationality of δ implies $\lfloor (n+1)\delta \rfloor = \lfloor (n+1)\delta \rfloor + 1$. Then

$$\lfloor (n+1)(1-\delta) \rfloor = n+1 - \lfloor (n+1)\delta \rfloor = n - Z_n.$$

Subtracting two consecutive identities completes the proof. $\square$

An event $z_n = 1$ adds a block to the history while maintaining capacity $\mathcal{C}_n$; an event $z_n = 0$ increments both counters. In any interval, capacity increments and vacancies sum exactly to its length. Reducing modulo q , for $q = 9$ or $q = 108$, gives

$$[\mathcal{C}_n]_q = [n]_q - [Z_n]_q.$$

Uniform block phase and capacity phase are related through integrated vacancy memory. In particular, a capacity hold and a complete winding of the connection have distinct mechanisms, although both may preserve a phase observation.

Monotone inversion of the clock gives a first-arrival time:

$$T_{\text{cap}}(a) = \min\{n : \mathcal{C}_n = a\} = \left\lceil \frac{a}{\rho} \right\rceil - 1, \quad a \geq 1.$$

With $\sigma = \rho^{-1} - 1$, the time required to gain b units of capacity is

$$T_{\text{cap}}(a+b) - T_{\text{cap}}(a) = b + \lfloor (a+b)\sigma \rfloor - \lfloor a\sigma \rfloor.$$

The difference of ceilings takes one of the two integers adjacent to $b\sigma$. Thus, 9 or 10 blocks are required to gain 9 capacity units, and 113 or 114 to gain 108. These are inverse-clock times measured from a first arrival; the winding counter is always interpreted in the index to which it belongs.

5.12 Symbolic suspension and aperiodic time crystal

Rotation coding brings together phase periodicity, local recurrence and global aperiodicity. Let $\mathcal{X}_\delta \subseteq \{0, 1\}^{\mathbb{Z}}$ be the closure, in the product topology, of the translates of the bilateral word $z_n(\theta)$. Denote the shift of its sequences by σ_{sh} . Preserving a nonadic phase, the step is

$$F(r, z) = ([r + 1]_9, \sigma_{\text{sh}} z), \quad (r, z) \in C_9 \times \mathcal{X}_\delta.$$

Definition 5.19 (Symbolic aperiodic time crystal). The unit-roof suspension of the preceding system is the space

$$\mathcal{S}_\delta = ((C_9 \times \mathcal{X}_\delta) \times \mathbb{R}) / \sim, \quad (y, t + 1) \sim (Fy, t),$$

with flow $\Phi_s[y, t] = [y, t + s]$. This model of temporal order, together with its vacancy coding and winding-counter representation, is called a *symbolic aperiodic time crystal*.

The adjective temporal designates the action of the flow and its return section; aperiodicity characterizes its itineraries. The frequency of ones in each sequence is δ : the uniform discrepancy bound strictly less than one, obtained by telescoping in any window, persists under closure. A periodic sequence would have rational frequency. Thus, $\mathcal{X}_\delta$ contains no periodic orbits. The suspension has no closed orbits either: a flow return would require an integer time and periodicity of the itinerary.

Finite words do recur. Length- m itineraries arise from a partition of the circle by $m + 1$ points; each interval is visited recurrently by the irrational rotation. Even when times are restricted to a fixed phase, the step 9δ remains irrational. This is why every word appears in all 9 phases, and explains the previously established complexities

$$p(m) = m + 1, \quad p_9(m) = 9(m + 1), \quad p_3(m) = 3(m + 1).$$

Recurrence of local configurations coexists with the absence of a global period. Its topological entropy is zero because these complexities grow linearly.

Realization through the observables V_9, W_9 and shift T expresses the same order through the covariance relations (5.17). Uniform-clock frequencies belong to $\frac{1}{9}\mathbb{Z} + \delta\mathbb{Z}$, modulo 1. The capacity clock additionally preserves the relation $[C_n]_9 = [n]_9 - [Z_n]_9$; it is not replaced by uniform phase when interpreting returns. The term *time crystal* is thus defined through a mathematical dynamical system. Its physical realization requires specifying energy dynamics, its temporal observables and stability, data distinct from the present symbolic construction.

5.13 Mirror symmetry of the regions and dual realization

Mirror interchange acts on the regional block $B = (u^\pi, u^e, u^\varphi) \in (\mathbb{F}_3^6)^3$ by

$$cB = (u^e, u^\pi, u^\varphi).$$

Its fixed axis and aggregate are, respectively, u^φ and $s = u^\pi + u^e$. This fixedness concerns the involution in a fibre; both coordinates may evolve during nonadic transport. With $q = u^\pi + u^e + u^\varphi$, $a = u^\pi + u^e - u^\varphi$ and $d = u^\pi - u^e$, one recovers

$$u^\varphi = 2(q - a), \quad s = q - u^\varphi, \quad u^\pi = 2(s + d), \quad u^e = 2(s - d).$$

The equalities hold componentwise in $\mathbb{F}_3$, where $2^{-1} = 2$. The mirror preserves q, a, s and reverses d ; this antisymmetric coordinate individualizes the allocation that the aggregate preserves without distinguishing. Time-index inversion $\mathcal{I}$ and channel involution c therefore act on different data.

Phase return has an explicit witness:

$$\begin{aligned} B_1 &= (012222 \mid 121221 \mid 112202), \\ B_{10} &= (101222 \mid 112221 \mid 112002). \end{aligned}$$

Both positions have phase 1; the block, axis and memory have changed. Likewise, synchronization of the censuses at phases 8 and 9 annihilates their relative differences, while the diagonal component changes from (59, 59, 59) to (43, 43, 43). Equality of a relative observable does not exhaust the state determining it.

The dual realization preserves that architecture. In Section 10.1, the same prefix determines nested cylinders on one side and a sequence of encoded words with their frames on the other. The first map passes to the limit by contraction of diameters; the second satisfies $r_{n,m}Q_n = Q_m p_{n,m}$ and defines Q_∞ . Figure 6 anticipates these two maps. The subsequent incidence route constructs codes, lattices and the Moonshine realization in their own domains; the automorphism group acts on that realization, not on the digits through an implicit identification.

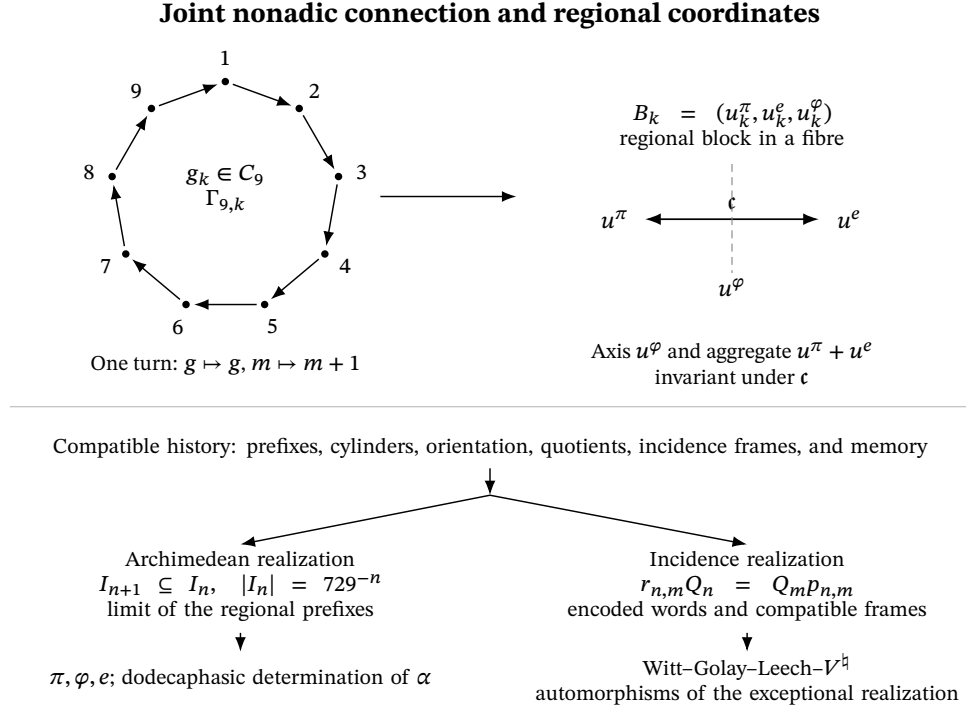


Figure 6: Nine phases of a connection and two realizations of the compatible history. The cycle represents the phase; each turn increments the counter. The regional diagram represents the involution $\mathfrak{c}$, which interchanges closure and propagation and fixes the self-scale axis in a fibre. This invariance does not require the axis to remain constant during transport. The two lower maps use the same prefix: one determines its numerical limit, and the other preserves words and incidence frames. The maps of this second realization are specified in Section 10.1; the exceptional route is developed afterwards.

5.13.1 Rectilinear representation of phases and return

The representation on the segment $[0, 10]$ preserves the positional origin of APP and orders its nine interior marks. In this diagram, the integers $1, \dots, 9$ are phase representatives. The horizontal position indicates the ordering of the charts; the upper labels specify regional operations. Figure 7 thus brings together the initial line, fixation of the self-scaling axis, the specular distribution of the two channels, and return with memory(see 5.10).

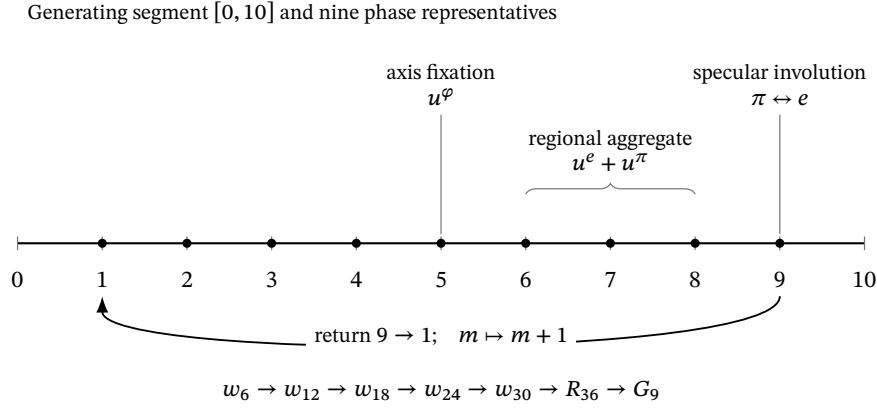


Figure 7: Rectilinear representation of the joint nonadic connection. The mark 5 locates the first strong saturation and the regional self-scaling axis; the brace 6–8 identifies the ternary aggregate fixed in each fibre; phase 9 selects the closure orientation. Return preserves phase and advances memory. The abscissae number operational positions, whereas the labels u^π , u^e , u^φ designate blocks of $\mathbb{F}_3^6$.

Axis fixation is formulated in the fibre of a given boundary. If $q = u^\pi + u^e + u^\varphi$ and $a = u^\pi + u^e - u^\varphi$, then

$$u^\varphi = 2(q - a), \quad s = u^\pi + u^e = q - u^\varphi \quad \text{in } \mathbb{F}_3^6.$$

The data (q, a) determine the axis and the aggregate; the ordered distribution of s retains the specular freedom resolved by prolongation. The fifth phase has one local solution and one extension, with boundary datum $c = 11111$ and residual signature $\rho_W = 000$. This is the meaning of its first strong saturation. The symbol φ above the mark 5 identifies the regional axis involved in that operation.

The intermediate phases preserve this decomposition within each fibre, while their boundary data evolve. The complete census of local multiplicities and extensions of the distinguished branch is

phase	1	2	3	4	5	6	7	8	9
N_{loc}	3	2	5	4	1	1	3	3	2
N_{ext}	2	5	4	1	1	3	3	2	1

Phase 6 combines local uniqueness with three prolongations; phase 7 resolves a triple specular fibre; phase 8 synchronizes the three regional censuses. Phase 9 retains two local distributions with the same axis and aggregate, one of which prolongs. These counts express different functions within the same connection and justify the marks on the line.

In particular, for phases 6, 7, 8, the regional sums are 012100, 101001, and 112000, respectively. The conservation represented by the brace is invariance under specular distribution within each fibre. Subsequent real evaluation applies the readouts of compatible cylinders to complete histories; the internal sum represented in the figure belongs to the ternary domain.

Finally, the arc $9 \rightarrow 1$ represents passage from one turn to the next. The formulas in Section 5.10 preserve the phase remainder and increment the turn counter. The rectilinear diagram and the cyclic diagram in Figure 6 therefore express two complementary aspects of the same holonomy: the ordering of positions and composition of return on the state with memory.

5.14 Temporal transport and tritic exponential composition

The Euler relations in Section 5.6 describe local realizations of the three regimes. For a fixed signature τ , the generator satisfies $J_\tau^2 = -\tau I$; algebraic conjugation takes J_τ to $-J_\tau$. Consequently,

$$\exp(sJ_\tau) \exp(tJ_\tau) = \exp((s+t)J_\tau), \quad \overline{\exp(tJ_\tau)} = \exp(-tJ_\tau),$$

and the product of an evolution with its conjugate is the unit. In the elliptic type this is realized by $\cos^2 t + \sin^2 t = 1$; in the hyperbolic type, by $\cosh^2 t - \sinh^2 t = 1$; in the parabolic type, $(I + tN)(I - tN) = I$, since $N^2 = 0$. The same conservation law admits three quadratic expressions.

The elliptic realization closes a winding; the nilpotent one preserves a linear displacement; the hyperbolic one separates expansion and contraction with unit determinant. At the regional parameters already generated,

$$\exp(\pi J_e) = -I, \quad \exp(J_p) = I + J_p, \quad \exp(2 \log \varphi J_h) = F_{av}^2.$$

The evaluation $\exp(I) = eI$ recognizes the unit of the common exponential law. The number e is not assigned exclusively to the parabolic regime.

Transport of a local realization by an isomorphism U preserves these relations: if $J' = UJU^{-1}$, the exponential series gives $U \exp(tJ) = \exp(tJ')U$. This identity transports the regime and its norm. A transition between different quadratic types belongs, by contrast, to the TPK selection of sheets and regimes; it is not a conjugation changing the value of J^2 . The temporal product therefore preserves the order of transitions and information about their endpoints.

This articulation distinguishes closure of a local representation from holonomy of a history. One may have $\exp(2\pi J_e) = I$ at the same time as $S|r, m\rangle = |r, m + 1\rangle$. The first equality restores regime orientation; the second records another winding. The discrete structure preserves both: local return and the accumulated provenance of the state realizing it.

5.15 Multiplicative irreducibility and return periods

The distinction between structure, evaluation, and phase also appears in the multiplicative sector. The local APP product contains both the residue and the quotient; its iteration makes it possible to construct normal forms and factorizations before examining their decimal periods. This development places the expression *prime mode* in a precise domain and relates irreducibility to the operations already defined, not to an isolated coincidence of periods (see 5.15).

Let

$$\mathcal{W}_9 = \{\emptyset\} \sqcup \bigsqcup_{\ell \geq 1} \{1, \dots, 9\}^\ell.$$

Sequences are ordered from the least significant digit. The evaluation $\text{ev}_9(u) = \sum_j u_j 9^j$, with $\text{ev}_9(\emptyset) = 0$, is a bijection onto the nonnegative integers: for a positive integer, dividing $n - 1$ by 9 uniquely determines the first digit and the continuation. This is bijective nonadic numeration; it preserves the boundary representative 9.

The sum $\boxplus_\#$ is obtained by applying the additive APP sheet to the digits at each position and propagating their quotients. If there is only one digit, it constitutes the provisional residue; a nonzero incoming quotient is normalized through a second additive operation. The sum of the quotients is transported to the next position. To multiply a sequence by a digit d , the multiplicative cell produces $u_j d = 9q_j + r_j$, and the incoming quotient c_j is incorporated through

$$(w_j, c_{j+1}) = \begin{cases} (r_j, q_j), & c_j = 0, \\ (R^+(r_j, c_j), q_j + q^+(r_j, c_j)), & c_j > 0. \end{cases}$$

The procedure begins with $c_0 = 0$ and incorporates the final quotient when it is nonzero. During addition, $c_j \leq 2$; during multiplication by a digit, $c_j \leq 9$. The local operations thus receive arguments within their domain. Denote this second procedure by $d \odot_\# u$, with $d \odot_\# \emptyset = \emptyset$.

For $v = (v_0, \dots, v_{m-1})$, the full product is constructed by the recurrence

$$a_m = \emptyset, \quad a_j = (9 \odot_\# a_{j+1}) \boxplus_\# (v_j \odot_\# u), \quad u \boxtimes_\# v = a_0.$$

The rule composes the two local sheets. Its evaluation satisfies

$$\text{ev}_9(u \boxtimes_\# v) = \text{ev}_9(u) \text{ev}_9(v). \quad (5.19)$$

Indeed, each transition preserves the partial sum plus the pending quotient weighted by the power of 9. The recurrence then gives $\text{ev}_9(a_j) = 9 \text{ev}_9(a_{j+1}) + v_j \text{ev}_9(u)$, whose iteration proves (5.19). Uniqueness of the normal form transports associativity and commutativity to the constructed product; its unit is (1).

Definition 5.20 (Reduced multiplicative coproduct). On the vector space of finite support generated by the nonempty normal forms, define

$$\tilde{\Delta}_\# e_w = \sum_{\substack{u \boxtimes_\# v = w \\ u, v \neq (1)}} e_u \otimes e_v.$$

A nonunit normal form is irreducible when its image under $\tilde{\Delta}_\#$ is zero.

The factorization fibres are finite by (5.19). In the completion $\mathcal{H}_\# = \ell^2(\mathcal{W}_9 \setminus \{\emptyset\})$, the images of distinct forms have disjoint supports. If $d_\#(w)$ counts the ordered nonunit factorizations, the domain of the closure is

$$\text{Dom}(\overline{\tilde{\Delta}_\#}) = \left\{ x \in \mathcal{H}_\# : \sum_w d_\#(w) |x_w|^2 < \infty \right\}.$$

Indeed, the squared norm of the image is that weighted sum. Therefore, the operator

$$A_\# = (\overline{\tilde{\Delta}_\#})^* \overline{\tilde{\Delta}_\#}$$

is diagonal, positive, and self-adjoint: its eigenvalue at e_w is the number of ordered nonunit factorizations of w . The projection

$$P_{\text{irr}} = (I - |e_{(1)}\rangle\langle e_{(1)}|) \mathbf{1}_{\{0\}}(A_\#)$$

selects exactly the irreducible forms. Through multiplicative evaluation, these correspond to the ordinary primes. Selection depends on the factorization structure and keeps the unit state separate.

For a value $n \geq 2$ coprime to 10, decimal division induces the residue permutation $r \mapsto 10r \pmod{n}$. Return of the residue 1 has period

$$\tau_n = \text{ord}_n(10).$$

If $n = \prod_p p^{a_p}$, the Chinese remainder theorem gives

$$\tau_n = \text{mcm}_{p^{a_p} \| n} \text{ord}_{p^{a_p}}(10).$$

The composite return synchronizes the returns of the prime powers. For a prime $p \notin \{2, 3, 5\}$, we have

$$\tau_p \mid p - 1, \quad M_p = \frac{10^{\tau_p} - 1}{p} \in \mathbb{N}, \quad M_p \equiv 0 \pmod{9}.$$

The last congruence expresses the nonadic closure of the decimal repetition: $10^{\tau_p} - 1$ is a multiple of 9, and p is invertible modulo 9. The same congruence holds for any n coprime to 90. In particular, 7, 13, 77, 91, 143 have decimal period 6, although only the first two are irreducible. The coproduct and the period therefore record different properties of the same constructed value.

This distinction makes precise the relationship with aperiodic refinement. The vacancy sequence determines the increments of the capacity index; factorization determines the multiplicative modes; decimal division determines their return times. The nonadic representative preserves phase closure, while the quotient and factorization preserve information that this closure does not distinguish. The link with the spectral constructions of the treatise resides in those operators and their intertwining. The proofs of their subsequent analytic results retain their own development; they are not replaced by the periodicity of a residual observation.

5.15.1 Vacancy discrepancy and Dirichlet readouts

The link also admits an exact analytic expression. The same encoding $z_n = z_n(0)$, for $n \geq 1$, is preserved, and two subsequent observables are considered: the series over all indices and the product over the indices selected by irreducibility. For $\operatorname{Re} s > 1$,

$$Z_{\text{vac}}^+(s) = \sum_{n \geq 1} \frac{z_n}{n^s}, \quad Z_{\text{vac}}^\times(s) = \prod_p (1 - p^{-s})^{-z_p}.$$

Both converge absolutely in that half-plane. Their common factor is the density δ , but their discrepancies differ.

Proposition 5.21 (Separation of vacancy density). *We have*

$$Z_{\text{vac}}^+(s) = \delta \zeta(s) + H_{\text{vac}}(s), \quad H_{\text{vac}} \in \mathcal{O}(\operatorname{Re} s > 0).$$

In $\operatorname{Re} s > 1$, with logarithms defined by the absolutely convergent Euler series,

$$Z_{\text{vac}}^\times(s) = \zeta(s)^\delta G_{\text{vac}}(s), \quad \log G_{\text{vac}}(s) = \sum_p (z_p - \delta) \log(1 - p^{-s})^{-1}.$$

Proof. The centred partial sums satisfy

$$A_N := \sum_{n=1}^N (z_n - \delta) = \lfloor (N+1)\delta \rfloor - N\delta, \quad |A_N| < 1.$$

Abel summation represents the centred term by $s \int_1^\infty A_{[x]} x^{-s-1} dx$. The integral converges locally uniformly in $\operatorname{Re} s > 0$, proving holomorphy. The second identity follows by separating $z_p = \delta + (z_p - \delta)$ in the logarithm of the product, within its domain of absolute convergence. $\square$

The decomposition makes visible what the readout on primes adds: it examines the same vacancy sequence on a subset determined by factorization. Control of the discrepancy over all integers gives the first holomorphic extension; the second observable preserves a discrepancy of its own on primes. These are two applications of the same discrete structure, with explicit analytic domains. This articulation explains their relevance to the arithmetic core without transferring to this article the proofs of the terminal results of the treatise.

6 Integral compatibility of regional generation, cylindrical refinement, and incidence

Regional prolongation produces ordered sequences of blocks together with their reading depths. The Archimedean projection and ternary incidence are calculated on these same sequences. The connection between the two realizations can be formulated entirely through integer operations: selection by rational intervals, Euclidean division, lifting of six trits, carry reading, and the Witt transformation. This formulation makes it possible to trace each coefficient from its generating block to its subsequent use.

The construction uses the three regional readers already obtained from the APP-TRIT-TPK state. We denote them by the indices $c \in \{P, E, F\}$, corresponding, respectively, to the closure, propagation, and autoscaling regions. Their real realization will be written as v_c . The subsequent identifications $v_P = \pi_{\text{HMT}}$, $v_E = e_{\text{HMT}}$, and $v_F = \varphi_{\text{HMT}}$ preserve this causal order. The operation that emits a block inspects the rational intervals of the regional reader; the real value enters the proof of correctness of that operation.

6.1 Positional emission through rational separation of cells

For each region, two rational sequences $\ell_c(j), u_c(j)$, constructed by its reader, are available and satisfy

$$\ell_c(j) \leq v_c \leq u_c(j).$$

The cell-separation property proved for these readers states that, for each integer $s > 0$, there exists J such that, for every $j \geq J$,

$$\lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1 = \lfloor sv_c \rfloor. \quad (6.1)$$

Avoidance of rational boundaries by the regional values is the hypothesis of the preceding theorems that guarantees this separation. Here we use their operational conclusion together with the readers and intervals that produce it.

Definition 6.1 (Separation index and emitted prefix). For $s > 0$, let

$$j_c(s) = \min\{j \in \mathbb{N} : \lfloor s\ell_c(j) \rfloor = \lfloor su_c(j) \rfloor - 1\}.$$

For an integer base $b \geq 2$ and a depth $n \geq 0$, define

$$P_c(b, n) = \lfloor b^n \ell_c(j_c(b^n)) \rfloor. \quad (6.2)$$

The prefix contains the integer part and the first n positions in base b ; consequently, $P_c(b, 0)$ is the regional integer part.

Theorem 6.2 (Termination and correctness of emission). *The index $j_c(s)$ exists for every $s > 0$. The emitted prefixes satisfy*

$$P_c(b, n) = \lfloor b^n v_c \rfloor, \quad P_c(b, n) \leq b^n v_c < P_c(b, n) + 1. \quad (6.3)$$

For each scale s , a single sufficiency index $J(s)$ can also be chosen that is valid simultaneously for all three regions.

Proof. The eventual equality in (6.1) makes the set defining $j_c(s)$ nonempty. Its minimum exists by the well-ordering of $\mathbb{N}$. At any index where the equality holds, the rational lower bound gives $\lfloor s\ell_c(j) \rfloor \leq \lfloor sv_c \rfloor$. The upper bound, together with avoidance of a cell boundary, gives $\lfloor sv_c \rfloor \leq \lfloor su_c(j) \rfloor - 1$. When the two integer endpoints coincide, the intervening integer coincides with them. This proves (6.3). For the joint statement, it suffices to take the maximum of the three sufficiency indices obtained at the same scale. The joint index controls all three emissions simultaneously, while each region retains its own intervals. $\square$

Proposition 6.3 (Exact compatibility of prefixes). *For $n, m \geq 0$,*

$$\left\lfloor \frac{P_c(b, n+m)}{b^m} \right\rfloor = P_c(b, n). \quad (6.4)$$

In particular, the next block $d_c(b, n) = P_c(b, n+1) \bmod b$ satisfies

$$P_c(b, n+1) = bP_c(b, n) + d_c(b, n), \quad 0 \leq d_c(b, n) < b. \quad (6.5)$$

Proof. Write $b^n v_c = N + \theta$, with $N = P_c(b, n)$ and $0 \leq \theta < 1$. Then $P_c(b, n+m) = b^m N + \lfloor b^m \theta \rfloor$, and the second summand belongs to $\{0, \dots, b^m - 1\}$. Euclidean division by b^m recovers N . The specialization $m = 1$ yields (6.5) and its residual bound. $\square$

6.2 Regional blocks of six trits at arbitrary depth

The integer equality $729 = 3^6$ determines the grouping of six ternary positions into a block. For $t \geq 0$, define

$$u_c(t) = P_c(729, t+1) \bmod 729, \quad B_{c,j}(t) = \left\lfloor \frac{u_c(t)}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6. \quad (6.6)$$

Thus $B_c(t)$ is an ordered word of six trits, and

$$u_c(t) = \sum_{j=0}^5 B_{c,j}(t) 3^{5-j}. \quad (6.7)$$

The word and its integer lift are mutually recoverable representations of the same emitted block.

Theorem 6.4 (Stability of extraction at any depth). *For every n ,*

$$P_c(729, n) = P_c(3, 6n).$$

If $t < n$, then

$$u_c(t) = \left\lfloor \frac{P_c(729, n)}{729^{n-t-1}} \right\rfloor \bmod 729. \quad (6.8)$$

Consequently, every extension of the prefix preserves all previously emitted blocks and all signatures calculated from them.

Proof. The first equality follows because both prefixes are separated at the same scale, $729^n = 3^{6n}$, and from Theorem 6.2. For the second, apply (6.4) at depths n and $t+1$, and then take the residue modulo 729. The ternary extraction in (6.6) is deterministic. Any explicit function of those eighteen coordinates therefore retains the same result when the prefix is extended. $\square$

The construction is quantified over $t \in \mathbb{N}$. A table materializing one hundred blocks is a finite evaluation of this construction; formula (6.8) determines its compatibility with every subsequent extension exactly.

Example 6.5 (First joint block). Evaluation of the selected regional readers gives the words

$$B_P(0) = 010211, \quad B_E(0) = 201101, \quad B_F(0) = 121200.$$

Their lifting by (6.7) produces

$$u_P(0) = 103, \quad u_E(0) = 523, \quad u_F(0) = 450.$$

For example, $201101_3 = 2 \cdot 243 + 27 + 9 + 1 = 523$. All six positions, including leading zeros, are retained in the band representation. The leading zero in 010211 has a defined position even though the decimal notation for 103 does not contain it.

6.3 Transition registers and reconstruction of linear lifts

Over $\mathbb{F}_3$, collect two consecutive emissions from each region in the matrix

$$R(t) = \begin{pmatrix} B_P(t) \\ B_P(t+1) \\ B_E(t) \\ B_E(t+1) \\ B_F(t) \\ B_F(t+1) \end{pmatrix}. \quad (6.9)$$

This definition fixes both the regional order and the temporal order. The first four registers are

$$X_0 = R(0), \quad Y_0 = R(1), \quad X_1 = R(2), \quad Y_1 = R(3).$$

Their values, obtained by extraction from the regional prefixes, are

$$\begin{aligned} X_0 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 2 & 0 & 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 0 \\ 1 & 1 & 2 & 2 & 0 & 2 \end{pmatrix}, & Y_0 &= \begin{pmatrix} 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \end{pmatrix}, \\ X_1 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \end{pmatrix}, & Y_1 &= \begin{pmatrix} 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 0 \end{pmatrix}. \end{aligned}$$

Theorem 6.6 (Unique reconstruction of the two initial links). *The equations $X_0 L_0 = Y_0$ and $X_1 L_1 = Y_1$ each have a unique solution in $M_6(\mathbb{F}_3)$. These solutions are*

$$\begin{aligned} L_0 &= \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, & L_1 &= \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}. \end{aligned}$$

Both matrices are invertible.

Proof. The matrices

$$\begin{aligned} A_0 &= \begin{pmatrix} 1 & 1 & 2 & 0 & 1 & 2 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 2 & 1 & 2 & 1 & 0 & 1 \\ 0 & 2 & 0 & 1 & 0 & 2 \\ 2 & 1 & 1 & 0 & 2 & 2 \\ 2 & 1 & 1 & 1 & 1 & 2 \end{pmatrix}, & A_1 &= \begin{pmatrix} 1 & 0 & 1 & 2 & 0 & 0 \\ 0 & 0 & 1 & 1 & 2 & 2 \\ 0 & 1 & 2 & 0 & 1 & 1 \\ 1 & 1 & 0 & 2 & 0 & 0 \\ 1 & 1 & 2 & 1 & 1 & 2 \\ 1 & 0 & 0 & 0 & 0 & 2 \end{pmatrix} \end{aligned}$$

satisfy $A_i X_i = X_i A_i = I$, by multiplication in $\mathbb{F}_3$. The solutions are therefore forced by $L_i = A_i Y_i$; multiplication produces the two displayed matrices. Their inverses are

$$L_0^{-1} = \begin{pmatrix} 1 & 2 & 0 & 2 & 0 & 2 \\ 2 & 0 & 2 & 2 & 0 & 0 \\ 2 & 1 & 2 & 0 & 2 & 0 \\ 1 & 0 & 0 & 0 & 2 & 0 \\ 0 & 2 & 1 & 1 & 2 & 0 \\ 2 & 2 & 2 & 2 & 2 & 2 \end{pmatrix}, \quad L_1^{-1} = \begin{pmatrix} 2 & 0 & 2 & 1 & 2 & 1 \\ 2 & 1 & 0 & 0 & 2 & 0 \\ 2 & 0 & 2 & 2 & 0 & 2 \\ 2 & 0 & 0 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 & 0 \\ 2 & 0 & 1 & 2 & 2 & 0 \end{pmatrix}.$$

The bilateral verification $L_i L_i^{-1} = L_i^{-1} L_i = I$ proves exact recovery of each transformed word. All products use residues 0, 1, 2 with their interpretation in $\mathbb{F}_3$. $\square$

The causal reading of this result is precise: the regional emissions produce $R(t)$; the registers $R(0), \dots, R(3)$ determine the two lifts above; their inverses recover the transformed words. Continuity for every t belongs to the coinductive readers and Theorem 6.4. The matrix reconstruction proved here has the two links specified in its statement as its domain.

6.4 Simultaneous refinement in ternary and decimal bases

A ternary emission of six trits and a decimal emission of three digits are readings in different bases. To preserve their relationship, record the state

$$s = (n, k, N, D),$$

where N is a prefix at depth n in base 729, and D is a prefix at depth k in base 1000. Their Archimedean cylinders are

$$I_{729}(s) = \left[\frac{N}{729^n}, \frac{N+1}{729^n} \right), \quad I_{1000}(s) = \left[\frac{D}{1000^k}, \frac{D+1}{1000^k} \right). \quad (6.10)$$

The half-open upper endpoint assigns each value to a unique cell at a given base and depth.

Definition 6.7 (Integer compatibility defects). Define

$$A(s) = N1000^k - D729^n, \quad (6.11)$$

$$B(s) = (N+1)1000^k - D729^n = A(s) + 1000^k. \quad (6.12)$$

The state is compatible when

$$A(s) < 729^n, \quad B(s) > 0. \quad (6.13)$$

Lemma 6.8 (Integer criterion for intersection). *The cylinders in (6.10) have a nonempty intersection if and only if (6.13) holds.*

Proof. Two half-open intervals $[a, b)$ and $[c, d)$, with $a < b$ and $c < d$, intersect exactly when $a < d$ and $c < b$. Applying this condition to (6.10) and multiplying by the positive denominators gives $N1000^k < (D+1)729^n$ and $D729^n < (N+1)1000^k$. These are the two inequalities in (6.13). $\square$

Definition 6.9 (Update of depths and prefixes). For $0 \leq u < 729$ and $0 \leq d < 1000$, define

$$T_u(n, k, N, D) = (n+1, k, 729N+u, D),$$

$$T_{u,d}(n, k, N, D) = (n+1, k+1, 729N+u, 1000D+d).$$

The first operation refines only the ternary cylinder. The second refines both readings and preserves both depths in the state.

Proposition 6.10 (Exact boundary recurrences). *The updates above satisfy*

$$A(T_u s) = 729A(s) + u1000^k, \quad (6.14)$$

$$A(T_{u,d} s) = 729000A(s) + u1000^{k+1} - d729^{n+1}. \quad (6.15)$$

Proof. For the first identity,

$$(729N + u)1000^k - D729^{n+1} = 729(N1000^k - D729^n) + u1000^k.$$

For the second,

$$\begin{aligned} (729N + u)1000^{k+1} - (1000D + d)729^{n+1} \\ = 729000(N1000^k - D729^n) + u1000^{k+1} - d729^{n+1}. \end{aligned}$$

Both are integer identities preceding any real approximation. $\square$

Theorem 6.11 (Refinement of the generated regional cylinders). *Let*

$$s_c(n, k) = (n, k, P_c(729, n), P_c(1000, k)).$$

Then $s_c(n, k)$ is compatible for all n, k . Moreover,

$$\begin{aligned} T_{u_c(n)}s_c(n, k) &= s_c(n + 1, k), \\ T_{u_c(n), d_c(1000, k)}s_c(n, k) &= s_c(n + 1, k + 1). \end{aligned}$$

Euclidean division of the refined prefixes recovers the preceding prefixes exactly. For every $a, b \geq 0$,

$$\frac{P_c(729, n + a) - P_c(729, n + a) \bmod 729^a}{729^a} = P_c(729, n),$$

and the analogous identity holds in base 1000.

Proof. Theorem 6.2 places v_c simultaneously in both intervals in (6.10); the intersection criterion gives compatibility. The update identities are (6.5) in each base. Recovery follows from (6.4). Thus the defect recurrence and prefix recovery are consequences of the same update. $\square$

Example 6.12 (First pair of regional cylinders). For $n = k = 1$, evaluation of the three regions gives

c	N	D	A	B
P	2290	3141	211	1211
E	1981	2718	-422	578
F	1179	1618	-522	478

In all three rows, $A < 729$ and $B > 0$. The first prefix column decomposes as $2290 = 3 \cdot 729 + 103$, $1981 = 2 \cdot 729 + 523$, and $1179 = 729 + 450$: exactly the previously emitted six-trit blocks reappear. The different signs of A express the relative positions of the two lower endpoints without altering the regional value's membership in both intervals.

6.5 Selection of the next block by the regional reader

Compatibility of two intervals expresses a relationship between readings. Determining the next block additionally uses the rational interval produced by the region. This operation can be formulated without discarding the depth history.

Definition 6.13 (Admissibility of regional prolongation). For fixed c, n , let $s = 729^{n+1}$ and $j = j_c(s)$. A residue $u \in \{0, \dots, 728\}$ is admissible when

$$[s\ell_c(j)] \leq 729P_c(729, n) + u \leq [su_c(j)] - 1. \quad (6.16)$$

Theorem 6.14 (Uniqueness of the prolongation compatible with the reader). *Condition (6.16) is equivalent to $u = u_c(n)$. There is therefore exactly one admissible prolongation of each region at each depth.*

Proof. At the separation index, both integer endpoints in (6.16) coincide with $P_c(729, n + 1)$. The condition is therefore equivalent to $729P_c(729, n) + u = P_c(729, n + 1)$. By (6.5), the right-hand side is $729P_c(729, n) + u_c(n)$. Cancelling the common term gives equality of the residues. Conversely, this equality satisfies the condition. $\square$

The effective dependence of this selection is fully specified: selected region, rational reader, depth, separation index, preceding prefix, and next residue. The defect A summarizes a boundary relationship; the state (n, k, N, D) and the regional intervals retain the data used by the update.

6.6 Ternary signature, carry, Witt incidence, and reading clock

At a fixed depth t , write $B_{c,j} = B_{c,j}(t)$. For each column j , calculate the following integers and residues:

$$S_j = B_{P,j} + B_{E,j} + B_{F,j}, \quad q_j = S_j \bmod 3, \quad c_j = \lfloor S_j/3 \rfloor, \quad (6.17)$$

$$a_j = (B_{P,j} + B_{E,j} - B_{F,j}) \bmod 3, \quad w_j = \sum_c \mathbf{1}_{B_{c,j} \neq 0}, \quad \bar{w}_j = w_j \bmod 3. \quad (6.18)$$

The residue of the axial expression is taken in the integers. An implementation using natural numbers calculates it as $(B_{P,j} + B_{E,j} + 3 - B_{F,j}) \bmod 3$, which preserves that residue exactly.

The ternary incidence matrix used by the reader is

$$W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}. \quad (6.19)$$

Two residual charges are preserved for each region:

$$v_c^{\text{res}} = \sum_j B_{c,j} \bmod 3, \quad v_c^W = \sum_j (B_c W)_j \bmod 3. \quad (6.20)$$

The reader's complete signature is $\mathcal{S}(B) = (q, a, c, w, \bar{w}, v^{\text{res}}, v^W)$. Its structure separately preserves the column residue, carry, axial contrast, support weight, and the two charge readings.

Theorem 6.15 (Exact recovery and bounds of the signature). *For each column and each region,*

$$S_j = q_j + 3c_j, \quad 0 \leq q_j, a_j < 3, \quad 0 \leq c_j \leq 2, \quad (6.21)$$

$$B_{F,j} = 2(q_j + 3 - a_j) \bmod 3, \quad 0 \leq w_j \leq 3, \quad 0 \leq \bar{w}_j < 3. \quad (6.22)$$

Moreover,

$$v_c^W = \left(\sum_j ((B_c W)_j \bmod 3) \right) \bmod 3. \quad (6.23)$$

The stated recoveries determine each column's total and its autoscaling coordinate; the regional register additionally preserves the three ordered bands.

Proof. Identity (6.21) is Euclidean division of S_j by 3. Each summand of S_j lies between 0 and 2, so $0 \leq S_j \leq 6$ and $c_j \leq 2$. In $\mathbb{F}_3$, subtracting the two congruences defining q_j and a_j gives $q_j - a_j = 2B_{F,j}$. Since the inverse of 2 is 2, this yields (6.22); the summand 3 permits an expression using natural numbers. The bounds on w_j follow by summing three indicators. Finally, reducing an integer sum modulo 3 is equivalent to reducing each summand and then reducing their sum, which proves (6.23). $\square$

Example 6.16 (Signature of the first joint block). The three initial bands produce

$$\begin{aligned} q &= (0, 0, 2, 2, 1, 2), & a &= (1, 2, 0, 1, 1, 2), \\ c &= (1, 1, 0, 1, 0, 0), & w &= (2, 2, 2, 3, 1, 2), \\ v^{\text{res}} &= (2, 2, 0), & v^W &= (2, 1, 1). \end{aligned}$$

The fourth column has total $2 + 1 + 2 = 5$; its signature records $q_3 = 2$ and $c_3 = 1$, so $q_3 + 3c_3 = 5$. Axial recovery gives $2(2 + 3 - 1) \bmod 3 = 2$, precisely the autoscaling coordinate. The Witt images, reduced componentwise, are

$$(2, 0, 2, 1, 1, 2), \quad (0, 0, 0, 2, 0, 2), \quad (2, 1, 1, 2, 1, 0).$$

Their residual sums by row yield the three charges $(2, 1, 1)$.

Definition 6.17 (Integer clock of decimal reading). The transition index t and decimal representation depth are related by

$$\kappa(t) = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\}. \quad (6.24)$$

Proposition 6.18 (Characterization and monotonicity of the clock). *The integer $\kappa(t)$ is uniquely determined by*

$$1000^{\kappa(t)} \leq 729^{t+1} < 1000^{\kappa(t)+1}. \quad (6.25)$$

The function κ is monotone and satisfies $\kappa(20) = \kappa(21) = 20$.

Proof. The powers of 1000 are strictly increasing and unbounded; there is therefore a unique interval between consecutive powers containing 729^{t+1} . Monotonicity of the latter expression proves that of κ . The final two evaluations are the integer comparisons

$$1000^{20} \leq 729^{21} < 1000^{21}, \quad 1000^{20} \leq 729^{22} < 1000^{21}.$$

These comparisons use exact integer powers. □

Equality of decimal depth at two distinct transitions explains why both indices must be retained: the reading clock can remain constant while the ternary band advances and its signature is updated. Chronological information resides in the ordered register of transitions.

6.7 Composition theorem for the block, cylinder, and signature

Theorem 6.19 (Joint compatibility of regional readings). *For each depth n , there exists a unique triple of blocks (u_P, u_E, u_F) satisfying regional admissibility (6.16) in all three regions. This triple is $(u_P(n), u_E(n), u_F(n))$. Simultaneously:*

1. *each block updates its cylinder through recurrences (6.14) and (6.15);*
2. *its six trits produce the signature $S(B_P(n), B_E(n), B_F(n))$;*
3. *the column totals are recovered as $q + 3c$, and the Witt charges are obtained from the same bands through (6.23);*
4. *every deeper prefix returns exactly these blocks, truncated cylinders, and signatures.*

Proof. Apply Theorem 6.14 to each of the three regional indices. Coordinatewise uniqueness gives joint uniqueness. Substitute those same residues into the operations $T_u, T_{u,d}$, which allows Theorem 6.11 to be applied. The deterministic application of (6.6) produces the bands used by (6.17)–(6.20). The recovery identities are those of Theorem 6.15. Finally, Theorem 6.4 and truncation in each base prove stability at greater depth. At every step the identity of the emitted block has been preserved; neither reading requires the selection of a different block. □

The conclusion is an operational compatibility between cylindrical contraction and preservation of incidence. Both use the same generated register, retain the depths, and admit integer recovery. This result is the explicit form of the connection

$$\text{emitted regional block} \longrightarrow \begin{cases} \text{cylinder and integer boundary,} \\ \text{band, signature, carry, and Witt charge.} \end{cases}$$

The subsequent development of incidence and fields uses the second reading while preserving its common provenance with the first.

6.8 Calendar census and simultaneous compatibility of the regional biographies

The preceding rows are assembled, without yet applying any decimal reader, into the three block biographies produced by the transitions:

$$\begin{aligned} B^{\text{cl}} &= 010211|012222|010211|002111|110221, \\ B^{\text{pr}} &= 201101|121221|102011|012222|102011, \\ B^{\text{au}} &= 121200|112202|121020|010210|010200. \end{aligned} \quad (6.26)$$

Indeed, the consecutive pairs of the first two segments are the rows of X_0, Y_0 , and those of the last two are the rows of X_1, Y_1 . Thus, the biographies are another representation of the record of twelve transitions, not a collection of conventional prefixes.

To verify the operative word of the calendar, let $c = c_1c_2c_3c_4 \in \{0, 1\}^4$ and, for each regime $\chi \in \{\text{cl}, \text{pr}, \text{au}\}$, define

$$u_0^\chi = b_0^\chi, \quad u_j^\chi = u_{j-1}^\chi L_{c_j} \quad (1 \leq j \leq 4).$$

To condense the census, denote by

$$N_{\text{ar}}(c) := \#\{(j, \chi) : u_j^\chi = b_j^\chi, 1 \leq j \leq 4\}, \quad N_{\text{bio}}(c) := \#\{\chi : (u_0^\chi, \dots, u_4^\chi) = B^\chi\}.$$

Exact multiplication in $\mathbb{F}_3$ for the sixteen calendars gives:

c	$N_{\text{ar}}(c)$	$N_{\text{bio}}(c)$
0000, 0001	6	0
0010	9	0
0011	12	3
0100, 0101, 0110, 0111	3	0
1000, 1001, 1010, 1011	0	0
1100, 1101, 1110, 1111	0	0

(D)

The table exhausts the $2^4 = 16$ possibilities, and each of its entries is obtained by four multiplications of a row by one of the two displayed matrices. Consequently, 0011 is the unique calendar that simultaneously reconstructs the three complete biographies. In operator notation,

$$(L_0, L_0, L_1, L_1) \quad (6.27)$$

is not an independent prescription: it is the unique binary reading compatible with the twelve edges already produced by U_t .

The calendar (6.27) produces four new six-component blocks. Their concatenations form

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30}.$$

The subscripts of w indicate cumulative length. The boundary R_{36} preserves the terminal structure and opens the prolongation relation; it is not renamed as a final periodic block that could be repeated indefinitely.

The identity $L = X^{-1}Y$ proves uniqueness once the transitions are fixed; it does not replace their production by U_t . Conversely, (D) proves within the article the uniqueness of the ordering of the four transports. The causal chain is thereby closed: the APP census produces the initial regions, TRIT fixes regime and orientation, TPK produces the transition record, and that record determines L_0, L_1 and 0011 before any Archimedean realization.

7 Regional determination of the dodecaphasic descriptor and selection of the coupling register

The determination of the coupling register combines two constructions that should be developed in their actual order. The first produces the inputs to the selector from the regional representations of the TPK: six-trit bands, column signatures, a conversion calendar, an oriented descriptor, and a functional panel. The second operates on these inputs and on the terminal eight-block repertoire, enumerates its admissible orderings, and selects a unique register through two simultaneous conditions: exceptional incidence and reading heterotypy. The value of the register appears at the end of this composition.

The present development presupposes the regional representations obtained in the preceding chapters from APP-TRIT-TPK. Its purpose is to make explicit every intermediate operation leading from those representations to the finite selector. The realization of the same register through signed incidences, Hadamard inversion, and the representation of the fine-structure constant are then incorporated. The three constructions retain their respective domains: selection of the register, recovery from its channels, and arithmetic evaluation of its representations.

7.1 Regional bands, encoding, and integer signatures

7.1.1 Domains and encoding of the regional bands

The channel order is fixed as

$$\chi \in \{\pi, e, \varphi\}, \quad B_\chi(t) \in \mathbb{F}_3^6, \quad t \in \mathbb{N}.$$

The channel symbols identify the regions already generated. Each $B_\chi(t)$ retains its place in its sequence of representations; the index t counts ternary blocks. The natural-number evaluation of a band $b = (b_0, \dots, b_5)$ is

$$\text{enc}_6(b) = 243b_0 + 81b_1 + 27b_2 + 9b_3 + 3b_4 + b_5, \quad 0 \leq \text{enc}_6(b) < 729. \quad (7.1)$$

Its fixed-width inverse is

$$\text{dig}_j(n) = \left\lfloor \frac{n}{3^{5-j}} \right\rfloor \bmod 3, \quad 0 \leq j < 6.$$

Leading zeros are thus retained: the words 002001 and 2001 could share an abbreviated integer notation, but the type of the former contains six positions. All subsequent concatenations use this width.

Within the hundred-block window, the six-hundred-trit representation of channel χ determines an integer $P_\chi^{(600)}$. Extraction is performed by

$$b_\chi(t) = \left\lfloor \frac{P_\chi^{(600)}}{729^{99-t}} \right\rfloor \bmod 729, \quad 0 \leq t < 100, \quad B_\chi(t) = \text{enc}_6^{-1}(b_\chi(t)). \quad (7.2)$$

The six-hundred-trit integer is a representation of the regional producer; the hundred-row table is the evaluation of (7.2). This separation allows a historical table to be replaced by its producer without changing the selector that consumes it.

7.1.2 Residues, quotients, occupancy, and orientation

For each column j and time t , introduce

$$T_j(t) = B_\pi(t)_j + B_e(t)_j + B_\varphi(t)_j, \quad (7.3)$$

$$q_j(t) = T_j(t) \bmod 3, \quad c_j(t) = \lfloor T_j(t)/3 \rfloor, \quad (7.4)$$

$$a_j(t) = (B_\pi(t)_j + B_e(t)_j - B_\varphi(t)_j) \bmod 3, \quad (7.5)$$

$$w_j(t) = \mathbf{1}_{B_\pi(t)_j \neq 0} + \mathbf{1}_{B_e(t)_j \neq 0} + \mathbf{1}_{B_\varphi(t)_j \neq 0}, \quad \bar{w}_j(t) = w_j(t) \bmod 3. \quad (7.6)$$

The subtraction in the third expression is performed in the integers and then reduced. In an implementation over natural numbers, the equivalent expression is $(B_\pi + B_e + 3 - B_\varphi) \bmod 3$; truncated subtraction would change the reader.

Proposition 7.1 (Reconstruction of the column sum). *For all admissible bands,*

$$T_j(t) = q_j(t) + 3c_j(t), \quad 0 \leq q_j(t) < 3, \quad 0 \leq c_j(t) \leq 2, \quad 0 \leq w_j(t) \leq 3.$$

Proof. Each column sums three representatives from $\{0, 1, 2\}$, so that $0 \leq T_j(t) \leq 6$. Euclidean division by three yields the stated residue and quotient. Occupancy counts three binary conditions; its bound follows directly from that definition. $\square$

This identity preserves the quotient of the local sum. Reconstructing the three individual bands requires the bands themselves or an additional reader that distinguishes them: a column sum has fewer coordinates than the triple that produces it. Accordingly, the temporal row used below jointly retains the three bands and the four fields $q, a, c, \bar{w}$.

7.2 Conversion clock and first zero advance

7.2.1 Ternary capacity and decimal depth

A representation of $t + 1$ bands contains $6(t + 1)$ trits. The associated conversion calendar is defined by the integer

$$d_t = \max\{k \in \mathbb{N} : 1000^k \leq 3^{6(t+1)}\} = \max\{k \in \mathbb{N} : 1000^k \leq 729^{t+1}\}. \quad (7.7)$$

The definition compares integer capacities. The counter t and the depth d_t record different quantities: a transition adds one six-trit band; the clock counts complete powers of the decimal base one thousand contained in the accumulated ternary capacity.

Lemma 7.2 (Integer characterization of the clock). *For $t, k \in \mathbb{N}$,*

$$d_t = k \iff 1000^k \leq 729^{t+1} < 1000^{k+1}.$$

Moreover, $d_t \leq d_{t+1}$.

Proof. The powers of one thousand form a strictly increasing, unbounded sequence. The power 729^{t+1} lies between two consecutive terms; the lower exponent is precisely the maximum in (7.7). The inequality $729^{t+1} < 729^{t+2}$ implies monotonicity of the maximum. $\square$

Proposition 7.3 (First zero advance). *The least time satisfying $d_{t+1} = d_t$ is*

$$t_* = 20.$$

In particular,

$$d_0 = 0, \quad d_{19} = 19, \quad d_{20} = 20, \quad d_{21} = 20, \quad d_{22} = 21.$$

Proof. The exact integer comparisons

$$1000^{20} \leq 729^{21}, \quad 729^{22} < 1000^{21}, \quad 1000^{21} \leq 729^{23} < 1000^{22}$$

give the values at times twenty, twenty-one, and twenty-two. For $0 \leq t \leq 20$, the expression $729^{t+1}/1000^t = 729(729/1000)^t$ is decreasing and its value at twenty is greater than or equal to one. Hence, $1000^t \leq 729^{t+1} < 1000^{t+1}$ and $d_t = t$. No time less than twenty can satisfy $d_{t+1} = d_t$; at twenty it does hold. All comparisons used involve integer powers. $\square$

The number twenty is therefore an output of the minimality criterion. The rule selecting the instant is “the first zero advance of the clock”; the algorithm and its proof can use this rule without receiving twenty as an external parameter.

7.2.2 Nonadic return and temporal antecedent

The length of the phase return determines

$$r = t_* - 9 = 11. \quad (7.8)$$

The descriptor queries are made at $t_* + 1$, $r + 1$, and r . This distribution preserves the relation between the pause instant, the nonadic translation, and the immediately associated phase reading. Times eleven, twelve, and twenty-one designate positions in the regional calendar; their temporal difference remains recorded even when two times produce the same decimal depth.

7.3 Oriented construction of the twenty-four-trit descriptor

7.3.1 Opposite half-band defects

Take the oriented defects of the regional chart

$$\delta_+ = (0, 0, 0, 1, 1, 1), \quad \delta_- = (0, 0, 0, 2, 2, 2) \in \mathbb{F}_3^6.$$

Their opposition is verified componentwise:

$$\delta_+ + \delta_- = 0 \quad \text{in } \mathbb{F}_3^6.$$

The representative 2 realizes -1 in the ternary chart. Concatenating the three-position tails in negative-positive order produces the word 222111. Adding the defects and concatenating their tails are different operations: the former has codomain $\mathbb{F}_3^6$, whereas the latter preserves the order of two segments of length three.

7.3.2 Dual charge and phase selection

In the regional chart, the six-coordinate transformation uses

$$A_{\text{reg}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

With row vectors, the dual charge of $b \in \mathbb{F}_3^6$ is the sum of the coordinates of bA_{reg} . The integer row sums of A_{reg} are $(5, 7, 7, 7, 7, 7)$; this gives the local expression

$$\eta^\vee(b) = 2b_0 + b_1 + b_2 + b_3 + b_4 + b_5 \pmod{3}. \quad (7.9)$$

The reading phase at instant r is $\theta_r = (r - 1) \bmod 3$. For each channel, define

$$\varepsilon_\chi(r) = \begin{cases} 1, & \eta^\vee(B_\chi(r)) = \theta_r, \\ 2, & \eta^\vee(B_\chi(r)) \neq \theta_r. \end{cases}$$

The phase-charge representation is

$$D(r) = \varepsilon_\pi(r)B_\pi(r) + \varepsilon_e(r)B_e(r) + \varepsilon_\varphi(r)B_\varphi(r) \quad \text{in } \mathbb{F}_3^6. \quad (7.10)$$

The coefficient is determined by an equality of charges before evaluating the word that will result from the combination.

7.3.3 Composition of the descriptor

Let $\parallel$ denote the concatenation of fixed-width words. The complete construction is

$$\Omega_{24} = (B_e(t_* + 1) + \delta_-) \parallel (\delta_-|_{3,4,5} \parallel \delta_+|_{3,4,5}) \parallel (B_\pi(r + 1) + B_e(r + 1)) \parallel D(r). \quad (7.11)$$

Each length-six summand is evaluated in $\mathbb{F}_3^6$; the final concatenation has length $6 + 6 + 6 + 6 = 24$. This construction distinguishes the descriptor Ω_{24} from the regional stages w_{24}^χ : the two objects share a length, but their producers and their places in the composition differ.

Proposition 7.4 (Regional evaluation of the descriptor). *The regional representations and the first-zero-advance criterion determine*

$$\Omega_{24} = 020022\ 222111\ 211201\ 021101, \quad [\Omega_{24}]_3 = 66241910521.$$

Proof. The required bands, preserving the order (π, e, φ) , are

t	$B_\pi(t)$	$B_e(t)$	$B_\varphi(t)$
11	022212	110001	100021
12	012012	202222	000120
21	221022	020100	211021

and come from extraction (7.2). At time twenty-one, $020100 + 000222 = 020022$ in $\mathbb{F}_3^6$. The defect tails provide 222111. At time twelve, $012012 + 202222 = 211201$.

At time eleven, the dual charges are 0, 1, 2, respectively; the phase is $(11 - 1) \bmod 3 = 1$. The resulting coefficients are (2, 1, 2) and

$$2(022212) + (110001) + 2(100021) = 021101 \quad \text{in } \mathbb{F}_3^6.$$

Concatenating these four results gives the stated word. Its integer evaluation is obtained by Horner's method in base three, or by three successive concatenations in base 729. $\square$

The preceding table serves a verification purpose: each of its entries is obtained from the regional producer. Construction rule (7.11) makes it possible to reproduce the descriptor without introducing the final twenty-four-trit string as an input.

7.4 Functional panel and terminal repertoire

7.4.1 Nine regional positions

The functional panel retains the first three bands of each of the three channels, in the declared regional order:

$$\mathcal{P}_9 = (b_\pi(0), b_\pi(1), b_\pi(2), b_e(0), b_e(1), b_e(2), b_\varphi(0), b_\varphi(1), b_\varphi(2)). \quad (7.12)$$

Its evaluation is

$$\mathcal{P}_9 = (103, 161, 103, 523, 457, 301, 450, 398, 438).$$

The nine positions retain their regional provenance, although the value 103 occurs twice. The panel has nine ordered incidences and eight distinct values. The subsequent enumeration uses the incidences; candidate deduplication takes place at a later stage.

7.4.2 Unordered eight-block repertoire

The terminal repertoire involved in this construction is

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\} \subset \mathbb{F}_3^6. \quad (7.13)$$

Its eight elements are distinct. The integer encoding, sorted by value solely to fix a computational representation, is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

The documentary provenance of this repertoire is the terminal segmentation of the seventy-two-trit word and its position-wise recovery through incidences of the phase-charge and affine readers. The selector developed below receives the repertoire without its historical ordering and tests its $8!$ orderings.

The exact formulation of the reassembled implementation maintains an explicit interface here: the regional producers materially replace the descriptor, the panel, and the temporal rows; repertoire (7.13) retains its earlier documentary producer. This distinction identifies where each input enters. Membership of eight words in a temporal reader repertoire, selection of those eight words, and determination of their terminal order are three different statements. The terminal-selection proof establishes the third for the declared repertoire.

Documentary recovery by position can be expressed exactly. Let $\mathcal{I}$ be the phase-charge incidence register and let $\mathcal{J}$ be the register of the affine extension. Each incidence retains the position ℓ in the seventy-two-trit word and the emitted word u . For $5 \leq \ell \leq 12$, form

$$\mathcal{S}_\ell = \{u : (\ell, u) \in \mathcal{I} \cup \mathcal{J}\}.$$

The recovery procedure checks that all eight positions occur and that each $\mathcal{S}_\ell$ is a singleton. If s_ℓ is its unique element, the output delivered to the selector is

$$\mathcal{S}_{\text{term}} = \{s_5, s_6, \dots, s_{12}\}.$$

The singleton proof for each position preserves the coherence of the historical witnesses; the subsequent export retains the eight words and removes only the historical ordering. The selector reconstructs the order through complete enumeration and the terminal conditions, instead of receiving it from the provenance register.

7.4.3 Temporal incidences of the terminal repertoire and recovery by position

The representation of the terminal repertoire uses the same regional blocks, the same carries, and the same calendar that produce the initial twenty-four-trit descriptor. Two operations in this composition should be made explicit separately. The first recovers eight words from recorded temporal incidences. The second presents those eight words to the dodecaphasic selector as an unordered set. The selector re-establishes the order through its incidence, cylinder, and closure conditions. The distinction preserves the provenance witnesses without turning the order of a historical segmentation into an input to the selector.

Let $B_t = (b_{t,0}, b_{t,1}, b_{t,2}) \in (\mathbb{F}_3^6)^3$ be the regional boundary already generated at time t , with $0 \leq t < 100$. In the chart used by the historical register, the incidence matrix is

$$A_{\text{hist}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 2 & 2 \\ 1 & 1 & 0 & 2 & 1 & 2 \\ 1 & 1 & 2 & 0 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 1 \\ 1 & 2 & 2 & 1 & 1 & 0 \end{pmatrix}.$$

The words are regarded as row vectors. Their two charges and the chronological phase are

$$r_{t,i} = \sum_{j=1}^6 b_{t,i,j}, \quad d_{t,i} = \sum_{j=1}^6 (b_{t,i} A_{\text{hist}})_j, \quad \vartheta_t = t - 1 \pmod{3}.$$

All operations in these three expressions take place in $\mathbb{F}_3$. The historical subscript identifies a specific chart of the Witt incidence; it keeps the coordinate transformation explicit when another presentation of that same incidence is used.

For $c \in \{r_t, d_t\}$, $\delta \in \mathbb{F}_3$ and $\epsilon \in \{1, -1\}$, define

$$\eta_i(c, \delta, \epsilon) = \begin{cases} \epsilon, & c_i = \vartheta_t + \delta, \\ -\epsilon, & c_i \neq \vartheta_t + \delta, \end{cases} \quad L_{t,c,\delta,\epsilon} = \sum_{i=0}^2 \eta_i(c, \delta, \epsilon) b_{t,i}.$$

The orientation -1 is written as 2 when printing the ternary coefficients. Alongside these twelve comparative readers per boundary, the register has the affine reader

$$L_t^{\text{af}} = \sum_{i=0}^2 (d_{t,i} + \vartheta_t) b_{t,i}.$$

Adding the phase in the affine reader and comparing charges in the preceding reader are different operations, both acting on data already produced by the TPK boundary.

t	$b_{t,0}$	$b_{t,1}$	$b_{t,2}$	r_t	d_t
8	200121	212212	110110	011	202
11	022212	110001	100021	001	012
30	210112	110202	211110	100	012
34	121022	000011	112110	220	021
37	120121	011202	112221	100	201
49	110212	200212	100122	110	201
58	111121	022121	122201	122	220
67	010012	001220	011120	122	122
75	220000	020111	100002	120	021
81	122201	020202	201122	202	001
90	022112	022110	121010	202	200

Table 6: Temporal boundaries realizing the printed incidences of the terminal repertoire. Each six-digit entry preserves the order of the six coordinates of the regional boundary.

The following table displays all comparative incidences of the register on the twelve blocks of the segmentation under consideration. The letters r and d denote the visible charge and the dual charge, respectively. Column j retains the position index of the historical witness; the operation delivering the data to the selector forgets this index after forming the set of words.

The tenth block has an affine witness. At $t = 30$, $d_{30} = (0, 1, 2)$ and $\vartheta_{30} = 2$, whence

$$(d_{30,0} + \vartheta_{30}, d_{30,1} + \vartheta_{30}, d_{30,2} + \vartheta_{30}) = (2, 0, 1)$$

and, using the bands in Table 6,

$$L_{30}^{\text{af}} = 2(210112) + 0(110202) + (211110) = (001001) \quad \text{in } \mathbb{F}_3^6.$$

The equality holds coordinatewise: before reduction modulo three, the sum vector is $(6, 3, 1, 3, 3, 4)$.

Proposition 7.5 (Complete recovery of the repertoire through incidences). *Consider the comparative-incidence register in Table 7, augmented by the affine witness $(j, t, L) = (10, 30, 001001)$. For each position $5 \leq j \leq 12$, the set of output words of its incidences is a singleton. Extracting these eight words and subsequently forgetting their order produces exactly*

$$\mathcal{S}_{\text{term}} = \{002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110\}.$$

Their encoding as base-three integers is

$$\{28, 55, 98, 175, 437, 498, 687, 714\}.$$

j	t	charge	δ	ϵ	$(\eta_0\eta_1\eta_2)$	$L_{t,c,\delta,\epsilon}$
4	11	d	0	1	212	021101
5	67	d	1	-1	211	002001
5	67	d	2	1	211	002001
5	67	r	1	-1	211	002001
5	67	r	2	1	211	002001
6	81	d	0	1	222	020111
6	81	r	2	1	222	020111
7	34	r	1	-1	111	200110
7	75	d	1	-1	211	200110
7	75	r	2	-1	211	200110
8	90	r	0	1	121	121012
8	90	r	1	-1	121	121012
9	49	d	0	-1	121	010122
11	11	d	2	-1	211	221110
11	37	d	0	-1	121	221110
12	8	d	0	-1	111	222110
12	8	r	1	-1	111	222110
12	58	d	1	-1	111	222110
12	58	r	0	-1	111	222110

Table 7: Complete phase-charge incidences on the terminal segmentation within the hundred boundaries of the register. Word matches between rows retain their times, orientations, and charge types.

Proof. The incidence multiplicities at positions $j = 5, 6, 7, 8, 9, 10, 11, 12$ are, respectively,

$$(4, 2, 3, 2, 1, 1, 2, 4).$$

At each position, all rows print the same word. Each row is verified by substituting its three bands and three coefficients into the formula for L ; the tenth case has been calculated explicitly through L_{30}^{af} . For example, at $t = 67$ and with coefficients $(2, 1, 1)$, one obtains

$$2(010012) + (001220) + (011120) = (002001) \pmod{3}.$$

The eight results are distinct as words of length six. Their encoding $\sum_{k=1}^6 w_k 3^{6-k}$ gives, in position order, $(55, 175, 498, 437, 98, 28, 714, 687)$. Forgetting the order yields the stated set of integers. Thus, extraction preserves each complete word, including its zero positions, and delivers eight distinct elements to the selector. $\square$

The preceding proof is an exact recovery from identified incidences. Its input comprises the incidence register with its positions and witnesses. The subsequent orbital selection receives only $\mathcal{S}_{\text{term}}$ and the initial regional descriptor; it traverses the $8!$ orderings and determines their compatibility with the functional boundaries and the oriented incidences of the dodecaphasic wheel. Both operations are thus presented contiguously, with explicit domains: temporal provenance of the eight words and constructive determination of their order.

The comparative table contains the nineteen incidences that its formula produces on the twelve-block segmentation over the hundred declared times. This finite exhaustiveness is checked by traversing $100 \cdot 2 \cdot 3 \cdot 2 = 1200$ evaluations. The affine reader extends coverage of the terminal positions from seven to eight. This count characterizes the local reader repertoire just defined; the other TPK operations retain the domains and constructive functions established in their respective derivations.

7.5 Comparative readers, affine reader, and temporal incidence

7.5.1 Witt transformation of the selector

The selector uses the six-coordinate chart

$$A_{\text{ter}} = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}.$$

Define $W(b) = bA_{\text{ter}}$ in $\mathbb{F}_3^6$, together with the charges

$$\eta(b) = \sum_{j=0}^5 b_j \pmod{3}, \quad \eta_{\text{ter}}^{\vee}(b) = \eta(W(b)).$$

The descriptor chart and the selector chart are related by the permutation $\sigma = (0, 1, 2, 4, 5, 3)$, written as a list of images:

$$(A_{\text{ter}})_{ij} = (A_{\text{reg}})_{\sigma(i), \sigma(j)}. \quad (7.14)$$

Lemma 7.6 (Compatibility of the charge functional). *For every band b ,*

$$\eta_{\text{ter}}^{\vee}(b) = \eta^{\vee}(b).$$

Proof. The integer sum of each corresponding row of the two matrices agrees: it is five in the first row and seven in the other five. By distributivity,

$$\sum_j \sum_i b_i (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{ter}})_{ij} = \sum_i b_i \sum_j (A_{\text{reg}})_{ij}.$$

Reduction modulo three gives the equality of charges. Matrix identity (7.14) is verified by substituting its thirty-six entries and also preserves the coordinate change. $\square$

Equality of the total charge functional allows its representations to be compared. The vector transformations remain identified by their matrices and the explicit permutation; the preceding proof preserves that coordinate information.

7.5.2 Twelve comparative readers per instant

Define the temporal phase

$$\theta(t) = (t + 2) \pmod{3}.$$

This expression realizes $(t - 1) \pmod{3}$ also at $t = 0$, where the phase equals two. For each charge type $\nu \in \{\eta, \eta_{\text{ter}}^{\vee}\}$, offset $o \in \{0, 1, 2\}$ and orientation $s \in \{1, 2\}$, take

$$\epsilon_{\chi}^{\nu, o, s}(t) = s \begin{cases} 1, & \nu(B_{\chi}(t)) = \theta(t) + o \pmod{3}, \\ 2, & \nu(B_{\chi}(t)) \neq \theta(t) + o \pmod{3}. \end{cases}$$

The corresponding comparative word is

$$C_{\nu, o, s}(t) = \sum_{\chi \in \{\pi, e, \varphi\}} \epsilon_{\chi}^{\nu, o, s}(t) B_{\chi}(t) \quad \text{in } \mathbb{F}_3^6. \quad (7.15)$$

The parameters generate twelve labelled representations per row. Two labels may produce the same word; the temporal witnesses and labels remain available in the complete calendar.

7.5.3 Affine phase–charge reader

The affine reader uses coefficients

$$a_{\chi}^{\text{af}}(t) = \eta_{\text{ter}}^{\vee}(B_{\chi}(t)) + \theta(t) \pmod{3},$$

and produces

$$F(t) = \sum_{\chi \in \{\pi, e, \varphi\}} a_{\chi}^{\text{af}}(t) B_{\chi}(t) \quad \text{in } \mathbb{F}_3^6. \quad (7.16)$$

The comparative rule distinguishes agreement and difference of charges; the affine rule adds charge and phase before combining the bands. This operational difference is what heterotypic selection uses.

7.5.4 Word–residue incidence

Let R be the cyclic rotation of the six positions of a band. In the row at time t , form the residual repertoire

$$\mathcal{R}(t) = \{R^j v : v \in \{q(t), a(t), c(t), \bar{w}(t)\}, 0 \leq j < 6\}.$$

The temporal incidence relations are

$$\mathcal{A}_{\text{cmp}} = \{(C_{\nu, o, s}(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t), \nu, o, s \text{ admissible}\}, \quad (7.17)$$

$$\mathcal{A}_{\text{af}} = \{(F(t), u) : 0 \leq t < 100, u \in \mathcal{R}(t)\}. \quad (7.18)$$

Each pair relates a read word to a residue of the same row. Time is quantified jointly across both coordinates of the pair. In the calendar used, times $0, \dots, 99$ are distinct; identifying the same row and identifying the same time are therefore equivalent operations.

Proposition 7.7 (Existential compression of the temporal repertoire). *The function*

$$M(v, u) = \mathbf{1}_{(v, u) \in \mathcal{A}_{\text{cmp}}} + 2\mathbf{1}_{(v, u) \in \mathcal{A}_{\text{af}}} \in \{0, 1, 2, 3\}$$

preserves exactly the existence of each reading type for the pair (v, u) .

Proof. The first bit equals one exactly when there exist a time, a residual field, a rotation, and a comparative label that produce the pair. The second bit expresses the corresponding condition for the affine reader. Union by bitwise disjunction preserves each existence condition as all rows are traversed. A pair may admit both types, in which case the mask equals three. $\square$

The function M is an existence reader. The witness times, their multiplicities, and the individual labels remain in (7.18) and in the calendar that produces them. The finite implementation uses M to decide heterotypy, whereas the genealogical archive retains the richer provenance data.

7.6 Enumeration of cylinders and decimal candidates

7.6.1 Concatenation of seventy-two and seventy-eight trits

For an ordering $\tau = (s_1, \dots, s_8)$ of $\mathcal{S}_{\text{term}}$, define

$$P_{72}(\tau) = [\Omega_{24} \| s_1 \| \cdots \| s_8 \|]_3.$$

Recursively, start from $p_0 = [\Omega_{24}]_3$ and compute $p_i = 729p_{i-1} + [s_i]_3$ for $1 \leq i \leq 8$. This recurrence preserves the leading zeros of each block through explicit multiplication by 729. For each position b of the functional panel,

$$P_{78}(\tau, b) = 729P_{72}(\tau) + b.$$

The lengths are $24 + 8 \cdot 6 = 72$ and $72 + 6 = 78$, respectively. The corresponding ternary cylinder is

$$I_{\tau, b} = \left[\frac{P_{78}(\tau, b)}{3^{78}}, \frac{P_{78}(\tau, b) + 1}{3^{78}} \right). \quad (7.19)$$

7.6.2 Exact intersection with the twelve-triad grid

The decimal reading uses twelve triads, that is, the grid $N/10^{36}$. Write $D = 10^{36}$ and $Q = 3^{78}$ for brevity.

Lemma 7.8 (Integer endpoints of the cylinder). *For an integer prefix p of length seventy-eight, the numerators of the grid points belonging to its cylinder are*

$$\left\{ N \in \mathbb{Z} : \left\lceil \frac{pD}{Q} \right\rceil \leq N < \left\lceil \frac{(p+1)D}{Q} \right\rceil \right\}. \quad (7.20)$$

Each cylinder contains at most one point of this grid.

Proof. The condition $N/D \in [p/Q, (p+1)/Q]$ is equivalent to $pD/Q \leq N < (p+1)D/Q$. For integer N , the first inequality is equivalent to $\lceil pD/Q \rceil \leq N$; the second is equivalent to $N < \lceil (p+1)D/Q \rceil$. The interval length in coordinate N is $D/Q < 1$, since $10^{36} < 3^{78}$. It can therefore contain at most one integer. $\square$

Ceilings are evaluated by integer division: $\lceil a/Q \rceil = \lfloor (a+Q-1)/Q \rfloor$ for $a \geq 0$. The entire enumeration uses exact integers; the choice of boundary points is fixed by the half-open right endpoint.

First retain the indexed list of incidences

$$\mathcal{C}_{\text{ind}} = \{(\tau, j, N) : N/D \in I_{\tau, \mathcal{P}_9(j)}\}, \quad 1 \leq j \leq 9,$$

and then obtain the set of numerators

$$\mathcal{C} = \{N : \exists \tau, j (\tau, j, N) \in \mathcal{C}_{\text{ind}}\}.$$

Passing to $\mathcal{C}$ removes repetitions of a numerator, retained in $\mathcal{C}_{\text{ind}}$ together with their provenance.

Lemma 7.9 (Recovery of the ordering from the cylinder point). *If $(\tau, j, N) \in \mathcal{C}_{\text{ind}}$, block number i of the seventy-eight-trit word is recovered by*

$$\left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13.$$

In particular, blocks five through twelve recover τ .

Proof. Membership in the cylinder means $p \leq QN/D < p+1$, where $p = P_8(\tau, \mathcal{P}_9(j))$. Hence, $\lfloor QN/D \rfloor = p$. Extraction of the earlier blocks agrees with successive division of the prefix p by powers of 729: truncating a positional expansion before extracting a block preserves that block. The displayed formula performs exactly this extraction. The first four blocks form Ω_{24} and the next eight are the ordering τ . $\square$

Proposition 7.10 (Exact candidate count). *For Ω_{24} , $\mathcal{S}_{\text{term}}$ and $\mathcal{P}_9$ defined above,*

$$\# \text{Ord}(\mathcal{S}_{\text{term}}) = 40320, \quad \# \mathcal{C}_{\text{ind}} = 21902, \quad \# \mathcal{C} = 19446.$$

Proof. The eight blocks are distinct, so their orderings are the $8! = 40320$ permutations. For each permutation and each of the nine panel positions, compute $p = P_8(\tau, b)$, apply endpoints (7.20), and add the unique integer when the integer interval is nonempty. The sum of the cardinalities of these intervals is 21902. The union of their integers contains 19446 distinct elements. The exhaustive evaluation of these operations is certified by the theorems `indexed_cylinder_count` and `candidate_count` of the finite selector. The algorithm performs exactly the enumeration described: permutations, nine panel incidences, bounds by integer division, and duplicate removal. $\square$

The numbers 21902 and 19446 count different objects. The former counts indexed incidences containing a decimal point; the latter counts distinct numerators. The total number of cylinder queries is $40320 \cdot 9 = 362880$, including cylinders whose intersection with the grid is empty.

7.7 Exceptional incidence and heterotypy condition

7.7.1 Exceptional face produced by the panel

A band is lifted to twelve coordinates by

$$\Gamma(b) = (b, W(b)) \in \mathbb{F}_3^{12}.$$

The support of a word is the set of positions of its nonzero coordinates, with human-readable indices $1, \dots, 12$. The first regional bands of the panel produce

$$\mathcal{B} = \text{supp } \Gamma(B_\pi(0)) \cap \text{supp } \Gamma(B_e(0)) = \{4, 6, 9, 10\}. \quad (7.21)$$

The operation uses the regional words 103 and 523 and the selector's Witt matrix. The face is computed before querying the candidate coordinates.

For $N \in \mathcal{C}$, its twelve triads are

$$K_i(N) = \left\lfloor \frac{N}{1000^{12-i}} \right\rfloor \bmod 1000, \quad 1 \leq i \leq 12. \quad (7.22)$$

The crown support is

$$\mathcal{H}(N) = \{i : K_i(N) \geq 729\}, \quad \mathcal{C}_{\mathcal{B}} = \{N \in \mathcal{C} : \mathcal{H}(N) = \mathcal{B}\}.$$

The boundary $729 = 3^6$ relates the six-trit band to the decimal triad. Exhaustive evaluation of the equality of supports gives

$$\#\mathcal{C}_{\mathcal{B}} = 79. \quad (7.23)$$

The support preserves positions, while the complete register also preserves the twelve magnitudes. The next stage uses this positional information together with relations between magnitudes.

7.7.2 Ternary words and oriented differences of the candidate

The i th ternary band of the point N/D is obtained by

$$T_i(N) = \left\lfloor \frac{N 729^i}{D} \right\rfloor \bmod 729, \quad 1 \leq i \leq 13. \quad (7.24)$$

The oriented edge $a \rightarrow b$ has residue

$$R_{a,b}(N) = (K_a(N) - K_b(N)) \bmod 729. \quad (7.25)$$

Subtraction is performed in $\mathbb{Z}$ before reduction. For this edge, the pair

$$(\text{enc}_6^{-1}(T_b(N)), \text{enc}_6^{-1}(R_{a,b}(N))).$$

is queried in relations (7.18). It is therefore a joint query of the destination word and the oriented difference of the triads.

Write $\text{Cmp}(a, b; N)$ when the pair belongs to $\mathcal{A}_{\text{cmp}}$ and $\text{Af}(a, b; N)$ when it belongs to $\mathcal{A}_{\text{af}}$.

7.7.3 Determination of the transverse orbit

The twenty-four-trit descriptor determines the first three decimal triads before the terminal filtrations. Indeed, its cylinder satisfies the exact inclusion

$$\left[\frac{66241910521}{3^{24}}, \frac{66241910522}{3^{24}} \right) \subseteq \left[\frac{234543140}{10^9}, \frac{234543141}{10^9} \right).$$

The two endpoint inequalities are verified by integer cross-products. Thus, every candidate shares the three reference positions 1, 2, 3 and their representations (234, 543, 140).

The step-four translation on $\mathbb{Z}/12\mathbb{Z}$ has the four orbits

$$(1, 5, 9), \quad (2, 6, 10), \quad (3, 7, 11), \quad (4, 8, 12).$$

The first three each contain one of the reference positions; the fourth is the only one disjoint from $\{1, 2, 3\}$. The oriented forest of steps three and four is fixed by the edges

$$\begin{aligned} 1 &\rightarrow 4, & 1 &\rightarrow 5, & 2 &\rightarrow 6, \\ 3 &\rightarrow 7, & 4 &\rightarrow 8, & 5 &\rightarrow 9, \\ 6 &\rightarrow 10, & 7 &\rightarrow 11, & 8 &\rightarrow 12. \end{aligned}$$

Its first edge has step three and the remaining edges have step four. The two internal edges of the distinguished orbit are precisely $4 \rightarrow 8$ and $8 \rightarrow 12$. The location of the reading condition is thus fixed by the geometry of the positions and by the initial prefix, before examining the candidates of the exceptional face.

7.7.4 Terminal heterotypy condition

The terminal condition is

$$\begin{aligned} \mathcal{J}(N) = & [\text{Cmp}(4, 8; N) \wedge \text{Af}(8, 12; N)] \\ & \vee [\text{Af}(4, 8; N) \wedge \text{Cmp}(8, 12; N)]. \end{aligned} \quad (7.26)$$

Both assignments of types to the two edges are admitted. The selection requires the existence of a realization with distinct types; the individual labels and their times remain available in the incidence witnesses.

Theorem 7.11 (Terminal selection of the coupling register). *On the candidate domain produced by Ω_{24} , $\mathcal{S}_{\text{term}}$, $\mathcal{P}_3$ and the hundred-row regional calendar, there exists a unique numerator simultaneously satisfying exceptional incidence and heterotypy:*

$$\{N \in \mathcal{C}_{\mathcal{B}} : \mathcal{J}(N)\} = \{234543140729659824621058914794146601\}.$$

Its twelve-triad register is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \quad (7.27)$$

Proof. The preceding count produces the 79 elements of $\mathcal{C}_{\mathcal{B}}$. For each one, its twelve coordinates are extracted by (7.22); the destination bands T_8 , T_{12} and residues $R_{4,8}$, $R_{8,12}$ are computed; the two masks of the temporal repertoire are queried; and disjunction (7.26) is applied. The resulting list contains exactly the stated integer, according to the exhaustive evaluation certified by `terminal_selection`. Equality of the list with a singleton implies existence and uniqueness in the declared domain. Base-one-thousand extraction from that integer gives (7.27).

The selection procedure receives the enumerated producers and rules. The final integer appears in the statement of the verified equality after selection has been executed. This logical position allows the result to be checked without using it as a selection criterion. $\square$

7.7.5 Arithmetic witness of membership in a cylinder

One of the incidences of the selected numerator uses the ordering

$$(002001, 020111, 200110, 121012, 010122, 001001, 221110, 222110)$$

and the boundary $438 = 121020_3$. For this incidence,

$$P_2 = 5283881584882673136514102926090957,$$

$$P_{78} = 3851949675379468716518781033120308091.$$

The integer-endpoint formula yields

$$\left\lceil \frac{P_{78} 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146601,$$

$$\left\lceil \frac{(P_{78} + 1) 10^{36}}{3^{78}} \right\rceil = 234543140729659824621058914794146602.$$

The unique integer in the interval is therefore N_K . The thirteen ternary bands of its decimal point are

020022	222111	211201	021101
002001	020111	200110	121012
010122	001001	221110	222110
121020			

This calculation exhibits a concrete witness of membership in the enumerated domain. Uniqueness of the register follows from the exhaustive examination of all candidates, in addition to this particular witness.

7.8 Regional composition and continuity of the realizations

7.8.1 Composition of the regional generators

Let $\text{Sel}(\Omega, \mathcal{S}, \mathcal{P}, \mathcal{N})$ be the selector defined by the enumeration and the two preceding filters. The regional producers satisfy the equalities

$$\Omega_{\text{reg}} = \Omega_{24}, \quad \mathcal{P}_{\text{reg}} = \mathcal{P}_9, \quad \mathcal{N}_{\text{reg}} = \mathcal{N}_{100},$$

where $\mathcal{N}_{\text{reg}}$ is computed by band extraction and signature reading. By congruence of functions,

$$\begin{aligned} \text{Sel}(\Omega_{\text{reg}}, \mathcal{S}_{\text{term}}, \mathcal{P}_{\text{reg}}, \mathcal{N}_{\text{reg}}) \\ = \text{Sel}(\Omega_{24}, \mathcal{S}_{\text{term}}, \mathcal{P}_9, \mathcal{N}_{100}) = \{N_K\}. \end{aligned} \tag{7.28}$$

The equality replaces three documentary interfaces with their actual regional construction. The terminal repertoire retains the provenance specified in (7.13).

The register is then formed by the twelve Euclidean divisions in (7.22). Each coordinate belongs to $\{0, \dots, 999\}$ and the length is twelve by construction. The implementation that takes the first element of the selected list is justified by equality with the singleton: the default value of a computational operation on empty lists never enters this result.

7.8.2 Relation to signed incidences and inversion

The present construction provides a prospective selection in an explicit finite domain. The subsequent development of the dodecaphasic register produces its oriented channels from the enriched terminal state and proves an integral inversion through Hadamard. That inversion recovers a register from compatible channels; here a register is selected from its regional and incidence-based inputs. Equality of their results allows the two realizations to be composed while preserving the origin of each operation.

Uniqueness of selection and uniqueness of inversion are distinct properties. The former quantifies over the candidates constructed in this chapter. The latter quantifies over the registers sharing a compatible signature. Their conjunction justifies transporting the selected register to the signed realization and recovering it from that realization, without turning a retrospective recovery into the producer of the register.

7.8.3 Relation between the terminal counts

The sequence

$$40320 \longrightarrow 21902 \longrightarrow 19446 \longrightarrow 79 \longrightarrow 1$$

records, successively, orderings, indexed incidences with the grid, distinct numerators, candidates of the exceptional face, and heterotypic survivors. The arrow notation indicates the order of the operations; the second number counts a different class of objects from the first.

The regional-catalogue count $196 \cdot 197 \cdot 196 \longrightarrow 331 \longrightarrow 2 \longrightarrow 1$, developed in its corresponding place, starts from another population: regional blocks from three catalogues and terminal conditions of Gram, occupancy, distance, and orientation. Both counts retain their domains. Their convergence to the same representation is expressed through the identification of their readers and outputs; the intermediate numbers describe different operations and remain documented separately.

7.8.4 Regional composition of the fine-structure constant

The output K subsequently enters the ordered regional composition and the base-one-thousand carry. Its later use preserves the twelve positions and the reading boundary. Coupling to the analytic root and representations at arbitrary depth uses that same output, together with the regional coordinates already generated and the rules of the analytic chart. This dependency prepares the chapter devoted to the two correlated realizations of the fine-structure constant.

The scope of this chapter is precise: production of the descriptor and its regional inputs, determination of the finite domain, unique selection on that domain, and material composition with the producers. The analytic recurrence at all orders and the compatibility of infinite carries are developed subsequently with their own proofs.

7.9 Executable certification and traceability of the operations

The formal development preserves the correspondence between the mathematical definitions and their executable realizations. The signature and clock proofs are found in `N69RegionalSignature.lean`; band extraction and reproduction of the hundred rows, in `GeneratedN69Rows.lean`; the minimality criterion and construction of Ω_{24} , in `RegionalW24.lean`. Compatibility of the two Witt matrices is assembled in `WittReaderCompatibility.lean`.

The temporal readers, rotations, oriented residues, and heterotypic condition are formalized in `TerminalReaders.lean`. Enumeration and counts are formalized in `TerminalOrbitalSelection.lean`. The regional panel is linked through `RegionalPanelBridge.lean`, and composition (7.28) is expressed in `SelectedFromRegionalInputs.lean`.

The finite enumeration checks use Lean’s native evaluation; their reports record `Lean.ofReduceBool`, in addition to the library’s customary logical axioms. The regional reconstruction of the descriptor uses kernel proofs and ordinary reduction. This distinction describes the verification mechanisms actually employed. The compilation receipts attest to their declared modules and dependencies; the mathematical proofs retain the domains, producers, and hypotheses made explicit in the body of the chapter.

7.10 Complete census of cylinders and selection of the terminal boundary

The terminal readout jointly uses two kinds of information: compatibility of the regional cylinders and an incidence profile with labelled channels and positions. This section constructs the complete catalogues, enumerates their triples, and proves the effect of each reader condition. The starting point consists of the regional outputs already produced in Section 5, not values introduced to define the APP-TRIT-TPK transport. Reduction to a matrix belongs to a subsequent readout of these outputs and preserves the order of the three channels.

The result is a census theorem with an explicit starting structure. The profile of Gram products, weights, distances, and marginals is declared as a datum of the terminal reader. The exhaustiveness of the enumeration proves what this profile selects on the specified catalogues; it does not, by itself, prove that the profile is a universal law or that it can be reconstructed from the unique triple that ultimately satisfies it. This distinction allows the complete finite proof to be incorporated without reversing its genealogy.

7.10.1 From regional outputs to ternary catalogues

Let $x_\pi, x_e, x_\varphi \in [0, 1)$ be the fractional parts of the three regional coordinates already constructed. We use their expressed cylinders of one thousand decimal places,

$$J_\chi^{(1000)} = \left[\frac{d_\chi}{10^{1000}}, \frac{d_\chi + 1}{10^{1000}} \right), \quad \chi \in \{\pi, e, \varphi\}.$$

The integers d_χ are finite output readouts. A length of one thousand is sufficient for the integers that follow, as verified from the two bounds of each interval; no infinite depth is assumed. We define

$$p = 30, \quad r = 1050, \quad L = p + r = 1080, \quad k = 171, \quad D = 3^L, \quad Q = 1000^k, \quad (7.29)$$

and

$$P_\chi = \lfloor 3^{30} x_\chi \rfloor, \quad T_\chi = \lfloor Q x_\chi \rfloor.$$

Here k counts decimal triads and does not denote the dodecaphasic register K . We have $1000^{171} \leq 3^{1080} < 1000^{172}$.

Lemma 7.12 (Stable extraction of an integer). *For $d, m, h \in \mathbb{Z}$, with $m, h > 0$, the value of $\lfloor mx \rfloor$ is constant on $[d/h, (d+1)/h)$ if and only if*

$$\left\lfloor \frac{md}{h} \right\rfloor = \left\lfloor \frac{m(d+1) - 1}{h} \right\rfloor. \quad (7.30)$$

When this holds, either side gives that value.

Proof. The minimum of mx is attained at the left endpoint. Its supremum is $m(d+1)/h$ and is not attained. The largest integer value that $\lfloor mx \rfloor$ can take is $\lceil m(d+1)/h \rceil - 1$, which equals the second side because the numerator and denominator are integers. Equality of the endpoints of the range of values proves the assertion. $\square$

The stability test holds in all three channels for $m = 3^{30}, Q, 3^{1080}$. The last choice will be used solely for a subsequent comparison, not to select the profile. The first thirty recovered positions are precisely the five six-trit stages of the regional transport:

Channel	Thirty-trit prefix	P_χ
π	01021101222201021100211110221	29152671743887
e	201101121221102011012222102011	147887858824447
φ	121200112202121020010210010200	127247717616687

The agreement preserves the prolongation already constructed; the decimal readout is not used to replace its transport matrices retrospectively.

For a fixed channel, a tail of r trits is represented by $0 \leq R < 3^r$. Its cylinder and the cylinder of the first k triads are

$$I_{\chi,R} = \left[\frac{P_\chi 3^r + R}{D}, \frac{P_\chi 3^r + R + 1}{D} \right), \quad J_{\chi,T} = \left[\frac{T_\chi}{Q}, \frac{T_\chi + 1}{Q} \right). \quad (7.31)$$

The catalogue includes all tails for which these two cylinders have a nonempty intersection.

Proposition 7.13 (Catalogue by exact intersection). *The compatible tails are the integers in the interval $[a_\chi, b_\chi]$, where*

$$a_\chi = \max\left(0, \left\lfloor \frac{T_\chi D}{Q} \right\rfloor - P_\chi 3^r\right), \quad (7.32)$$

$$b_\chi = \min\left(3^r - 1, \left\lfloor \frac{(T_\chi + 1)D - 1}{Q} \right\rfloor - P_\chi 3^r\right). \quad (7.33)$$

For the specified regional cylinders, no clipping applies and we obtain

χ	$a_\chi \bmod 729$	$b_\chi \bmod 729$	$b_\chi - a_\chi + 1$
π	67	262	196
e	584	51	197
φ	117	312	196

(7.34)

Proof. Write $N = P_\chi 3^r + R$. The two half-open intervals in (7.31) intersect exactly when

$$NQ < (T_\chi + 1)D, \quad (N + 1)Q > T_\chi D.$$

The second inequality is equivalent to $N \geq \lfloor T_\chi D/Q \rfloor$; the first, to $N \leq \lfloor ((T_\chi + 1)D - 1)/Q \rfloor$. Subtracting $P_\chi 3^r$ and retaining $0 \leq R < 3^r$ proves the formulae. Evaluation by integer division of the T_χ extracted using (7.30) gives the table. Since $r \geq 6$, $P_\chi 3^r \equiv 0 \pmod{729}$, so the residue of the first endpoint can also be obtained directly from $\lfloor T_\chi D/Q \rfloor \bmod 729$. $\square$

The last six trits of a tail form the word $\nu_6(R \bmod 729)$, where ν_6 denotes the ternary representation of length six, including its leading zeros. By the table, each catalogue contains fewer than 729 consecutive elements, so this reduction is injective on it. Thus, the three boundary catalogues are completely described, without choosing the final triples:

$$\begin{aligned} \mathcal{B}_\pi &= \{\nu_6(67 + i) : 0 \leq i < 196\}, \\ \mathcal{B}_e &= \{\nu_6((584 + j) \bmod 729) : 0 \leq j < 197\}, \\ \mathcal{B}_\varphi &= \{\nu_6(117 + \ell) : 0 \leq \ell < 196\}. \end{aligned} \quad (7.35)$$

Requiring the left endpoint of $I_{\chi,R}$ to belong to $J_{\chi,T}$ would impose a different condition: it would lose the first cylinder that intersects the boundary and would give 195, 196, 195. The census uses the intersection of the complete cylinders, in accordance with its definition.

7.10.2 Declared profile and compatibility indicators

For $u, v \in \mathbb{F}_3^6$, let $g(u, v) = \sum u_i v_i \in \mathbb{F}_3$, $n(u) = g(u, u)$, $w(u) = \#\{i : u_i \neq 0\}$, and $h(u, v) = \#\{i : u_i \neq v_i\}$. The last two functions take integer values; the first two take values in $\mathbb{F}_3$. The reader's local profile, in the order (π, e, φ) , is

$$\begin{aligned} (g_{\pi e}, g_{\pi \varphi}, g_{e \varphi}) &= (2, 2, 0), & (n_\pi, n_e, n_\varphi) &= (0, 0, 0), \\ (w_\pi, w_e, w_\varphi) &= (3, 3, 3), & (h_{\pi e}, h_{\pi \varphi}, h_{e \varphi}) &= (2, 5, 3). \end{aligned} \quad (7.36)$$

The words are ordered as in (7.35). We use $[E] \in \{0, 1\}$ for the indicator of a condition E . For each pair of channels χ, ψ , the rectangular matrix $A_{\chi\psi}$ has entry $[g(u, v) = g_{\chi\psi}]$. The matrix $A_{\chi\psi}^h$ multiplies that entry by $[h(u, v) = h_{\chi\psi}]$. Let D_χ^n be the diagonal matrix of $[n(u) = 0]$, and D_χ^{nw} that of $[n(u) = 0][w(u) = 3]$.

Theorem 7.14 (Exhaustive enumeration by finite sums). *The product of the three catalogues contains 7 567 952 triples. The successive conditions in (7.36) leave*

$$7\,567\,952 \longrightarrow 275\,794 \longrightarrow 6235 \longrightarrow 3127 \longrightarrow 331. \quad (7.37)$$

These four counts are, respectively,

$$\begin{aligned} N_g &= \text{tr}(A_{\pi e} A_{e\varphi} A_{\varphi\pi}), \\ N_{gn} &= \text{tr}(D_{\pi}^n A_{\pi e} D_e^n A_{e\varphi} D_{\varphi}^n A_{\varphi\pi}), \\ N_{gnw} &= \text{tr}(D_{\pi}^{nw} A_{\pi e} D_e^{nw} A_{e\varphi} D_{\varphi}^{nw} A_{\varphi\pi}), \\ N_{gnwh} &= \text{tr}(D_{\pi}^{nw} A_{\pi e}^h D_e^{nw} A_{e\varphi}^h D_{\varphi}^{nw} A_{\varphi\pi}^h). \end{aligned} \quad (7.38)$$

All products and traces in this formula are calculated over the integers, not over the field with three elements.

Proof. The size of the product is $196 \cdot 197 \cdot 196$. When a trace in (7.38) is expanded, its indices run through exactly one triple of words. The summand equals one if it satisfies all the represented conditions and zero otherwise. Each triple is counted once. The diagonal matrices add the individual conditions; the rectangular factors add the pairwise conditions. This proves that the traces are precisely the stated cardinalities.

To make their evaluation explicit, fix the indices i, j of the first two channels and form the sets of indices ℓ of the third that satisfy each pairwise condition. Their indicators are intersected, the relevant individual indicators are added, and the cardinality is summed. The final sum runs over $0 \leq i < 196, 0 \leq j < 197$. Substitution of the words in (7.35) and the profile (7.36) gives, in that order, 275794, 6235, 3127, 331. The supplement records the four contributions of each of the 196 indices i and the complete list of 331 triples; its program evaluates these sums with exact binary indicators and checks each survivor by scalar evaluation. No sample or tolerance is involved. $\square$

7.10.3 Labelled incidence and orientation charge

For a surviving triple, let $B \in \mathbb{F}_3^{3 \times 6}$ be its matrix of ordered rows. The second profile requires

$$\text{rank}_{\mathbb{F}_3} B = 3, \quad w_{\text{col}}(B) = (1, 1, 2, 2, 3, 0), \quad q_{\partial}(B) = (1, 2, 2, 1, 2, 0), \quad (7.39)$$

where w_{col} counts the nonzero elements in each column and q_{∂} sums its entries modulo three. This is a profile of labelled columns; permuting them without transporting the profile defines a different reader. Denote by $\mathcal{F}$ the set of 331 triples from the preceding theorem.

Proposition 7.15 (Reduction of the census to the terminal pair). *The sums of indicators over $\mathcal{F}$, successively imposing rank, column weights, and column sums, are*

$$\sum_{B \in \mathcal{F}} [\text{rank } B = 3] = 331, \quad N_{\text{rango,pesos}} = 18, \quad N_{\text{rango,pesos,sumas}} = 2. \quad (7.40)$$

With catalogue indices starting at one, the two final triples are (120, 197, 157) and (129, 197, 148), and their matrices are

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

Proof. For each triple in the already enumerated set, the rank is computed by elimination over $\mathbb{F}_3$, the nonzero elements of the six columns are counted, and their entries are summed modulo three. The products of the indicators in (7.39) give (7.40). Substituting the two final indices into (7.35) gives the displayed matrices. The complete list of survivors and the exact filters in the supplement allow the elimination to be reproduced without presupposing these two matrices as the domain. $\square$

The orientation $\sigma = (1, 1, -1) = (1, 1, 2) \in \mathbb{F}_3^3$ selects the line $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$. The row charges of B_0, B_1 are (0, 2, 0) and (2, 2, 1); only the latter belongs to this line. This recovers the last step of the selector in Section 7.15, now preceded by the construction and enumeration of its complete domain. The stable readout of $[3^{1080} x_{\chi}]$, performed after the filters, gives the boundaries 021020, 001220, 100210, with the same ranks (129, 197, 148). This subsequent comparison does not define the profile or enter the sums of indicators.

7.10.4 Preserved information and scope of the result

The procedure preserves the thirty-trit prefix, the compatible tail, the regional order, the position of each trit, and the orientation. Reduction modulo 729 preserves the identity of each candidate here because the interval of tails has length less than 729; this injectivity is not asserted for an arbitrary history. The census of 331 also proves that the local profile alone does not select a single matrix: the labelled marginals and the charge supply additional effective conditions.

Thus, a complete finite enumeration and its selector have been recovered on output cylinders, with an explicit readout profile. Construction of the signed register additionally uses its memory reader and its channels, as specified below; the visible matrix is not identified with that register by equality of the number of components. Nor does this count prove the universality of the profile, generate the regional coordinates anew, or decide a claim about alpha or about the complete continuum.

8 Dodecaphase record and the rational phase constant

Generation of the regional coordinates does not exhaust the information in the APP-TRIT-TPK state. One Archimedean readout determines a value through compatible cylinders; another readout preserves the opposition of the sheets, the orientation of the route, and the record of the incidences traversed. The dodecaphase record belongs to this second readout before entering the closure of α . It is therefore not defined by subtracting a physical constant from a combination of previously known values. It is reconstructed from a signed coordinate of the same arithmetic state with trajectory memory that produces the regional coordinates.

The distinction is necessary even when two objects have twelve coordinates. A twelve-block catalogue word, a list of twelve oriented events, a signed integer vector, and a base-thousand dodecaphase record are different objects. Comparing them requires the maps that relate them. Equality of length does not provide these maps, and loss of signs is not merely a simplification of notation.

This chapter preserves the causal cutoff established in the preceding chapters: the nine APP positions on the horizontal and vertical axes produce the arithmetic sheets; TRIT types regime and orientation; TPK selects, transports, updates, and prolongs, preserving residue, quotient, carry, sheet, route, boundary, and memory. The coordinates used here are obtained from the joint discrete structure of the continuum. The chain $w_6 \rightarrow w_{12} \rightarrow w_{18} \rightarrow w_{24} \rightarrow w_{30} \rightarrow R_{36} \rightarrow G_9$ is not replaced by a finite schedule. Nor is the order $U_t \rightarrow \Gamma_9 \rightarrow K_{\text{ph}}$ reversed: the last map is a reduced-memory readout, not the generator of the signed record.

Reversible coordinatization and incidence determination. The result organizing this chapter is the equivalence of three coordinatizations of a register previously produced by APP-TRIT-TPK. The Hadamard transformation relates the signed register to the integer register; the difference extractor relates the latter to its transverse variations and its uniform component:

$$U \xrightleftharpoons[\mathcal{H}_{12}]{\mathcal{H}_{12}/4} K \xrightleftharpoons{(D_3, D_4, Q)} (D_3 K, D_4 K, Q(K)).$$

The first equivalence is established on the integral sublattice $\mathcal{L}_H$; the second, on the image of the extractor. With base 1000, the 12 positions and the phase origin fixed, the evaluation $K \mapsto \kappa_{\text{per}}$ adds a fourth reversible coordinatization: multiplying by $1000^{12} - 1$ and performing the Euclidean divisions recovers the full register. The proofs in §8.8, §8.9 and §8.10 establish these equivalences in their respective domains.

Two compositions subsequently act on this register. The positional composition combines its components with the closure, propagation and self-scaling coordinates, and its normalization determines α_A . The incidence composition applies the cubic threshold and yields

$$K \mapsto B_K = \{j : K_j \geq 729\} = \{4, 6, 9, 10\}.$$

The full register allows both compositions to be reconstructed. The support B_K specifically retains membership in the class selected by the threshold. The identity $B_K = H_e \cap P_\pi$ in (10.12) connects this integer extraction to the incidence of the regional codes. The negative support of their balance with K completes the flag $B_K \subset H_\alpha^-$ developed in the exceptional prolongation. Thus the role of K is defined by an invertible transformation and by precise subsequent maps; the additional information on depth and transport remains in the state from which the register originates.

8.1 Cyclic extension and memory grading

The observable schedule contains one hundred and eight positions, organized into twelve nonadic windows. With positive indices, let

$$E_m = \{9(m-1) + 1, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

The ordered support $E_1 \sqcup \cdots \sqcup E_{12}$ preserves position and window membership. The group $C_{108} = \mathbb{Z}/108\mathbb{Z}$, by contrast, preserves the law of cyclic translation. For the representative $0 \leq \tilde{\theta} < 108$, the coordinates

$$\beta(\theta) = \left[\left[\frac{\tilde{\theta}}{9} \right] \right]_{12}, \quad r(\theta) = 1 + (\tilde{\theta} \bmod 9)$$

separate the dodecaphase sector from the interior nonadic position. Neither by itself preserves the total number of turns.

Proposition 8.1 (The carry of the schedule). *The maps*

$$\iota : C_{12} \longrightarrow C_{108}, \quad [b] \longmapsto [9b], \quad p : C_{108} \longrightarrow C_9, \quad [\theta] \longmapsto [\theta]_9$$

form a nonsplit exact sequence

$$0 \longrightarrow C_{12} \xrightarrow{\iota} C_{108} \xrightarrow{p} C_9 \longrightarrow 0.$$

For the set-theoretic section choosing the representatives $0, \dots, 8$, the failure of additivity is the carry

$$c(r, r') = \left[\left[\frac{\tilde{r} + \tilde{r}'}{9} \right] \right]_{12}.$$

Proof. The kernel of p consists of the multiples of nine and coincides with the image of ι . Reduction is surjective, and multiplication by nine is injective on the stated domain. If the sequence split, the middle group would be isomorphic to $C_{12} \times C_9$, whose exponent is $\text{mcm}(12, 9) = 36$; the exponent of C_{108} is 108. This contradiction proves nonsplitting. Finally, Euclidean division of $\tilde{r} + \tilde{r}'$ by nine gives the stated carry and verifies the identity for the section. $\square$

The result explains a distinction that will reappear in alpha: returning to the phase is not equivalent to returning to the state. If q_{mem} counts nonadic turns, the schedule preserves only $\beta = [q_{\text{mem}}]_{12}$. After one hundred and eight steps, β returns, but q_{mem} increases by twelve. The record also preserves the cylinder, the incidence ledger, and the vertical memory. Closure of a finite chart does not turn this history into a memoryless cycle.

8.2 The oriented record and the integral $1 + 5 + 6$ chart

For a history produced by TPK, let τ_m be the subroute over E_m , σ_m its orientation, κ_m the boundary carry, and Λ_m the local incidence ledger. The record is

$$\mathcal{R}_{12}(x) = ((E_m, \tau_m, \sigma_m, \kappa_m, \Lambda_m))_{m=1}^{12}.$$

Its domain is the space of arithmetic histories with trajectory memory, not the set of phase labels. Projection onto the schedule forgets precisely some of the data required by the signed readout.

An initial chart of the record can be written in terms of oriented events. The reference construction (Subsection 8.2) fixes the event code set $\{2, 4, 6, 8\}$ and the sector orientation

$$\Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -).$$

If d_t is the direction code at step t , this gives

$$e_i = \Sigma_{12}(i+1) \# \{t \in E_{i+1} : d_t \in \{2, 4, 6, 8\}\}, \quad 0 \leq i < 12.$$

The formula keeps the nonnegative event count separate from orientation. It does not identify a cardinal direction with a TRIT sign. The vector e , whose components are signed local counts, is also not yet the vector U that will appear below.

Opposite positions determine, for $0 \leq i < 6$,

$$p_i = e_i + e_{i+6}, \quad a_i = e_i - e_{i+6}.$$

With

$$s_C = \sum_{i=0}^5 p_i, \quad M_i = p_i - p_5 \quad (0 \leq i < 5),$$

the integral chart is defined by

$$\mathcal{L}_{12}(e) = (s_C, M_0, \dots, M_4, a_0, \dots, a_5).$$

The decomposition $1 + 5 + 6$ is not an arbitrary allocation of coordinates: one coordinate preserves the charge, five compare the pairs with a reference pair, and six preserve the opposition of the half-crowns.

Lemma 8.2 (Inversion of the integral chart). *An integer tuple $(s_C, M_0, \dots, M_4, a_0, \dots, a_5)$ belongs to the image of $\mathcal{L}_{12}$ if and only if*

$$s_C - \sum_{i=0}^4 M_i \equiv 0 \pmod{6}, \quad p_i \equiv a_i \pmod{2} \quad (0 \leq i < 6),$$

where

$$p_5 = \frac{s_C - \sum_{i=0}^4 M_i}{6}, \quad p_i = p_5 + M_i \quad (i < 5).$$

In that case its preimage is unique and is given by

$$e_i = \frac{p_i + a_i}{2}, \quad e_{i+6} = \frac{p_i - a_i}{2}.$$

Proof. The sum of the five margins is $s_C - 6p_5$, which gives the division by six. Once the p_i are reconstructed, each pair of equations $p_i = e_i + e_{i+6}$, $a_i = e_i - e_{i+6}$ has the indicated solution. This solution is integral exactly when the parity congruence holds. All operations are inverted, so no residual freedom remains. $\square$

This reversibility shows what information is preserved; it does not authorize discarding the rest of the ledger. The complete record has a larger domain than this count chart. Extraction of the channels feeding U uses that complete record and cannot be deduced from the list (e_i) by identifying names.

8.3 Evaluation of incidences and determination of the transverse coordinates

The dodecaphasic register preserves a history of visits and oriented traces. Its transverse coordinates are obtained through a composition of operations: arithmetic evaluation of the visits, counting by windows, decimal normalization, and cyclic differences. Reconstructing the register from those differences is another operation, whose existence and uniqueness will be proved below. This separation makes it possible to specify both the data used by the extractor and the information it transmits to the harmonic readout.

Let $\mathcal{L}$ be an incidence register produced by the selection, transport, and updating of an APP-TRIT-TPK history. Each visit preserves its route, APP position, sheet, time, multiplicity, and boundary class; each trace preserves its members, orientation, multiplicity, and length in the unit declared by its readout. The windows are

$$E_m = \{9(m-1) + 1, \dots, 9m\}, \quad 1 \leq m \leq 12.$$

Evaluation of the sum or product sheet determines the integer $n(v)$ of each visit and its positive division

$$n(v) = r(v) + 9q(v), \quad 1 \leq r(v) \leq 9.$$

The residue is calculated from that evaluation. The boundary class of a visit with residue 9 is preserved by $\epsilon_\partial(v) = 1$ on the corona and -1 in the interior.

The incidence presentation of the register uses the three counts

$$A_m = \sum_{v \in E_m} w(v) \mathbf{1}_{\{1,3,5,7\}}(r(v)), \quad (8.1)$$

$$C_m = \sum_{j \geq 0} (\nu_{120,m}^-(j) - \nu_{120,m}^+(j)), \quad (8.2)$$

$$V_m = \sum_{v \in E_m} w(v) \mathbf{1}_{\{9\}}(r(v)) \epsilon_{\partial}(v). \quad (8.3)$$

Here $w(v)$ is the incidence multiplicity and $\nu_{120,m}^{\pm}(j)$ counts the oriented traces of the indicated channel with reduced length $120(1 + 3j)$, according to the enumeration of the register. The collection of traces and its multiplicity are data of the evaluation, together with their corresponding production rule. The integer sum requires finite support or a construction that determines its meaning. The reduced length of a trace and the number of times in the window are differently typed quantities; the readout unit specifies their relationship. The symbols A_m, C_m, V_m here denote integer counts, distinct from the angular pair and the constitutive operator.

Let $[x]_{10} \in \{0, \dots, 9\}$ be the Euclidean representative. Normalization also preserves the quotient in

$$x = [x]_{10} + 10[x/10].$$

The block is formed as

$$z_m = 100[A_m]_{10} + 10[C_m]_{10} + [V_m]_{10}, \quad 0 \leq z_m \leq 999. \quad (8.4)$$

With cyclic indices modulo 12, let

$$(D_a z)_m = z_m - z_{m+a}, \quad a \in \{3, 4\}, \quad Q(z) = \sum_{m=1}^{12} z_m.$$

The incidence composition is expressed by

$$\mathcal{E}_{\text{cnt}}(\mathcal{L}) = \mathcal{D}(z) := (D_3 z, D_4 z, Q(z)) \in \mathbb{Z}^{25}. \quad (8.5)$$

Its arguments are the visits and traces with their counting rules. The evaluation does not receive K, U, α , or the expected transverse coordinates. To identify it with $\mathcal{E}_{108}^{90,120}$, the composition with the actual collection and incidences of the terminal state is also preserved. The invertibility of $\mathcal{D}$ determines a subsequent reconstruction; by itself, it does not select that collection of incidences.

Proposition 8.3 (Residual information determined by the extractor). *Let $N, N' \in \mathbb{Z}^{12 \times 3}$ be two count registers, ordered by (A, C, V) . For the operations (8.4)–(8.5),*

$$\mathcal{E}_{\text{cnt}}(N) = \mathcal{E}_{\text{cnt}}(N') \iff N - N' \in 10\mathbb{Z}^{36}.$$

The 36 sectoral residues determine exactly the output of 25 coordinates. The quotients and provenance constitute additional information in the register.

Proof. Equality modulo 10 produces the same blocks and the same transverse coordinates. Conversely, equality of the coordinates gives $D_3(z - z') = D_4(z - z') = 0$. Steps 3 and 4 generate the cycle of length 12, so $z - z'$ is constant. Equality of Q makes that constant zero. The expansion of each integer from 0 to 999 into three decimal digits is unique; hence the three residues of each window are recovered. $\square$

The proposition is valid on the entire indicated integer domain. It does not postulate that the 36 classes are independent on the subset of admissible registers produced by TPK.

8.4 Integral image and inversion of the transverse coordinates

To express the inversion, the blocks are reindexed by $i \in \mathbb{Z}/12\mathbb{Z}$, from 0 to 11. Let $(Sz)_i = z_{i+1}$, $D_3 = I - S^3$, and $D_4 = I - S^4$. Given an integer triple (b, c, q) , define

$$w_i = c_i - b_{i+1}, \quad P_0 = 0, \quad P_i = \sum_{j=0}^{i-1} w_j \quad (1 \leq i \leq 12), \quad T(w) = \sum_{i=0}^{11} P_i. \quad (8.6)$$

Theorem 8.4 (Characterization of the integral image). *A triple $(b, c, q) \in \mathbb{Z}^{25}$ belongs to $\text{im } \mathcal{D}$ if and only if*

$$b_i = w_i + w_{i+1} + w_{i+2}, \quad 0 \leq i < 12, \quad (8.7)$$

$$\sum_{i=0}^{11} w_i = 0, \quad (8.8)$$

$$q + T(w) \equiv 0 \pmod{12}. \quad (8.9)$$

Its unique integer preimage is

$$z_i = \frac{q + T(w)}{12} - P_i, \quad 0 \leq i < 12. \quad (8.10)$$

The twelve relations (8.7) and relation (8.8) are linearly independent over $\mathbb{Q}$.

Proof. For an output of $\mathcal{D}$,

$$c_i - b_{i+1} = (z_i - z_{i+4}) - (z_{i+1} - z_{i+4}) = z_i - z_{i+1}.$$

Three consecutive increments sum to b_i , and their total cyclic sum is zero. Moreover, $P_i = z_0 - z_i$, whence $q = 12z_0 - T(w)$. This gives the conditions and the inverse formula. Conversely, the congruence makes the coordinates of (8.10) integers, and closure allows them to be extended cyclically. Their adjacent differences are w_i , so $D_3 z = b$ and

$$(D_4 z)_i = w_i + w_{i+1} + w_{i+2} + w_{i+3} = w_i + b_{i+1} = c_i.$$

Summation gives $Q(z) = q$. The formula also determines uniqueness. Finally, the invertible integer change $(b, c) \mapsto (b, w = c - Sb)$ transforms the first twelve relations into $b = (I + S + S^2)w$: each solves for a different coordinate of b . Closure $\sum_i w_i = 0$ adds an independent relation. $\square$

The rational image has dimension 12. The integral condition adds the congruence of order 12 in (8.9). The eleven increments $w_0, \dots, w_{10}$ and the charge q are sufficient coordinates, with $w_{11} = -\sum_{i=0}^{10} w_i$. The 25 coordinates preserve those 12 degrees of freedom through 13 linear relations. If the triple also comes from (8.4), the inverse belongs to $\{0, \dots, 999\}^{12}$ and recovers its three residues per window.

8.4.1 Agreement with the harmonic transform

The transform $\mathcal{H}_{12}$ acts through H_4 on the orbits $(r, r+3, r+6, r+9)$, $r = 0, 1, 2$, where

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad H_4^2 = 4I_4.$$

For a compatible triple, let

$$B_r = \sum_{j=0}^3 c_{r+3j}, \quad a_0 = \frac{q + 2B_0 + B_1}{3}, \quad a_1 = a_0 - B_0, \quad a_2 = a_1 - B_1.$$

The harmonic readout has blocks

$$(\Pi_H(b, c, q))^{(r)} = (a_r, b_{r+3} - b_{r+9}, b_r + b_{r+6}, b_r - b_{r+6}).$$

Proposition 8.5 (Agreement of reconstructions). *On $\text{im } \mathcal{D}$,*

$$\Pi_H \mathcal{D} = \mathcal{H}_{12}, \quad \mathcal{D}^{-1} = \frac{1}{4} \mathcal{H}_{12} \Pi_H.$$

The image of Π_H restricted to that domain is

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12} u \equiv 0 \pmod{4}\}.$$

Proof. For $z = \mathcal{D}^{-1}(b, c, q)$, let $h_r = \sum_{j=0}^3 z_{r+3j}$. Shifting by four positions takes each orbit to the next; therefore $B_r = h_r - h_{r+1}$. The identity $q = h_0 + h_1 + h_2$ gives $a_r = h_r$. Within an orbit, write (x_0, x_1, x_2, x_3) for its coordinates and $d_j = x_j - x_{j+1}$. Then

$$\begin{aligned} d_1 - d_3 &= x_0 + x_1 - x_2 - x_3, \\ d_0 + d_2 &= x_0 - x_1 + x_2 - x_3, \\ d_0 - d_2 &= x_0 - x_1 - x_2 + x_3. \end{aligned}$$

These are the remaining three rows of $H_4 x$. Inversion and the integral characterization follow from $\mathcal{H}_{12}^2 = 4I$. $\square$

The harmonic congruence condition controls the output of Π_H ; membership of the channels in $\text{im } \mathcal{D}$ also requires (8.7)–(8.9). An exact negative control is obtained by taking $b = 0$, $q = 0$, and $c = e_0 - e_3$. The three orbital sums of c vanish, so $\Pi_H(b, c, q) = 0$, but the image relations fail. The harmonic readout does not replace the compatibility check for the channels.

8.5 Conjugation, covariance, and memory of the incidence register

Let $\rho(m) = 13 - m$. If incidence conjugation reverses the order of the windows, preserves the arithmetic and boundary classes of the visits, and interchanges the trace channels, then

$$(A_m^\dagger, C_m^\dagger, V_m^\dagger) = (A_{\rho(m)}, -C_{\rho(m)}, V_{\rho(m)}).$$

With $c_m = [C_m]_{10}$, decimal normalization gives

$$z_m^\dagger = z_{\rho(m)} + 100\mathbf{1}_{\{c_{\rho(m)} \neq 0\}} - 20c_{\rho(m)}. \quad (8.11)$$

Indeed, $[-C]_{10} = 0$ if $[C]_{10} = 0$, and is $10 - [C]_{10}$ otherwise. The channel negation is applied before reconstructing the decimal block.

Application to a specific involution also preserves the window endpoints. For a route $\gamma = (x_0, \dots, x_n)$ read before advancement,

$$\sum_{k=0}^{n-1} f(x_{n-k}) = \sum_{k=0}^{n-1} f(x_k) + f(x_n) - f(x_0). \quad (8.12)$$

Both sums contain the same interior terms; the difference is the final endpoint minus the initial one. On closed routes this correction vanishes. On open windows it is incorporated before reduction modulo 10.

Proposition 8.6 (Consequence of a periodic reduction). *If the observable of a fixed collection of routes satisfies $o(t + 54) = o(t)$, its multiplicities are preserved under that return, and the same local readout acts on windows of nine times, then*

$$(A, C, V)_{m+6} = (A, C, V)_m, \quad z_{m+6} = z_m.$$

Proof. Translation $t \mapsto t + 54$ takes each window bijectively to the one six positions later and preserves each counted incidence and its multiplicity. Equality is transmitted to the counts and their decimal normalizations. $\square$

A presentation with twelve distinct blocks therefore preserves information that does not factor through that observable of period 54: incidence selection, orientation, endpoints, multiplicity, or memory. The proposition delimits the effect of that reduction; it does not assign period 54 to the complete history.

8.5.1 Determination on orbits and accumulation of events

Fixing only $Q(z) = q$ leaves an affine class of the lattice

$$\mathcal{Z}_0 = \{v \in \mathbb{Z}^{12} : \sum_i v_i = 0\} = \bigoplus_{i=0}^{10} \mathbb{Z}(e_i - e_{11}).$$

For example, $100\mathbf{1} + e_0$ and $100\mathbf{1} + e_1$ have charge 1201 and satisfy the image checks through their respective channels. They are witnesses to the compatibility equations, without any assertion that they belong to HMT terminal histories.

Proposition 8.7 (Cyclic covariance). *Let ρ be the window translation of a register R , of minimal period $p \mid 12$, and let S be the translation of its coordinates. A covariant readout on that orbit is determined by $z \in \text{Fix}(S^p)$. The condition $Q(z) = q$ admits integer solutions if and only if $12/p$ divides q ; when they exist, they form an affine class of rank $p - 1$.*

Proof. The rule $F(\rho^j R) = S^j z$ is well defined exactly when $S^p z = z$. These vectors consist of $12/p$ repetitions of a block of length p , and their charge is $\frac{12}{p} \sum_{i=0}^{p-1} z_i$. The formula proves the divisibility and rank. Covariance of the channels follows from $D_a S = S D_a$ and $Q S = Q$. $\square$

In the harmonic chart, the corresponding action is $\hat{S}_H = \mathcal{H}_{12} S \mathcal{H}_{12} / 4$. If the origin is marked, the action also transports that mark. The period of the orbit indicated here is not identified with that of the enriched history. For an initial readout F_{108} already defined, naturality under truncation is expressed by $F_N = F_{108} \circ \tau_{N,108}$; this identity preserves the initial evaluation without selecting its production rule.

Proposition 8.8 (Additive composition of incidence contributions). *Suppose that the initial state and each event a_t produce, through previously fixed rules on the register, contributions $(\delta w_t, \delta q_t)$ such that*

$$\sum_i \delta w_{t,i} = 0, \quad \delta q_t + T(\delta w_t) \equiv 0 \pmod{12}.$$

With compatible initial conditions $(w^{(0)}, q^{(0)})$, updating by summation determines a unique register at each stage. Its increment is

$$\delta z_{t,i} = \frac{\delta q_t + T(\delta w_t)}{12} - P_i(\delta w_t).$$

The construction respects the concatenation of segments and preserves the readout of a prefix under subsequent extensions.

Proof. The compatibility conditions are preserved by summation, and the inverse (8.10) is additive in (w, q) . Induction fixes each state. Summation over concatenated segments is the sum of their contributions; a prefix receives the same summands before and after extension. $\square$

This proposition gives a sufficient criterion for additive accumulation, not a requirement that every TPK readout be linear. When events are transported by nontrivial actions, their contributions are expressed in the terminal chart through the corresponding affine cocycle. In both cases, the event rules and the initial state preserve their incidence provenance.

The preceding development determines the passage

$$\mathcal{L} \mapsto (A, C, V) \mapsto z \mapsto (D_3 z, D_4 z, Q(z)) \mapsto \mathcal{H}_{12} z$$

with the conditions of each map. Its domain differs from the regional register in Appendix B: there, the equations $X_a L_a = Y_a$ determine the lifts from ternary transitions. Here, the incidences of visits and traces determine integer blocks and their channels. The uniqueness of one reconstruction does not replace the production of the register used by the other. Both preserve their primitives, operations, and proofs in the APP–TRIT–TPK chain.

8.6 Two readouts of the terminal state

Let x_{term} denote the state produced by the composition of selection, transport, and update up to the terminal cutoff. The operator expression is

$$x_{\text{term}} = (\mathcal{U}_{108} \circ \cdots \circ \mathcal{U}_1)(x_{\text{seed}}), \quad \mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

Its two relevant readouts are

$$x_{\text{term}} \xrightarrow{(\pi_{\text{term}}, R_{\text{sgn}})} (B_{\text{term}}, U).$$

The matrix B_{term} retains the visible ternary readout; the vector U preserves another coordinate of the same history. No arrow $B_{\text{term}} \rightarrow U$ is inserted between them.

The terminal selector of the reference construction (Subsection 8.6) acts on catalogues of sizes 196, 197, and 196. The local signatures reduce their $196 \cdot 197 \cdot 196 = 7\,567\,952$ triples to 331, and the column margin $q_{\partial} = (1, 2, 2, 1, 2, 0)$ leaves two matrices:

$$B_0 = \begin{pmatrix} 0 & 2 & 0 & 2 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 1 & 0 & 1 & 0 \end{pmatrix}, \quad B_1 = \begin{pmatrix} 0 & 2 & 1 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 & 2 & 0 \\ 1 & 0 & 0 & 2 & 1 & 0 \end{pmatrix}.$$

The row orientation is $\sigma = (1, 1, -1) = (1, 1, 2)$ in $\mathbb{F}_3^3$. The row charges of the candidates are $(0, 2, 0)$ and $(2, 2, 1)$. Only the second belongs to $\langle \sigma \rangle = \{(0, 0, 0), (1, 1, 2), (2, 2, 1)\}$; hence $B_{\text{term}} = B_1$. This check proves the final step of the selector from the stated census. It does not turn the column margin into a consequence of the row charge, nor replace the preceding enumeration with inspection of these two matrices.

The signed readout has the typed composition

$$R_{\text{sgn}} = \Pi_H \circ \mathcal{E}_{108}^{90,120} \circ \mathcal{R}_{12} : \mathcal{X}_{\text{TPK}}^{\text{term,mem}} \longrightarrow \mathbb{Z}^{12}. \quad (8.13)$$

The channel record determines

$$\mathcal{E}_{108}^{90,120}(\mathcal{R}_{12}(x_{\text{term}})) = (b^{90}, b^{120}, Q_{\text{TPK}}),$$

with coordinates

$$\begin{aligned} b^{90} &= (-495, -116, -684, 108, 601, -90, \\ &\quad -173, -88, 313, 560, -397, 461), \\ b^{120} &= (-425, -281, -481, 671, -255, 30, \\ &\quad 475, -543, 680, 251, 6, -128), \\ Q_{\text{TPK}} &= 6263. \end{aligned}$$

These quantities are used as coordinates of the preceding record, not as parameters fitted through alpha. The reconstruction proved in the following subsections begins exactly at this typed output. The material trace of the preceding extractor and the scope of its self-contained transcription are preserved in the technical provenance note; the subsequent arithmetic is not presented as a substitute for that trace.

8.7 From the channel record to the signed vector

The harmonic readout Π_H can be written without using either K or alpha. To avoid confusing its orbit sums with the alpha blocks, we denote them by s_1, s_2, s_3 . Let

$$B_r = \sum_{j=0}^3 b_{r+3j}^{120}, \quad d_{r,j} = b_{r+3j}^{90}, \quad 1 \leq r \leq 3,$$

and

$$s_1 = \frac{Q_{\text{TPK}} + 2B_1 + B_2}{3}, \quad s_2 = s_1 - B_1, \quad s_3 = s_2 - B_2. \quad (8.14)$$

Division by three belongs to the readout of the three orbits. Its integrality is checked on compatible outputs of the record; it is not assumed for an arbitrary element of $\mathbb{Z}^{25}$.

The three signed blocks are

$$U^{(r)} = (s_r, d_{r,1} - d_{r,3}, d_{r,0} + d_{r,2}, d_{r,0} - d_{r,2}). \quad (8.15)$$

Integer substitution gives

$$(B_1, B_2, B_3) = (972, -1073, 101), \quad (s_1, s_2, s_3) = (2378, 1406, 2479),$$

and therefore

$$\begin{aligned} U^{(1)} &= (2378, -452, -668, -322), \\ U^{(2)} &= (1406, 998, -204, -28), \\ U^{(3)} &= (2479, -551, -371, -997). \end{aligned}$$

Interleaving by columns yields

$$U = (2378, 1406, 2479, -452, 998, -551, -668, -204, -371, -322, -28, -997). \quad (8.16)$$

The presence of negative values and components greater than 999 shows that U is not a base-thousand digit word. Nor is it $U_6 \parallel U_6$, a catalogue concatenation of period six. The distinction concerns domain and preserved information, not merely values.

The constructive order thus far is

$$x_{\text{term}} \longrightarrow \mathcal{R}_{12} \longrightarrow (b^{90}, b^{120}, Q_{\text{TPK}}) \longrightarrow U.$$

The inversion that follows produces the dodecaphase record. The fact that the channels can subsequently be recovered from the dodecaphase record is a reversibility check; it does not reverse this causal order.

8.8 Orbitwise Hadamard transformation and integer reconstruction

In the twelve-position cycle, step three determines three orbits:

$$(1, 4, 7, 10), \quad (2, 5, 8, 11), \quad (3, 6, 9, 12).$$

The separation of four sectors in each orbit is what, once origin and orientation have been fixed, admits the ninety-degree angular readout. The transformation does not act on three consecutive tetrads in the natural ordering.

Let

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}.$$

If P_{orb} reorders by the three indicated orbits, define

$$\mathcal{H}_{12} = P_{\text{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\text{orb}}.$$

Lemma 8.9 (Hadamard inversion and integrality). *We have*

$$H_4^2 = 4I_4, \quad H_4^{-1} = \frac{1}{4}H_4, \quad \det H_4 = -16.$$

For $u \in \mathbb{Z}^4$, the preimage $H_4^{-1}u$ is integral if and only if $H_4 u \equiv 0 \pmod{4}$.

Proof. The rows are orthogonal and have squared norm four; the matrix is symmetric. This gives $H_4^2 = 4I_4$ and the inverse. Integer elimination, or expansion of the determinant, gives the negative sign and the value -16 . The integrality condition is exactly divisibility by four of all four coordinates of H_4u . $\square$

Definition 8.10 (Integral dodecaphase transformation). The sublattice of signed records and its integral coordinatization are

$$\mathcal{L}_H = \{u \in \mathbb{Z}^{12} : \mathcal{H}_{12}u \equiv 0 \pmod{4}\}, \quad \mathcal{K}(u) = \frac{1}{4}\mathcal{H}_{12}u.$$

The map $\mathcal{K} : \mathcal{L}_H \rightarrow \mathbb{Z}^{12}$ is an isomorphism of abelian groups, with inverse $k \mapsto \mathcal{H}_{12}k$. The canonical record K is the image of the signed record generated by the preceding operations, with phase origin and orientation fixed.

Proof. The congruence defines precisely the integrality of $\mathcal{K}(u)$. The identity $\mathcal{H}_{12}^2 = 4I$ proves that both compositions are inverse. In particular, every $k \in \mathbb{Z}^{12}$ has preimage $\mathcal{H}_{12}k \in \mathcal{L}_H$. $\square$

Proposition 8.11 (Canonical integral record). *The inversion $K^{(r)} = H_4U^{(r)}/4$ of the blocks in (8.15) produces*

$$\begin{aligned} K^{(1)} &= (234, 729, 621, 794), \\ K^{(2)} &= (543, 659, 58, 146), \\ K^{(3)} &= (140, 824, 914, 601). \end{aligned}$$

In the natural ordering, the dodecaphase record is

$$K = (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601). \quad (8.17)$$

It is the unique rational preimage of U and belongs to $\{0, \dots, 999\}^{12}$.

Proof. The three products before division are

$$\begin{aligned} H_4U^{(1)} &= (936, 2916, 2484, 3176), \\ H_4U^{(2)} &= (2172, 2636, 232, 584), \\ H_4U^{(3)} &= (560, 3296, 3656, 2404). \end{aligned}$$

All coordinates are divisible by four. Dividing and undoing the orbit permutation gives (8.17). Uniqueness follows from rational invertibility, not from a search among candidates close to a physical value. $\square$

Integrality can also be formulated in lattice terms. The determinantal divisors of H_4 are 1, 2, 4, 16: the entries have greatest common divisor one; the minors of order two are even and one has absolute value two; those of order three are multiples of four and one has absolute value four. Hence

$$\text{SNF}(H_4) = \text{diag}(1, 2, 2, 4),$$

and for the global transformation,

$$\text{coker } \mathcal{H}_{12} \cong (\mathbb{Z}/2\mathbb{Z})^6 \oplus (\mathbb{Z}/4\mathbb{Z})^3.$$

The index of the image is $16^3 = 4096$. Thus, not every signed integer vector admits an integer preimage; the vector obtained satisfies the precise congruences required by inversion. The factor $1/4$ is forced by the matrix and does not have the status of a free coefficient.

8.9 Cyclic differences and the uniform component

With cyclic indices modulo twelve, let

$$(D_3 z)_m = z_m - z_{m+3}, \quad (D_4 z)_m = z_m - z_{m+4}, \quad Q(z) = \sum_{m=1}^{12} z_m.$$

The first operator compares sectors separated by three positions; the second, by four. The channel names 90 and 120 refer here to these separations in the oriented cycle, not to identification with a physical operator from another domain.

Proposition 8.12 (Completeness of the transverse system). *The operator*

$$\mathcal{D} = (D_3, D_4, Q) : \mathbb{Z}^{12} \longrightarrow \mathbb{Z}^{25}$$

has rank twelve and nonzero invariant factors

$$\text{diag}(1, \dots, 1, 12),$$

with eleven units. In particular, its outputs determine a unique dodecaphase record.

Proof. If $D_3 z = D_4 z = 0$, the vector is constant along both steps. Since 3 and 4 generate the cycle modulo twelve, $\ker D_3 \cap \ker D_4 = \langle \mathbf{1} \rangle$. The total charge eliminates this last freedom because $Q(\mathbf{1}) = 12$.

For the integral statement, the rows of (D_3, D_4) are oriented incidences of a connected graph. A spanning tree provides eleven unit invariant factors. Every minor of order twelve must contain the row Q . Adding the first eleven columns to the last makes the latter zero in the incidence rows and twelve in row Q . All maximal minors are divisible by twelve, and the tree produces one of absolute value exactly twelve. The final factor is therefore twelve. $\square$

Checking (8.17) gives

$$D_3 K = b^{90}, \quad D_4 K = b^{120}, \quad Q(K) = 6263.$$

Formulas (8.14) and (8.15), in turn, reconstruct U from those differences. This closes a triangle of exact representations of the same object: transverse record, signed vector, and dodecaphase record. Equality of results does not turn their codomains into a single one; it proves that the specified maps recover the relevant information.

8.10 Two rational readouts: κ_{36} and κ_{per}

Once K has been constructed, the base-thousand chart allows two different operations. The first reads a single turn; the second prolongs the dodecaphase record periodically. Let $B = 1000$ and define its concatenation integer

$$N_K = \sum_{m=1}^{12} K_m B^{12-m} = 234543140729659824621058914794146601. \quad (8.18)$$

The leading zeros of a block, as in 058, are part of its three-digit representation. They do not change its integer value, but they make it possible to concatenate the twelve blocks unambiguously.

Definition 8.13 (Finite and periodic readouts of the dodecaphase record). The one-turn readout is

$$\kappa_{36} = \sum_{m=1}^{12} K_m B^{-m} = \frac{N_K}{B^{12}}.$$

The periodic prolongation is

$$K_j^{\text{per}} = K_{1+((j-1) \bmod 12)}, \quad \kappa_{\text{per}} = \sum_{j \geq 1} K_j^{\text{per}} B^{-j}.$$

Proposition 8.14 (Exact value and period of the prolonged dodecaphase record). *We have*

$$\kappa_{\text{per}} = \frac{N_K}{B^{12} - 1}, \quad \kappa_{\text{per}} - \kappa_{36} = \frac{N_K}{B^{12}(B^{12} - 1)} = B^{-12}\kappa_{\text{per}}. \quad (8.19)$$

The infinite word of the dodecaphase record has minimum period twelve in base thousand and minimum period thirty-six in base ten.

Proof. The sum of the first turn is $N_K B^{-12}$; each new turn multiplies its contribution by B^{-12} . The geometric series gives $N_K/(B^{12} - 1)$, and subtraction gives the second identity.

The twelve components of K are distinct. A proper period of the cycle would identify two of them; the minimum block period is therefore twelve. The decimal expansion is canonical, since the dodecaphase record has no tail of 999 blocks. If its minimum decimal period were d , it would divide 36. Grouping digits in threes would produce a block period of $d/\gcd(d, 3)$. No proper divisor of 36 gives twelve under this operation. Hence the minimum decimal period is 36. $\square$

Both readouts are rational, but they are not the same rational number. The first terminates after twelve blocks; the second repeats the turn. Moreover, $0 < \kappa_{\text{per}} - \kappa_{36} < B^{-12}$. Their proximity does not allow one to be substituted for the other in an exact identity. In particular, the finite closure of alpha uses κ_{36} , whereas the sum of the prolonged section uses κ_{per} .

8.11 Reversibility of the rational constant and phase change

The rational constant κ_{per} determines the register K completely when the base, length, and readout origin are preserved. Its numerical and structural functions are related through an exact inversion.

Proposition 8.15 (Integral recovery and cyclic transformation). *Let $I(K) = \sum_{j=1}^{12} K_j 1000^{12-j}$. Then*

$$I(K) = (1000^{12} - 1)\kappa_{\text{per}}.$$

The integer expansion of $I(K)$ into twelve base-1000 blocks recovers every K_j , including the leading zeros of each block. If $\rho(K) = (K_2, \dots, K_{12}, K_1)$, then

$$\kappa(\rho K) = 1000\kappa(K) - K_1. \quad (8.20)$$

Proof. The first identity inverts the definition of periodic evaluation. Iterated Euclidean division by 1000, with twelve fixed positions, recovers the vector. On the other hand,

$$I(\rho K) = 1000I(K) - K_1(1000^{12} - 1).$$

Dividing by $1000^{12} - 1$ gives (8.20). $\square$

Recovery therefore composes $\kappa \mapsto K \mapsto U$ and the transformations D_3K, D_4K, Q already constructed. Rational evaluation preserves the signed register and the cyclic differences that depend on it. The total depth and transport operators of a history remain in their own coordinates: the domain of the preceding inversion is the integral dodecaphasic register.

Write $\kappa_j = \kappa(\rho^j K)$, with indices modulo 12. The same identity gives

$$K_{j+1} = 1000\kappa_j - \kappa_{j+1}, \quad \sum_{j=0}^{11} \kappa_j = \frac{\sum_{j=1}^{12} K_j}{999} = \frac{6263}{999}.$$

Phase acts exactly on the constant: its value is fixed in the canonical readout, while its rotations follow an affine law. The orbit sum is invariant under the choice of origin.

Subsequent incidence also has an expression in these coordinates. The cubic threshold determines

$$B_K = \{j + 1 : 1000\kappa_j - \kappa_{j+1} \geq 729\} = \{4, 6, 9, 10\}.$$

The tetrad is recovered from the rational orbit because the latter first recovers the integer components. The signed support associated with α additionally uses its regional registers and boundary normalization. The relationship between coordinate and incidence is thus expressed through identifiable operations.

This is an operational definition of the role of K : it provides a reversible integral coordinatization of the oriented register, whose periodic evaluation and incidence functions participate in subsequent constructions. Value, transformation, and domain are preserved jointly.

8.12 Periodicity and the cokernel of the transport operator

The periodicity just proved belongs to the dodecaphase record. It does not imply periodicity of the regional coordinates, of the closure carries, or of α . It is useful to make this separation precise through a second, purely integral quotient.

Let S be the cyclic shift of twelve coordinates, with fixed orientation, and let $b \geq 2$ be an integer base. The periodic carry operator is $\partial_b = S - bI$. If a periodic identity is written

$$A = P + E - \Phi - K + \partial_b C,$$

then the transformations

$$K \mapsto K + \partial_b h, \quad C \mapsto C + h$$

preserve its right-hand side. This invariance compares representatives of a carry class; it is not authorization to modify the dodecaphase record already selected by the record.

Proposition 8.16 (Periodic carry quotient). *With the preceding orientation,*

$$\text{coker}(S - bI) \cong \mathbb{Z}/(b^{12} - 1)\mathbb{Z}.$$

Proof. The cyclic module is presented as $\mathbb{Z}[t]/(t^{12} - 1)$. Imposing $S = bI$ amounts to imposing $t = b$, leaving the relation $b^{12} - 1 = 0$. Evaluating the coefficients at b , with the chosen concatenation order, represents the resulting class. $\square$

The denominator $b^{12} - 1$ of κ_{per} and this quotient come from the same cyclic length, but their objects remain different: one is a rational evaluation and the other a quotient of a carry module. The finite closure of α will be an oriented chain with a right boundary. Its prolongation will transport a compatible remote boundary, not a carry identified by decree between the first and thirteenth positions.

This prepares the next chapter. The dodecaphase record has been fixed before solving the α recurrence; its Hadamard reconstruction is integral and unique; its transverse differences recover the record; and its two rational evaluations have separate domains and uses. None of these facts uses α as an input, and none yet determines the arithmetic type of the output of the complete closure.

9 Derivation of the fine-structure constant

In this development, the fine-structure constant receives an arithmetic determination that brings together the three regional coordinates and the integral dodecaphase record. The operation combines their blocks with fixed orientation and transports the quotients of positional normalization. The resulting parameter expresses compatibility of that oriented difference with cylinder prolongation. Its value and the incidence data used in its construction will remain related throughout the exceptional development.

The internal order is important. The coordinates π , e , and φ , already constructed in the preceding section, are prior coordinates of the same arithmetic state with trajectory memory and may legitimately act as inputs to a subsequent reader. They are not conventional constants injected to select alpha. Correlated generation of the four coordinates shares origin and coinductive compatibility without requiring that all four components be evaluated in a single step.

The two routes express different mathematical roles of the same parameter. The first realizes an integer closure by Euclidean division in base 1000. The second studies the analytic equilibrium of vacancies through the resolvent $E_{21/22}$ and its graded prolongations. One route determines blocks and quotients; the other determines a root through an equation and a uniqueness domain. Their identification is formulated afterward in the common cylinders at every depth. The exposition distinguishes these three results: positional construction, analytic determination, and compatibility of the two.

9.1 First route: positional determination through integer closure

9.1.1 Domain and orientation of integer closure

Let $P_j, E_j, \Phi_j \in \{0, \dots, 999\}$ be the canonical blocks of the three regional coordinates, and prolong the dodecaphase record by

$$K_j = K_{1+((j-1) \bmod 12)}.$$

The channel orientation is $(1, 1, -1)$. The contribution of the dodecaphase record is subtracted according to the already typed dodecaphase closure. For $B = 1000$, the local operation distinguishes the signed balance

$$Z_j := P_j + E_j - \Phi_j - K_j$$

from the sum incorporating the incoming quotient. Normalization is

$$s_j = Z_j + c_{j+1}, \quad A_j = s_j - B \left\lfloor \frac{s_j}{B} \right\rfloor, \quad c_j = \left\lfloor \frac{s_j}{B} \right\rfloor. \quad (9.1)$$

Thus $A_j \in \{0, \dots, B-1\}$, even when s_j is negative. The carry entering from the right belongs to the domain of the operation; it cannot be omitted because the table is printed from left to right.

Definition 9.1 (Finite-depth closure). For a depth N and an integer boundary t , the map $\mathcal{C}_N$ receives $(P_{1:N}, E_{1:N}, \Phi_{1:N}, K_{1:N}, c_{N+1} = t)$ and applies (9.1) in the order $N, N-1, \dots, 1$. Its output is the pair of words

$$(A_{1:N}, c_{1:N+1}).$$

Euclidean division has no residual sign choice: the remainder is fixed in $\{0, \dots, 999\}$, and the quotient is determined. For example, $-431 = 569 - 1000$ and $-1200 = 800 - 2 \cdot 1000$. The negative carries of closure are not anomalies; they are the integer information needed to maintain the oriented equality.

9.1.2 Dodecaphase window and carry conditions

The first twelve triads obtained are

$$\begin{aligned} P &= (141, 592, 653, 589, 793, 238, 462, 643, 383, 279, 502, 884), \\ E &= (718, 281, 828, 459, 045, 235, 360, 287, 471, 352, 662, 497), \\ \Phi &= (618, 033, 988, 749, 894, 848, 204, 586, 834, 365, 638, 117). \end{aligned}$$

In the closure channel, the five regions determining coordinate P preserve their distinct antecedents. Sharing this reading does not identify them as regions or histories.

With boundary $c_{13} = 0$, the recurrence produces the following table. Each input, incoming carry, total before normalization, obtained block, and outgoing carry is shown. The table is a reproducible integer computation, not a list of blocks selected for their resemblance to a target.

j	P_j	E_j	Φ_j	K_j	c_{j+1}	s_j	A_j	c_j
1	141	718	618	234	0	7	007	0
2	592	281	033	543	0	297	297	0
3	653	828	988	140	-1	352	352	0
4	589	459	749	729	-1	-431	569	-1
5	793	045	894	659	-2	-717	283	-1
6	238	235	848	824	-1	-1200	800	-2
7	462	360	204	621	0	-3	997	-1
8	643	287	586	058	-1	285	285	0
9	383	471	834	914	-1	-895	105	-1
10	279	352	365	794	0	-528	472	-1
11	502	662	638	146	0	380	380	0
12	884	497	117	601	0	663	663	0

Table 8: Initial closure with zero right boundary. The dependency propagates from the last row to the first.

The carry word is

$$(c_1, \dots, c_{13}) = (0, 0, 0, -1, -1, -2, -1, 0, -1, -1, 0, 0, 0),$$

and the window reading is the rational number

$$\alpha_{12}^{(0)} = \frac{7297352569283800997285105472380663}{10^{36}} = 0.007297352569283800997285105472380663. \quad (9.2)$$

The superscript recalls the chosen boundary. This window is exact in its finite domain; it is not simply identified with the truncation of the infinite section. The difference between the two readings will become visible when the carry of the remote tail is transported.

9.1.3 Telescoping and reversibility in a boundary fiber

The local form of the equality is

$$P_j + E_j - \Phi_j - K_j - A_j = Bc_j - c_{j+1}. \quad (9.3)$$

The carry term is a link between adjacent positions. Eliminating it coordinate by coordinate gives a false equality. Only an evaluation preserving its endpoints allows it to telescope.

Proposition 9.2 (Integer identity at arbitrary depth). *For $\iota_N(z) = \sum_{j=1}^N z_j B^{N-j}$, every execution of $\mathcal{C}_N$ satisfies*

$$\iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(K) - \iota_N(A) = B^N c_1 - c_{N+1}. \quad (9.4)$$

Proof. Multiplying (9.3) by B^{N-j} and summing produces on the right-hand side

$$\sum_{j=1}^N (Bc_j - c_{j+1}) B^{N-j} = B^N c_1 - c_{N+1}.$$

Each interior carry appears twice with opposite signs; only the two endpoints survive. $\square$

For Table 8, both endpoints are zero. If p_{12}, e_{12}, f_{12} are the three truncated fractional sums,

$$\alpha_{12}^{(0)} = p_{12} + e_{12} - f_{12} - \kappa_{36}.$$

The dodecaphase-record constant appearing here is κ_{36} , not κ_{per} . The domain of this identity contains truncations and a finite boundary; its form does not authorize erasing both conditions when passing to the complete section.

Local inversion is exact as well:

$$K_j = P_j + E_j - \Phi_j + c_{j+1} - A_j - Bc_j.$$

Given P, E, Φ , blocks A , and the complete carries, it recovers the dodecaphase record. Concatenation additionally recovers

$$\iota_N(K) = \iota_N(P) + \iota_N(E) - \iota_N(\Phi) - \iota_N(A) - B^N c_1 + c_{N+1}.$$

This reversibility is a property of the fiber with fixed data and boundaries. It must not be turned into a new causal recipe starting from alpha and manufacturing the record afterward. The direction of construction remains signed record, dodecaphase record, closure, and output.

9.1.4 Prolongation and canonical normalization of the complete section

Prolongation preserves the periodic dodecaphase record and receives the complete regional sequences. Let

$$p = \sum_{j \geq 1} P_j B^{-j}, \quad e_0 = \sum_{j \geq 1} E_j B^{-j}, \quad f = \sum_{j \geq 1} \Phi_j B^{-j}.$$

Each series is a realization of the corresponding internal character. The integer parts already obtained are 3, 2, and 1, so $p = \pi - 3$, $e_0 = e - 2$, and $f = \varphi - 1$.

Proposition 9.3 (Existence and uniqueness of complete normalization). *The series*

$$y = p + e_0 - f - \kappa_{\text{per}} = \sum_{j \geq 1} (P_j + E_j - \Phi_j - K_j) B^{-j} \quad (9.5)$$

converges absolutely. On the domain of the obtained blocks, $0 < y < 10^{-2}$. There is a unique canonical expansion (A_j) of y , with no eventually constant 999 tail, and a unique bounded integer sequence (c_j) satisfying (9.3) with $c_1 = 0$. These carries belong to the invariant set $\mathcal{C} = \{-2, -1, 0, 1, 2\}$.

Proof. The coefficients of the series are bounded in absolute value by $2(B - 1)$, so a geometric series dominates it. If $Z_j = P_j + E_j - \Phi_j - K_j$ and $R_j(Z) = \sum_{r \geq 0} Z_{j+r} B^{-r-1}$, each tail is the difference of two sums of two canonical tails and belongs to $(-2, 2)$. The first coefficient is $Z_1 = 141 + 718 - 618 - 234 = 7$; hence $y = (7 + R_2(Z))/1000 \in (0.005, 0.009)$.

Let (A_j) be the canonical expansion of y . Define

$$c_j = R_j(Z) - R_j(A).$$

Multiplying equality of the two series by B^{j-1} and separating their first $j - 1$ terms gives

$$c_j = \sum_{r=1}^{j-1} (A_r - Z_r) B^{j-1-r} \in \mathbb{Z},$$

and $c_1 = 0$. Furthermore, $R_j(A) \in [0, 1)$, so $-3 < c_j < 2$, placing all carries in $\mathcal{C}$. The tail-shift identities give $Bc_j = Z_j - A_j + c_{j+1}$, which is the required recurrence.

Conversely, any bounded solution with canonical output and $c_1 = 0$ telescopes in the limit to the same series y . The canonical expansion is unique, and once it is fixed, the tail formula determines all carries. Finally, if a finite execution is applied with $c_{j+1} \in \mathcal{C}$, then $-2000 \leq s_j \leq 2000$, so its Euclidean quotient again belongs to $\mathcal{C}$. $\square$

The proposition makes explicit the normalization of the subsequent reader: it does not replace generation of the three sequences with a conventional series introduced as data. Once those sequences have been produced by the coinductive state, their combination and the normalization boundary are determined without consulting alpha.

The finite procedure preserved in the corpus propagates the five boundaries of $\mathcal{C}$ and retains only the prefix common to all of them. If two depths have coalesced before a boundary, the deterministic right-to-left recurrence gives the same blocks to its left. This is compatibility with truncation. A boundary is not chosen because it gives favorable digits, nor is all-depth compatibility inferred from observing a long run.

The initial comparison shows why this precaution matters. Stabilized prolongation gives $c_{13}^\infty = -1$, whereas the isolated window imposed $c_{13} = 0$. Thus,

$$884 + 497 - 117 - 601 - 1 = 662$$

in the last block of the first turn. The first eleven triads do not change, but the prolongation begins

$$(007, 297, 352, 569, 283, 800, 997, 285, 105, 472, 380, 662, 999, \dots).$$

An occurrence of block 999 does not violate the canonical convention: what is excluded is an eventually constant 999 tail. The zero-boundary window ends in 663; the truncation of the complete section ends in 662. Both statements describe different and exact operations. They must not be merged into a single projection notation.

9.1.5 Operative definition and complete additive equation

Definition 9.4 (Alpha coordinate of closure). The integer route determines the coordinate

$$\alpha_A = \sum_{j \geq 1} A_j B^{-j} = p + e_0 - f - \kappa_{\text{per}}, \quad (9.6)$$

where (A_j, c_j) is the compatible canonical normalization of (9.1). In the scalar realization of the HMT coordinates,

$$\boxed{\alpha_A = \pi + e - \varphi - 4 - \kappa_{\text{per}}.} \quad (9.7)$$

The integer four is not calibrated: it is $3 + 2 - 1$, the combination of the integer parts of the three coordinates. Base thousand arises from the cubic chart, and period twelve from the dodecaphase record. The signs are those of the oriented channel and closure, and the remote boundary is that imposed by the compatible section. These are the effective origins of the coefficients of the equation.

At this level, alpha can be described as the scalar coordinate of the closure defect between the three regional realizations and the rational dodecaphase return, computed with compatible carry. The term *defect* denotes a determinate algebraic difference, not an arbitrary loss or an imperfection of the system. This description does not suppress the genealogy: without the regions, signed record, orientation, and carry law, the scalar equality would not be a sufficient exposition of the construction.

Three objects that alpha notation may accompany in subsequent readers must also be kept separate: A is a word of blocks; the negative pre-carry support is obtained from the signs of the totals before normalization; and an exceptional radial vector belongs to an incidence codomain. None is interchangeable with the real value α_A .

Signed balance and sum with incoming quotient. Positional division consists of two consecutive stages. With $K_j^{\text{per}} = K_{1+((j-1) \bmod 12)}$, denoting the periodic prolongation also written K_j in (9.1), the balance before quotient transport is

$$Z_j := R_j + E_j - \Phi_j - K_j^{\text{per}}.$$

The variable s_j in (9.1) denotes the sum already including the incoming contribution, so that

$$s_j = Z_j + c_{j+1}, \quad A_j = Z_j + c_{j+1} - 1000c_j.$$

The incidence configuration H_{α}^- uses the signs of the balance Z_j in the initial window; the determination of the digits also uses the linking quotients c_{j+1} and c_j . This distinction jointly preserves the signed support and the normalization of the expansion.

Exact dependence on the truncation boundary. The regional coordinates and the dodecaphasic register come from the common APP-TRIT-TPK generation. Once produced, their positional composition has an explicit dependence on readout depth. Let $B = 1000$,

$$Z_j = P_j + E_j - \Phi_j - K_j^{\text{per}}, \quad y_N = \sum_{j=1}^N Z_j B^{-j}, \quad R_{N+1}(Z) = \sum_{r \geq 0} Z_{N+1+r} B^{-r-1}.$$

The complete series in (9.5) satisfies exactly

$$\alpha_A = y_N + B^{-N} R_{N+1}(Z), \quad -2 < R_{N+1}(Z) < 2.$$

This identity combines the contribution of the window and that of its extension, both expressed in the same coordinates.

Proposition 9.5 (Boundary transport and canonical cylinder). *Let $A_j^{(N,t)}$ be the output of finite positional division with $c_{N+1} = t \in \mathcal{C} = \{-2, -1, 0, 1, 2\}$, and let $a_N^{(t)} = \sum_{j=1}^N A_j^{(N,t)} B^{-j}$. For the registers in the domain considered, we have*

$$\alpha_A - a_N^{(t)} = B^{-N}(R_{N+1}(Z) - t), \quad |\alpha_A - a_N^{(t)}| < 4B^{-N}. \quad (9.8)$$

The quotient of the complete extension is $c_{N+1}^{\infty} = \lfloor R_{N+1}(Z) \rfloor$. With this boundary,

$$0 \leq \alpha_A - a_N^{(c_{N+1}^{\infty})} < B^{-N},$$

so the resulting window is the canonical prefix of order N .

Proof. Telescoping (9.3) gives $a_N^{(t)} = y_N + tB^{-N} - c_1$. Since $Z_1 = 7$, every incoming state belongs to $\mathcal{C}$, and the first dividend lies between 5 and 9, we have $c_1 = 0$. Substituting this equality into the series decomposition proves (9.8). The bound follows from $R_{N+1}(Z) \in (-2, 2)$ and $t \in \mathcal{C}$.

For complete canonical normalization, $c_{N+1}^{\infty} = R_{N+1}(Z) - R_{N+1}(A)$ and $R_{N+1}(A) \in [0, 1)$. Integrality of the quotient implies $c_{N+1}^{\infty} = \lfloor R_{N+1}(Z) \rfloor$. The error formula then becomes $\alpha_A - a_N^{(c_{N+1}^{\infty})} = B^{-N} R_{N+1}(A)$, proving membership in the canonical cylinder. $\square$

The proposition identifies the meaning of the right boundary: it is the integer part of the signed tail expressed at the scale of the next position. Calculation over the five boundaries preserves this information as a finite collection of possibilities until the blocks to the left stabilize. The quantitative contraction comes from depth and tail transport; its object is the normalization of coordinates already generated.

Graded domain of the analytic resolvent. The analytic composition displayed in §9.2.2 acts on $\text{span}\{e_7, e_8, e_9\}$ and satisfies $Ne_9 = 0$. Its evaluation produces exactly $T_{7:9}$; degree 9 is the last component of that domain. The identity $N^3 = 0$ explains why the resolvent is expressed through three summands. Describing an extension to higher degrees additionally requires specification of the transport between successive graded domains and its action on the initial component of each domain. Local nilpotence and global compatibility thus serve distinct mathematical functions.

9.1.6 Constructive determination and role of the dodecaphasic register

The preceding definition specifies what it means to determine α from structure. The operation's domain contains four registers of known provenance: the three regional sequences and the integral phase register. Normalization also incorporates a boundary condition compatible with prolongation. Once these data are fixed, Euclidean division determines each remainder and its quotient. Proposition 9.3 extends this determination to the complete system. Independence from a target physical value is therefore a property of the construction order: the value compared at the end does not enter the selection of any argument of this map.

This causal independence must be distinguished from algebraic independence between numbers. The former is established by identifying the domain, operations and provenance of coefficients; the latter would be a claim about polynomial relations. The result used here is the former. The construction determines a coordinate through registers produced by APP-TRIT-TPK and allows subsequent verification of the relations it satisfies. That order makes it possible to use an already generated coordinate in a later operation without converting it into an external parameter.

The register K has a more precise function than that of a numerical correction added to a formula. Its twelve positions come from the reversible transformation of the signed register. The equation $U = \mathcal{H}_{12}K$ preserves the possibility of returning to that register, while the transverse differences and total sum permit another reconstruction. In periodic evaluation, the order of positions determines the positional weight of each component. The same set of components, arranged in a different order, generally produces a different rational coordinate. The phase origin and orientation therefore belong to the definition of the number entering (9.7).

Normalization of α composes this information with the regions. Its local identity distinguishes two successive operations. First, the oriented block balance is formed; that balance is then represented by a canonical remainder and a quotient linking adjacent positions. The first operation preserves the sign of the balance and allows subsequent extraction of a support. The second allows prefixes of the real coordinate to be formed. Both readouts remain related because the state preserves the data before and after normalization. The signed support and positional expansion are thus different realizations of an explicit arithmetic composition.

The telescoping identity expresses the global content of this composition. Interior quotients cancel in pairs when a prefix is evaluated, while their endpoints remain in the equality. This cancellation is a property of the weighted sum; it neither removes quotients from the state nor allows them to be suppressed before summation. The right boundary of a window receives the contribution of the continuation. For this reason, evaluation of an isolated window and restriction of a prolonged section are distinct operations. The change of the final block from 663 to 662, proved above, gives an explicit realization of that difference without altering the blocks already stabilized to its left.

In this sense, bidirectionality has a concrete arithmetic formulation. Generation provides compatible prolongations; normalization transmits their boundary's integer information from lower-weight to higher-weight positions. The equation preserves the contribution of both directions through its indices and boundary conditions. Reversibility in the fixed fibre is checked by reconstructing K from the remaining registers and the complete quotients. This inversion certifies consistency of the original operation; the genealogical direction remains the one that first produces the register and then the α coordinate.

The second determination has complementary content. There, the immediate object is a vacancy balance, and the parameter appears as its root in an isolating domain. The positional route provides a canonical normalization of registers; the analytic route studies the vanishing of an operator. Presenting both requires preserving precisely this difference of functions. Their relations with the common state must be expressed through refinement maps and comparison of their complete realizations, as developed in the corresponding subsection. Agreement of a finite approximation constitutes a localized check; identification of the complete constructions has the domain and scope of its own statement.

Finally, this organization explains alpha's position in the article. Its equation appears after regional generation and the construction of K , because it uses both results. The electronic section subsequently uses that coordinate as a factor of its composition. The exceptional prolongation receives the registers and their supports, where incidence information is preserved. These two continuations take up the same provenance and develop distinct mathematical functions: evaluation of a central section and construction of a boundary configuration. The development of each makes it possible to follow the structure through the equation, instead of limiting comparison to its scalar value.

9.2 Second route: analytic determination through vacancies

9.2.1 Transport invariants and graded resolvent

The analytic route receives invariants of the oriented state, not the output α_A . Its sixth-order truncation is

$$P_6(x) = 2\pi x - \frac{7}{4}x^2 + \frac{x^3}{2\pi} + \frac{x^4}{20} - \frac{2x^5}{21} - \frac{x^6}{46}, \quad (9.9)$$

and the vacancy condition uses

$$\delta = \log_{10} \frac{10}{9}.$$

The ratio 10/9 compares the decimal chart with the nine APP positions of the nonadic construction. Its logarithm belongs to this subsequent compactification reading; it is not a target value of alpha.

The genealogy of the coefficients must be preserved in full, but with its exact logical strength. Recent developments bring together a marked signature of the state. The synchronization matrix and associated census are

$$D_{\text{syn}} = \begin{pmatrix} 85 & 112 \\ 41 & 52 \end{pmatrix}, \quad N_9 = 43.$$

From its determinant one extracts

$$a = -\frac{\det D_{\text{syn}}}{N_9} = \frac{172}{43} = 4.$$

The clock induced on the short gaps has secondary spectrum $\{6, 7\}$; its upper endpoint gives $b = 7$. The tritic cardinality $m = 3$ determines

$$t_* = mb = 21, \quad k_* = mb - 1 = 20.$$

The vacancy reading in the thousand chart gives

$$v_- = \lfloor 1000\delta \rfloor = 45, \quad v_+ = \lceil 1000\delta \rceil = 46.$$

These integers are typed antecedents of the truncation's evaluation. They are not obtained from its root or from a metrological approximation to it.

The lower branch additionally coincides with the determinant of the APP block

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \det B_c = 45.$$

The transverse rank of the preceding chapter's record gives $r_{\text{dod}} = 11$, and the phase semicrown provides $f_{\text{ph}} = 8\binom{6}{2} = 120$. In particular, $45r_{\text{dod}} = 495$.

With $\omega = 2\pi$, the signature

$$\mathcal{I}_9 = (\omega, m, a, b, k_*, t_*, v_-, v_+, f_{\text{ph}}, r_{\text{dod}})$$

feeds the graded reader

$$\begin{aligned} \mathfrak{G}_9(\mathcal{I}_9)(x) = & \omega x - \frac{b}{a}x^2 + \omega^{-1}x^3 + \frac{x^4}{k_*} - \frac{2x^5}{t_*} - \frac{x^6}{v_+} \\ & - \frac{x^7}{f_{\text{ph}}} + \frac{x^8}{v_-} + \frac{2x^9}{v_-r_{\text{dod}}}. \end{aligned} \quad (9.10)$$

Substitution recovers the coefficients of P_6 and its ninth-order prolongation. The pair 45/46 crosses the separation between the two: the upper branch occupies degree six, and the lower reappears in degrees eight and nine.

The factorization of (9.10) is formulated for a state preserving origin, channel order, and orientation. On that marked domain, each coefficient is fixed by $(\omega, m, a, b, k_*, t_*, v_-, v_+, f_{\text{ph}}, r_{\text{dod}})$. Two readouts preserving the signature and applying the same graded evaluation agree term by term through degree nine. This uniqueness does not extend to readouts that forget those data.

9.2.2 Nilpotent resolvent in degrees 7, 8, and 9

The first prolongation admits an especially explicit finite operator construction. On the space with basis e_7, e_8, e_9 , let

$$Ne_7 = -\frac{8}{3}e_8, \quad Ne_8 = \frac{2}{11}e_9, \quad Ne_9 = 0, \quad v_7 = -\frac{1}{120}e_7.$$

The symbol N here denotes a nilpotent operator, not a truncation depth. Its domain and grading are fixed before the polynomial is evaluated at a number.

Lemma 9.6 (Exact nilpotent prolongation). *One has $N^3 = 0$ and*

$$(I - N)^{-1}v_7 = -\frac{e_7}{120} + \frac{e_8}{45} + \frac{2e_9}{495}.$$

Evaluation $e_n \mapsto x^n$ produces

$$T_{7:9}(x) = -\frac{x^7}{120} + \frac{x^8}{45} + \frac{2x^9}{495}. \quad (9.11)$$

Proof. The operator raises degree and vanishes on the last basis vector, so its cube is zero. The inverse is the finite resolvent $I + N + N^2$. Moreover,

$$Nv_7 = \frac{1}{45}e_8, \quad N^2v_7 = \frac{2}{495}e_9.$$

Summing the three terms and evaluating gives the formula. $\square$

Here the denominators 120, 45, and 495 are not decimal fits: they arise from the semicrown, APP determinant, and its product with the dodecaphase rank. The transfers $-8/3$ and $2/11$, together with the sign and degree of the seed, belong to the typed reader. Preserving only cycle symmetry and replacing this operator by another symmetric operator does not necessarily preserve the coefficients.

9.2.3 Multiplicity-one root of the ninth-order truncation

Define

$$R_9 = P_6 + T_{7:9}, \quad E_{21/22}^{(9)}(x) = R_9(x) - \delta, \quad I_\alpha = (0, 10^{-2}).$$

This finite operator has its own existence and uniqueness result.

As a subsequent check of the characters already constructed, the rational isolation can be reproduced without a decimal table. Let

$$A_n(t) = \sum_{j=0}^n \frac{(-1)^j t^{2j+1}}{2j+1}, \quad L_n(t) = 2 \sum_{j=0}^n \frac{t^{2j+1}}{2j+1} \quad (0 < t < 1).$$

The alternating-series criterion and the geometric tail sum give

$$A_{2n+1}(t) < \arctan t < A_{2n}(t), \quad (9.12)$$

$$L_n(t) < \log \frac{1+t}{1-t} < L_n(t) + \frac{2t^{2n+3}}{(2n+3)(1-t^2)}. \quad (9.13)$$

Apply (9.12) to Machin's identity $\pi = 16 \arctan(1/5) - 4 \arctan(1/239)$, and (9.13) to

$$\log(10/9) = \log \frac{1 + 1/19}{1 - 1/19}, \quad \log 10 = \log(10/9) + 2 \log \frac{1 + 1/2}{1 - 1/2}.$$

Write

$$\delta := \frac{\log(10/9)}{\log 10}.$$

Truncation orders 80, 20, 40, 120, applied respectively to the four preceding series, give rational intervals for π and δ , both of width less than 10^{-70} . These bounds check a subsequent realization; they neither select the APP-TRIT-TPK state nor receive a value of alpha.

Proposition 9.7 (Multiplicity-one root of the truncation). *The operator $E_{21/22}^{(9)}$ has a unique root ξ_9 in I_α , and that root is simple. It also has the rational isolation*

$$\frac{729735256928380099}{10^{20}} < \xi_9 < \frac{729735256928380100}{10^{20}}. \quad (9.14)$$

Proof. The internal-coordinate bounds $3.14 < \pi < 3.15$ and $0.045 < \delta < 0.046$ suffice for existence. At zero, $E_{21/22}^{(9)}(0) = -\delta < 0$. At 10^{-2} , the linear contribution minus the negative terms exceeds 0.0626, so the operator value is positive.

For $0 \leq x \leq 10^{-2}$, discarding the positive terms of the derivative gives

$$(E_{21/22}^{(9)})'(x) \geq 2\pi - \frac{7}{2} \cdot 10^{-2} - \frac{10}{21} \cdot 10^{-8} - \frac{6}{46} \cdot 10^{-10} - \frac{7}{120} \cdot 10^{-12} > 6.24.$$

Continuity gives existence; the derivative bound gives uniqueness and simplicity. For the left endpoint l and the right endpoint u of (9.14), rational substitution of the bounds (9.12)–(9.13) in the polynomial gives, with directed rounding,

$$-4.6 \cdot 10^{-20} < E_{21/22}^{(9)}(l) < -4.5 \cdot 10^{-20} < 0, \quad 0 < 1.6 \cdot 10^{-20} < E_{21/22}^{(9)}(u) < 1.8 \cdot 10^{-20}.$$

The monotonicity already proved places the unique root between these endpoints. The interval is checked after constructing the operator and does not enter the selection of its coefficients. $\square$

The result concerns ξ_9 . It does not authorize writing $\xi_9 = \alpha_A$ as an exact equality of complete numbers. A truncation root may share a prefix with the root of the prolonged operator and differ at a later position. Simplicity allows control of root movement under perturbations; it does not annul those perturbations.

9.3 Identification criterion for compatible readouts

Identification of two complete readouts requires their cylinders and restriction maps to be preserved. For a compatible graded collection,

$$E_{21/22}^{(6)} \longleftarrow E_{21/22}^{(9)} \longleftarrow E_{21/22}^{(12)} \longleftarrow \cdots, \quad E_{21/22} = \varprojlim_m E_{21/22}^{(6+3m)},$$

the limit denotes the readout together with its transports, not a freely selected tail. The following criterion separates the consequence of compatibility from the effective construction establishing it below.

Proposition 9.8 (Identification of compatible limits). *If the values of the integer and analytic routes belong to the same cylinders C_N^α for every N , and $\text{diam } C_N^\alpha \rightarrow 0$, then*

$$\alpha_A = \alpha_B = \alpha.$$

Proof. For every N , the distance between the two values is bounded by $\text{diam } C_N^\alpha$. Letting N tend to infinity gives zero distance. Existence of compatible sections ensures that this is not the empty intersection of an approximation system. $\square$

The difference can be expressed algebraically. Let $F = \mathbb{Q}(\pi, \delta)$, and let $j_9 : F[[x]] \rightarrow F[x]/(x^{10})$ be truncation. Then

$$\ker j_9 = x^{10}F[[x]].$$

All operators

$$E_h(x) = E_{21/22}^{(9)}(x) + x^{10}h(x)$$

have the same ninth-order truncation. However, taking $h = 0$ and $h = 1/729$,

$$E_{1/729}(\xi_9) = \frac{\xi_9^{10}}{729} > 0,$$

so they do not share that root. The observation does not permit choosing a tail foreign to HMT transport; it demonstrates precisely why the generated tail and its compatibility must be preserved. Nor does an arbitrary formal series suffice for speaking of analytic roots without its convergence domain or the realization of the cylinder system.

9.4 Compatibility between the dodecaphasic and analytic representations of alpha

Compatibility is established at every depth. The reversible dodecaphasic representation and the analytic representation of the alpha coordinate $E_{21/22}$ are two internal representations with distinct codomains, produced from the same enriched APP-TRIT-TPK state. The jet $E_{21/22}^{(9)}$ is an exact finite witness, but does not by itself determine the tail. The completion constructed below is not deduced from the bare jet: it receives the shared coordinate of the state and expresses its analytic image through an explicit recurrence. It therefore does not constitute an independent causal selection; it constitutes a typed analytic representation of the same generated point.

This separation preserves the causal order

$$\text{APP} \longrightarrow \text{TRIT} \longrightarrow \text{TPK} \longrightarrow \mathcal{X}_\alpha^{\text{enr}} \longrightarrow (P, E, \Phi, U_{12}^{\text{sgn}}, K) \longrightarrow \alpha_{\text{HMT}}.$$

The coordinates P, E, Φ and the register K are earlier outputs of the same enriched state. Their use in the closure of alpha is therefore a subsequent internal composition, not an injection of conventional constants.

For $r \in 6\mathbb{N}$, $r \geq 36$, let $\hat{\mathcal{X}}_r^{\text{enr}, \alpha}$ be the sector of truncated states that extend the internal alpha seed fixed at R_{36} . Write

$$x_r^\alpha = (\tilde{x}_r, \chi_\alpha, \varepsilon_r^\alpha, N_r^\alpha),$$

where χ_α is the internal character of the sector, not the expressed scalar value. Each state preserves remainder, quotient, carry, sheet, orientation, phase, route, cylinder, boundary, signature, register, memory, survival, provenance and depth. If

$$d_r^\alpha = \text{Dig}_{729}(\varepsilon_r^\alpha), \quad \varepsilon_{r+6}^\alpha = 729\varepsilon_r^\alpha - d_r^\alpha, \quad N_{r+6}^\alpha = 729N_r^\alpha + d_r^\alpha,$$

the effective transition is

$$\mathcal{U}_r^\alpha = \text{Upd}_r^\alpha \circ \text{Tra}_r^\alpha \circ \text{Sel}_r^\alpha, \quad \text{Ext}_{r+6, r}^{\alpha, \rightarrow} = \text{graph}(\mathcal{U}_r^\alpha).$$

The single-valuedness of these extensions, with the seed R_{36} fixed, produces a unique compatible section: the phase returns while memory and depth advance. The global domain of both determinations is

$$\mathcal{X}_\alpha^{\text{enr}} := \hat{\Omega}_\alpha := \varprojlim_{r \geq 36} (\hat{\mathcal{X}}_r^{\text{enr}, \alpha}, \text{Ext}_{r+6, r}^{\alpha, \rightarrow}). \quad (9.15)$$

The notation κ_{per} retains the periodic readout of the register already constructed in the base-one-thousand chart fixed in (9.7).

9.4.1 The dodecaphasic closure as a complete section

For $N \geq 1$, set

$$\mathcal{W}_N := \{0, \dots, 999\}^N, \quad \mathsf{T}_{\alpha, N}(\mathbf{s}) := (A_1, \dots, A_N), \quad \mathbf{s} \in \mathcal{X}_{\alpha}^{\text{enr}},$$

where the blocks A_j are obtained through (9.1) with the compatible boundary $c_{N+1}^{\infty} = [R_{N+1}(Z)]$. If $N' \geq N$, let $\rho_{N', N} : \mathcal{W}_{N'} \rightarrow \mathcal{W}_N$ be the restriction to the first N blocks.

Theorem 9.9 (Complete dodecaphasic output and projective compatibility). *The collection $(\mathsf{T}_{\alpha, N})_{N \geq 1}$ is compatible:*

$$\rho_{N', N} \circ \mathsf{T}_{\alpha, N'} = \mathsf{T}_{\alpha, N} \quad (N' \geq N). \quad (9.16)$$

Its canonical cylinders

$$C_N^A(\mathbf{s}) := \left[\sum_{j=1}^N A_j 1000^{-j}, \sum_{j=1}^N A_j 1000^{-j} + 1000^{-N} \right) \cap I_{\alpha} \quad (9.17)$$

are nested, have diameter at most 1000^{-N} and satisfy

$$\bigcap_{N \geq 1} C_N^A(\mathbf{s}) = \{\alpha_A(\mathbf{s})\}, \quad \alpha_A = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}. \quad (9.18)$$

Consequently, the first route expresses a unique infinite section, which we denote by $\alpha_{\text{HMT}} := \alpha_A$.

Proof. Proposition 9.3 provides the unique canonical expansion and the unique bounded sequence of carries. The local identity (9.3) and its telescoping (9.4) preserve the remote boundary; Proposition 9.5 proves that restricting an extended execution recovers exactly the preceding prefix. This gives (9.16). Membership of the full sum in each cylinder follows from the tail identity, and $\text{diam } C_N^A \leq 1000^{-N} \rightarrow 0$; the intersection contains a single point. Finally, (9.7) identifies that point with the expression in (9.18). All its coefficients and signs are fixed before the representation of alpha. $\square$

The window $\alpha_{12}^{(0)}$, closed with $c_{13} = 0$, remains an exact finite calculation; it must not be confused with the restriction of the complete section, whose compatible boundary satisfies $c_{13}^{\infty} = -1$. In particular,

$$\rho_{36, 12}(\alpha_{A, 36}) = \alpha_{A, 12} \quad (9.19)$$

is an equality of compatible words, not an equality between words of different lengths or an identification of the isolated window with the limit.

9.4.2 Analytic representation of the alpha coordinate

The analytic extension is specified through all its coefficients. Let $\mathbf{s} \in \mathcal{X}_{\alpha}^{\text{enr}}$. Before decimal normalization, the same state already expresses the pre-coordinate

$$a(\mathbf{s}) := p(\mathbf{s}) + e_0(\mathbf{s}) - f(\mathbf{s}) - \kappa_{\text{per}}(\mathbf{s}) \in I_{\alpha}, \quad (9.20)$$

with $p = \pi_{\text{HMT}} - 3$, $e_0 = e_{\text{HMT}} - 2$ and $f = \varphi_{\text{HMT}} - 1$. Equivalently, $a = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \kappa_{\text{per}}$. No conventional α is introduced here: p, e_0, f are the three regional characters already generated, and κ_{per} is the rational reading of the terminal register. The integer closure applies to (9.20) the normalization in (9.1); the chart below expresses a compatible analytic representation on the same pre-coordinate. Its dependency graph receives $(p, e_0, f, \kappa_{\text{per}})$ directly: it does not contain an edge $\alpha_A \rightarrow \lambda$, although the equality proved below allows both representation names to be identified subsequently.

Write $q = 1/729$, let $F_{\mathbf{s}}$ be the ordered field generated by the coordinates of $\mathbf{s}$, and set $E_{9, \mathbf{s}}(X) = E_{21/22}^{(9)}(\mathbf{s}; X)$. We define the linking coefficient

$$\lambda(\mathbf{s}) := -\frac{E_{9, \mathbf{s}}(a(\mathbf{s}))(1 - q a(\mathbf{s})^3)}{a(\mathbf{s})^{10}}. \quad (9.21)$$

All its arguments belong to the state before metrological recognition. In particular, λ is not obtained by fitting a target decimal string. Equation (9.21) must be read with its exact type: once the first representation of the state has produced $a(s)$, it fixes the unique extension within the geometric class defined below. It is not a second independent selection of $a(s)$, but a correlated analytic representation of the same enriched state.

Indeed, for $\mu \in F_s$ set

$$E_{s,\mu}(X) := E_{9,s}(X) + \mu \frac{X^{10}}{1 - qX^3}.$$

Since $0 < a(s) < 10^{-2}$ and $qa(s)^3 < 1$, we have

$$E_{s,\mu}(a(s)) = 0 \iff \mu = \lambda(s). \quad (9.22)$$

Thus, the coefficient seed contains no freedom once the state and the class of q -geometric extensions have been fixed. This uniqueness is relative to the declared class; it does not assert that the ninth-order jet by itself selects a tail among all possible analytic series.

The local action on blocks of three coefficients is

$$\mathcal{C}_\alpha : F_s^3 \longrightarrow F_s^3, \quad \mathcal{C}_\alpha(u, v, w) = q(u, v, w), \quad c^{(0)} = (\lambda(s), 0, 0). \quad (9.23)$$

Therefore, for every $n \geq 0$,

$$b_{10+3n} = q^n \lambda(s), \quad b_{11+3n} = b_{12+3n} = 0. \quad (9.24)$$

This is the rule that computes each coefficient beyond degree nine. No free parameter remains at each extension.

For $M \geq 0$, let

$$E_{s,M}(X) := E_{9,s}(X) + \lambda(s) \sum_{n=0}^M q^n X^{10+3n}. \quad (9.25)$$

The complete function is

$$E_{s,\infty}(X) := E_{9,s}(X) + \lambda(s) \frac{X^{10}}{1 - X^3/729}. \quad (9.26)$$

To formulate the projective system without implicit types, we fix $d_M = 12 + 3M$ and define the jet fibration

$$\mathfrak{F}_M := \coprod_{s \in \mathcal{X}_\alpha^{\text{nr}}} \{s\} \times F_s[X]/(X^{d_M+1}). \quad (9.27)$$

If $M' \geq M$, the restriction is

$$\tau_{M',M}(s, [P]_{d_{M'}}) := (s, [P]_{d_M}). \quad (9.28)$$

Proposition 9.10 (Naturality of the analytic recurrence). *Let*

$$\Gamma_M(s) = (s, [E_{s,M}]_{d_M}).$$

If $M' \geq M$, the truncation $\tau_{M',M}$ satisfies

$$\tau_{M',M} \circ \Gamma_{M'} = \Gamma_M, \quad \Gamma_{E21}(s) = \varprojlim_M \Gamma_M(s). \quad (9.29)$$

The projection onto the three new degrees is precisely $\mathcal{C}_\alpha^{M+1}(c^{(0)})$.

Proof. Equation (9.24) fixes the block of degrees $10 + 3n, 11 + 3n, 12 + 3n$. Passing from M to $M + 1$ adds only the next block; truncating it recovers the preceding polynomial literally. Compatibility for $M' \geq M$ is obtained by composition. Thus, the inverse limit is not a section chosen from the bare jet, but the unique orbit of $c^{(0)}$ under $\mathcal{C}_\alpha$. $\square$

9.4.3 Uniform convergence, simple root and rational intervals

The regional cylinders and the periodic register provide bounds through rational interval arithmetic. For each coordinate

$$c \in \{\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}\},$$

let P_c be its certified thousand-digit nonadic prefix. The datum used is not the rational number P_c treated as though it were the complete value, but the half-open cylinder and its outer closure

$$\mathcal{C}_c^{(1000)} = [P_c, P_c + \varepsilon), \quad \overline{\mathcal{C}}_c^{(1000)} = [P_c, P_c + \varepsilon], \quad \varepsilon = 10^{-1000}.$$

Directed arithmetic operates conservatively with these closures. If P_π, P_e, P_φ denote the three prefixes, we obtain

$$\begin{aligned} a_- &:= P_\pi + P_e - (P_\varphi + \varepsilon) - 4 - \kappa_{\text{per}}, \\ a_+ &:= (P_\pi + \varepsilon) + (P_e + \varepsilon) - P_\varphi - 4 - \kappa_{\text{per}}, \\ A_{1000} &:= [a_-, a_+], \quad a(s) \in [a_-, a_+] = A_{1000}, \quad \text{diam } A_{1000} = 3\varepsilon. \end{aligned} \quad (9.30)$$

The natural interval extension of (9.21) then transports A_{1000} , the cylinder of π_{HMT} and the directed rational bound for δ to a closed interval $\Lambda_{1000} \ni \lambda(s)$; none of the three tails is frozen. In particular,

$$\frac{7297352569283800997285105472380662}{10^{36}} < a(s) < \frac{7297352569283800997285105472380663}{10^{36}}, \quad (9.31)$$

$$-8.711 \times 10^{-6} < \lambda(s) < -8.709 \times 10^{-6}.$$

These inequalities use certified prefixes with their rational tail bound; they do not receive a table of alpha values. The package verifier reproduces them from the three nonadic outputs and from the terminal chart.

Let

$$r := \frac{10^{-6}}{729} < 1.372 \times 10^{-9}.$$

For $0 \leq x \leq 10^{-2}$, the tail beyond M satisfies

$$|E_{s,\infty}(x) - E_{s,M}(x)| \leq 8.711 \times 10^{-26} \frac{r^{M+1}}{1-r}. \quad (9.32)$$

Moreover,

$$\left| \frac{d}{dx} \left(\lambda \frac{x^{10}}{1 - x^3/729} \right) \right| \leq 8.711 \times 10^{-6} \left(\frac{10 \cdot 10^{-18}}{1-r} + \frac{3 \cdot 10^{-24}/729}{(1-r)^2} \right) < 8.712 \times 10^{-23}. \quad (9.33)$$

The tail of the derivatives beyond block M also admits the depth-dependent estimate

$$\sup_{0 \leq x \leq 10^{-2}} |E'_{s,\infty}(x) - E'_{s,M}(x)| \leq 8.711 \times 10^{-24} r^{M+1} \left(\frac{13+3M}{1-r} + \frac{3r}{(1-r)^2} \right) \xrightarrow{M \rightarrow \infty} 0. \quad (9.34)$$

Theorem 9.11 (Limit function and unique simple root of the correlated chart). *The sequence $E_{s,M}$ converges uniformly, together with its derivatives, to $E_{s,\infty}$ on $\bar{I}_\alpha = [0, 10^{-2}]$. We have*

$$E_{s,\infty}(a(s)) = 0, \quad \inf_{x \in \bar{I}_\alpha} E'_{s,\infty}(x) > 6.239. \quad (9.35)$$

Consequently, $a(s)$ is the unique simple root of the complete function in I_α .

Proof. The ratio of two consecutive terms of the tail is bounded by $r < 1$, which gives (9.32); term-by-term differentiation and summation of the two geometric series give (9.33) and (9.34). The proof for the jet establishes $E'_{9,s} > 6.24$ on $\bar{I}_\alpha$; hence, the complete perturbation leaves the derivative above 6.239. Finally, substituting (9.21) into (9.26), the two summands evaluated at $a(s)$ cancel exactly. The function is strictly increasing; its unique root is therefore simple and coincides with that pre-coordinate. $\square$

To make the identification effective at every depth, directed endpoint evaluations first certify

$$E_{s,\infty}(0) < 0 < E_{s,\infty}(10^{-2}), \quad E'_{s,\infty} \geq \mu := 6239/1000 > c := 6 \quad \text{on } [0, 10^{-2}].$$

By continuity there is a root in the interval and, by the second inequality, it is unique. Only after this existence, monotonicity and the invariant that the bracket contains the root have been established is the mean-value-theorem fallback applied. We start from $D_0 = [0, 10^{-2}]$. If $D_j = [r_j, s_j]$, let $m_j = (r_j + s_j)/2$, and let $[F_j^-, F_j^+]$ be a directed rational enclosure of $E_{s,\infty}(m_j)$, obtained by propagating the cylinders of $\pi_{\text{HMT}}, e_{\text{HMT}}, \varphi_{\text{HMT}}, \delta$ and λ , with

$$0 \leq F_j^+ - F_j^- \leq \frac{c}{4} (s_j - r_j). \quad (9.36)$$

When $F_j^+ < 0$, the right half is retained; when $F_j^- > 0$, the left half is retained. If $F_j^- \leq 0 \leq F_j^+$, no attempt is made to decide whether the midpoint is exactly the root. Let $w_j = s_j - r_j$. The uniform bound $E'_{s,\infty} \geq c = 6$ and the mean value theorem imply that any root contained in D_j belongs to

$$D_{j+1} = D_j \cap \left[m_j - \frac{w_j}{4}, m_j + \frac{w_j}{4} \right]. \quad (9.37)$$

Indeed, let ξ_j be the root and let $y_j = E_{s,\infty}(m_j) \in [F_j^-, F_j^+]$. Since the enclosure contains zero and satisfies (9.36), we have $|y_j| \leq cw_j/4$. The mean value theorem gives $|m_j - \xi_j| \leq |y_j|/c \leq w_j/4$, proving (9.37). The argument includes the case $y_j = 0$: the root is then at the midpoint, but the algorithm does not need to recognize this equality in order to retain it.

If the available enclosure does not satisfy (9.36), the source cylinders are first refined and the evaluation is repeated; no sign is declared and no insufficient contraction is accepted. In each of the three branches the new interval retains the root and has diameter at most $w_j/2$. For each finite target depth, directed refinement allows the width bound to be met without an uncertified comparison of a real number with zero. Monotonicity proves inductively that

$$a(s) \in D_{j+1} \subseteq D_j, \quad \text{diam } D_{j+1} \leq \frac{1}{2} \text{diam } D_j, \quad \text{diam } D_j \longrightarrow 0, \quad (9.38)$$

where the inequality is a contraction bound, not an imposed equality. The executable certificate applies this rule to twenty-four base-one-thousand levels and checks that the complete outer closure A_{1000} , not only its lower endpoint, lies inside each expressed cylinder. Its negative tests reject an excessively wide enclosure, an incorrectly ordered enclosure, a point other than the midpoint, a domain without a sign change, and a degenerate bracket. Coinductive extension allows one thousand to be replaced by any required depth.

Specifically, for $S_N = 1000^N$ the verifier computes

$$j_N^- := \lfloor S_N a_- \rfloor, \quad j_N^+ := \lfloor S_N a_+ \rfloor \quad (9.39)$$

and expresses the level only if $j_N^- = j_N^+$. This equality, which includes the upper endpoint of the outer closure, proves $A_{1000} \subset [j_N^-/S_N, (j_N^- + 1)/S_N)$ and avoids assigning to the left cell an endpoint lying exactly on its right boundary.

Let $C_N^A(s)$ be the integer cylinder in (9.17). We recursively choose $m(N) > m(N-1)$ so that $\text{diam } D_{m(N)} < 1000^{-N}/4$, and define

$$I_{B,N} := D_{m(N)} \cap C_N^A(s). \quad (9.40)$$

Define $\ell_N := \inf I_{B,N}$ and $u_N := \sup I_{B,N}$. Both are rational; the interval is nonempty because both factors contain $a(s)$. Since the cylinder is half-open, u_N may not belong to $I_{B,N}$. Evaluation at that endpoint is understood through the continuous extension of $E_{s,\infty}$ to the outer closure. Thus,

$$I_{B,N+1} \subseteq I_{B,N} \subseteq C_N^A(s), \quad \text{diam } I_{B,N} \longrightarrow 0, \quad E_{s,\infty}(\ell_N) \leq 0 \leq E_{s,\infty}(u_N). \quad (9.41)$$

The bound (9.32) allows a truncation to be chosen that preserves the sign at each endpoint where the complete function does not vanish. If an endpoint coincides with the root, the identity of the

complete function and the membership already proved are used; that same sign is not attributed to its truncations. Indeed, the exact remainder is

$$E_{s,\infty}(x) - E_{s,M}(x) = \lambda(s)x^{10} \frac{(qx^3)^{M+1}}{1 - qx^3}.$$

In the positive domain under consideration, where $\lambda(s) < 0$, we obtain at the root

$$E_{s,M}(a(s)) = -\lambda(s)a(s)^{10} \frac{(qa(s)^3)^{M+1}}{1 - qa(s)^3} > 0 \quad (M < \infty).$$

Therefore, the certification of the weak signs in (9.41) refers to $E_{s,\infty}$. Its evaluation uses the complete function or the truncation accompanied by the signed remainder. This covers both the included infimum and the supremum that may be excluded from the cylinder.

Proposition 9.12 (Compatibility squares and common cylinders). *The restrictions of words and of analytic representations commute:*

$$\begin{array}{ccc} \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{T_{\alpha,N'}} & \mathcal{W}_{N'} \\ \parallel & & \downarrow \rho_{N',N} \\ \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{T_{\alpha,N}} & \mathcal{W}_N \end{array} \quad \begin{array}{ccc} \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_{M'}} & \mathfrak{F}_{M'} \\ \parallel & & \downarrow \tau_{M',M} \\ \mathcal{X}_\alpha^{\text{enr}} & \xrightarrow{\Gamma_M} & \mathfrak{F}_M. \end{array}$$

The collection $I_{B,N}$ provides the effective connection between the two squares and satisfies concretely $I_{B,N} \subseteq C_N^A$ for every N .

Proof. The first square is the compatibility of integer normalization; the second is Proposition 9.10. The inclusion follows from the definition (9.40), and nonemptiness and the signs follow from Theorem 9.11. No step identifies the root of a truncation with that of an arbitrary tail. $\square$

We now define, without leaving the prefix reader implicit,

$$\alpha_B(s) := \arg \text{zero}_{x \in I_\alpha} E_{s,\infty}(x), \quad \text{pref}_N(x) := \frac{\lfloor 1000^N x \rfloor}{1000^N}, \quad \alpha_{B,N} := \text{pref}_N(\alpha_B). \quad (9.42)$$

Theorem 9.13 (Projective equality of the two correlated representations). *For every N , the integer and analytic routes express the same prefix:*

$$\boxed{\alpha_{A,N} = \text{pref}_N(\alpha_A) = \text{pref}_N(\alpha_B) = \alpha_{B,N}.} \quad (9.43)$$

In the limit,

$$\boxed{\alpha_A = \arg \text{zero}_{x \in I_\alpha} E_{s,\infty}(x) = \alpha_B = \alpha_{\text{HMT}}.} \quad (9.44)$$

Proof. Integer normalization expresses exactly $a(s)$. Theorem 9.11 proves that the analytic chart represents that same point as its unique simple root. The half-open cylinders C_N^A fix a canonical prefix, and $I_{B,N} \subseteq C_N^A$ contains $\alpha_B = \alpha_A$. Therefore, their prefixes coincide for each N ; nesting and diameters tending to zero give equality of the limits. Metrological recognition occurs afterwards. $\square$

The proved identity connects two representations of the same state through the coefficient recurrence and compatible cylinders. The analytic function is a correlated representation of the already generated precoordinate, not a second causally independent selection. The jet, geometric class, and state coordinates retain their respective roles; conventional recognition comes afterwards.

10 Exceptional prolongation: from dodecaphasic incidence to Moonshine

The numerical expression does not exhaust the state that generated it. In the preceding construction, APP preserves both sheets and division with remainder and quotient; the TRIT determines the local regime and its orientation; the TPK composes selection, transport, carry, memory, and return. The Archimedean readout contracts compatible cylinders. The readout that we now study instead preserves the incidence of positions and its transport between frames. Both start from the same arithmetic state with trajectory memory. No exceptional geometry is reconstructed from the final decimals of π , φ , e , or α .

This point allows us to specify the meaning of the prolongation $\alpha/K \longrightarrow$ exceptional incidence. The arrow does not assert that an isolated real number determines a code, a lattice, or a group. It transports the dodecaphasic register and the signed data that entered the determination of α , together with the words and frames arising from APP–TRIT–TPK. In particular, the ordered register K can be recovered from its rational constant κ with the base, length, and origin conventions already fixed. Projection onto a support, by contrast, preserves only the selected incidences. Meanwhile, K_{ph} denotes another readout, with reduced memory, subsequent to the holonomy Γ_9 .

The exposition separates three mathematical operations: constructing the finite objects and their maps; prolonging them to a lattice and a vertex operator algebra; and subsequently recognizing the objects obtained by the relevant classical theorems. The names Paley, Witt, Mathieu, Golay, Leech, and Monster designate these recognitions and the explicit realizations that make them verifiable. They are not retrospective selectors of the seeds or coefficients of the generation.

The argument begins with compatibility of prefixes and frames, constructs the ternary code, and determines the flag through K and the signed record of α . The marked incidence then selects the lattice origin and the Leech neighbour. The final step realizes the vertex algebra and its automorphisms. At each stage, the structure preserved will be specified: linear combinations, supports, discriminant classes, bilinear form, or graded product. Explanations of the role of K , the normalization, and the origin of norm 54 accompany their respective constructions in Sections 10.3 and 10.8.

10.1 Compatibility of the Archimedean and incidence realizations

Let X_n^{cal} be the space of admissible prefixes of length n of the arithmetic state with trajectory memory, with truncations $p_{n,m} : X_n^{\text{cal}} \rightarrow X_m^{\text{cal}}$ for $m \leq n$. The superscript recalls that the declared initial frame and its transport are preserved, not that a parameter is fitted to an experimental value. We write (w_j, f_j) for the visible six-trit word and the frame at step j . The frames satisfy a causal law

$$f_{j+1} = g_j(x)f_j,$$

where $g_j(x)$ depends on the prefix already constructed. Consequently, a prolongation does not alter earlier frames.

The chain

$$w_6 \longrightarrow w_{12} \longrightarrow w_{18} \longrightarrow w_{24} \longrightarrow w_{30} \longrightarrow R_{36} \longrightarrow G_9$$

is preserved in this readout. The return of the phase resets neither the frame nor the state memory. Likewise, the census of 468 decimal emissions, the 243 visible six-trit words, and the ambient space $\mathbb{F}_3^6$, with 729 words, remain distinct. The ambient space will be used to construct a linear code; it does not replace the admissible domain of emissions.

For a block b of six trits, the Archimedean readout takes its integer address $\nu(b) \in \{0, \dots, 728\}$. A prefix determines

$$q_n = m + \sum_{j=0}^{n-1} \frac{\nu(b_j)}{729^{j+1}}, \quad I_n = [q_n, q_n + 729^{-n}].$$

Prefix compatibility gives nested cylinders whose diameter tends to zero. Their Cauchy class first belongs to the internal completion $\mathbb{R}_{\text{HMT}}$; its subsequent identification with the usual real line does not enter the choice of blocks.

For the incidence readout we introduce, on the same prefix, the graph map

$$\gamma_A : \mathbb{F}_3^6 \longrightarrow \mathbb{F}_3^{12}, \quad w \longmapsto (w, wA). \quad (10.1)$$

Words are written as row vectors. If $A^f = fAf^{-1}$, the change of frame satisfies the concrete identity

$$\gamma_{A^f}(wf^{-1}) = \gamma_A(w)(f^{-1} \oplus f^{-1}). \quad (10.2)$$

The calibrated codomain retains f together with the encoded word. We then define

$$Q_n(x) = ((f_j, \gamma_{f_j A_w f_j^{-1}}(w_j)))_{0 \leq j < n}. \quad (10.3)$$

Proposition 10.1 (Compatibility and prolongation of the incidence readout). *If $r_{n,m}$ truncates the encoded sequences, then*

$$r_{n,m} Q_n = Q_m p_{n,m}.$$

Thus the Q_n determine a unique map Q_∞ between the respective inverse limits. In each frame, the map preserves exactly the visible word, since $\text{pr}_1 \gamma_A = \text{id}$.

Proof. Two compatible prefixes have the same words and the same transport operators through index $m - 1$. By induction, they have the same frames at those indices. The first m components of (10.3) coincide, proving the square. The universal property of the inverse limit gives Q_∞ . The recovery assertion is projection onto the first six coordinates of each graph. $\square$

This local injectivity has a precise domain. It does not prove that Q_∞ recovers all the deep memory of the TPK state if only the sequence of visible words and frames has been retained. Still less does it prove injectivity after passing from words to supports. Moreover, a general change of basis in $\text{GL}_6(\mathbb{F}_3)$ does not itself preserve ordinary Hamming weight. In a general chart, the incidence and metric of the initial chart are transported as well. The support calculations below are performed in the declared Paley chart; monomial relabelings directly preserve their coordinate interpretation.

10.2 The finite realization and the ternary code

The elliptic operator of the TRIT admits the following finite realization in the six-position chart. These positions coordinatize the six units $\{1, 2, 4, 5, 7, 8\}$ modulo nine, preserved by modular transport. On $\mathbb{P}^1(\mathbb{F}_5)$, after declaring the order $\infty, 0, 1, 2, 3, 4$, let C be the matrix with zero diagonal, $C_{\infty,a} = C_{a,\infty} = 1$, and $C_{a,b} = \chi(b - a)$ in the five finite positions. Here $\chi(1) = \chi(4) = 1$, $\chi(2) = \chi(3) = -1$, and $\chi(0) = 0$. This is a coordinate realization of the quadratic structure already transported; identifying the six positions with the projective line is a chart datum, not a new numerical input to the generator.

The quadratic-sum identities give $C^2 = 5I_6$. Reducing modulo three, we obtain

$$A_W = \begin{pmatrix} 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 2 & 2 & 1 \\ 1 & 1 & 0 & 1 & 2 & 2 \\ 1 & 2 & 1 & 0 & 1 & 2 \\ 1 & 2 & 2 & 1 & 0 & 1 \\ 1 & 1 & 2 & 2 & 1 & 0 \end{pmatrix}, \quad A_W^2 = 2I_6 = -I_6. \quad (10.4)$$

The quadratic equality realizes the elliptic regime; it does not identify the field $\mathbb{F}_3$ with an Archimedean realization of the imaginary unit. These are different realizations of a common operator relation, with their respective domains.

Definition 10.2 (Code of the dodecaphasic chart). The ternary code associated with (10.4) is

$$C_W = \{(w, wA_W) : w \in \mathbb{F}_3^6\} \subset \mathbb{F}_3^{12}.$$

Its set of hexads is

$$\mathcal{H} = \{\text{supp}(c) : c \in C_W, \text{wt}(c) = 6\}.$$

Proposition 10.3 (Self-duality, distance, and Witt design). *The code C_W is self-dual with parameters $[12, 6, 6]_3$. Its weight enumerator is*

$$\sum_{c \in C_W} z^{\text{wt}(c)} = 1 + 264z^6 + 440z^9 + 24z^{12}. \quad (10.5)$$

The set $\mathcal{H}$ has 132 elements and forms an $S(5, 6, 12)$ system: every set of five positions is contained in a unique hexad.

Proof. The generator matrix $[I_6 \mid A_W]$ has rank six and, by symmetry of A_W , satisfies

$$[I_6 \mid A_W][I_6 \mid A_W]^T = I_6 + A_W^2 = 0.$$

The code is self-orthogonal with dimension half its length and is therefore self-dual. Finite evaluation of the 3^6 words gives (10.5); this is an enumeration over the explicit matrix, without real values or external targets. In particular, the minimum distance is six.

If two words of weight six have the same support, among the six ratios of their nonzero coordinates one of the signs 1, -1 occurs at least three times. Subtracting the corresponding multiple would yield a nonzero word of weight at most three, which is impossible. Hence a support corresponds to exactly the pair $\{c, -c\}$, and there are $264/2 = 132$ supports. If two different supports shared five positions, the same argument would cancel at least three coordinates in a union of size seven, producing weight at most four. Thus a five-element set belongs to at most one hexad. Finally,

$$132 \binom{6}{5} = 792 = \binom{12}{5},$$

so it belongs to one and only one. □

The subsequent recognition is the extended ternary Golay code and the small Witt design [13]. The preceding proof explains which object is recognized. It does not replace the APP-TRIT-TPK genealogy with a search among known codes.

10.3 Dodecaphasic flag associated with K and α

Articulation of regional coordinates and closure incidence. The function of this prolongation is understood by distinguishing three operations: preserving a record, evaluating a coordinate, and extracting an incidence relation. All three act on information produced by APP-TRIT-TPK. Their articulation makes explicit the structural thesis of the article: a numerical realization has an operator provenance, and certain data of that provenance additionally admit a combinatorial and lattice realization. The equations allow both paths to be followed, together with their domains and the information retained in each.

The starting point is the terminal state with its compatible prolongations. It retains the residues and quotients of the two APP sheets, the TRIT regime and orientation, and TPK transport with its memory. The closure, propagation, and self-scaling regions are determined in that domain before their readouts establish the coordinates π, e, φ . At the same time, the oriented record of the twelve windows determines the signed vector U . The subsequent appearance of K comes from the integral transformation proved in §8.8:

$$U = \mathcal{H}_{12}K, \quad K = \frac{1}{4}\mathcal{H}_{12}U, \quad \mathcal{H}_{12}^2 = 4I_{12}. \quad (10.6)$$

The congruence defining $\mathcal{L}_H$ guarantees the integrality of the inverse. Thus K is a reversible coordinatization of the signed record in the relevant sublattice. Its positions retain the phase order; its differences D_3K, D_4K retain the transverse relations; the sum $Q(K)$ determines the uniform component. This structure explains why the record participates both in numerical composition and in the selection of a boundary configuration.

The periodic evaluation

$$K \mapsto \kappa_{\text{per}} = \frac{\sum_{m=1}^{12} K_m 1000^{12-m}}{1000^{12} - 1}$$

keeps the twelve blocks recoverable when the base, length, and phase origin are fixed. Extracting a support performs a different operation: it preserves the membership of each position in a class defined by a threshold and groups records having the same support. The distinction between these two maps is decisive. The scalar precision of κ_{per} and the incidence of B_K are two specific mathematical contents, related through the complete record from which they arise.

In the frame inherited from the dodecaphasic register, the data are

$$\begin{aligned} K &= (234, 543, 140, 729, 659, 824, 621, 058, 914, 794, 146, 601), & (10.7) \\ w_\pi &= 010211, & w_e &= 201101, \\ w_\varphi &= 121200, & w_{\text{APP}} &= 022020. \end{aligned}$$

Applying γ_{A_w} yields the following incidences:

$$\begin{aligned} P_\pi &= \{2, 4, 5, 6, 7, 8, 9, 10, 11\}, \\ H_e &= \{1, 3, 4, 6, 9, 10\}, \\ H_\varphi &= \{1, 2, 3, 4, 7, 9\}, \\ H_{\text{APP}} &= \{2, 3, 5, 10, 11, 12\}. \end{aligned}$$

The first is the support of a word of weight nine; the other three are hexads. Through its high-block readout, the register provides

$$B_K = \{i : K_i \geq 729\} = \{4, 6, 9, 10\} = H_e \cap P_\pi. \quad (10.8)$$

The integer 729 here marks a boundary of the declared block format. It is not identified merely by cardinality with the ternary ambient space used in (10.2). In this specific register, all integer thresholds between 660 and 729 produce the same support: the support recovers neither the threshold nor the twelve blocks.

Before carry, the arithmetic route to α preserves the signed vector

$$Z_0 = (7, 297, 353, -430, -715, -1199, -3, 286, -894, -528, 380, 663). \quad (10.9)$$

Its negative support is

$$H_\alpha^- = \{4, 5, 6, 7, 9, 10\}. \quad (10.10)$$

These signs are not extracted from the decimal expansion of the scalar α ; they are preserved in the state preceding its numerical expression.

The normalization of alpha and its signed configuration. The coordinate of α is determined along the positional route by composing the regional blocks and the dodecaphasic register. Before positional division, the oriented combination has components $Z_j = P_j + E_j - \Phi_j - K_j^{\text{per}}$, and the balance with the incoming quotient is $s_j = Z_j + c_{j+1}$. The normalization satisfies

$$A_j = s_j - 1000c_j = Z_j + c_{j+1} - 1000c_j, \quad 0 \leq A_j < 1000. \quad (10.11)$$

This identity preserves both the normalized block and the contribution of the transported quotient. The limit of the compatible prefixes determines the scalar coordinate of the closure. The signed

configuration, in turn, retains which positions of the combination before normalization exhibit a negative defect. In the initial window, this configuration is the vector Z_0 in (10.9), and its negative support is H_α^- .

This defines two kinds of content of a single operation. Positional normalization determines a real coordinate through its compatible remainders and quotients; the negative support determines a configuration of positions in the twelve-component cycle. The sign refers to the register balance before normalization. Its incidence readout remains available because the state retains that balance together with the normalized result. This articulation belongs to the proved positional route; the comparison with the analytic vacancy operators retains the precise scope established in the chapter on α .

Proposition 10.4 (Agreement of the two flag selectors). *Within $\mathcal{H}$, the support (10.10) is the unique hexad H satisfying*

$$B_K \subset H \subset P_\pi, \quad (|H \cap H_e|, |H \cap H_\varphi|, |H \cap H_{\text{APP}}|) = (4, 3, 2).$$

Thus the signed readout and the incidence readout select the same flag $B_K \subset H_\alpha^-$.

Proof. The four hexads containing B_K are obtained by adding, respectively, the pairs

$$\{1, 3\}, \quad \{2, 11\}, \quad \{5, 7\}, \quad \{8, 12\}.$$

Only the second and third pairs are contained in P_π . For the second, the intersection with H_φ has size three and that with H_{APP} has size three; it fails the last condition. For the third, the three intersections have sizes 4, 3, 2. The latter produces precisely the negative support of (10.9). $\square$

It is useful to preserve both the flag and its marked frame:

$$\mathcal{J}_* = (B_K \subset H_\alpha^-), \quad \tilde{\mathcal{J}}_* = (\mathcal{J}_*; P_\pi, H_e, H_\varphi, H_{\text{APP}}).$$

Union, intersection, and difference are equivariant under simultaneous relabeling of the design. The marked class, not the position numbers taken individually, is the transported object.

Lemma 10.5 (Origin selected by marked incidence). *For the preceding frame,*

$$o_* = (H_e \cap H_\varphi) \setminus (H_\alpha^- \cup H_{\text{APP}}) = \{1\}.$$

The rule is equivariant under every automorphism applied simultaneously to the data of $\tilde{\mathcal{J}}_$.*

Proof. The first intersection is $\{1, 3, 4, 9\}$. The union being subtracted contains 3, 4, 9 and does not contain 1. Equivariance follows because a bijection commutes with set operations. $\square$

This origin will select a direction of the lattice neighbor when the A_2 metric and the positive chamber are declared. Incidence alone does not contain that metric. The lattice step will explicitly preserve both data.

From ternary coding to the marked flag. The regions also provide the codes w_π, w_e, w_φ and w_{APP} in the inherited frame. The map

$$\gamma_{A_W} : w \mapsto (w, wA_W)$$

preserves the six-component word as its first coordinate and adds its transform by A_W . The relation $A_W^2 = -I_6$ fixes the quadratic compatibility of this realization. The next step, $c \mapsto \text{supp}(c)$, retains the nonzero positions. For words of weight six, the support identifies exactly the pair $\{c, -c\}$, as proved by the minimum distance of the code. Thus, the quotient by sign produces the hexads, while the complete code remains available for operations requiring its symbols.

In this chart, the two constructions of the tetrad coincide:

$$\{i : K_i \geq 729\} = H_e \cap P_\pi = B_K = \{4, 6, 9, 10\}. \quad (10.12)$$

The left-hand side comes from an integer comparison of the components of K ; the right-hand side, from the intersection of the coded propagation and closure supports. The threshold belongs to the declared block chart and remains explicit in the definition of the readout. The complete record preserves the values on which it acts, even when different thresholds produce the same subset. The equality therefore expresses compatibility between specified maps.

The support H_α^- completes this tetrad to a hexad. The condition of containment in the support P_π , together with intersection sizes $(4, 3, 2)$ relative to $H_e, H_\varphi, H_{\text{APP}}$, selects a unique hexad. The following proofs bring together the two determinations: that obtained from the sign of Z_0 and that obtained from incidence. The result is organized as

$$\tilde{J}_* = (B_K \subset H_\alpha^-; P_\pi, H_e, H_\varphi, H_{\text{APP}}). \quad (10.13)$$

The inclusion is the flag; the four additional supports constitute its marked frame. Simultaneous permutations transport the frame and preserve its inclusions and intersections. The relevant information resides in this transported configuration and its isomorphism class.

The five regions of π recover their internal differentiation here. They share w_π as their visible ternary code, but retain different emissions, signatures, and multiplicities. In the binary realization on a marked dodecad, the transverse region corresponds to the octad whose intersection with the dodecad is the tetrad; the other four regions correspond to lifts of the four hexads of its star. The negative region is distinguished by H_α^- , and the chart ordering individuates the three positive regions. This correspondence preserves the regional roles and the distribution $432 + 4 \cdot 144 = 1008$, as well as realizing the incidence partition $1 + 4$.

10.4 Isometric intertwining of the incidence representations

Incidence defines a linear map whose action can be checked position by position:

$$I : \mathbb{R}^{12} \longrightarrow \mathbb{R}^{\mathcal{H}}, \quad (Iv)(H) = \sum_{p \in H} v_p. \quad (10.14)$$

Let $V_{11} = \mathbf{1}^\perp \subset \mathbb{R}^{12}$. The two Euclidean realizations are here subsequent languages for the same finite design.

Proposition 10.6 (Isometry of the dodecaphasic screen). *We have*

$$I^\top I = 36I_{12} + 30J_{12},$$

where J_{12} is the all-ones matrix. Consequently,

$$U_W = \frac{1}{6}I|_{V_{11}} : V_{11} \xrightarrow{\sim} E_{\text{inc}} := I(V_{11})$$

is an equivariant isometry for the permutation actions of the design automorphisms.

Proof. Each point belongs to $r = \binom{11}{4}/\binom{5}{4} = 66$ hexads, and each pair to $\lambda_2 = \binom{10}{3}/\binom{4}{3} = 30$. These are, respectively, the diagonal and off-diagonal entries of $I^\top I$. On $\mathbf{1}^\perp$, the J_{12} term vanishes, leaving $36I$. Finally, if g is an automorphism, $p \in H$ is equivalent to $g(p) \in g(H)$; hence $I\rho_{12}(g) = \rho_{\mathcal{H}}(g)I$. $\square$

The incidence information also has a concrete spectral readout. Define

$$G_{H,H'} = \binom{|H \cap H'|}{4}.$$

Lemma 10.7 (Action of the intersection kernel). *If $J_{132,12}$ is the rectangular all-ones matrix,*

$$GI = 30I + 15J_{132,12}.$$

Thus G acts by the scalar 30 on E_{inc} .

Proof. The (H, p) entry of GI counts pairs (T, H') with $T \subset H \cap H'$, $|T| = 4$, and $p \in H'$. If $p \in H$, there are ten four-element sets T containing p , each contained in four hexads, and five not containing it, each with a unique extension of the five-element set $T \cup \{p\}$. The total is $10 \cdot 4 + 5 = 45$. If $p \notin H$, the fifteen four-element subsets of H each give one extension: the total is 15. This proves the identity. The rectangular term vanishes on V_{11} . $\square$

Here a material intertwiner does exist: its domain, codomain, entries, and normalization are determined. For example, any orthogonal projector P already constructed on V_{11} is transported to $Q = U_W P U_W^*$ on E_{inc} , with $U_W P = Q U_W$. This assertion does not identify all operators acting on those spaces solely by their trace.

10.5 The Witt star and the generation of M_{12}

Definition 10.8 (Star over a four-element set). For $B \in \binom{[12]}{4}$, the Witt star is the collection of the four hexads containing B . Its petals are the pairs $H \setminus B$. The permutation σ_B fixes B and interchanges the two points of each petal.

The existence of four hexads follows from

$$\lambda_4 = \frac{\binom{8}{1}}{\binom{2}{1}} = 4.$$

Two petals cannot share a point, since two hexads would then contain the same five-element set. They are therefore four disjoint pairs partitioning $[12] \setminus B$. This description is neither a five-pointed star nor a geometric metaphor: it is a four-to-one incidence of the design.

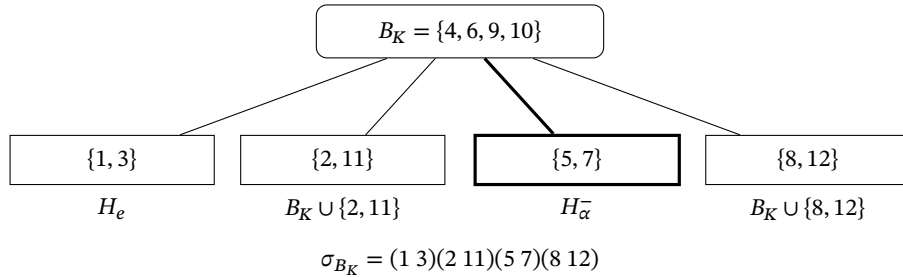


Figure 8: The concrete star of the dodecaphasic flag. Each line indicates the union of the central four-element set with the pair at its endpoint; the thick line marks the hexad also selected by the signs preceding the carry of α . The four pairs exhaust the eight outer positions.

Proposition 10.9 (The collection of stars). *The $\binom{[12]}{4} = 495$ permutations σ_B preserve $\mathcal{H}$. The group they generate has order 95040 and is the Mathieu group M_{12} in its action on the small Witt design.*

Proof. Preservation is a finite check on the 132 hexads of (10.2): for each four-element set, the four petals are constructed and one verifies $\{\sigma_B(H) : H \in \mathcal{H}\} = \mathcal{H}$. Closure under composition of these permutations produces 95040 elements. In lexicographic enumeration, it suffices to successively add six stars not yet contained in the group; the intermediate orders are 2, 6, 18, 360, 7920, 95040, and the 495 stars belong to the final group. The procedure and the six generators are made explicit in the provenance technical note. Finally, one applies the recognition of the automorphism group of $S(5, 6, 12)$ as M_{12} , whose order is $12 \cdot 11 \cdot 10 \cdot 9 \cdot 8 = 95040$. $\square$

An individual star generates a cyclic group of order two. It does not by itself generate M_{12} , much less the Monster. Nor does the existence of 495 generators of a permutation group prove that each is the transport of a specific TPK loop. That assertion requires the corresponding loop map; the result established here is the finite incidence action.

Remark 10.10 (Equality of traces does not identify operators). On V_{11} , a star has spectrum $+1$ with multiplicity seven and -1 with multiplicity four. Its normalized trace is $3/11$. A rank-three projector on the same space also has normalized trace $3/11$, but spectrum $1^3, 0^8$. They are not conjugate: their minimal polynomials and spectra differ. The intertwiner (10.14) transports defined operators; it does not turn this scalar coincidence into an equivalence of operators.

10.6 The binary residue and the five regions of π

The next recognition uses the extended binary Golay code G_{24} , with parameters $[24, 12, 8]_2$. Its weights are 0, 8, 12, 16, 24, and the weight-eight supports form the design $S(5, 8, 24)$. Fix a dodecad D , the support of a word of weight twelve. This datum and the incidence isomorphism to be declared below belong to the realization chart; they are not used to generate or select the decimals of π .

Proposition 10.11 (Residue of a dodecad and lifting of hexads). *The set*

$$\mathcal{H}(D) = \{O \cap D : O \text{ is an octad, } |O \cap D| = 6\}$$

is an $S(5, 6, 12)$ system on D . Each of its hexads has a unique lift to an octad. For any four-element set $B \subset D$, the five octads containing it split into four residual lifts and a unique octad $O_B^\perp$ with $O_B^\perp \cap D = B$.

Proof. A five-element set $A \subset D$ belongs to a unique octad O . If $j = |O \cap D|$, the binary word given by the sum of O and D has weight $20 - 2j$. Among the allowed weights, and with $0 \leq j \leq 8$, only $j = 2, 4, 6$ are possible. Since $A \subset O \cap D$, we have $j \geq 5$ and therefore $j = 6$. This gives existence and uniqueness of the residual hexad, as well as uniqueness of its lift by choosing five of its points. Moreover,

$$\lambda_4(S(5, 8, 24)) = \frac{\binom{20}{1}}{\binom{4}{1}} = 5, \quad \lambda_4(S(5, 6, 12)) = 4.$$

The four residual hexads over B give four distinct octads. The fifth cannot meet D in six points, since it would give another residual hexad; because it contains B , its intersection has size four and equals B . $\square$

A chart $\ell : [12] \rightarrow D$ preserving hexads identifies $\mathcal{H}$ with $\mathcal{H}(D)$. Its existence is the recognition of the design already constructed as the small Witt design; a concrete choice of ℓ fixes the relabeling. The lift

$$\iota_D : \mathcal{H} \longrightarrow \{\text{octads of } G_{24}\}, \quad H \longmapsto O \text{ such that } O \cap D = \ell(H),$$

is injective because intersection with D recovers $\ell(H)$. This injectivity has hexads as its domain, not the register K .

The relation to the five regions of π preserves richer data than the number five. The emissions and their signatures are:

Region	Multiplicity	Emission U_6	Signature
$\perp$	432	(501, 614, 498, 169, 272, 272)	555555
$++_1$	144	(810, 923, 870, 169, 272, 272)	339555
$++_2$	144	(810, 983, 810, 169, 272, 272)	393555
$++_3$	144	(870, 923, 810, 169, 272, 272)	933555
$--$	144	(870, 923, 810, 418, 674, 521)	933717

In each three-digit decimal block, the signature uses the modular difference of the first two digits. The five emissions have the same visible trit word $w_\pi = 010211$, but not the same regional genealogy. The

three positive regions preserve the tail 169|272|272; the negative region changes it to 418|674|521; the transverse region has constant signature and greater multiplicity.

The marked extension of the regional chart sends the transverse region to $O_{\ell(B_K)}^\perp$, the negative one to $\iota_D(H_\alpha^-)$, and the three positive ones to the other three lifts of the star. This realizes the $4 + 1$ partition of the five octads over $\ell(B_K)$. The individual correspondence between the three positive names and the three remaining petals requires a chart ordering; incidence does not determine it without that datum. Thus the precise assertion is a marked correspondence of regions, compatible with their roles and the flag, not a canonical bijection deduced from cardinality five.

The identity $432 + 4 \cdot 144 = 1008$ preserves the regional multiplicities. Also, $1008 = 36 \cdot 28$, and an octad has $\binom{8}{2} = 28$ pairs. These equalities are arithmetic catalogue checks, not substitutes for the map ι_D or the regional chart.

10.6.1 Pentadic representation of the microfibre and its incidence

The arrangement of the five regions displays simultaneously the unity of the ternary quotient and the differentiation retained by the catalogue. The relevant map is the restriction

$$\Psi|_{\mathcal{R}_\pi^{\text{micro}}} : \mathcal{R}_\pi^{\text{micro}} \longrightarrow \{010211\}, \quad \Psi(U) = (-U) \bmod 3.$$

Its five preimages are the U_6 vectors in the preceding table. Figure 9 recovers their radial organization, preserving the chart order, signatures and multiplicities of the catalogue(see 9).

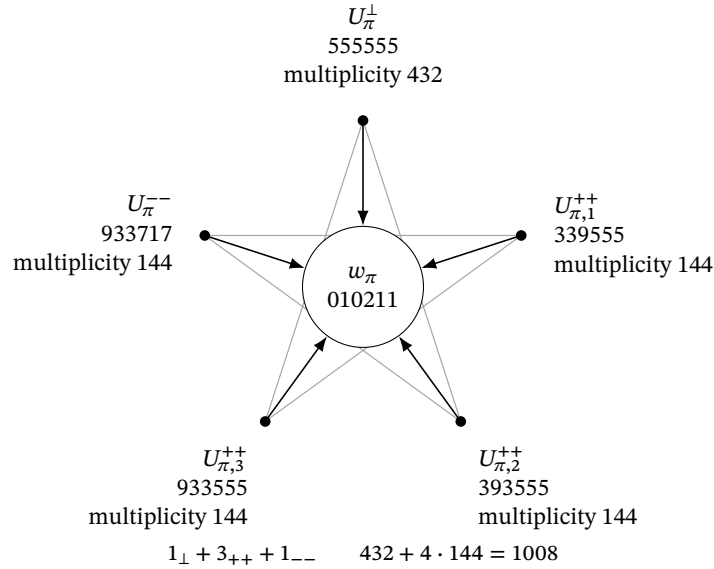


Figure 9: The five regions of π in a pentadic arrangement. Each position preserves its multiplicative signature and multiplicity; the radial arrows represent the common projection $\Psi(U) = 010211$. The star-shaped outline organizes the five positions of the chart. Their realization in the five Witt octads is effected through the regional alignment and the $4 + 1$ decomposition in the text.

Projection to the centre identifies the five ternary images; the register of outer positions retains the regional information that makes the incidence realization possible. The distribution $1_\perp + 3_{++} + 1_{--}$ specifies one transverse region, three positive regions and one negative region. Their normalized weights are $3/7$ and $1/7$ for each of the other four: the transverse multiplicity is threefold, even though the five images under Ψ coincide.

The connection with the Witt star is formulated through the flag $B_K \subset H_\alpha^-$, the alignment ℓ and the lift ι_D already constructed. The five regional positions are realized in the five octads over $\ell(B_K)$:

$$\begin{aligned} U_\pi^\perp &\longmapsto O_{\ell(B_K)}^\perp, \\ U_\pi^{--} &\longmapsto \iota_D(H_\alpha^-), \\ \{U_{\pi,1}^{++}, U_{\pi,2}^{++}, U_{\pi,3}^{++}\} &\longmapsto \{\iota_D(B_K \cup P) : P \in \mathcal{P}_+\}, \\ \mathcal{P}_+ &= \{\{1, 3\}, \{2, 11\}, \{8, 12\}\}. \end{aligned}$$

The three positive positions are assigned to the three pairs in an ordered chart. The transverse and negative rows are distinguished by their regional predicates and by the flag.

The two figures are thereby articulated: Figure 8 displays the four hexads of the small Witt design over B_K ; Figure 9 displays the five regions that, under the alignment, occupy the four residual octads and the transverse octad of the binary design. The $4 + 1$ law, proved in Proposition 10.11, specifies the change of incidence. In this composition, K determines the marked tetrad and the signature of α distinguishes one of its hexads. In addition to their common ternary value, the regions retain the structure required for this transport to the exceptional realization.

10.7 From the ternary code to the Niemeier lattice

The passage to dimension twenty-four need not first go through the binary code. It is performed directly from C_W by discriminant gluing. Two compatible prolongations are thus distinguished: one preserves the residue $W_{12} \subset W_{24}$; the other constructs the lattice on which the vertex operator algebra will be realized.

Let $A_2 = \mathbb{Z}a_1 \oplus \mathbb{Z}a_2$ have Gram matrix

$$\begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}, \quad \rho = a_1 + a_2, \quad \rho^2 = 2.$$

We represent the three classes of A_2^*/A_2 by

$$\varpi_0 = 0, \quad \varpi_1 = \frac{2a_1 + a_2}{3}, \quad \varpi_2 = \frac{a_1 + 2a_2}{3}.$$

The map $j \mapsto \varpi_j + A_2$ additively identifies $\mathbb{F}_3$ with that discriminant group. For $c \in C_W$, write $\phi(c) = (\varpi_{c_1}, \dots, \varpi_{c_{12}})$ and define

$$N(C_W) = \bigcup_{c \in C_W} (A_2^{12} + \phi(c)). \quad (10.15)$$

Theorem 10.12 (Ternary gluing). *The object (10.15) is a positive, even, unimodular lattice of rank 24. Its root system is exactly A_2^{12} ; it is therefore recognized as the Niemeier lattice with that root system.*

Proof. Linearity of the code makes the union of classes closed under addition. Self-orthogonality annihilates the discriminant pairing: this is a multiple of $c \cdot d/3$ modulo $\mathbb{Z}$. Thus the lattice is integral. The norm of a glue class is $2 \text{ wt}(c)/3$ modulo $2\mathbb{Z}$. The weights in (10.5) are multiples of three, so it is even.

The index of A_2^{12} in $N(C_W)$ is $|C_W| = 3^6$. Consequently,

$$\det N(C_W) = \frac{3^{12}}{(3^6)^2} = 1.$$

Positivity comes from the orthogonal sum of the twelve A_2 metrics. In a nonzero class of A_2^*/A_2 , the minimum norm is $2/3$; every nonzero word has at least six nonzero positions. Thus a vector in a nontrivial glue class has norm at least $6 \cdot 2/3 = 4$. No roots of norm two occur outside A_2^{12} . Within it, the roots are precisely those of its twelve components. $\square$

10.8 Incidence selection of the rootless unimodular neighbor

From the incidence origin to lattice modification. The marked flag determines a position through an equivariant set-theoretic operation:

$$o_* = (H_e \cap H_\varphi) \setminus (H_\alpha^- \cup H_{\text{APP}}) = \{1\}.$$

Its function is to distinguish one component among the twelve involved in the lattice realization. The complete ternary code acts as a glue code in the discriminant module $(A_2^*/A_2)^{12}$. Its preimage defines $N(C_W)$, and the self-duality, weights, and distance of the code translate, respectively, into integrality and determinant properties, parity, and norm bounds for the glue classes. Here the symbolic information of the code serves a function that the isolated support no longer performs.

The A_2 metric and the positive radial chamber are fixed in the lattice development. Within that chamber, the congruence required for an even index-three neighbour has twelve realizations of minimum norm 54, permuted by changing the distinguished component. The origin o_* selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v_\alpha^2 = 2(4^2 + 11) = 54.$$

The coefficient 4 belongs to that solution of the congruence in the stated chamber. The subscript α records the incidence provenance of the selection: its content is the frame that participated in the closure and now singles out a lattice direction.

Pairing with v_α determines the index-three sublattice $N_{\alpha,0}$, and adjoining the class $v_\alpha/3$ produces the neighbour N_α of (10.17). This modification preserves rank and recovers unimodularity and evenness; congruence selection eliminates the previous roots, and local bounds exclude roots in the new classes. Identification with Leech follows these verifications. The binary realization of the five regions and this lattice construction are two prolongations of the common frame: the latter directly receives the ternary code as glue data.

The flag label $o_* = 1$ is now realized as one of the twelve A_2 components. This operation declares the preceding metric and the positive radial chamber

$$v(a) = \sum_{i=1}^{12} a_i \rho_i, \quad a_i = 1 + 3b_i, \quad b_i \in \mathbb{Z}_{\geq 0}.$$

The evenness requirement for the index-three neighbor is $v(a)^2 \equiv 0 \pmod{18}$. Since

$$v(a)^2 = 2 \sum_i (1 + 3b_i)^2,$$

this congruence is equivalent to $\sum_i b_i \equiv 1 \pmod{3}$. The smallest norm in the chamber is attained with a single $b_i = 1$ and all others zero; it is 54. The marked origin selects

$$v_\alpha = 4\rho_1 + \rho_2 + \cdots + \rho_{12}, \quad v_\alpha^2 = 54. \tag{10.16}$$

The word “smallest” refers to this positive chamber. It does not assert minimality throughout the residue class: the vector $(a_1 - 2a_2, \rho, \dots, \rho)$ represents the same class modulo three and has norm $14 + 11 \cdot 2 = 36$. The chamber restriction is part of the construction data, not a conclusion that can be omitted.

Let $N = N(C_W)$. Define

$$N_{\alpha,0} = \{x \in N : \langle x, v_\alpha \rangle \equiv 0 \pmod{3}\}, \quad N_\alpha = N_{\alpha,0} + \mathbb{Z} \frac{v_\alpha}{3}. \tag{10.17}$$

Theorem 10.13 (Leech neighbor determined by the marked flag). *With the declared metric and chamber, N_α is a positive, even, unimodular lattice of rank 24, with no vectors of norm two. By the corresponding classification theorem, it is isometric to the Leech lattice Λ_{24} .*

Proof. The character $x \mapsto \langle x, v_\alpha \rangle \bmod 3$ is nonzero: a simple root of a component pairs with $a_i \rho_i$ to give $a_i \equiv 1 \pmod{3}$. Hence $[N : N_{\alpha,0}] = 3$ and $\det N_{\alpha,0} = 9$. Moreover, $v_\alpha/3 \in N_{\alpha,0}^*$ and $v_\alpha \in N_{\alpha,0}$. If $v_\alpha/3$ belonged to N , integrality of N would make all pairings with v_α divisible by three, contradicting the nonzero character. The class of $v_\alpha/3$ therefore has order three. It follows that

$$[N_\alpha : N_{\alpha,0}] = 3, \quad \det N_\alpha = 9/3^2 = 1.$$

The extension is integral, and for $x \in N_{\alpha,0}$,

$$\left(x + m \frac{v_\alpha}{3}\right)^2 = x^2 + 2m \frac{\langle x, v_\alpha \rangle}{3} + 6m^2 \in 2\mathbb{Z}.$$

It is therefore even.

It remains to exclude roots, which is the decisive step. The roots of N belong to A_2^{12} . Their pairings with v_α are congruent to ± 1 or ± 2 modulo three; none belongs to $N_{\alpha,0}$. For the other two classes of the neighbor, each component belongs to one of the local classes

$$A_2 + \varpi_j \pm \rho/3, \quad j = 0, 1, 2.$$

Since $4\rho/3 \equiv \rho/3 \pmod{A_2}$, this description also holds in the distinguished component. Writing a local vector as $(ua_1 + va_2)/3$, its norm is $2(u^2 - uv + v^2)/9$. The six preceding classes have nonzero residues and minimum $2/9$: representatives minimizing the quadratic form are $(\pm 1, 0)$, $(0, \pm 1)$, $(1, 1)$, $(-1, -1)$ in the appropriate residues. A vector in a new class therefore has norm at least $12 \cdot 2/9 = 8/3 > 2$. There are no roots in any of the three classes. The classification of positive even unimodular lattices of rank twenty-four identifies the unique rootless case with Λ_{24} . $\square$

The argument does not recognize Leech from a coincidence of dimension. It constructs the gluing, proves integrality, evenness, and determinant, and eliminates roots by a local bound. Only then does it apply the classification theorem. Dependence on α is preserved in the flag selecting o_* and in the vector (10.16); the scalar value of the constant is not converted into a lattice length.

10.9 The vertex operator algebra and the Monster

Let $\Lambda = N_\alpha$, already recognized as Leech. The next step uses its integral lattice, not a decimal expansion. The lattice vertex operator algebra construction is written

$$V_\Lambda = M(1) \otimes \mathbb{C}_\varepsilon[\Lambda].$$

Here $M(1)$ is the rank-twenty-four Heisenberg oscillator space; $\mathbb{C}_\varepsilon[\Lambda]$ is the group algebra twisted by the cocycle of the lattice central extension. The cocycle and the lifting of isometries are part of the construction: a bare vector-space sum does not determine the vertex products.

Negation $\lambda \mapsto -\lambda$ admits the standard lift θ . With its twisted module and the compatible parity choice, the Frenkel–Lepowsky–Meurman construction produces

$$V_{\text{HMT}}^\natural = V_\Lambda^+ \oplus (V_\Lambda^T)^+. \quad (10.18)$$

The sign $+$ denotes the even part of the lifted action. The expression includes the orbifold vertex product; it is not merely a sum of graded spaces.

Theorem 10.14 (Prolongation to the Moonshine module). *The algebra (10.18), constructed on the neighbor (10.17), is a realization of the Moonshine algebra $V^\natural$. It has central charge 24, zero weight-one component, and automorphism group isomorphic to the Monster $\mathbb{M}$. Its graded character is*

$$\text{Tr}_{V_{\text{HMT}}^\natural} q^{L_0-1} = J(\tau) = j(\tau) - 744 = q^{-1} + 196884q + 21493760q^2 + \cdots, \quad q = e^{2\pi i \tau}. \quad (10.19)$$

Proof. Theorem 10.13 verifies the lattice hypotheses: positivity, evenness, unimodularity, rank twenty-four, and absence of roots. One then applies the Frenkel–Lepowsky–Meurman construction and theorem for the Leech lattice. These identify the orbifold with $V^\natural$ and establish its character and automorphism group. Composing the flag, the neighbor, and the VOA construction functor gives the realization stated here. The group is not deduced from the coefficient 196884. $\square$

The theorem used is a subsequent classical recognition with verified hypotheses, not a new proof in this article of the FLM construction [11]. Furthermore, the notation $q = e^{2\pi i\tau}$ is the conventional Fourier variable used to recognize the character. It introduces neither π nor e as inputs to the coinductive generator that has already determined them.

Part of the trace mechanism can be seen before the orbifold. Only the zero lattice vector is fixed by negation; each oscillator contributes alternating sign. Hence

$$\mathrm{Tr}_{V_\Lambda} \theta q^{L_0-1} = t_2(\tau) := q^{-1} \prod_{n \geq 1} (1 + q^n)^{-24}.$$

This trace in the lattice algebra must not be confused with the character without insertion of $V^\natural$. Nor does it by itself identify a Monster conjugacy class: the space, insertion, and lift must be specified before functions are compared.

Grading, unit and automorphisms. The Frenkel–Lepowsky–Meurman construction is applied to the lattice N_α once its properties have been verified. It incorporates the oscillator space, the central extension of the lattice, the vertex product and the sector twisted by the lift of negation. The result is the algebra $V_{\mathrm{HMT}}^\natural$ of (10.18), whose automorphism group is isomorphic to the Monster. Its domain of action is thus fixed by an algebraic structure equipped with a grading and a product. Borcherds’ Moonshine theorem subsequently determines the graded traces with insertions of these automorphisms [11, 12].

At this level the unit acquires a precise realization: the weight-zero component is the vacuum line and supplies the coefficient 1 of q^{-1} in the graded character. At the cylindrical level, refinement maps preserve the constant function by precomposition. Each preservation statement therefore has its own object and map. The development of the proofs makes this continuity of construction explicit, from arithmetic operations and memory to incidence, the metric and vertex products.

This articulation places K and the closure of α within the main argument. K reversibly preserves an integral coordinate of the history; the normalization of α composes the regional registers with that coordinate; the signed configuration determines a flag; and the marked flag selects the lattice modification on which Moonshine is realized. The following proofs develop these maps, distinguishing reversible equivalences, information quotients and realizations that add a metric or an algebraic structure. This expresses a demonstrable relationship between structure, equation and value, preserving the attribution of each construction and the scope of its theorems.

10.10 Graded Moonshine traces and the Monster representation

For $g \in \mathrm{Aut}(V_{\mathrm{HMT}}^\natural)$, the trace with insertion is

$$T_g(\tau) = \mathrm{Tr}_{V_{\mathrm{HMT}}^\natural} g q^{L_0-1}.$$

The monstrous Moonshine theorem identifies these series as Hauptmoduln of the corresponding genus-zero groups [12, Theorem 1.1]. The character J , the group $\mathbb{M}$, and the collection $g \mapsto T_g$ are realizations of the algebra already constructed. They do not form a causal sequence in which J is first guessed and the Monster then deduced.

The unit has a precise algebraic location here. The weight-zero subspace is the vacuum line, $V_0^\natural = \mathbb{C}\mathbf{1}$, and the coefficient of q^{-1} in J is therefore 1. In weight one, the dimension is zero, accounting for the absence of the constant term in $J = j - 744$. A vertex algebra morphism preserves the vacuum vector. This property supplies the technical meaning of preservation of the unit at this level; the unit of the cylindrical observables is preserved by the precomposition proved earlier. The relation between the two levels must follow the construction maps with their units.

In weight two, the equality

$$196884 = 1 + 196883 = 196560 + 324 = 54(5 \cdot 729 + 1).$$

appears. The first decomposition corresponds to the trivial conformal line and the irreducible component of dimension 196883 of the Monster representation. The other expressions have lattice or arithmetic readouts. Their equality does not automatically identify their summands with regions, words, or TPK routes. Doing so would require presenting the maps between those spaces and proving that they respect the relevant actions.

In particular, an isometry $N_\alpha \simeq \Lambda_{24}$ induces a realization of (10.18), and a choice of isomorphism with $V^\natural$ transports its automorphism group. This has not constructed a Monster action on the digits of π , on the blocks of K , or on all TPK histories. Such an additional assertion would require a representation ρ_{TPK} and a map F with an explicit square

$$F\rho_{\text{TPK}}(g) = \rho_{V^\natural}(g)F,$$

including domain, codomain, and preserved information. The absence of that step from the present assertion does not withdraw the incidence intertwiner U_W , the flag, or the neighbor already constructed; it only delimits a different strengthening, concerning action on the source state.

The chain established in this section can be summarized, keeping its transport data visible, as

$$\begin{aligned} &\text{APP-TRIT-TPK and arithmetic state with trajectory memory} \longrightarrow (w_j, f_j; K, Z_0) \\ &\longrightarrow (C_W, \mathcal{H}; B_K \subset H_\alpha^-, P_\pi, H_e, H_\varphi, H_{\text{APP}}) \\ &\longrightarrow (N(C_W), o_*, \text{metric and chamber}) \longrightarrow N_\alpha \longrightarrow V_{\text{HMT}}^\natural. \end{aligned}$$

At the incidence level, the stars generate M_{12} and the residual chart realizes the regional $4 + 1$ structure in W_{24} . At the lattice and VOA level, the construction gives Leech and the Moonshine module. These are prolongations of the same marked source, with different objects and maps. Causal continuity does not require identifying their codomains or erasing the explicit choices of frame, chamber, and lift.

The result of the section is therefore a consolidated derivation: the finite and lattice proofs make effective the passage from dodecaphasic incidence to the hypotheses of the recognition theorems. This result concerns the specified constructions and hypotheses; it does not add experimental validation to the comparison of constants.

11 Dodecaphasic update and expressed memory

Let d_t be the integer direction code preserved by a TPK trace. This code and the cardinal direction of the cursor are distinct coordinates. The twelve windows $E_m = \{9m + 1, \dots, 9m + 9\}$, $0 \leq m < 12$, partition the 108 steps of the register. The orientation is

$$\Sigma_{12} = (+, +, +, -, -, -, +, +, +, -, -, -).$$

The event of a window is the output

$$a_m = \Sigma_{12}(m) \sum_{t \in E_m} \mathbf{1}_{\{2,4,6,8\}}(d_t). \quad (11.1)$$

The notation a_m avoids confusing this event with Euler's number. The source register also preserves the TRIT orientation, signature, carry and corresponding history segment (see 8).

Proposition 11.1 (Increment produced by transport). *Let $\mathbf{e}_m$ be the basis of $\mathbb{R}^{12}$ and $\mathbf{S}\mathbf{e}_m = \mathbf{e}_{m+1 \pmod{12}}$. If $z_{m+1} = z_m + a_m \mathbf{e}_m$ and $y_m = S^{-m} z_m$, then*

$$y_{m+1} = S^{-1} y_m + \eta_m, \quad \eta_m = a_m S^{-1} \mathbf{e}_0. \quad (11.2)$$

After a complete winding,

$$y_{12} = y_0 + \sum_{m=0}^{11} a_m \mathbf{e}_m. \quad (11.3)$$

Proof. Multiplying the update of z_m by $S^{-(m+1)}$ gives

$$y_{m+1} = S^{-1} y_m + a_m S^{-(m+1)} \mathbf{e}_m = S^{-1} y_m + a_m S^{-1} \mathbf{e}_0.$$

Upon iteration, the coefficient of event a_m is $S^{-(12-m)} \mathbf{e}_0 = \mathbf{e}_m$. Since $S^{12} = I$, (11.3) follows. $\square$

Thus, η_m is not a free force chosen to impose a conservation law: it is a representation of the TPK event. Equality of phase after a winding does not cancel the accumulated events. This readout preserves the counts, whereas the complete register also preserves their order and provenance. We distinguish η_m , a vector increment, from η_{ret} , the scalar action mantissa, and from η_{el} , the shape parameter to be introduced in Section 17.

11.1 Quadratic balance

We define

$$\nabla_3 = I - S^3, \quad \nabla_4 = I - S^4, \quad B = \nabla_3^T \nabla_3 + \nabla_4^T \nabla_4, \quad \mathcal{M}(y) = \frac{1}{2} y^T B y.$$

The oriented readouts of the signed register satisfy $(D_j z)_m = z_m - z_{m+j}$; with the convention $\mathbf{S}\mathbf{e}_m = \mathbf{e}_{m+1}$, one has $D_j = I - S^{-j} = \nabla_j^T$. Both directions produce the same quadratic form B , since powers of S commute. We have $S^T B S = B$ and $B_{00} = 4$. Substituting (11.2) yields exactly

$$\mathcal{M}(y_{m+1}) - \mathcal{M}(y_m) = a_m (B y_m)_0 + 2a_m^2. \quad (11.4)$$

Likewise, the charge $q(y) = \sum_j y_j$ satisfies $q(y_{m+1}) - q(y_m) = a_m$. The proof consists of expanding the quadratic polynomial after removing the permutation S^{-1} . No metrological value enters this calculation. The form $\mathcal{M}$ measures this register; its interpretation as a specific physical energy would require the corresponding dimensional readout.

11.2 Transport resolvents and preservation of memory

The extended history determines the windows $I_m = \{9m+1, \dots, 9m+9\}$, for $m \geq 0$. With the numbering in this section, $I_m = E_m$ during the first cycle. Let $\sigma_m \in \{-1, 1\}$ be the orientation preserved by that history; for $0 \leq m < 12$, it coincides with $\Sigma_{12}(m)$. Define

$$a_m = \sigma_m \sum_{t \in I_m} \mathbf{1}_{\{2,4,6,8\}}(d_t), \quad |a_m| \leq 9.$$

The register and the moving frame are extended by

$$z_{m+1} = z_m + a_m \mathbf{e}_{m \bmod 12}, \quad y_m = S^{-m} z_m.$$

The initial state z_0 preserves the previous history; it is zero for an initially empty prefix. In this subsection, write $R = S^{-1}$ for frame transport. Update (11.2) is maintained in every window. Subsequent orientation and events are obtained from the extended history: return of the frame does not presuppose periodicity of the event sequence.

Proposition 11.2 (Truncation identity with terminal state). *For $N \geq 1$, let*

$$Y_N(u) = \sum_{m=0}^N y_m u^m, \quad H_{\eta,N}(u) = \sum_{m=0}^{N-1} \eta_m u^m.$$

The exact polynomial identity holds:

$$(I - uR)Y_N(u) = y_0 + uH_{\eta,N}(u) - u^{N+1}Ry_N. \quad (11.5)$$

Proof. In degrees $1 \leq m \leq N$, the coefficient on the left-hand side is $y_m - Ry_{m-1} = \eta_{m-1}$, by (11.2). Its coefficients in degrees zero and $N+1$ are y_0 and $-Ry_N$, respectively. Comparison of coefficients proves the equality. $\square$

The terminal term preserves the transported memory when a history is cut. The identity allows a finite evaluation to be compared with its extension while keeping explicit the state left at the boundary.

11.2.1 Correspondence with the common memory development

Changing the window numbering does not change the register. In the first cycle, the conventions of the two developments are related by

$$E_{m+1}^{(I)} = I_m = E_m^{(II)}, \quad 0 \leq m < 12.$$

The signs and events in each window coincide after this index change. The transport convention also coincides: $S\mathbf{e}_j = \mathbf{e}_{j+1 \bmod 12}$ and $R = S^{-1}$. Therefore, update (11.2) is the same identity as equation (2.24) in the common core, with the same preserved previous state.

Proposition 2.18 proves the generating-series identity, its convergence for $|u| < 1$, and the uniform tail bound (2.29). The finite identity (11.5) adds the boundary term $-u^{N+1}Ry_N$ here: it allows the history to be cut without replacing its terminal state by zero. In both cases, the parameter u counts windows of nine transitions.

Elimination of memory components preserves the operator, source, and readout according to Proposition 2.19 and equations (2.30) and (2.31). The example of the cycle of twelve windows (2.32) distinguishes compression of the resolvent from the resolvent of the compression. Composition of three segments and transitivity of reduction retain their proofs alongside identities (2.33) and (2.35). These references belong to the complete development included in this same article. Balance (11.4) and the terminal state in (11.5) preserve, in their present application, the increments and the boundary of the register; they do not by themselves determine additional analytic coefficients.

12 Dodecaphasic seal and reversible rational readout

The terminal register has a visible coordinate and a signed coordinate. The seal is obtained from the latter through the orbitwise Hadamard transformation; a subsequent scalar readout recovers the seal when its representation framework is preserved. This property specifies what a structural definition of a constant means here: the production and recovery maps are made explicit, in addition to its value.

Let $U \in \mathbb{Z}^{12}$ be the signed coordinate produced by the register, H_4 the Hadamard matrix developed in Section 14, and P_{orb} the permutation grouping

$$(1, 4, 7, 10), \quad (2, 5, 8, 11), \quad (3, 6, 9, 12).$$

Define

$$\mathcal{H}_{12} = P_{\text{orb}}^{-1}(H_4 \oplus H_4 \oplus H_4)P_{\text{orb}}.$$

Then $\mathcal{H}_{12}^2 = 4I$. When $\mathcal{H}_{12}U \equiv 0 \pmod{4}$, the integral seal is

$$K = \frac{1}{4}\mathcal{H}_{12}U, \quad U = \mathcal{H}_{12}K.$$

The transverse differences D_3K, D_4K and charge $Q(K)$ recover the channels specified by the register (see 8.12). The seal is thus a reconstructible coordinate of that oriented register; depth and complete history remain distinct fields.

Proposition 12.1 (Recovery of blocks and transport of the origin). *Fix an integer base $b \geq 2$ and twelve ordered positions, and let $K_j \in \{0, \dots, b-1\}$. Define*

$$I(K) = \sum_{j=1}^{12} K_j b^{12-j}, \quad \kappa(K) = \frac{I(K)}{b^{12}-1}.$$

The exact readout κ uniquely determines the twelve blocks. For the rotation $\rho K = (K_2, \dots, K_{12}, K_1)$,

$$\boxed{\kappa(\rho K) = b\kappa(K) - K_1.}$$

Proof. The integer $I = (b^{12} - 1)\kappa$ decomposes uniquely into twelve base- b positions, padding the required initial positions with zeros. Moreover,

$$I(\rho K) = bI(K) - K_1(b^{12} - 1).$$

Dividing by $b^{12} - 1$ gives the transformation law. For the constant block $K_j = b - 1$, the readout is $\kappa = 1$; the fixed periodic representation type likewise preserves its recoverability. $\square$

In the register's base 1000, the three-digit blocks remain recoverable, including leading zeros. The periodic rational readout preserves more than a truncated decimal approximation. Changing the phase origin generally changes its value according to the preceding identity; after twelve rotations it returns. Memory advance in the TPK state is an additional transformation: periodicity of the readout does not identify accumulated histories.

Remark 12.2 (Complexity, entropy and transport). Word complexity and its transport through carry towards $e + \pi$ are distinct from the entropy of a symbolic dynamics, reduced-state entropy and thermodynamic production. Each requires its own definition and a specific transport map. The register reversibility proved in this section preserves the blocks and their phase origin; by itself, it establishes neither an arithmetic classification of $e + \pi$ nor an identification of the three notions of entropy.

13 Cyclic decomposition and determinantal representations

The cyclic projectors

$$P_3 = \frac{1}{4}(I + S^3 + S^6 + S^9), \quad P_4 = \frac{1}{3}(I + S^4 + S^8)$$

have images of dimensions three and four. If $E = P_3 - P_4$, then

$$\frac{\text{Tr } E}{12} = -\frac{1}{12}. \quad (13.1)$$

The scalar comes from a normalized rank difference, not from an ordinary sum of positive integers.

Theorem 13.1 (Determinantal representation of the channel readout). *Let $\mathcal{H}_3 = \text{Fix}(S^3)$ and $\mathcal{H}_4 = \text{Fix}(S^4)$. For $0 < s < 1$,*

$$F(s) = \frac{\det_{\mathcal{H}_3}(I - sS)}{\det_{\mathcal{H}_4}(I - sS)} = \frac{1 - s^3}{1 - s^4} = \frac{1 + s + s^2}{1 + s + s^2 + s^3}. \quad (13.2)$$

Its continuous extension at $s = 1$ equals $3/4$.

Proof. The restriction of S to each image is a cycle of three or four positions, respectively. Their characteristic polynomials give $\det(I - sS) = 1 - s^r$ for $r = 3, 4$. Cancelling the common factor $1 - s$ leaves the final expression, which is regular at $s = 1$. $\square$

The substitution $s = q^{30}$ expresses $F(q^{30}) = (1 - q^{90})/(1 - q^{120})$. The same pair of spaces that produces (13.1) produces the channel quotient. This equality does not identify a twelve-coordinate matrix with the entire TPK state.

13.1 Oriented information and the complex sector

In the antipodal sector $V_- = \text{Ran}((I - S^6)/2)$, the operator $J = S^3$ satisfies $J^2 = -I$ and $J^\top = -J$. With $Q = P_4(I - S^6)/2$, we have

$$\text{rank } Q = 2, \quad E|_{V_-} = -Q, \quad B|_{V_-} = 5I - 3Q.$$

On $\text{Ran } Q$, $S^2 = -I$ and the determinant is $1 + s^2$. Thus,

$$F(s) = \frac{1 + s + s^2}{1 + s} \frac{1}{1 + s^2}.$$

Reversing orientation changes the sign of the form $\omega(u, v) = \langle Ju, v \rangle$, but preserves B and F . Scalar evaluation does not distinguish all oriented transports.

The transported trace also produces

$$a_n^{\text{tr}} = \text{Tr}(S^n E) = 3\mathbf{1}_{3|n} - 4\mathbf{1}_{4|n},$$

and, for $|s| < 1$,

$$-\log F(s) = \sum_{n \geq 1} \frac{a_n^{\text{tr}}}{n} s^n.$$

This collection of moments connects the present block with the collected spectral readouts. Its complete arithmetic development, including Catalan, will have a separate location; it is not assumed here that an identity of functions by itself determines its arithmetic nature.

14 Planck constants and the determinantal structure of action

14.1 Determinantal action scale and decimal valuation

The absolute scale and the relative correction originate in different objects. The former uses the local block of the dodecaphasic register with step three,

$$H_4 = \begin{pmatrix} 1 & 1 & 1 & 1 \\ 1 & 1 & -1 & -1 \\ 1 & -1 & 1 & -1 \\ 1 & -1 & -1 & 1 \end{pmatrix}, \quad G_H = \mathbb{Z}^4 / H_4 \mathbb{Z}^4. \quad (14.1)$$

The operation uses a local quotient. The three blocks of the complete register are not replaced by a tensorization that would multiply the number of sheets of this readout again.

Lemma 14.1 (Number of sheets of the local quotient). *The Smith normal form of H_4 is $\text{diag}(1, 2, 2, 4)$. Consequently,*

$$G_H \simeq \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/2\mathbb{Z} \oplus \mathbb{Z}/4\mathbb{Z}, \quad |G_H| = 16.$$

Proof. The greatest common divisor of the entries is one. The minors of order two are even, and one has value -2 , so their greatest common divisor is two. Since $H_4 H_4^T = 4I$ and $\det H_4 = -16$, the cofactors are ± 4 ; the third determinantal divisor is four. The fourth is 16. Successive ratios of the divisors 1, 2, 4, 16 are 1, 2, 2, 4. Their product gives the quotient order. $\square$

Definition 14.2 (Determinantal action readout). The multiplicative norm of the scalar section α , constant on the sheets of G_H , and one application of self-scaling φ produce

$$\Sigma_H = \varphi N_{G_H}(\alpha), \quad N_{G_H}(\alpha) = \prod_{g \in G_H} \alpha = \alpha^{16}. \quad (14.2)$$

The associated decimal chart determines $\text{ord}_{10}(\Sigma_H) = \lfloor \log_{10} \Sigma_H \rfloor$.

The sixteenth power thus arises from the number of sheets of the quotient, once this local readout is fixed. The factor φ acts once on the basis of the action line. The Smith calculation alone would not select an arbitrary functional on the quotient: the multiplicative norm and self-scaling form part of the declared operation.

Theorem 14.3 (Decade of the action section). *The readout (14.2), applied to the HMT coordinates already generated, satisfies*

$$10^{-34} < \Sigma_H < 10^{-33}, \quad \text{ord}_{10}(\Sigma_H) = -34. \quad (14.3)$$

Proof. The following rational intervals, much coarser than the available cylinders, suffice:

$$\frac{1618}{1000} < \varphi < \frac{1619}{1000}, \quad \frac{7297}{10^6} < \alpha < \frac{7298}{10^6}.$$

The function $(x, a) \mapsto x a^{16}$ is increasing in both positive arguments. The two integer inequalities

$$1618 \cdot 7297^{16} > 10^{65}, \quad 1619 \cdot 7298^{16} < 10^{66}$$

imply

$$10^{-34} < \frac{1618}{1000} \left(\frac{7297}{10^6} \right)^{16} < \Sigma_H < \frac{1619}{1000} \left(\frac{7298}{10^6} \right)^{16} < 10^{-33}.$$

The definition of decimal order completes the proof. No SI unit or reference mass enters these inequalities. $\square$

The normalized section begins with $\Sigma_H = 1.0462438381 \dots \times 10^{-34}$. Its mantissa is not identified with the action coordinate before or after return. The determinantal readout fixes the decimal order; the two return coordinates are obtained through the following readouts.

14.2 Action sections and the dimensionless return quotient

Let $\mathcal{L}_S$ be an oriented action line and $\mathcal{U}_S$ its positive basis. The fifth-order character and the regional return term have the expressions

$$H_5 = \frac{1}{2} \sqrt{\frac{1000\alpha}{\varphi}} - \alpha + \frac{9}{16}\alpha^2 - \frac{5}{9}\alpha^3 + \frac{7}{48}\alpha^4 - \frac{1}{54}\alpha^5, \quad (14.4)$$

$$D_A = \exp\left(-\frac{100\pi\alpha}{9}\right), \quad (14.5)$$

$$\mathcal{C}_\pi(\alpha) = 169\alpha^6 + \frac{D_A\alpha^7}{1 - D_A\alpha}, \quad (14.6)$$

$$\eta_{\text{ret}} = H_5 - \frac{90}{\pi} \mathcal{C}_\pi(\alpha). \quad (14.7)$$

These are subsequent readouts of dodecaphasic closure and its regional incidence. The integer 169 is retained as the regional selector already constructed, and the coefficients of H_5 as those of its local character; they are not replaced by an interpolation of the Planck constant. The formulas make explicit exactly which readouts are needed for the electron stage. There is no need to introduce other functionals of mass theory here.

For $0 < \alpha < 1$, one has $0 < D_A < 1$, so $D_A\alpha < 1$. The rational term in (14.6) preserves the complete sum of the geometric tail:

$$\frac{D_A\alpha^7}{1 - D_A\alpha} = \sum_{n=0}^{\infty} D_A^{n+1} \alpha^{n+7}. \quad (14.8)$$

It is not a truncation that may be silently modified when precision changes. After the term of index N , the remaining tail is $D_A^{N+2}\alpha^{N+8}/(1 - D_A\alpha)$, allowing evaluation to be controlled without recourse to a target quantity.

Definition 14.4 (Action sections and return readout). The sections before and after return are

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S, \quad \hbar_{\text{ret}} = \eta_{\text{ret}} 10^{-34} \mathcal{U}_S. \quad (14.9)$$

Whenever $\eta_{\text{ret}} > 0$, define

$$\delta_{\text{ret}} = H_5 - \eta_{\text{ret}}, \quad R_{\text{act}} = \frac{\hbar_{\text{pre}}}{\hbar_{\text{ret}}} = \frac{H_5}{\eta_{\text{ret}}}. \quad (14.10)$$

Proposition 14.5 (Positivity and invariance of the ratio). *On the intervals used in the proof of (14.3), $0 < \eta_{\text{ret}} < H_5$, and therefore $R_{\text{act}} > 1$. The ratio does not change when the basis $\mathcal{U}_S$ is replaced by any other common positive basis.*

Proof. The terms of $\mathcal{C}_\pi$ are positive. For an elementary bound, it suffices to use $\alpha < 0.008$, $\varphi < 1.619$, $\alpha > 0.007$, $D_A < 1$, and $\pi > 3$. The root term in (14.4) exceeds one, while the sum of the negative terms is less than 0.008001; in particular, $H_5 > 0.99$. Likewise,

$$0 < \frac{90}{\pi} \mathcal{C}_\pi < 30 \left(169(0.008)^6 + \frac{(0.008)^7}{1 - 0.008} \right) < 10^{-6}.$$

Thus $\eta_{\text{ret}} > 0.989999 > 0$ and $\eta_{\text{ret}} < H_5$. Finally, the basis change divides both action coordinates by the same positive factor, which cancels in their quotient. $\square$

Subsequent evaluation gives

$$\begin{aligned} H_5 &= 1.0545718176461544696 \dots, \\ \eta_{\text{ret}} &= 1.0545718169150390611 \dots, \\ R_{\text{act}} &= 1.0000000006932817630 \dots. \end{aligned}$$

The difference and quotient are two readouts of the same sections, one additive and one multiplicative. They do not represent two independent Planck constants. The passage from angular action to action

per complete winding is expressed, in each section, by $h = 2\pi\hbar$. The subsequent chart $\mathcal{U}_S \mapsto \mathbb{J} s$ assigns a unit, but does not act retroactively on the decimal order already proved.

Regional return may also be written in an angular chart. This reparametrization is not an indispensable additional dependency: (14.4)–(14.7) directly supply H_5 , η_{ret} , and R_{act} . If angle and counterangle are introduced, both must be transported in the same chart; degrees and radians are not mixed within a difference.

14.2.1 Action character and provenance of its coefficients

The reduced Planck constant, $\hbar$, is the physical object motivating the realization of the angular action section. Its position in the electronic dependency must be specified at two levels. The determinantal norm Σ_H determines the decade of the scale. The character H_5 and regional return then determine the coordinates of the action sections. This separation presents the result required for the electron without anticipating the complete developments of other sectors of dimensional mechanics.

The provenance of the fifth-order character can be made explicit through invariants already fixed on the support. In the central chart we consider

$$B_c = \begin{pmatrix} 7 & 2 \\ 2 & 7 \end{pmatrix}, \quad \lambda_+ = 9, \quad \lambda_- = 5.$$

The calendar has length 12, the step-3 orbit has length 4, and the relevant census quotient is $104976/1944 = 54$. The character coefficients satisfy

$$\frac{9}{16} = \frac{\lambda_+}{4^2}, \quad \frac{5}{9} = \frac{\lambda_-}{\lambda_+}, \quad \frac{7}{48} = \frac{\text{tr}(B_c)/2}{4 \cdot 12}, \quad \frac{1}{54} = \frac{1944}{104976}. \quad (14.11)$$

The assignment of these invariants to degrees 2, 3, 4, 5, with the alternating signs of (14.4), belongs to the definition of the analytic character used. Making it explicit is essential: the set of cardinalities, considered without the character and without the order of its degrees, would admit other expressions. The constructive content lies in the composition of the support, orientation and declared analytic rule.

14.2.2 Regional continuation and renewal equation

The selector 169 determines the degree-six impulse. Subsequent prolongation is organized by the transfer D_A and a unit normalization of the determinantal line. These rules admit a formulation in formal series before substituting $z = \alpha$, with D_A already constructed. Write

$$\mathcal{C}(z) = 169z^6 + R(z), \quad R(z) \in z^7\mathbb{R}[[z]].$$

Concatenating a prolongation increases the degree by one and multiplies its weight by D_A . The first strict prolongation has weight D_A . Its renewal equation is therefore

$$R(z) = D_A z^7 + D_A z R(z). \quad (14.12)$$

Proposition 14.6 (Formal uniqueness of the normalized continuation). *Equation (14.12) determines a unique series, and*

$$\mathcal{C}(z) = 169z^6 + \frac{D_A z^7}{1 - D_A z} = 169z^6 + \sum_{n=7}^{\infty} D_A^{n-6} z^n.$$

The realization is analytic on $|D_A z| < 1$.

Proof. The element $1 - D_A z$ has constant term one and is invertible in the formal power series ring. Solving the equation produces the displayed series. Equivalently, the degree-seven coefficient is D_A and each subsequent coefficient is D_A times its predecessor. The geometric series converges absolutely in the indicated disk. $\square$

Normalization of the first prolongation is an indispensable operational datum. If only the degree-six impulse and concatenation ratio were preserved, the series

$$169z^6 + \lambda \frac{D_A z^7}{1 - D_A z}$$

would still share both data for any λ . The unit rule fixes $\lambda = 1$ before evaluation. This locates each rule's contribution to uniqueness of the readout, and the rational expression preserves the complete continuation.

14.2.3 Reduced Planck constant and return of the action section

In the dimensional chart of (14.9), the pre-return section is the model's construction intended for comparison with the reduced Planck constant:

$$\hbar_{\text{pre}} = H_5 10^{-34} \mathcal{U}_S.$$

The subsequent section also preserves the regional compensation $90\mathcal{C}(\alpha)/\pi$. Both belong to the same action line, and their quotient R_{act} measures the multiplicative effect of return. Passage to $\hbar$ through 2π uses the relation between angular parametrization and a complete winding.

This construction effectively enters $R_{\text{act}}^{80-54\alpha+6\Delta_4}$. The electron preserves its central coordinate and trajectory; the chronological scale does not replace that structure with an undifferentiated quantum. Nor is a mass already expressed in MeV multiplied again by 10^{-34} : the decade belongs to the action section, cancels in its dimensionless quotient and reappears only where the corresponding dimensional realization acts.

The numerical comparison of H_5 , the returned section and $\hbar/(2\pi)$ is presented in Section 17.6. This distinguishes the name of the physical object, the equation the model assigns to it and the precision with which that equation reproduces it.

14.3 Angular coordinates, winding normalization, and action units

The action quantity and the angular coordinate belong to different spaces. The section $\hbar_{\text{ret}}$ occupies the action line $\mathcal{L}_S$; η_{ret} is its coefficient in the declared basis. Angular representation acts on the dimensionless pair $(\alpha, \eta_{\text{ret}})$, after its construction:

$$\mathcal{P}(\alpha, \eta_{\text{ret}}) = ([1000\alpha]_{360}, [2(\eta_{\text{ret}} + \alpha)]_{360}). \quad (14.13)$$

Nonadic completion determines the factor 1000; bidirectional closure determines the factor two. The circular chart of 360 units makes the partitions $12 \cdot 30 = 4 \cdot 90 = 3 \cdot 120$ visible. On the positive branch of the article, the representatives are

$$A_{\text{deg}} = 1000\alpha, \quad C_{\text{deg}}^* = 2(\eta_{\text{ret}} + \alpha).$$

Proposition 14.7 (Shift of the base-thousand representation). *Let $\alpha = \sum_{n \geq 1} b_n 1000^{-n}$, with $b_n \in \{0, \dots, 999\}$, be the compatible expansion already generated. The lifted coordinate $A_{\text{deg}} = 1000\alpha$ satisfies*

$$A_{\text{deg}} = b_1 + \sum_{n \geq 1} b_{n+1} 1000^{-n}.$$

Representation preserves the block sequence from the second block onward; the initial block moves to the integer part.

Proof. The series converges absolutely. Multiplying each term by 1000 and separating the initial term gives the identity. The operation acts on the constructed expansion, with its blocks and memory; subsequent circular representation takes its class modulo 360. $\square$

Definition 14.8 (Angular-coordinate charts). The degree chart expresses one complete winding in 360 units. The radian chart expresses it as 2π . The coordinate changes and their inverses are

$$x = \frac{\pi}{180} A_{\text{deg}}, \quad y = \frac{\pi}{180} C_{\text{deg}}^*, \quad A_{\text{deg}} = \frac{180}{\pi} x, \quad C_{\text{deg}}^* = \frac{180}{\pi} y.$$

On a traversal with retained winding number, the lifted coordinates are used; the circular quotient expresses their class modulo the corresponding winding.

The definition characterizes degrees and radians through their exact relation to a winding. The register additionally retains orientation, windings, and transport memory. Representation of its angular class preserves phase, while the lift distinguishes traversals with the same final phase. The physical action readout must specify which of these coordinates enters the integral.

Proposition 14.9 (Covariance of the action readout). *For $h_a = 2\pi\hbar_a$, the action of a lifted coordinate satisfies*

$$S_a(\theta) = \hbar_a \theta_{\text{rad}} = h_a \frac{\theta_{\text{deg}}}{360}. \quad (14.14)$$

A positive winding expresses h_a .

Proof. Substituting $\theta_{\text{rad}} = \pi\theta_{\text{deg}}/180$ gives $\hbar_a\pi\theta_{\text{deg}}/180 = (2\pi\hbar_a)\theta_{\text{deg}}/360$. In one winding, $\theta_{\text{deg}} = 360$, yielding h_a . $\square$

The periodicity of a representation is declared separately. If the transported representation satisfies $U(2\pi) = -I$, its operator restoration occurs at 4π . This property of the lift coexists with the per-winding normalization in (14.14). Each preserves its object: integrated quantity, circular phase, and represented state.

14.4 Covariance of the channels and the incidence functional

The channels can equivalently be written as

$$q_{\pm} = \exp[-(x \pm y)] = \exp\left[-\frac{\pi}{180}(A_{\text{deg}} \pm C_{\text{deg}}^*)\right]. \quad (14.15)$$

Proposition 14.10 (Recovery of the even and odd coordinates). *In the lifted real chart, the positive channels uniquely determine both coordinates:*

$$A_{\text{deg}} = -\frac{90}{\pi} \log(q_+ q_-), \quad C_{\text{deg}}^* = \frac{90}{\pi} \log \frac{q_-}{q_+}.$$

The interchange $(q_+, q_-) \mapsto (q_-, q_+)$ preserves A_{deg} and reverses C_{deg}^* .

Proof. The sum and difference of the logarithms in (14.15) are, respectively, $-\pi A_{\text{deg}}/90$ and $-\pi C_{\text{deg}}^*/90$. The real logarithm is single-valued on positive channels. The product is symmetric and the quotient is inverted upon interchange. $\square$

In the twelve-position register, the antipodal involution $r \mapsto r + 6 \pmod{12}$ forms six pairs. Uniform distribution of the odd coordinate over these pairs is $C_{\text{deg}}^*/6$: adding its six contributions recovers C_{deg}^* . This sectorial coordinate is distinguished from the Planck constant h , which belongs to the action line.

The conversion is applied exactly once to each argument. The quantity α remains dimensionless and the quantity $\hbar$ continues to belong to the action line; their preceding coordinates enter the map $\mathcal{P}$.

In addition to the channels, the Barbero–Immirzi functional contains a bilinear component in the angular representatives. Its transport is

$$\frac{\pi}{180} A_{\text{deg}} C_{\text{deg}}^* = \frac{180}{\pi} xy. \quad (14.16)$$

The identity is obtained by replacing both degree representatives with their radian expressions. Thus the complete formula in analytic coordinates is

$$\gamma_{\text{HMT}} = \frac{180}{\pi} xy + 12[S_{90}(e^{-x+y}) - S_{90}(e^{-x-y})] - [S_{120}(e^{-x+y}) - S_{120}(e^{-x-y})].$$

Equivalence of the two charts transports the bilinear component and exponential functions simultaneously. This transformation makes explicit the precise role of the degree in the construction and allows evaluation to be reproduced without ambiguity in angular units.

15 Dodecaphase fundamental chain and hexad–octad functional

The active incidence is constructed before any gravitational realization. Let

$$H_{\text{act}} = \{1, 2, 4, 5, 7, 8\}, \quad \mathcal{B}_2(H_{\text{act}}) = \{A \subset H_{\text{act}} : |A| = 2\}.$$

Then

$$|H_{\text{act}}| = 6, \quad |\mathcal{B}_2(H_{\text{act}})| = 15,$$

and the incidences before and after enlargement have cardinalities

$$\mathcal{F}_{90} = H_{\text{act}} \times \mathcal{B}_2(H_{\text{act}}), \quad |\mathcal{F}_{90}| = 90,$$

$$\mathcal{F}_{120} = \{1, \dots, 8\} \times \mathcal{B}_2(H_{\text{act}}), \quad |\mathcal{F}_{120}| = 120.$$

These two cardinalities are outputs of the phase rule; they are neither angles nor parameters supplied by the physical realization.

Denote by $C_{12} = \mathbb{Z}/12\mathbb{Z}$ the temporal record of TPK and by ∂_{ret} its unique return seam per turn. In the free module of oriented events, consider the fundamental chain

$$c_{12}^+ = \sum_{j \in C_{12}} [j, \mathcal{F}_{90}] - [\partial_{\text{ret}}, \mathcal{F}_{120}]. \quad (15.1)$$

The sum contains exactly the twelve sectors of the wheel; the second term is unique because one turn has a single closure seam. Its sign is the boundary sign. The bidirectional involution sends c_{12}^+ to $c_{12}^- = -c_{12}^+$: it changes the orientation of the chain without altering its absolute multiplicities or its incidence types.

For $m \in \{90, 120\}$, let

$$S_m(q) = \frac{q^m}{1 - q^{3m}}.$$

The return readout $\mathcal{R}$ is defined on the generators of (15.1) by

$$\mathcal{R}([j, \mathcal{F}_{90}]) = S_{90}, \quad \mathcal{R}([\partial_{\text{ret}}, \mathcal{F}_{120}]) = S_{120}.$$

The first equality is constant on C_{12} by covariance under phase rotation; the second has a different domain and acts only at the enlarged seam. By linearity,

$$\boxed{\mathcal{R}(c_{12}^+) = 12S_{90} - S_{120}}. \quad (15.2)$$

Thus the coefficients $12 : -1$ are the image of the fundamental chain: twelve phase cells and one oriented boundary. They are not selected through the conventional value of the Barbero–Immirzi parameter, through a decimal comparison, or by prescribing in advance the pole coefficient to be calculated next.

Theorem 15.1 (Pole coefficient of the dodecaphase chain). *The functional $F(q) = \mathcal{R}(c_{12}^+)(q) = 12S_{90}(q) - S_{120}(q)$, determined by the twelve sectors and their oriented seam, has coefficient $1/24$ for $(1 - q)^{-1}$. Its complex residue at $q = 1$ is $-1/24$. The pole coefficient is an output of the already constructed functional.*

Proof. For a functional with a simple pole at $q = 1$, write

$$p_1(F) := \lim_{q \uparrow 1} (1 - q)F(q).$$

The identity

$$1 - q^{3m} = (1 - q) \sum_{j=0}^{3m-1} q^j$$

implies

$$p_1(S_m) = \lim_{q \uparrow 1} \frac{q^m}{\sum_{j=0}^{3m-1} q^j} = \frac{1}{3m}.$$

By linearity,

$$p_1(\mathcal{R}(c_{12}^+)) = \frac{12}{270} - \frac{1}{360} = \frac{48-3}{1080} = \frac{1}{24}.$$

These equalities are taken in $\mathbb{Q}$. The complex residue is the coefficient of $(q-1)^{-1}$, so that

$$\text{Res}_{q=1} \mathcal{R}(c_{12}^+)(q) = -p_1(\mathcal{R}(c_{12}^+)) = -\frac{1}{24}.$$

□

The Diophantine equation

$$\frac{a}{270} - \frac{b}{360} = \frac{1}{24} \iff 4a - 3b = 45$$

remains a subsequent arithmetic certificate. By itself, it admits the collection $(a, b) = (12+3t, 1+4t)$; the chain (15.1) eliminates this ambiguity before calculation, since changing $(12, 1)$ alters the multiplicity of the orbit or its seam and therefore no longer represents a fundamental turn of TPK.

With the conjugate channels $q_{\pm}$ already generated by the angular branch, one finally obtains

$$\mathcal{R}(c_{12}^+)(q_-) - \mathcal{R}(c_{12}^+)(q_+) = 12\Delta S_{90} - \Delta S_{120}.$$

This is the internal input of the hexad–octad functional. Its evaluation as the Barbero–Immirzi parameter and its insertion into the area operator are subsequent representations: they do not participate in the selection of c_{12}^+ , 90/120, 12 : −1, or 1/24.

Falsifiers. The construction is refuted if the phase census does not yield twelve sectors, if one turn contains more than one seam, if the incidences do not have cardinalities 90 and 120, if the involution does not reverse the entire chain, or if the exact coefficient of $(1-q)^{-1}$ in its image differs from 1/24.

16 Hexadic–octadic incidence and the Barbero–Immirzi functional

Let $H \hookrightarrow O$ be a marked inclusion, $|H| = 6$, $|O| = 8$. The incidence objects

$$\mathcal{F}_6 = H \times \binom{H}{2}, \quad \mathcal{F}_8 = O \times \binom{H}{2}$$

have cardinalities 90 and 120. Their normalized difference gives

$$\frac{|\mathcal{F}_6| - |\mathcal{F}_8|}{24\binom{6}{2}} = -\frac{1}{12}.$$

This is a combinatorial realization of the same scalar. Numerical equality with (13.1) does not erase the distinction between the spaces on which it has been calculated.

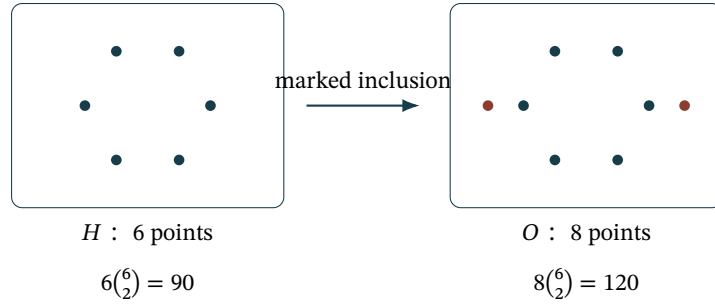


Figure 10: Marked incidence from six to eight points. The drawing represents the sets and the inclusion; it does not purport to turn the faces or vertices of a cube into Steiner blocks without their incidence map.

16.1 The weights precede the pole coefficient

In the free module of oriented events, the source fixes the fundamental chain of the turn (see 15.2)

$$c_{12}^+ = \sum_{j \in \mathbb{Z}/12\mathbb{Z}} [j, \mathcal{F}_6] - [\partial_{\text{ret}}, \mathcal{F}_8].$$

Multiplicity twelve corresponds to the sectors, and multiplicity one to the oriented seam. The involution changes the orientation of the complete chain. Let

$$S_m(q) = \frac{q^m}{1 - q^{3m}}, \quad 0 < q < 1.$$

The linear readout assigns S_{90} to each sector and S_{120} to the seam; consequently,

$$\mathcal{R}(c_{12}^+) = 12S_{90} - S_{120}.$$

Theorem 16.1 (Pole coefficient and conjugate evaluation). *The image of the fundamental chain has coefficient $1/24$ relative to $(1 - q)^{-1}$ and residue $-1/24$ relative to $(q - 1)^{-1}$. Its evaluation in the previously generated channels determines*

$$\gamma_{\text{HMT}} = \frac{\pi AC^*}{180} + 12[S_{90}(q_-) - S_{90}(q_+)] - [S_{120}(q_-) - S_{120}(q_+)]. \quad (16.1)$$

Proof. The factorization $1 - q^{3m} = (1 - q) \sum_{j=0}^{3m-1} q^j$ implies $\lim_{q \uparrow 1} (1 - q)S_m(q) = 1/(3m)$. By linearity,

$$\frac{12}{270} - \frac{1}{360} = \frac{1}{24}.$$

The coordinate $q - 1$ reverses the sign of the residue. The conjugate evaluation and the angular component of the source produce (16.1). $\square$

The Diophantine certificate $4a - 3b = 45$ admits the pairs $(12 + 3t, 1 + 4t)$, $t \geq 0$; it certifies minimality of $(12, 1)$, but does not replace the origin of the multiplicities in the chain. The evaluation preserved in the source owner is

$$\gamma_{\text{HMT}} = 0.27400767269445927474725526077398041940 \dots$$

This number is an output of the source functional; it is not an external value used to select A , C^* , or their coefficients.

16.2 Parity and structural sensitivity

Reversing C^* interchanges q_+ and q_- and changes the sign of the angular component. Therefore,

$$\gamma_{\text{HMT}}(A, -C^*) = -\gamma_{\text{HMT}}(A, C^*).$$

The functional preserves orientation, whereas other readouts, such as q_+q_- , are even. This difference precedes its physical interpretation.

16.3 Structural dependence of the functional and areal normalization

The preceding construction locates the Barbero–Immirzi parameter at the intersection of three determinations: marked incidence, dodecaphasic orientation and angular evaluation. Incidence supplies the event spaces; orientation fixes the chain on those spaces; evaluation transports the chain to a dimensionless quantity. This decomposition preserves the origin of each coefficient when the functional is subsequently used in area geometry.

The inclusion $\mathcal{F}_6 \hookrightarrow \mathcal{F}_8$ preserves the marked pair of the hexad. Its complement has cardinality $2\binom{6}{2} = 30$. This increment reappears as the determinant of the map that reconstructs boundary occupations from the area and the weighted observable. The equality of these two integers has a specific explanation here: the sector weights differ precisely by the number of events added by the marked inclusion.

Proposition 16.2 (Incidence and recoverability of occupations). *Let n_6, n_8 be the aggregate occupations of two sectors, and let $r_6 = |\mathcal{F}_6|$, $r_8 = |\mathcal{F}_8|$. The map*

$$(n_6, n_8) \longmapsto (n_6 + n_8, r_6 n_6 + r_8 n_8)$$

is invertible on its image whenever the inclusion adds at least one event. For the constructed inclusion, its determinant is 30.

Proof. The matrix of the map is $\begin{pmatrix} 1 & 1 \\ r_6 & r_8 \end{pmatrix}$. Its determinant is $r_8 - r_6 = |\mathcal{F}_8 \setminus \mathcal{F}_6|$. The inclusion under consideration adds $2\binom{6}{2} = 30$ events. Linear inversion recovers both occupations and preserves the integer constraints of its image. $\square$

The result recovers aggregate occupations. Information about each link and the order of events remains in the full state and in its additional readouts. This hierarchy gives a precise formulation of the distinction between area value and boundary genealogy: area preserves a sum; the pair of observables preserves the decomposition of that sum by sectors; the register additionally preserves its distribution and history.

The parameter γ_{HMT} transports this incidence to the areal scale. Its structural interpretation is that of an oriented coefficient of the hexadic–octadic return functional. The gravitational realization subsequently recognizes it in the Holst action. Its intrinsic definition, value and role in the physical equation are thus distinguished.

17 Action ellipse and modular reciprocity

With the outputs $A, C^*, \hbar > 0$ fixed in the chart $|C^*/A| < 1$, we define

$$\eta_{\text{el}} = \text{artanh}(C^*/A), \quad a_S = \sqrt{2\hbar} e^{\eta_{\text{el}}/2}, \quad b_S = \sqrt{2\hbar} e^{-\eta_{\text{el}}/2}.$$

The balanced canonical coordinates have units of the square root of action; position and momentum are not added here without a dimensional chart. They satisfy

$$a_S b_S = 2\hbar, \quad \tau_{\text{el}} = i \frac{b_S}{a_S}, \quad R_{\text{el}} = \sqrt{\frac{b_S}{a_S}} = \left(\frac{A - C^*}{A + C^*} \right)^{1/4}.$$

The positive root determines the composition $\tau_{\text{el}} = iR_{\text{el}}^2$ and allows the angular ratio to be recovered:

$$\frac{C^*}{A} = \frac{1 - R_{\text{el}}^4}{1 + R_{\text{el}}^4}.$$

Shape inversion sends R_{el} to R_{el}^{-1} and τ_{el} to $-1/\tau_{\text{el}}$ (see 17.1).

Proposition 17.1 (Coincident energy, distinguishable action). *Let*

$$D(\eta) = \begin{pmatrix} e^\eta & 0 \\ 0 & e^{-\eta} \end{pmatrix}, \quad S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad J_2 = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}.$$

Both maps satisfy $T^\top D(-\eta)T = D(\eta)$. However, for $\omega = dQ \wedge dP$,

$$S_2^* \omega = -\omega, \quad J_2^* \omega = \omega.$$

If $\mathcal{I}(\gamma) = \oint_\gamma P dQ$ along a closed curve,

$$\mathcal{I}(S_2 \gamma) = -\mathcal{I}(\gamma), \quad \mathcal{I}(J_2 \gamma) = \mathcal{I}(\gamma).$$

Proof. Conjugation interchanges the diagonal entries of D . The determinants of S_2, J_2 are $-1, 1$, respectively, which proves the transformation of ω . For $\lambda = P dQ$,

$$S_2^* \lambda = d(PQ) - \lambda, \quad J_2^* \lambda = \lambda - d(PQ).$$

Integrating the exact differential along the closed curve completes the proof. $\square$

The quadratic energy and the modulus do not suffice to distinguish the two transports. The orientation of the action does distinguish them, even in the circular form. Geometric marking is thus part of the recoverable readout, not an embellishment added to the value.

17.1 Semiaxes, foci, and interior radii

The figure is obtained from the angular and action outputs of the preceding sections. APP preserves the sheets and their residues and quotients; TRIT fixes the orientation; TPK transports boundary, carry, and memory to the representations of the same enriched state and the joint discrete structure of the continuum. Here the pair (A, C^*) and the positive section $\hbar$ are received after that construction. The geometry makes explicit what information their readouts preserve, without using an observed length to select the source state (see 17.1).

Let $\xi = C^*/A \in (-1, 1)$, with $A \neq 0$ and both angles in the same unit. Write $\eta = \text{artanh } \xi$ and

$$a_S = \sqrt{2\hbar} e^{\eta/2}, \quad b_S = \sqrt{2\hbar} e^{-\eta/2}, \quad \gamma(\theta) = (a_S \cos \theta, b_S \sin \theta).$$

The semiaxes are labelled by their coordinates: if $\eta < 0$, the vertical axis is the major one. Both belong to the square-root-of-action line of the balanced chart, not to the line of spatial lengths.

Proposition 17.2 (Focal geometry and radial interior). *Let $a = \max(a_S, b_S)$, $b = \min(a_S, b_S)$. The focal half-distance and eccentricity are*

$$f_S = \sqrt{a^2 - b^2} = \sqrt{4\hbar \sinh |\eta|}, \quad e_f = \frac{f_S}{a} = \sqrt{1 - e^{-2|\eta|}}. \quad (17.1)$$

For $\eta > 0$, the foci are $(\pm f_S, 0)$; for $\eta < 0$, they are $(0, \pm f_S)$. At $\eta = 0$ they coincide with the centre. The boundary radius in polar direction ϕ is

$$\rho_\partial(\phi) = \left(\frac{\cos^2 \phi}{a_S^2} + \frac{\sin^2 \phi}{b_S^2} \right)^{-1/2}. \quad (17.2)$$

The interior segment in that direction consists of $r(\cos \phi, \sin \phi)$, with $0 \leq r < \rho_\partial(\phi)$. In particular, $b \leq \rho_\partial(\phi) \leq a$.

Proof. The squares of the major and minor semiaxes are $2\hbar e^{|\eta|}$ and $2\hbar e^{-|\eta|}$. Their difference gives (17.1). To verify the focus, suppose $\eta > 0$. At $X = \gamma(\theta)$, the distances to $F_+ = (f_S, 0)$ and $F_- = (-f_S, 0)$ satisfy

$$|X - F_+|^2 = (a_S - f_S \cos \theta)^2, \quad |X - F_-|^2 = (a_S + f_S \cos \theta)^2.$$

Since $a_S > f_S$, their positive roots sum to $2a_S$. For $\eta < 0$, interchanging the coordinates gives the same proof. In the circular case, $f_S = 0$. Finally, substituting $(Q, P) = r(\cos \phi, \sin \phi)$ into $Q^2/a_S^2 + P^2/b_S^2 = 1$ gives (17.2). The coefficient of r^2 lies between a^{-2} and b^{-2} , proving the bounds and the characterization of the interior. $\square$

17.2 Parametric phase, polar angle, and action area

Proposition 17.3 (Areal law and two angular coordinates). *The phase θ of γ and its continuous polar angle ϕ , lifted from zero, satisfy*

$$\phi(\theta) = \arg(a_S \cos \theta + ib_S \sin \theta), \quad \frac{d\phi}{d\theta} = \frac{a_S b_S}{a_S^2 \cos^2 \theta + b_S^2 \sin^2 \theta} > 0. \quad (17.3)$$

In a chart where both tangents are defined, $\tan \phi = (b_S/a_S) \tan \theta$. The swept oriented area is

$$\mathcal{A}(\theta) = \frac{1}{2} \int_0^\theta (QP' - PQ') dt = \hbar \theta, \quad \mathcal{A}(1) = \hbar, \quad \mathcal{A}(2\pi) = h. \quad (17.4)$$

Proof. The derivative of the argument of a nonzero vector is $(QP' - PQ')/(Q^2 + P^2)$. Here the numerator is $a_S b_S = 2\hbar$, proving the derivative and monotonicity. The ratio P/Q gives the tangent relation; the continuous lift fixes the quadrants and the number of turns. The areal integral equals $a_S b_S \theta/2 = \hbar \theta$, and $h = 2\pi \hbar$ gives the final identity. $\square$

The parametric phase does not generally coincide with the polar angle. One radian of phase sweeps out $\hbar$, but is not thereby identified with a circular arc or a sector of unit polar angle. In polar coordinates, the same law reads $d\mathcal{A} = \rho_\partial(\phi)^2 d\phi/2$. With the orientation of γ used here, $\oint P dQ = -h$: the positive area is h , while the integral of that one-form preserves its sign. This agrees with the distinction between the symplectic and antisymplectic transports of the preceding section.

For the control case in the figure, $\xi = 1/3$, the normalized coordinates are exactly

$$\eta = \frac{1}{2} \log 2, \quad \frac{a_S}{\sqrt{2\hbar}} = 2^{1/4} = 1.1892071150 \dots, \quad \frac{b_S}{\sqrt{2\hbar}} = \frac{f_S}{\sqrt{2\hbar}} = R_{\text{el}} = 2^{-1/4} = 0.8408964153 \dots$$

These are regression numbers for the drawing, not values assigned to the angular generator. The local certificate also checks the circular form and the orientations $\xi = \pm 1/3, \pm 3/5$.

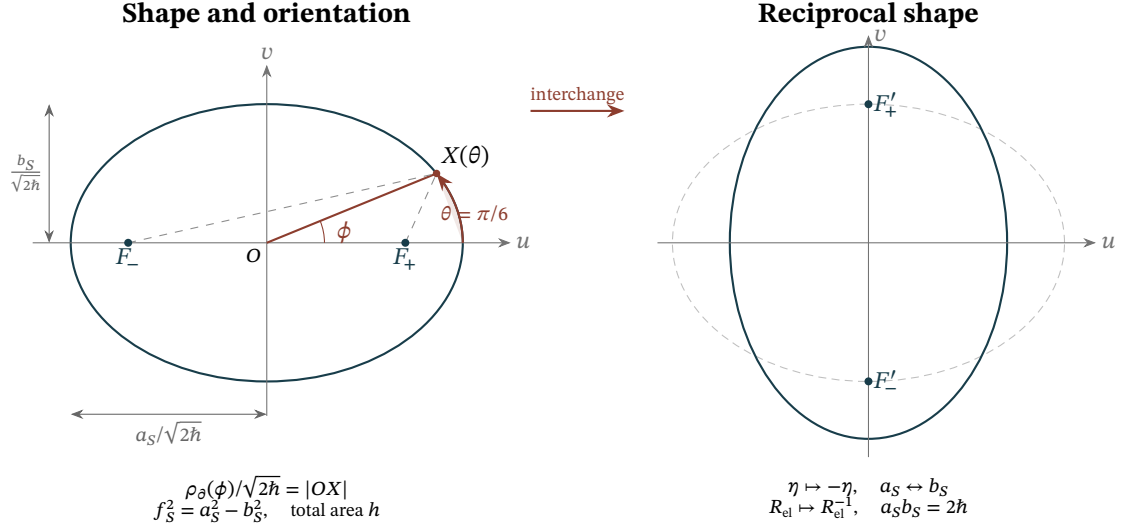


Figure 11: Semiaxes, foci, and reciprocity of the action ellipse in the normalized coordinates $u = Q/\sqrt{2\hbar}$, $v = P/\sqrt{2\hbar}$. The marked sector has parametric phase $\theta = \pi/6$, distinct from its polar angle ϕ ; its action area is $\hbar\pi/6$. The figure uses the algebraic case $\xi = 1/3$ to make the relationships legible: it does not assign that value to the HMT angular output. The foci are calculated from the semiaxes. On the right, interchanging the axes preserves the total area h and inverts the dimensionless modular radius.

17.3 Shape reciprocity and interior distance

The rectangular representation recovers

$$R_{\text{el}} = \sqrt{b_S/a_S} = e^{-\eta/2}, \quad R_{\text{el}}^4 = \frac{1-\xi}{1+\xi}, \quad \xi = \frac{1-R_{\text{el}}^4}{1+R_{\text{el}}^4}. \quad (17.5)$$

The proof consists in substituting $2\eta = \log((1+\xi)/(1-\xi))$. Interchanging the axes sends η to $-\eta$, R_{el} to R_{el}^{-1} , and $\tau = iR_{\text{el}}^2$ to $-1/\tau$. The product $a_S b_S$ is preserved. The radius R_{el} is dimensionless; the polar radius ρ_∂ has units of square root of action.

The interior also admits the Hilbert projective distance. If z_- , x , y , z_+ are ordered along a chord of the ellipse, define

$$d_H(x, y) = \frac{1}{2} \log \frac{|y - z_-| |x - z_+|}{|x - z_-| |y - z_+|}.$$

This distance is dimensionless: units cancel in the cross-ratio. The normalization $(Q, P) \mapsto (Q/a_S, P/b_S)$ takes the ellipse to the unit disk and preserves that ratio, since lengths on the same line are multiplied by a common factor. It thus transports the metric to the Beltrami–Klein model. Neither this interior distance nor the distance between the foci is identified with an atomic length.

17.4 Bohr radius: a consequence of the action scale

In the electrodynamic realization, the Bohr scale is expressed as $a_B = \hbar/(m_e c \alpha)$ [10]. In the chain of the present article, that readout follows the constructions of action, vacuum, fine structure, and electronic mass. Here $\hbar, m_e, c, \alpha > 0$ are received in the same dimensional realization.

Proposition 17.4 (Areal expression for the Bohr scale). *In that realization,*

$$a_B = \frac{a_S b_S}{2m_e c \alpha} = \frac{\mathcal{A}(2\pi)}{2\pi m_e c \alpha} = \frac{\bar{\lambda}_C(m_e)}{\alpha}, \quad \bar{\lambda}_C(m_e) = \frac{\hbar}{m_e c}. \quad (17.6)$$

Proof. Substituting the identities $a_S b_S = 2\hbar$ and $\mathcal{A}(2\pi) = 2\pi\hbar$ into the Bohr expression gives the first two members. The definition of $\bar{\lambda}_C(m_e)$ gives the third. $\square$

The numerator has the dimension of action and $m_e c$ that of momentum; the quotient is a length. This composition uses the product of the semiaxes and the other three quantities: it does not assert that the Bohr radius is a semiaxis, a focus, a Hilbert distance, or the modular radius. Nor does it equate the electronic mass with the circular quantum of order 108. The notation a_B avoids confusing this length with the normalization a_0 of the area operator.

17.5 Inductive–capacitive realization and constitutive continuity

The electrodynamic bridge of the ellipse preserves an additional chart: a clock $\omega > 0$ and a reference impedance $Z_* > 0$. The construction recovered from the corpus defines

$$L_\eta = \frac{Z_*}{\omega} e^{-\eta}, \quad C_\eta = \frac{1}{Z_* \omega} e^\eta, \quad Z_\eta = \sqrt{L_\eta / C_\eta}. \quad (17.7)$$

Here C_η is capacitance, distinct from the counterangle C^* .

Proposition 17.5 (Dynamic product and constitutive ratio). *In the preceding chart,*

$$L_\eta C_\eta = \omega^{-2}, \quad \frac{Z_\eta}{Z_*} = e^{-\eta} = \frac{b_S}{a_S} = R_{\text{el}}^2. \quad (17.8)$$

With $Q = \sqrt{Z_*} q$, $P = \Phi / \sqrt{Z_*}$, the energy $H_{LC} = q^2 / (2C_\eta) + \Phi^2 / (2L_\eta)$ takes the form

$$H_{LC} = \frac{\omega}{2} (e^{-\eta} Q^2 + e^\eta P^2).$$

On $\gamma(\theta)$, its electric and magnetic components are $U_E = \hbar \omega \cos^2 \theta$ and $U_B = \hbar \omega \sin^2 \theta$; their sum is $\hbar \omega$.

Proof. The product in (17.7) cancels the exponents and Z_* ; its ratio is $Z_*^2 e^{-2\eta}$. Taking the positive root and using (17.5) gives (17.8). Substituting $q = Q / \sqrt{Z_*}$ and $\Phi = \sqrt{Z_*} P$ into H_{LC} gives its balanced form. Finally, the semiaxes of γ cancel the factors $e^{\mp\eta}$, and $\cos^2 \theta + \sin^2 \theta = 1$. $\square$

This bridge distinguishes shape, clock, and transducer. The choice of dynamical direction must agree with the symplectic convention: for the equations $\dot{Q} = \partial_P H$, $\dot{P} = -\partial_Q H$, the preceding parametrization satisfies $\dot{\theta} = -\omega$. The energy identities do not depend on reversing that traversal.

The subsequent article on the vacuum will extend the channels already produced, $q_\pm = e^{-(A_{\text{rad}} \pm C_{\text{rad}}^*)}$, and the 90/120 incidence through its resolvent readout. Their responses $r_\pm = (1 - q_\pm^{90}) / (1 - q_\pm^{120})$ express $\hat{Z} = r_- / r_+$ and $\hat{c} = (r_+ r_-)^{-1}$. These are spectral responses, not geometric radii. Identity (17.8) fixes the LC realization here; identifying it with the constitutive impedance of the vacuum preserves the map between the two readouts and their charts. Their values are not equated merely because they share the angular antecedent.

17.6 Comparison of the reduced Planck constant

The International System in force since 20 May 2019 fixes $h = 6.62607015 \times 10^{-34}$ J s exactly [14]. Its reduced constant is therefore

$$\hbar_{\text{SI}} = \frac{6.62607015}{2\pi} 10^{-34} \text{ J s} = 1.0545718176461563912 \dots \times 10^{-34} \text{ J s}.$$

The ellipsis denotes the expansion of an exact expression, not an experimental uncertainty in h in the current SI. This reference value is used here after evaluation of the HMT readouts; its role is comparative.

The two return coordinates in this article are

$$H_5 = 1.0545718176461544696 \dots, \\ \eta_{\text{ret}} = 1.0545718169150390611 \dots$$

Under the identification of $\mathcal{U}_S$ with J_s , the relative difference of the earlier section with respect to $\hbar_{SI}$ is approximately -1.82×10^{-15} . That of the later section is approximately -6.93×10^{-10} . The second difference includes the effect of the return used by the electronic corrector; it is not equivalent to the evaluation error of the first section.

The precision of these comparisons establishes the quantitative proximity of the construction of H_5 while distinguishing proximity from exact identity. In the present manuscript, the displayed expressions do not provide an exact equality with $\hbar/(2\pi)$. The result proved for Σ_H is its decade -34 ; the analytic result for H_5 is the evaluation of the stated character. The physical identification is examined in this chart and with the explicit differences just given. The SI value enters neither the definition of the determinantal norm nor the regional recurrence.

18 Discrete polarization and the tritic mode of arithmetic projections

Refinement and comparison between the APP sheets provide two complementary observations. The first follows the accumulated discrepancy between the number of transitions and the capacity acquired; the second preserves the mode of the tritic selector in the pairing of the additive and multiplicative fibres. Discrete polarization and the electronic prefactor are obtained from these respective operations. Their articulation with the vacuum requires preserving both the temporal information and the spaces on which the operators act.

18.1 Refinement discrepancy and endpoint convention

The cardinalities of the ternary block and its decimal representation fix

$$\rho = \log_{1000} 729 = \log_{10} 9, \quad \delta = 1 - \rho = \log_{10}(10/9).$$

We have $0 < \delta < 1$. Moreover, δ is irrational: an equality $\delta = p/q$, with $q > 0$, would imply $10^{q-p} = 9^q$, which is incompatible with prime factorization. This mismatch already enters the encoding of refinement. Here its centred readout is made explicit, preserving the phase origin and the endpoint convention.

Definition 18.1 (Lower and upper encodings). For $\theta \in \mathbb{R}$ and $n \in \mathbb{Z}$, define

$$z_n^\downarrow(\theta) = [(n+1)\delta + \theta] - [n\delta + \theta], \quad (18.1)$$

$$z_n^\uparrow(\theta) = \lceil (n+1)\delta + \theta \rceil - \lceil n\delta + \theta \rceil. \quad (18.2)$$

Both take values in $\{0, 1\}$. The arrow distinguishes only the endpoint convention for the intervals; tritic orientation retains its earlier definition.

The lower encoding coincides with the one used in the development of refinement. The upper encoding recovers the formulation of polarization using the ceiling function. The two expressions are related by an exactly determined boundary term.

Proposition 18.2 (Change of convention and boundary term). Let $b_n(\theta) = \mathbf{1}_{\mathbb{Z}}(n\delta + \theta)$. Then

$$z_n^\uparrow(\theta) - z_n^\downarrow(\theta) = b_n(\theta) - b_{n+1}(\theta). \quad (18.3)$$

In particular, for $N \geq 0$,

$$\sum_{n=0}^{N-1} (z_n^\uparrow - z_n^\downarrow) = b_0 - b_N.$$

Proof. For every real x , $\lceil x \rceil = \lfloor x \rfloor + 1 - \mathbf{1}_{\mathbb{Z}}(x)$. Applying this identity to both endpoints of each increment gives (18.3); its sum telescopes. The irrationality of δ implies that, for fixed θ , at most one integer index satisfies $n\delta + \theta \in \mathbb{Z}$. $\square$

With $\theta = 0$, the first upper symbol is 1 and the lower one is 0; for $n \geq 1$ they coincide. The initial term belongs to the register and is preserved when the convention is changed. The densities and discrepancy bounds are transported with that term, without modifying the TPK update or restarting the history at each phase return.

18.2 Bounded polarization and exact temporal balance

Definition 18.3 (Centred discrete polarization). The polarization of the upper encoding, relative to the density δ , is

$$P_N^\uparrow(\theta) = \sum_{n=0}^{N-1} (z_n^\uparrow(\theta) - \delta), \quad P_0^\uparrow = 0. \quad (18.4)$$

Its temporal increment is

$$J_N^\uparrow(\theta) = P_{N+1}^\uparrow(\theta) - P_N^\uparrow(\theta) = z_N^\uparrow(\theta) - \delta. \quad (18.5)$$

Centring separates uniform growth of density δ from phase information. Thus, $P_N^\uparrow$ measures an accumulated imbalance, not the total number of vacancies. The increment $J_N^\uparrow$ preserves the local pulse and the background relative to which it is evaluated.

Theorem 18.4 (Uniform bound and discrete temporal continuity). *For $N \geq 0$ and every phase θ ,*

$$P_N^\uparrow(\theta) = [N\delta + \theta] - [\theta] - N\delta, \quad |P_N^\uparrow(\theta)| < 1. \quad (18.6)$$

For any integers $0 \leq a < b$,

$$\sum_{n=a}^{b-1} J_n^\uparrow(\theta) = P_b^\uparrow(\theta) - P_a^\uparrow(\theta), \quad \left| \sum_{n=a}^{b-1} J_n^\uparrow(\theta) \right| < 1. \quad (18.7)$$

The density of the vacancy sequence is δ .

Proof. Summing the ceiling increments in (18.2) yields the first identity. Writing $r(x) = [x] - x \in [0, 1)$ gives $P_N^\uparrow = r(N\delta + \theta) - r(\theta)$; its absolute value is less than one. On the interval $[a, b)$, one similarly obtains $r(b\delta + \theta) - r(a\delta + \theta)$, proving the second bound, which is sharper than the sum of two individual bounds. Finally, $N^{-1} \sum_{n=0}^{N-1} z_n^\uparrow = \delta + N^{-1} P_N^\uparrow \rightarrow \delta$. $\square$

The lower encoding admits the same proofs with $r(x)$ replaced by $[x] - x \in (-1, 0]$. In particular, $P_N^\uparrow - P_N^\downarrow = b_0 - b_N$. The bounded discrepancy previously used in the Dirichlet readouts and the present polarization therefore preserve an explicit relationship of origin and endpoints. Nor does aperiodicity introduce secular drift in this imbalance: the history can extend indefinitely while the centred discrepancy remains uniformly bounded.

18.3 Realization on the charge and time lines

To realize the balance dimensionally, specify a charge line $\mathcal{L}_Q$ with a nonzero section u_Q and an increasing sequence of times t_N on an oriented time line. With $\Delta t_N = t_{N+1} - t_N > 0$, the centred charge and the mean current of each interval are defined by

$$Q_N = P_N^\uparrow u_Q, \quad I_N = \frac{Q_{N+1} - Q_N}{\Delta t_N} = \frac{u_Q}{\Delta t_N} (z_N^\uparrow - \delta). \quad (18.8)$$

Thus, $Q_N \in \mathcal{L}_Q$ and $I_N \in \mathcal{L}_Q \otimes \mathcal{L}_T^{-1}$. The section u_Q and the time intervals are data of this realization; the preceding arithmetic construction has already determined $P_N^\uparrow$ and $J_N^\uparrow$.

Proposition 18.5 (Preservation of balance under dimensional realization). *For $a < b$,*

$$Q_b - Q_a = \sum_{n=a}^{b-1} \Delta t_n I_n, \quad \left| \frac{Q_b - Q_a}{u_Q} \right| < 1. \quad (18.9)$$

If all intervals have length t_0 , the two possible currents are $-\delta u_Q/t_0$ and $(1 - \delta)u_Q/t_0$.

Proof. Multiplying each identity in (18.8) by Δt_n and summing telescopes the charges. The bound is (18.7); the two local values follow from $z_n^\uparrow \in \{0, 1\}$. $\square$

The continuity proved is temporal and refers to the centred charge. A spatial realization also incorporates the localization of charges, incidence between cells, and identification of boundary fluxes. In a constitutive description by channels, the coupling is specified through an excitation map $\iota : \mathcal{L}_Q \rightarrow \mathcal{H}_{\text{ch}} \otimes \mathcal{L}_Q$ and an observation map $\ell : \mathcal{H}_{\text{ch}} \otimes \mathcal{L}_Q \rightarrow \mathcal{L}_Q$, where $\mathcal{H}_{\text{ch}}$ is a real realization of the channel space. If both are linear and independent of the time index, the response of a fixed operator $\mathcal{V}$ is transported by

$$Q_N^{\text{resp}} = \ell(\mathcal{V} \otimes I_{\mathcal{L}_Q})\iota(Q_N), \quad I_N^{\text{resp}} = \frac{Q_{N+1}^{\text{resp}} - Q_N^{\text{resp}}}{\Delta t_N}.$$

Linearity preserves the telescoping balance. This composition makes explicit the data of a constitutive identification: the scalar sequence determines the pulse, while ι , $\mathcal{V}$, and ℓ determine its excitation, transport, and observation through channels. Identities (18.4)–(18.9) establish the first construction independently of that choice of realization. The vacuum operator is constructed in Section 20 on its own channels and projectors.

18.4 Pairing the additive and multiplicative fibres

The second observation comes from the three-dimensional realization of APP, $\mathcal{A}_3 = D_9^3$, with uniform measure and $D_9 = \{1, \dots, 9\}$. The observables preserved are

$$\Sigma(x) = \rho_9(x_1 + x_2 + x_3), \quad \Pi(x) = \rho_9(x_1 x_2 x_3),$$

where $\rho_9(n) = 1 + [(n-1) \bmod 9]$ for $n > 0$. In $\ell^2(\mathcal{A}_3)$, let $R_{\Sigma,3}$ and $R_{\Pi,3}$ be the conditional expectations on their fibres. For example,

$$(R_{\Sigma,3}f)(x) = \frac{1}{|\Sigma^{-1}(\Sigma(x))|} \sum_{y: \Sigma(y)=\Sigma(x)} f(y).$$

They are orthogonal projections defined on the same support. The incidence matrix $N_{ab} = |\Sigma^{-1}(a) \cap \Pi^{-1}(b)|$ results from enumerating the 729 triples, adding and multiplying their coordinates before applying ρ_9 :

$$N = 9 \begin{pmatrix} 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 1 & 0 & 2 & 0 & 1 & 2 & 0 & 0 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 0 & 2 & 1 & 0 & 2 & 0 & 1 & 3 \\ 0 & 1 & 1 & 0 & 1 & 1 & 0 & 1 & 4 \\ 1 & 0 & 1 & 1 & 0 & 1 & 1 & 0 & 4 \\ 0 & 1 & 2 & 0 & 0 & 2 & 1 & 0 & 3 \end{pmatrix}. \quad (18.10)$$

The rows sum to 81, and the column sums are

$$(m_1, \dots, m_9) = (36, 36, 108, 36, 36, 108, 36, 36, 297).$$

Therefore, the pairing of the normalized indicator functions of the fibres is $C_{ab} = N_{ab}/\sqrt{81m_b}$. Normalization incorporates the multiplicities produced by APP, not a choice of singular values.

Proposition 18.6 (Singular mode of the tritic selector). *In the canonical chart, the selector vector is $m_{\text{ph}} = (1, -1, 0)^3$. We have*

$$Cm_{\text{ph}} = C^T m_{\text{ph}} = -\frac{1}{2}m_{\text{ph}}. \quad (18.11)$$

Thus, the corresponding singular value is $s = 1/2$.

Proof. In the six columns where m_{ph} is nonzero, $m_b = 36$ and $C_{ab} = N_{ab}/54$. Multiplying the rows of (18.10) by $(1, -1, 0)^3$ yields $-m_{\text{ph}}/2$. For the transposed product, the signed sums of columns 3, 6, 9 are zero; the other six give the same values $-m_{\text{ph},b}/2$. This proves both identities and, by composition, $CC^T m_{\text{ph}} = m_{\text{ph}}/4$. $\square$

18.5 Commutator norm and electronic prefactor

Theorem 18.7 (Exact norm of arithmetic incompatibility). *The two projections satisfy*

$$\| [R_{\Sigma,3}, R_{\Pi,3}] \| = \frac{\sqrt{3}}{4}. \quad (18.12)$$

The maximum is attained in the principal plane corresponding to the tritic mode of (18.11).

Proof. The rational matrix $M = CC^\top$ has entries $M_{ac} = \sum_b N_{ab}N_{cb}/(81m_b)$. Its complete spectrum is verified through the following independent subspaces: the vectors $(1, \dots, 1)$, $(1, -1, 0)^3$, and $(1, 1, -2)^3$ have eigenvalues 1, $1/4$, and $7/132$, respectively; the subspace of vectors supported on $\{3, 6, 9\}$ whose sum is zero has dimension 2 and eigenvalue $1/18$; the zero-sum vectors supported on $\{1, 4, 7\}$ or $\{2, 5, 8\}$ form a kernel of dimension 4. Substitution into the matrix verifies these actions, and the dimensions sum to 9. Consequently,

$$\text{spec } M = \{1, 1/4, 7/132, 1/18, 1/18, 0, 0, 0, 0\}.$$

For two orthogonal projections, a singular value $s \in (0, 1)$ of the pairing determines a principal plane in which their matrices are

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad Q = \begin{pmatrix} s^2 & s\sqrt{1-s^2} \\ s\sqrt{1-s^2} & 1-s^2 \end{pmatrix}.$$

This form is obtained by taking a unit vector u in the first image with $PQu = s^2u$ and completing it with $(Qu - s^2u)/(s\sqrt{1-s^2})$. Their commutator is $s\sqrt{1-s^2} \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$. The sectors with singular value 0 or 1 contribute zero. Therefore, the total norm is the maximum of $\sqrt{\lambda(1-\lambda)}$ over the spectrum of M . That maximum is $\sqrt{3/16}$, attained at $\lambda = 1/4$. $\square$

The norm is the prefactor entering the electronic composition. Its derivation preserves the three-dimensional support, the fibres, and the selector mode; scalar evaluation does not replace that structure. In particular, knowing $\sqrt{3}/4$ gives the maximum magnitude of the incompatibility, while the projectors preserve the directions in which it acts.

18.6 Quadratic defect and response of the 90/120 sectors

The tritic mode simultaneously determines

$$1 - s^2 = \frac{3}{4} = \frac{90}{120}, \quad s = \frac{1}{2}. \quad (18.13)$$

The first expression is the quadratic defect of the APP principal plane. The last preserves the lengths of the two calendar sectors. The constitutive response uses those sectors on another object: a contractive channel $q \in (0, 1)$. Its evaluation is

$$r(q) = \frac{1 - q^{90}}{1 - q^{120}}.$$

The relationship between the sector ratio and this response can be made precise without identifying their operators.

Proposition 18.8 (Limit of the response ratio). *For $0 < q < 1$,*

$$\frac{3}{4} < r(q) < 1, \quad \lim_{q \uparrow 1} r(q) = \frac{3}{4} = 1 - s^2. \quad (18.14)$$

Proof. With $u = q^{30} \in (0, 1)$, factoring $1 - u^3$ and $1 - u^4$ gives $r(q) = (1 + u + u^2)/(1 + u + u^2 + u^3)$. The limit at $u = 1$ is $3/4$. Moreover, $4(1 + u + u^2) - 3(1 + u + u^2 + u^3) = (1 - u)(3u^2 + 2u + 1) > 0$, and the denominator exceeds the numerator by $u^3 > 0$. This yields both inequalities. $\square$

The exact coincidence concerns the defect of the singular mode and the limit of the sector response. In the contractive interior, the response depends on q and is strictly greater than $3/4$. The constitutive development preserves its two channels $q_{\pm}$ and its projectors $P_{\pm}$; the projectors $R_{\Sigma,3}$ and $R_{\Pi,3}$, for their part, preserve the arithmetic fibres that determined the prefactor. An operator equivalence between the two realizations requires a map between their spaces that intertwines the corresponding actions. Equations (18.11), (18.12), and (18.14) specify the shared information and keep the data of each realization visible.

19 Sheet return, areal–volumetric incidence and constitutive orientation

The sheet structure enters before electromagnetic quantities are evaluated. Its function is to distinguish closure of a projection, restoration of an orientation and evolution of the transported register. These operations act on related objects of different types: a projected trajectory may have closed while its lift remains on the opposite sheet; orientation may have been restored while memory preserves the two windings performed. The constitutive construction must receive this information together with the structure determining it.

19.1 Projected return and restoration of the oriented cover

Let ℓ be a return loop of a surviving extension, and let $\varepsilon : \langle \ell \rangle \rightarrow \{+1, -1\}$ be the character of its lift, with $\varepsilon(\ell) = -1$. The associated interval bundle has the presentation

$$\mathcal{M} = ([0, 1] \times [-1, 1]) / (0, u) \sim (1, -u). \quad (19.1)$$

The decisive datum is the sign of transport around the specified loop. The identification of endpoints geometrically realizes that character.

Proposition 19.1 (Two closure orders). *One winding of the loop closes the base and reverses the fibre orientation. Two windings restore orientation. In the angular parametrization of the boundary readout, these returns correspond, respectively, to the areal closure 2π and the volumetric closure 4π of the oriented cover.*

Proof. The equivalence relation of (19.1) identifies the terminal fibre with the initial one by $u \mapsto -u$. Composing two transports gives $u \mapsto -(-u) = u$. Parametrizing one winding of the base by 2π assigns 4π to its square. The statement concerns this lift and this readout. $\square$

TPK memory also preserves the trajectory traversed and the register updates. The square of the sign character is the identity, but it does not impose vanishing of these updates. The spinorial representation, when used, must transport this datum through its own map.

19.2 Descent of incidence from the tritic calendar

In the primitive block $(+1, -1, 0)$, the TPK updates the arithmetic mode through $+1 : m \mapsto \Sigma$, $-1 : m \mapsto \Pi$ and $0 : m \mapsto m$. Its cyclic orbit is (Σ, Π, Π) . The sheet readout assigns Σ to the areal sheet and Π to the volumetric sheet. In the basis order $(e_{\text{ar}}, e_{\text{vol}})$, edge incidence is represented by

$$F_{\text{av}} = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}. \quad (19.2)$$

The three consecutive pairs are $\Sigma \rightarrow \Pi$, $\Pi \rightarrow \Pi$ and $\Pi \rightarrow \Sigma$, each with multiplicity one. The calendars of lengths 9, 27 and 108 contain 3, 9 and 36 copies of the primitive block. Their incidence matrices are therefore these multiples of F_{av} .

This construction preserves the distinction between chronological frequency and the measure of the continuation space. The former assigns weights $(1/3, 2/3)$ to the mode cycle. The latter is defined on the minimal memory-one symbolic envelope admitting exactly the preceding pairs. Its adjacency matrix is F_{av} ; trajectory counts belong to that envelope.

Proposition 19.2 (Return weights of the two sheets). *Let $F_0 = 0$, $F_1 = 1$ and $F_{n+1} = F_n + F_{n-1}$. For $n \geq 1$,*

$$F_{\text{av}}^n = \begin{pmatrix} F_{n-1} & F_n \\ F_n & F_{n+1} \end{pmatrix}. \quad (19.3)$$

The Parry measure of this envelope assigns the sheets weights proportional to $(1, \varphi^2)$, and the ratio of diagonal counts converges to φ^{-2} .

Proof. The power formula follows by induction using the recurrence. The square of F_{av} is strictly positive. Its positive Perron vector is $(1, \varphi)$, with eigenvalue φ , since $\varphi^2 = \varphi + 1$. As the matrix is symmetric, the Parry weights are proportional to the squares of that vector's components. The other root of the characteristic polynomial is $-\varphi^{-1}$; its modulus is strictly less than φ , giving the limit of the diagonal ratios. $\square$

Spectral identification of this incidence follows its construction from the calendar. The regional coordinate φ retains its earlier coinductive genealogy: recognition of the eigenvalue does not select the calendar.

19.3 Boundary marking and areal–volumetric ratio

Let $\Pi_{\text{HMT}}(X_\infty)$ be the value of the regional closure readout already constructed. With the diagonal sheet projectors, define

$$D_\pi = \Pi_{\text{HMT}}(X_\infty)P_{\text{ar}} + P_{\text{vol}}, \quad \Theta_n = \frac{\langle e_{\text{ar}}, D_\pi F_{\text{av}}^n e_{\text{ar}} \rangle}{\langle e_{\text{vol}}, F_{\text{av}}^n e_{\text{vol}} \rangle}. \quad (19.4)$$

Volumetric normalization is unity; regional closure marks the areal return only once. Applying (19.3),

$$\Theta_n = \Pi_{\text{HMT}}(X_\infty) \frac{F_{n-1}}{F_{n+1}} \longrightarrow \frac{\pi}{\varphi^2}. \quad (19.5)$$

The return ratio is determined by incidence, measure and marking. The quotient $2\pi/4\pi$, by contrast, describes the cover's two closure orders. Both relations participate in the sheet readout without replacing each other.

19.4 Transport of orientation to the constitutive channels

In a two-channel representation, let $\mathcal{R}$ be a self-adjoint involution and $\mathcal{S}$ a unitary interchange operator satisfying $\mathcal{S}\mathcal{R}\mathcal{S}^{-1} = -\mathcal{R}$. Write the angular coordinates already constructed as $x = A_{\text{rad}}$ and $y = C_{\text{rad}}^*$, with $x > |y|$. Then

$$B = xI + y\mathcal{R}, \quad T = e^{-B}, \quad \mathcal{S}T(y)\mathcal{S}^{-1} = T(-y). \quad (19.6)$$

Indeed, conjugation transforms $B(y)$ into $B(-y)$ and commutes with its convergent exponential series. The eigenvalues $q_\pm = e^{-x \mp y}$ are interchanged. To fix the response order, the chamber $x > y > 0$ is used, where $0 < q_+ < q_- < 1$. The map $(x, y) \mapsto (x, -y)$ transports it to the opposite chamber $x > -y > 0$, where $0 < q_- < q_+ < 1$. Both belong to the contractive domain $x > |y|$. On the boundary $y = 0$ the two channels coincide. Conjugation interchanges the chambers; it does not preserve the strict ordering of their labels. It acts on the projectors through $\mathcal{S}P_\pm\mathcal{S}^{-1} = P_\mp$, with $P_\pm = (I \pm \mathcal{R})/2$. This concrete map transmits orientation to the constitutive operator developed next.

20 Constitutive operator and reconstruction of the vacuum channels

The constitutive quantities are obtained by an operation on angular transport. The starting information comprises the channels, their projectors, and the orientation distinguishing them. This organization makes it possible to follow separately the construction of the operator, the evaluation of its spectral functions, and its subsequent dimensional realization.

20.1 Positive response on two sheets

In the oriented chamber $x > y > 0$ of Section 19, let $P_{\pm} = (I \pm \mathcal{R})/2$ and $T = q_+P_+ + q_-P_-$, with $0 < q_+ < q_- < 1$. The incidences constructed by the calendar fix the sectors 90 and 120. The constitutive response is

$$\mathcal{V} = (I - T^{90})(I - T^{120})^{-1}. \quad (20.1)$$

This definition receives the transport arising from the TPK and preserves the sheet assignments. The response rule and its domain are an explicit part of the construction.

The quotient arises from comparing the two calendar incidences on the same transport, not from fitting four quantities separately. If $S = T^{30}$ and $\Sigma_j(S) = I + S + \dots + S^{j-1}$, factoring $I - S^j$ gives

$$\mathcal{V}\Sigma_4(S) = \Sigma_3(S), \quad \mathcal{V} = \Sigma_3(S)\Sigma_4(S)^{-1}. \quad (20.2)$$

The factors Σ_3 and Σ_4 express, respectively, the sectors $90 = 3 \cdot 30$ and $120 = 4 \cdot 30$. Since the spectrum of S is contained in $(0, 1)$, $\Sigma_4(S)$ is invertible: the completion equation determines a unique response for the fixed transport. This uniqueness concerns the declared constitutive rule and its incidences. Electromagnetic identification adds the oriented assignment of sheets and the dimensional lines specified below.

Proposition 20.1 (Decomposition and positivity). *The response is well-defined and*

$$\mathcal{V} = r_+P_+ + r_-P_-, \quad r_{\pm} = \frac{1 - q_{\pm}^{90}}{1 - q_{\pm}^{120}}. \quad (20.3)$$

Its eigenvalues satisfy $3/4 < r_- < r_+ < 1$ in this chamber.

Proof. Each coefficient $1 - q_{\pm}^{120}$ is positive. The inverse of $I - T^{120}$ is the sum of the projectors divided by those coefficients, proving (20.3). With $s = q^{30}$, one obtains

$$f(s) = \frac{1 + s + s^2}{1 + s + s^2 + s^3}, \quad f'(s) = -\frac{s^2(3 + 2s + s^2)}{(1 + s + s^2 + s^3)^2} < 0. \quad (20.4)$$

The limits at 0 and 1 are 1 and 3/4, respectively. Monotonicity and the channel ordering complete the proof. $\square$

20.2 Four coordinates of a common response

Define the normalized evaluations

$$\hat{\varepsilon} = r_+^2, \quad \hat{\mu} = r_-^2, \quad \hat{Z} = \frac{r_-}{r_+}, \quad \hat{c} = \frac{1}{r_+r_-}. \quad (20.5)$$

The first pair preserves the quadratic responses assigned to each sheet. The product captures the common mode; the quotient distinguishes their relative orientation. The following hold exactly:

$$\hat{\mu}\hat{\varepsilon} = \hat{c}^{-2}, \quad \frac{\hat{\mu}}{\hat{\varepsilon}} = \hat{Z}^2, \quad r_+ = \frac{1}{\sqrt{\hat{Z}\hat{c}}}, \quad r_- = \sqrt{\frac{\hat{Z}}{\hat{c}}}. \quad (20.6)$$

The positive roots fix the inversion. Under sheet conjugation, read relative to the fixed markings $P_{\pm}$, $\hat{\varepsilon}$ and $\hat{\mu}$ are interchanged, $\hat{Z}$ is inverted, and $\hat{c}$ is preserved. The response is then transported to

the chamber $x > -y > 0$: the formulas remain valid, with the orderings $q_- < q_+$ and $r_+ < r_-$ reversed. Consequently, common mode and orientation are data recoverable through complementary observables.

Proposition 20.2 (Determination of the quadratic evaluations). *Let $W = r_+P_+ + r_-P_-$ be a positive response on the two sheets. With the product and quotient readouts fixed,*

$$\hat{c}^{-1} = \det W = r_+r_-, \quad \hat{Z} = r_-/r_+,$$

there exists a unique positive pair $(\hat{\varepsilon}, \hat{\mu})$ satisfying $\hat{\mu}\hat{\varepsilon} = \hat{c}^{-2}$ and $\hat{\mu}/\hat{\varepsilon} = \hat{Z}^2$. It is precisely (r_+^2, r_-^2) .

Proof. Positivity permits taking roots unambiguously:

$$\hat{\mu} = \frac{\hat{Z}}{\hat{c}} = \frac{r_-}{r_+}r_+r_- = r_-^2, \quad \hat{\varepsilon} = \frac{1}{\hat{Z}\hat{c}} = \frac{r_+}{r_-}r_+r_- = r_+^2.$$

These expressions satisfy both relations and are their only positive roots. $\square$

Thus the squares are not two independent choices added to the channels: they are determined by the multiplicative and oriented readouts once the constitutive relations are fixed. This proposition establishes uniqueness within that system of rules. Interpreting the rules as an electromagnetic response retains its additional physical-representation content.

The determinantal readout of transport and the constitutive response must also preserve their order. For $T = e^{-xI-yR}$, one has $\det T = e^{-2x}$, whereas

$$\det \mathcal{V} = f(q_+^{30})f(q_-^{30}) = \hat{c}^{-1}.$$

Both arise from the same two-sheet operator but evaluate different functions of its spectrum. In general, applying the determinant first and then a rational function is not equivalent to taking the determinant of that function of the operator. Keeping both eigenvalues until the response is completed preserves orientation information that an isolated determinant no longer distinguishes.

20.3 Cubic inversion and angular recovery

Theorem 20.3 (Constitutive inversion in the contractive domain). *For each $r \in (3/4, 1)$, there exists a unique $s \in (0, 1)$ such that $f(s) = r$. This s is the unique root in that interval of*

$$rs^3 + (r-1)(1+s+s^2) = 0. \quad (20.7)$$

The labeled response reconstructs $s_{\pm}$, $q_{\pm} = s_{\pm}^{1/30}$, and

$$x = -\frac{1}{2} \log(q_+q_-), \quad y = \frac{1}{2} \log \frac{q_-}{q_+}. \quad (20.8)$$

Proof. The derivative and limits in (20.4) prove that f is a decreasing bijection between the stated intervals. Multiplying $f(s) = r$ by its denominator produces (20.7). The positive root of order 30 recovers the channel. Finally, adding and subtracting $\log q_{\pm} = -x \mp y$ gives (20.8). $\square$

Reconstruction preserves the sheet labels. The complete operator $\mathcal{V}$ also preserves the projectors, and the inverse function is applied by spectral calculus. Recovering two angular coordinates is an operation defined on this representation; the complete historical TPK trace preserves additional data.

The interval $r \in (3/4, 1)$ and the inverse function of this theorem belong to the internal normalization of (20.5). If coordinates expressed in another unit chart are available, first recover this normalization through the transformation in Section 27; then apply cubic inversion.

20.4 Transported composition of constitutive responses

The angular transport and the incidences 90 and 120 originate in the same APP-TRIT-TPK structure: the sheets retain evaluations and their carries; the tritic regime retains orientation; the TPK calendar transports phase and memory to the angular channels. The constitutive response applies the function f of (20.4) to those channels. Its inversion expresses, in response coordinates, the composition of transports sharing a common frame. The constructive order is thereby preserved: transport, composition and spectral evaluation. The resulting operation acts on this representation of the enriched state; its inverse recovers the channels, while the complete memory history remains in the underlying structure.

Let $s = q^{30}$ and $\psi = f^{-1}$. The exponent 30 is the greatest common divisor of the incidences 90 and 120. Cubic inversion determines $\psi : (3/4, 1) \rightarrow (0, 1)$. Adjoining the boundary response $f(1) = 3/4$ extends the bijection to

$$\psi : D \rightarrow (0, 1], \quad D = [3/4, 1).$$

At this boundary the rational function f is used; the quotient $(1 - q^{90})/(1 - q^{120})$ has a removable singularity there.

Proposition 20.4 (Response monoid and additive coordinate). *The operation*

$$r \odot t = f(\psi(r)\psi(t)), \quad r, t \in D, \quad (20.9)$$

makes D a commutative cancellative monoid with identity $3/4$. The map

$$\ell(r) = -\frac{1}{30} \log \psi(r) \quad (20.10)$$

is an isomorphism from this monoid to $(\mathbb{R}_{\geq 0}, +)$. The interval $(3/4, 1)$ is closed under $\odot$; the identity belongs to the boundary extension. The identity is the only invertible element within D .

Proof. The product of two elements of $(0, 1]$ belongs to the same interval, and

$$\psi(r \odot t) = \psi(r)\psi(t).$$

For three responses, either association has image $\psi(r)\psi(t)\psi(u)$; injectivity of ψ proves associativity. Commutativity follows from scalar multiplication. If $r \odot t = r \odot u$, positivity of $\psi(r)$ permits cancellation, giving $\psi(t) = \psi(u)$ and hence $t = u$. The parameter 1 corresponds to $3/4$ and provides the identity. The logarithm transforms products into sums, and its range on $(0, 1]$ shows that ℓ is a bijection onto $\mathbb{R}_{\geq 0}$. In particular, the response of a channel q satisfies $\ell(r) = -\log q$. If both parameters are strictly less than 1, so is their product. Finally, two numbers in $(0, 1]$ have product 1 only when both are 1. $\square$

Proposition 20.5 (Admissible cancellation and inverse extension). *For $r, u \in D$, the equation $r \odot t = u$ has a solution $t \in D$ exactly when $u \geq r$. This solution is unique and is given by*

$$t = f\left(\frac{\psi(u)}{\psi(r)}\right). \quad (20.11)$$

The same function f is a bijection from $(0, \infty)$ onto $(0, 1)$. With this extension of ψ , the law (20.9) transports the multiplicative group of positive parameters to the interval $(0, 1)$. Group inversion satisfies

$$r^\ominus = \psi(r)r, \quad r \odot r^\ominus = \frac{3}{4}. \quad (20.12)$$

Proof. The equation is equivalent to $\psi(t) = \psi(u)/\psi(r)$. This quotient belongs to $(0, 1]$ exactly when $\psi(u) \leq \psi(r)$, which, by decreasing monotonicity, is equivalent to $u \geq r$. The bijection proves existence and uniqueness. The derivative in (20.4) is negative for every $s > 0$; the limits of f at 0 and infinity are 1 and 0, respectively. This establishes the extended bijection. Multiplying the numerator and denominator by s^3 gives

$$f(s^{-1}) = \frac{s^3 + s^2 + s}{s^3 + s^2 + s + 1} = sf(s).$$

The multiplicative inverse of $s = \psi(r)$ therefore has response $\psi(r)r$. Its composition with r corresponds to the parameter 1. $\square$

The extension to $s > 1$, equivalently to $q > 1$, is the algebraic extension of the reader. The chapter's physical contractive sector retains its declared domain. The coordinate ℓ extends to an isomorphism with $(\mathbb{R}, +)$ and changes sign under group inversion.

20.4.1 Operator realization in a common sheet frame

Proposition 20.6 (Composition with marked projectors). *Let $\mathcal{R} = \mathcal{R}^*$, $\mathcal{R}^2 = I$ and $P_{\pm} = (I \pm \mathcal{R})/2$. For $j = a, b$, let*

$$T_j = q_{j,+}P_+ + q_{j,-}P_-, \quad \mathcal{V}_j = f(T_j^{30}), \quad 0 < q_{j,\pm} < 1.$$

The composition $T_{a \circ b} = T_a T_b$ satisfies

$$\mathcal{V}_{a \circ b} = f(\psi(\mathcal{V}_a)\psi(\mathcal{V}_b)). \quad (20.13)$$

Its two responses are $r_{a,\pm} \odot r_{b,\pm}$. If $T_j = \exp[-(x_j I + y_j \mathcal{R})]$ with $x_j > |y_j|$, then

$$T_a T_b = \exp[-((x_a + x_b)I + (y_a + y_b)\mathcal{R})], \quad (20.14)$$

and $x_a + x_b > |y_a + y_b|$.

Proof. Orthogonality $P_+ P_- = 0$ gives

$$T_a T_b = q_{a,+}q_{b,+}P_+ + q_{a,-}q_{b,-}P_-.$$

In particular, T_a and T_b commute. Functional calculus on these projectors yields $\psi(\mathcal{V}_j) = T_j^{30}$ and $(T_a T_b)^{30} = T_a^{30} T_b^{30}$; applying f proves (20.13). More explicitly, for every function g defined on the two eigenvalues,

$$P_{\pm} g(T_j) = g(q_{j,\pm}) P_{\pm}.$$

Since $q_{j,\pm} = \exp[-(x_j \pm y_j)]$, their products prove (20.14). The triangle inequality concludes $x_a + x_b > |y_a| + |y_b| \geq |y_a + y_b|$. $\square$

The recovered coordinates can be written directly as

$$x = \frac{\ell(r_+) + \ell(r_-)}{2}, \quad y = \frac{\ell(r_+) - \ell(r_-)}{2}.$$

Composition adds these pairs. Simultaneous sheet exchange permutes the two channels of each transport, acts as $(x, y) \mapsto (x, -y)$ and is an automorphism of the componentwise law. Inversion of both transports in the algebraic extension acts as $(x, y) \mapsto (-x, -y)$. The two involutions commute and retain distinct functions: one exchanges orientation markings; the other inverts the transport.

To represent exchange by conjugation, use an operator L satisfying $L\mathcal{R}L^{-1} = -\mathcal{R}$. In the representation

$$\mathcal{R} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad L = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix},$$

direct multiplication gives $LP_{\pm}L^{-1} = P_{\mp}$, and hence $LT_jL^{-1} = q_{j,-}P_+ + q_{j,+}P_-$. Conjugation by $\mathcal{R}$ preserves its own projectors. A trace combining the channels likewise cannot replace the projectors in this functional identity: their labels contribute to the composition.

The common-frame hypothesis is substantive. For example, the positive definite operators

$$A_0 = \begin{pmatrix} 1/2 & 0 \\ 0 & 1/3 \end{pmatrix}, \quad B_0 = \begin{pmatrix} 1/2 & 1/6 \\ 1/6 & 1/2 \end{pmatrix}$$

have positive eigenvalues, but

$$A_0 B_0 = \begin{pmatrix} 1/4 & 1/12 \\ 1/18 & 1/6 \end{pmatrix}$$

is not self-adjoint. The proposition specifies transport composition on common projectors; its domain does not encompass an arbitrary mixture of material media. Realizing a concatenation of HMT routes additionally retains their memory data: algebraic closure of the parameter cone describes the representation and does not by itself select new primary routes.

20.4.2 Channel product and transport composition

The normalized speed of one state remains determined by

$$\hat{c} = (r_+ r_-)^{-1}.$$

Here the ordinary product combines the two responses of that same state. The law $\odot$ evaluates a different operation: composition of two transports on each sheet. The following exact calculation distinguishes their arguments:

$$f(1/2) = \frac{14}{15}, \quad \frac{14}{15} \odot \frac{14}{15} = f(1/4) = \frac{84}{85} \neq \frac{196}{225} = \left(\frac{14}{15}\right)^2. \quad (20.15)$$

The fractions in this example are algebraic test parameters. The expression for $\hat{c}$ and the preceding constitutive identities remain unchanged. The added content is an explicit law transporting composition and recovering its additive angular coordinate.

20.5 Logarithmic coordinates and refinement

Let $\mathcal{E} = \log(r_+ r_-)$ and $\mathcal{B} = \log(r_-/r_+)$. Then

$$\log \mathcal{V} = \frac{1}{2}\mathcal{E}I - \frac{1}{2}\mathcal{B}\mathcal{R}. \quad (20.16)$$

This decomposition displays the invariant and oriented components. In the amplification $\mathcal{V}_m = \mathcal{V} \otimes I_{9m}$, normalized partial expectations satisfy $E_m(\mathcal{V}_{m+1}) = \mathcal{V}_m$. For every polynomial p , $p(\mathcal{V}_m) = p(\mathcal{V}) \otimes I_{9m}$; hence $E_m(p(\mathcal{V}_{m+1})) = p(\mathcal{V}_m)$. Uniform continuity extends the identity to continuous functions on the spectrum. Thus projectors, powers, and the logarithm are preserved in the stated tower.

20.6 Dimensional types of the four evaluations

With the action, charge, and velocity lines, equipped with sections u_S, u_Q, u_C , the dimensional realization reads

$$Z_{\text{vac}} = \hat{Z}u_S u_Q^{-2}, \quad c_{\text{vac}} = \hat{c}u_C, \quad (20.17)$$

$$\mu_{\text{vac}} = \hat{\mu}u_S u_C^{-1} u_Q^{-2}, \quad \varepsilon_{\text{vac}} = \hat{\varepsilon}u_S^{-1} u_C^{-1} u_Q^2. \quad (20.18)$$

Multiplying and dividing these expressions preserves $\mu_{\text{vac}}\varepsilon_{\text{vac}} = c_{\text{vac}}^{-2}$ and $\mu_{\text{vac}}/\varepsilon_{\text{vac}} = Z_{\text{vac}}^2$. The SI chart subsequently evaluates the sections. Physical comparison requires presenting their construction and normalization alongside evaluation; the four normalized coordinates in (20.5) retain their dimensionless type.

20.7 Reconstruction of the action shape from the constitutive response

The constitutive response and the inductive–capacitive realization receive the same angular pair constructed from APP–TRIT–TPK. The former evaluates rational functions of the two channels; the latter uses the ratio of their exponents to determine the shape of the action ellipse. Cubic inversion allows the two readouts to be composed without identifying their numerical ratios.

Let $x = A_{\text{rad}}$ and $y = C_{\text{rad}}^*$, with $x > |y|$, be the lifted coordinates already constructed, and preserve their sheet labels. Write

$$q_{\pm} = e^{-(x \pm y)}, \quad s_{\pm} = q_{\pm}^{30}, \quad r_{\pm} = f(s_{\pm}), \quad f(s) = \frac{1 + s + s^2}{1 + s + s^2 + s^3}. \quad (20.19)$$

The contractive domain gives $s_{\pm} \in (0, 1)$ and $r_{\pm} \in (3/4, 1)$. In the chamber $x > y > 0$, we have $s_+ < s_-$ and $r_+ > r_-$; sheet interchange transports this chamber to the opposite orientation.

The image of the two normalized constitutive coordinates is the domain

$$\mathcal{U} = \{(z, v) \in (0, \infty)^2 : (zv)^{-1/2} \in (3/4, 1), \quad (z/v)^{1/2} \in (3/4, 1)\}. \quad (20.20)$$

Here $(z, v) = (\hat{Z}, \hat{c})$. The restrictions express membership of the reconstructed channels in the domain of the inversion theorem; they do not constitute a subsequent choice of their values.

Proposition 20.7 (Constitutive composition of the shape and LC impedance). *For $(\hat{Z}, \hat{c}) \in \mathcal{U}$, let $s_{\pm} \in (0, 1)$ be the unique roots of*

$$r_{\pm} s_{\pm}^3 + (r_{\pm} - 1)(1 + s_{\pm} + s_{\pm}^2) = 0, \quad r_+ = \frac{1}{\sqrt{\hat{Z}\hat{c}}}, \quad r_- = \sqrt{\frac{\hat{Z}}{\hat{c}}}. \quad (20.21)$$

The angular ratio, the anisotropy of the action ellipse, and the relative LC impedance are reconstructed exactly by

$$\frac{y}{x} = \frac{\log s_+ - \log s_-}{\log s_+ + \log s_-}, \quad (20.22)$$

$$\eta_{\text{el}} = \frac{1}{2} \log \frac{\log s_+}{\log s_-}, \quad (20.23)$$

$$\frac{Z_{\text{LC}}}{Z_*} = e^{-\eta_{\text{el}}} = \sqrt{\frac{\log s_-}{\log s_+}}. \quad (20.24)$$

In the last identity, $Z_{\text{LC}} = Z_{\eta_{\text{el}}}$ is the impedance of the LC chart in (17.7).

Proof. The cubic inversion in (20.7) uniquely determines $s_{\pm}$, because f is a decreasing bijection from $(0, 1)$ onto $(3/4, 1)$. Their positive roots of order 30 recover $q_{\pm}$. The identities

$$\log s_+ = -30(x + y), \quad \log s_- = -30(x - y)$$

prove (20.22) by difference and sum. Both logarithms are negative; their ratio is positive and their sum is nonzero. Therefore,

$$\frac{1}{2} \log \frac{\log s_+}{\log s_-} = \frac{1}{2} \log \frac{x + y}{x - y} = \text{artanh}(y/x) = \eta_{\text{el}}.$$

The relation $Z_{\eta_{\text{el}}}/Z_* = e^{-\eta_{\text{el}}}$, proved in (17.8), gives (20.24). $\square$

The composition preserves the information needed for each reconstruction. With $\hbar$, one recovers $a_S = \sqrt{2\hbar} e^{\eta_{\text{el}}/2}$ and $b_S = \sqrt{2\hbar} e^{-\eta_{\text{el}}/2}$; with the chart (ω, Z_*) , one recovers the inductance and capacitance already defined. Action, clock, and dimensional reference accompany the shape as typed data. Inverting two dimensionless coordinates does not reconstruct their units or the entire memory of the state.

Normalization enters before the cubic equations are solved. If Z_{vac} and c_{vac} are received in a dimensional chart, relations (20.18) first determine

$$\hat{Z} = \frac{Z_{\text{vac}}}{u_S u_Q^{-2}}, \quad \hat{c} = \frac{c_{\text{vac}}}{u_C}.$$

These are the coordinates to which (20.21) applies. The reference Z_* belongs to the LC realization and is explicitly preserved in (20.24).

The two ratios are related through the channels, but evaluate different functions:

$$\hat{Z} = \frac{f(s_-)}{f(s_+)}, \quad \frac{Z_{\text{LC}}}{Z_*} = \sqrt{\frac{\log s_-}{\log s_+}}. \quad (20.25)$$

The preceding reconstruction provides the map between the two readouts. In particular, interchanging the sheets inverts both ratios, changes η_{el} to $-\eta_{\text{el}}$, and preserves $\hat{c}$. In the symmetric limit $y = 0$, the channels coincide and both ratios equal 1. Outside that case, a common origin does not impose equality of the two functions. The action shape is recoverable from the constitutive response while preserving orientation and normalization.

20.8 Constitutive speed and kinematic realization of the clock

The constitutive response and the kinematic readout realize the same speed section. Their identification preserves the order of construction: the APP sheets, their TRIT orientation, and the angular transport expressed by TPK precede the speed readouts and the choice of units. In the two-sheet representation of Section 19, with $x > |y|$ and rank-one projectors $P_{\pm}$, the transport is

$$T = e^{-xI-y\mathcal{R}} = q_+P_+ + q_-P_-, \quad q_{\pm} = e^{-x \mp y}, \quad 0 < q_{\pm} < 1.$$

Its temporal blocks of lengths 90 and 120 produce the response $\mathcal{V}$ in (20.1). The coordinates x, y and the channels are outputs of the preceding angular development; this composition applies its readouts to that already constructed transport.

The symmetric logarithmic readout of temporal transport is

$$\mathcal{E} = \log[(1 - q_+^{90})(1 - q_-^{90})] - \log[(1 - q_+^{120})(1 - q_-^{120})]. \quad (20.26)$$

On the dimensional speed line, with positive section u_C , the kinematic readout expresses $c_{\text{int}} = \exp(-\mathcal{E})u_C$. The constitutive response expresses $c_{\text{vac}} = \hat{c}u_C$ through (20.5).

Proposition 20.8 (Constitutive and kinematic identity). *In the same dimensional chart,*

$$c_{\text{int}} = c_{\text{vac}} = \hat{c}u_C, \quad \hat{c} = (r_+r_-)^{-1} = \exp(-\mathcal{E}). \quad (20.27)$$

The identification preserves sheet interchange.

Proof. All factors in (20.26) are positive. The logarithm law and (20.3) give

$$\mathcal{E} = \log(r_+r_-) = \log \det \mathcal{V}, \quad e^{-\mathcal{E}} = (\det \mathcal{V})^{-1}.$$

The determinant corresponds to the realization with two rank-one sheets. Both definitions therefore express the same section. Sheet interchange permutes r_+ and r_- and preserves their product. The determinant $\det T = e^{-2x}$ corresponds to elementary transport, whereas $\det \mathcal{V} = r_+r_-$ corresponds to its temporal response; (20.27) uses the latter. $\square$

20.8.1 Path length and elementary duration

The positive dimensional frame (u_S, u_C, u_T, u_Q) induces the length basis $u_L = u_C u_T$. For an enriched path γ with $n = |\gamma|$ transitions and constant channel, the additive readouts are

$$T_{\text{int}}(\gamma) = nu_T, \quad L_{\text{int}}(\gamma) = n\hat{c}u_L. \quad (20.28)$$

Concatenation adds the numbers of transitions and their realizations. Enriched routes also preserve their orientation and memory.

Corollary 20.9 (Normalization of the elementary length). *For $n > 0$, $L_{\text{int}}(\gamma)/T_{\text{int}}(\gamma) = c_{\text{vac}}$. In the normalization $t_0 = u_T$,*

$$\ell_0 = c_{\text{int}}t_0 = \hat{c}u_L. \quad (20.29)$$

In a temporal chart $t_0 = \tau_0 u_T$, with $\tau_0 > 0$, one obtains $\ell_0 = \tau_0 \hat{c}u_L$.

Proof. Dividing the two expressions in (20.28) cancels n and uses $u_L/u_T = u_C$. Multiplying the speed section by t_0 proves both elementary formulas. $\square$

When the channel depends on the edge, the readout retains the additive form $L_{\text{int}}(\gamma) = \sum_{e \in \gamma} \hat{c}(e)u_L$. Its quotient by nu_T is the mean of the speeds of those edges. The constant-channel expression is the homogeneous specialization of this readout.

20.8.2 Change of chart and adapted normalization

Proposition 20.10 (Covariance of the identification). *Let $u'_C = du_C$ and $u'_T = \eta u_T$, with $d, \eta > 0$. The coordinates of the same sections satisfy*

$$\hat{c}' = d^{-1}\hat{c}, \quad \tau'_0 = \eta^{-1}\tau_0, \quad u'_L = d\eta u_L. \quad (20.30)$$

If another representation writes $c' = \hat{c}'u'_C$ with $u'_C = du_C$, equality of sections is exactly equivalent to $d\hat{c}' = \hat{c}$.

Proof. Equality of each section in the two bases gives $\hat{c}'u'_C = \hat{c}u_C$ and $\tau'_0u'_T = \tau_0u_T$. In particular,

$$\tau'_0\hat{c}'u'_L = \frac{\tau_0}{\eta} \frac{\hat{c}}{d} d\eta u_L = \tau_0\hat{c}u_L = \ell_0.$$

The final equivalence follows from the same substitution for speed. □

The adapted choice $d = \hat{c}$ gives $u'_C = c_{\text{vac}}$ and $\hat{c}' = 1$: it expresses the constructed section in an adapted basis. The coefficient in the internal chart remains $(r_+r_-)^{-1}$, and that is the normalization used by cubic inversion. Keeping the action and charge bases, (20.18) yields

$$\hat{\mu}' = d\hat{\mu}, \quad \hat{\varepsilon}' = d\hat{\varepsilon}, \quad \hat{\mu}'\hat{\varepsilon}' = (\hat{c}')^{-2}.$$

In the adapted chart, $\hat{\mu}' = r_-/r_+$ and $\hat{\varepsilon}' = r_+/r_-$, by substitution of $\hat{\mu} = r_-^2$ and $\hat{\varepsilon} = r_+^2$.

20.8.3 Preservation of the readout under amplification

For the refinement $\mathcal{V}_m = \mathcal{V} \otimes I_{9^m}$, the normalized trace $\tau_m = \text{Tr} / (2 \cdot 9^m)$ satisfies

$$\exp[-2\tau_m(\log \mathcal{V}_m)] = \exp[-\log r_+ - \log r_-] = \hat{c}. \quad (20.31)$$

Indeed, each eigenvalue appears 9^m times; that multiplicity cancels against the denominator of the normalized trace. The ordinary amplified determinant is $(r_+r_-)^{9^m}$. The two-sheet normalization, equivalently expressed by (20.31), preserves the constitutive readout during amplification. This realization preserves the enriched transport from which it originates.

21 Joint reconstruction of gauge couplings, Higgs and the constitutive response

The vacuum response and the electroweak sector admit complementary readings of the angular transport already constructed. The former retains both marked channels; the determinant retains their common contraction; the angular Higgs component also retains the orientation that disappears from that determinant. Composing these readings reconstructs their spectral antecedent and expresses impedance through the scalar self-coupling and the gauge couplings within one and the same tree-level realization.

21.1 Angular provenance and route signatures

The APP-TRIT-TPK chain is preserved: the surviving prolongation determines the route, the route produces its oriented register, and the register reader expresses the angular signature. The coordinates $\pi = \pi_{\text{HMT}}$ and $\alpha = \alpha_{\text{HMT}}$ are preceding outputs of the same generation; A and C^* are their subsequent angular representations. Radians are used in this block:

$$x = \frac{\pi A_{\text{deg}}}{180}, \quad y = \frac{\pi C_{\text{deg}}^*}{180}, \quad A_{\text{deg}} = 1000\alpha, \quad 0 < y < x. \quad (21.1)$$

Thus $x = 50\pi\alpha/9$. The channels and sheet markings are those of Section 19:

$$\begin{aligned} q_+ &= e^{-x-y}, & q_- &= e^{-x+y}, & D &= q_+q_- = e^{-2x}, \\ P_{\pm} &= \frac{I \pm \mathcal{R}}{2}, & T &= q_+P_+ + q_-P_-. \end{aligned} \quad (21.2)$$

Here $\mathcal{R}$ is the marked self-adjoint involution, and each projector has rank one in the basic two-sheet realization.

The duration–persistence angular reader is

$$\chi_{\text{dur}}(n, s) = \exp\left(nx + \frac{s}{6}y\right). \quad (21.3)$$

To preserve the finite provenance of the coefficients used below, we assemble the nominal fingerprints of the three routes of the length-108 schedule. The cardinal-displacement columns are read as (N, E, S, W) ; the following pair retains the sheet and residue expressed by that route. For each identified route, the table preserves the fingerprint and angular signature expressed by the preceding chain.

Sector	Route and axis	(N, E, S, W)	Sheet/residue	(n, s)
W	15/37, E–W	(23, 44, 11, 30)	(41, 8)	(94, –1)
Z	20/23, E–W	(33, 45, 10, 20)	(57, 7)	(95, –1)
H	24/29, N–S	(40, 34, 22, 12)	(47, 10)	(97, 9)

The three typed fingerprints preserve the route identifier, orientation and provenance register of the signatures used. Angular evaluation follows extraction from the enriched register; the table assembles its outputs to compose the characters in this realization. The markings $W_{\pm}$ below denote winding coordinates, distinct from the species name W :

$$\begin{aligned} W_{\pm} &= 6n \pm s, & n &= \frac{W_+ + W_-}{12}, & s &= \frac{W_+ - W_-}{2}, \\ (n, s)_W &= (94, -1), & (W_+, W_-)_W &= (563, 565), \\ (n, s)_Z &= (95, -1), & (W_+, W_-)_Z &= (569, 571), \\ (n, s)_H &= (97, 9), & (W_+, W_-)_H &= (591, 573). \end{aligned} \quad (21.4)$$

The matrix of the first map is $\begin{pmatrix} 6 & 1 \\ 0 & -1 \end{pmatrix}$, with determinant -12 and image $\{(u, v) \in \mathbb{Z}^2 : u + v \in 12\mathbb{Z}\}$. Indeed, the inverse formulas above are integral precisely on this image. The character retains both orientations through

$$\chi_{\text{dur}}(n, s) = \left(q_+^{-W_+} q_-^{-W_-}\right)^{1/12}. \quad (21.5)$$

Taking logarithms proves the identity: the numerator is $W_+(x+y) + W_-(x-y) = 12nx + 2sy$. This identifies the twelfth root and the divisor 6 of the reader.

Proposition 21.1 (Angular ratios determined by the signatures). *On a common scale section, with the shared refinement factors that permit their cancellation in the ratios,*

$$\frac{m_W^<}{m_Z^<} = e^{-x}, \quad \frac{m_H^<}{m_W^<} = e^{3x+5y/3}, \quad \left(\frac{m_H^<}{m_W^<}\right)^2 = e^{6x+10y/3}. \quad (21.6)$$

Proof. Dividing the W and Z characters subtracts their signatures: $(94, -1) - (95, -1) = (-1, 0)$. For H/W , the difference is $(97, 9) - (94, -1) = (3, 10)$. Equation (21.3) gives $3x + (10/6)y$; squaring multiplies this exponent by two. The common scale and shared refinements cancel before these operations. $\square$

These ratios retain the identifier of the angular chart. Application to masses with distinct sectorial refinements retains their additional factors; the cancellation condition is part of the statement.

21.2 Rationalized tree-level realization

The electroweak reading of this chart defines $c_W^2 = D$, $s_W^2 = 1 - D$ and $e_g^2 = 4\pi\alpha$. The positive root fixes $g = e_g/s_W$ and $g' = e_g/c_W$. Hence

$$g^2 = \frac{4\pi\alpha}{1-D}, \quad g'^2 = \frac{4\pi\alpha}{D}. \quad (21.7)$$

The quantity e_g is a dimensionless coupling in the rationalized normalization, distinct from a dimensional charge section. The scalar-potential convention is

$$\begin{aligned} V(H) &= -\mu_H^2 H^\dagger H + \lambda_H (H^\dagger H)^2, \\ m_H^2 &= 2\lambda_H v^2, \quad m_W^2 = \frac{g^2 v^2}{4}. \end{aligned} \quad (21.8)$$

The same v occurs in both relations. Division cancels its scale and yields $m_H^2/m_W^2 = 8\lambda_H/g^2$. Together with (21.6), this gives

$$\boxed{\lambda_H = \frac{g^2}{8} \exp\left(6x + \frac{10}{3}y\right)}. \quad (21.9)$$

The factor 8 arises from the two quadratic normalizations in (21.8); the coefficients 6 and 10/3 arise from the signatures. This distinction preserves the operation producing each factor. The composition receives the preceding HMT outputs and uses this tree-level physical realization at a common scale; a change of field normalization, scale or scheme is also transported in its mass and coupling relations.

21.3 Joint inverse and exact domain

Theorem 21.2 (Angular reconstruction from gauge couplings and Higgs). *For fixed $\pi > 0$, the map defined by (21.2), (21.7) and (21.9), on $\alpha > 0$, $x > y > 0$, is injective and has the inverse*

$$\begin{aligned} D &= \frac{g^2}{g^2 + g'^2}, & x &= -\frac{1}{2} \log D, & \alpha &= \frac{g^2 g'^2}{4\pi(g^2 + g'^2)}, \\ y &= \frac{3}{10} \left[\log\left(\frac{8\lambda_H}{g^2}\right) - 6x \right]. \end{aligned} \quad (21.10)$$

Its exact image consists of the triples $g, g', \lambda_H > 0$ such that, with D and x reconstructed above,

$$\frac{g^2}{8} e^{6x} < \lambda_H < \frac{g^2}{8} e^{28x/3}. \quad (21.11)$$

Proof. The two gauge equations give $g^2/(g^2 + g'^2) = D$ and $g^2 g'^2/(g^2 + g'^2) = 4\pi\alpha$. The real logarithm recovers x , and the scalar equation recovers y . The conditions $g, g' > 0$ imply $0 < D < 1$ and $x > 0$. Under the strictly increasing logarithm, inequality (21.11) is equivalent to $0 < \log(8\lambda_H/g^2) - 6x < 10x/3$, namely $0 < y < x$. Conversely, the reconstructed expressions give $4\pi\alpha/(1 - D) = g^2$, $4\pi\alpha/D = g'^2$ and $g^2 e^{6x+10y/3}/8 = \lambda_H$. Positive roots recover the original couplings. This proves necessity, sufficiency and uniqueness. $\square$

The preceding inversion describes this map between readers exactly. The section generated by HMT also preserves the relation in (21.1), which becomes

$$-\frac{1}{2} \log\left(\frac{g^2}{g^2 + g'^2}\right) = \frac{25}{18} \frac{g^2 g'^2}{g^2 + g'^2}. \quad (21.12)$$

Substituting the inverse expression for α in $x = 50\pi\alpha/9$ proves the identity. The counter-angle likewise retains its preceding return equation. Thus the reader image and the section produced by the generator are distinct objects: the preceding compatibilities select the section inside the reconstructible domain. Inversion acts after generation and checks the outputs that have been retained.

21.4 Higgs-impedance reconstruction

To express the oriented content while preserving the channels, define

$$\rho_{\text{or}} = \frac{q_-}{q_+} = e^{2y}, \quad \Xi_H = \frac{8D^3 \lambda_H}{g^2}. \quad (21.13)$$

Equation (21.9) and $D^3 = e^{-6x}$ give

$$\begin{aligned} \Xi_H &= e^{10y/3} = \rho_{\text{or}}^{5/3}, & \rho_{\text{or}} &= \Xi_H^{3/5}, \\ q_- &= \sqrt{D} \Xi_H^{3/10}, & q_+ &= \sqrt{D} \Xi_H^{-3/10}. \end{aligned} \quad (21.14)$$

The powers are the positive real roots. The exact domain of the oriented chamber is

$$0 < D < 1, \quad 1 < \Xi_H < D^{-5/3}. \quad (21.15)$$

Indeed, $0 < y < x$ is equivalent to $1 < e^{10y/3} < e^{10x/3}$. The lower endpoint gives $q_+ = q_-$; the upper endpoint gives $q_- = 1$. For the two orientations together, $|y| < x$ is equivalent to $D^{5/3} < \Xi_H < D^{-5/3}$.

Let $F(q) = f(q^{30}) = (1 - q^{90})/(1 - q^{120})$ be the response already proved in (20.3)–(20.4).

Corollary 21.3 (Confluence of scalar and constitutive readings). *On the domain (21.15),*

$$\hat{Z} = \frac{F(\sqrt{D} \Xi_H^{3/10})}{F(\sqrt{D} \Xi_H^{-3/10})}. \quad (21.16)$$

The complete responses are

$$\begin{aligned} r_- &= F(\sqrt{D} \Xi_H^{3/10}), & r_+ &= F(\sqrt{D} \Xi_H^{-3/10}), \\ \hat{\mu} &= r_-^2, & \hat{\varepsilon} &= r_+^2, & \hat{c} &= (r_- r_+)^{-1}. \end{aligned} \quad (21.17)$$

The two reconstructions also satisfy

$$\frac{F^{-1}(\sqrt{\hat{\mu}})}{F^{-1}(\sqrt{\hat{\varepsilon}})} = \rho_{\text{or}} = \Xi_H^{3/5}. \quad (21.18)$$

Proof. Substituting the channels of (21.14) into the four readings of (20.5) proves the first identities. The constitutive inverse of (20.7) recovers q_- and q_+ ; their ratio is ρ_{or} . Its equality to $\Xi_H^{3/5}$ was proved in (21.14). $\square$

On the section compatible with $A_{\text{deg}} = 1000\alpha$, the coupling $Q(D) := -9 \log D/25$ satisfies $Q(D) = 4\pi\alpha$ and $g^2 = Q(D)/(1 - D)$. Thus the same Higgs coordinate takes the form

$$\Xi_H = \frac{8(1-D)D^3\lambda_H}{Q(D)}, \quad \frac{1}{g^2} + \frac{1}{g'^2} = \frac{1}{Q(D)}. \quad (21.19)$$

These expressions relate the total coupling, its rationalized partition and the scalar orientation on the same section.

21.5 Retained information and transport of orientation

The gauge pair recovers α and x ; the ratio λ_H/g^2 then recovers y . Together with the markings $P_{\pm}$, it reconstructs the complete operators T and $\mathcal{V} = r_+P_+ + r_-P_-$. The three scalars determine the eigenvalues; the marked projectors also determine their operator realization. The enriched route history remains preceding information of higher resolution.

Under $y \mapsto -y$ relative to the fixed markings, D, g, g' remain invariant and $\Xi_H \mapsto \Xi_H^{-1}$. The constitutive readings are exchanged, $\hat{Z} \mapsto \hat{Z}^{-1}$, and $\hat{c}$ is preserved. If λ_H^+ and λ_H^- denote the evaluations in the two conjugate orientations, then

$$\lambda_H^+ \lambda_H^- = \frac{g^4}{64} D^{-6}. \quad (21.20)$$

Multiplying (21.9) at y and $-y$ proves the identity: the oriented contributions cancel. This comparison retains the signature reader and changes its angular argument. A simultaneous conjugation of routes and reference markings must also transport the signatures.

For the amplification $T_m = T \otimes I_{9^m}$, with $\tau_m = \text{Tr} / (2 \cdot 9^m)$ and $\mathcal{R}_m = \mathcal{R} \otimes I_{9^m}$, the appropriate coordinates are

$$D = \exp(2\tau_m(\log T_m)), \quad \rho_{\text{or}} = \exp(-2\tau_m(\mathcal{R}_m \log T_m)). \quad (21.21)$$

Indeed, $\log T_m = (\log T) \otimes I_{9^m}$, so the two traces are $-x$ and $-y$, respectively. Normalization removes the amplification multiplicity. The gauge, Higgs and vacuum readings are therefore compatible with these refinements; the ordinary amplified determinant would be D^{9^m} and has a different role.

The composition has a precise conclusion: the even gauge invariant and the oriented scalar coordinate jointly reconstruct the spectral pair governing the vacuum. When λ_H is evaluated using the same formula, the confluence is a consistency identity. An independent reading of λ_H , transported to identical scheme and scale, makes the equality an additional test. Both uses retain the signatures, sheet markings and tree-level normalization explicitly.

22 Cyclic decomposition, quarter-turn, and oriented Catalan moment

The dodecaphase record relates the constitutive response to other transport invariants. The link is established on determined subspaces and maps: the rank defect measures an incidence difference, the determinant preserves a scalar response, and a coefficient of the resolvent retains orientation. The collection of moments culminating in Catalan's constant enters at this last level.

22.1 Two subspaces of the dodecaphase cycle

In $\mathbb{R}^{12}$, let $Ce_j = e_{j+1 \bmod 12}$ and let

$$P_3^{\text{cyc}} = \frac{I + C^3 + C^6 + C^9}{4}, \quad P_4^{\text{cyc}} = \frac{I + C^4 + C^8}{3}, \quad H_d = \text{Ran } P_d^{\text{cyc}}. \quad (22.1)$$

The averages over cyclic subgroups are orthogonal projectors. H_3 consists of the periodic sequences whose period divides 3, and H_4 of those whose period divides 4. Their dimensions are 3 and 4. The restriction of C to each space realizes the corresponding cycle. The superscript cyc identifies the average over the dodecaphase cycle: P_3^{cyc} is the projector onto sequences whose period divides 3. Its domain and action differ from those of the three-dimensional isotypic projector of the icosahedral realization; equality of rank does not identify the two operators.

Proposition 22.1 (Rank defect and determinantal response). *The following identities hold:*

$$\frac{\text{Tr}(P_3^{\text{cyc}} - P_4^{\text{cyc}})}{12} = -\frac{1}{12}, \quad \frac{\det_{H_3}(I - sC)}{\det_{H_4}(I - sC)} = \frac{1 - s^3}{1 - s^4} = f(s), \quad 0 < s < 1. \quad (22.2)$$

Proof. The trace of an orthogonal projector is its rank, so the numerator of the first expression is $3 - 4$. The characteristic polynomial of a cycle of length d is $t^d - 1$; hence its determinant $\det(I - sC)$ is $1 - s^d$. Their quotient yields the second identity. $\square$

The substitution $s = q^{30}$ transports this representation to the $90/120$ quotient. At $s = 1$, the common constant mode is cancelled before taking the limit, which equals $3/4$. This realization of $-1/12$ as a trace defect has an explicit finite domain. The regularized and modular realizations of the same scalar retain their own operators and domains.

22.2 Faithful representation of the quarter-turn

Define the antipodal projector $P_{\text{ant}} = (I - C^6)/2$ and $Q = P_4^{\text{cyc}} P_{\text{ant}}$. This projector acts on the twelve-phase cycle and differs from the constitutive projector P_- in Section 20. The projectors P_4^{cyc} and P_{ant} commute, and Q has rank 2. In its image, fix the orthonormal basis

$$u = \frac{1}{\sqrt{6}}(1, 0, -1, 0, 1, 0, -1, 0, 1, 0, -1, 0)^T, \quad v = Cu. \quad (22.3)$$

Componentwise verification gives $Cu = v$ and $Cv = -u$. Consequently, the isometry $W(a, b) = au + bv$ satisfies

$$CW = WJ, \quad J = \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \quad J^2 = -I. \quad (22.4)$$

The complex structure of the plane is realized by the effective action of the cycle. The sign depends on the chosen orientation: in the same basis, $C^3W = -WJ$. This precision allows the plane to be composed with the oriented readout.

22.3 Resolvent and unlimited depth

Since $C^2 = -I$ on $\text{Ran } Q$, for $0 \leq s < 1$ one has

$$(I - sC)^{-1}|_{\text{Ran } Q} = \frac{I + sC}{1 + s^2}, \quad k_4(s) := \langle v, (I - sC)^{-1}u \rangle = \frac{s}{1 + s^2}. \quad (22.5)$$

The sequence $\langle v, C^n u \rangle$ is $0, 1, 0, -1, \dots$. The depth readout preserves this character through the moment

$$\mathcal{C}_2(s) = \sum_{k \geq 0} \frac{(-1)^k s^{2k+1}}{(2k+1)^2}, \quad 0 \leq s \leq 1. \quad (22.6)$$

The series converges absolutely and uniformly on this interval by comparison with $\sum (2k+1)^{-2}$. For $0 < s < 1$, the differentiated series converge uniformly on every compact subinterval of the interior. With $\mathcal{D} = s d/ds$, one obtains

$$\mathcal{D}^2 \mathcal{C}_2(s) = k_4(s), \quad \mathcal{C}_2(s) = \int_0^s \frac{dt}{t} \int_0^t \frac{d\xi}{\xi} k_4(\xi). \quad (22.7)$$

The conditions $\mathcal{C}_2(0) = 0$ and $\lim_{s \downarrow 0} \mathcal{D} \mathcal{C}_2(s) = 0$ fix the two integration constants. Indeed, the difference between two solutions of $\mathcal{D}^2 F = k_4$ has the form $a \log s + b$; the second condition annuls a and the first annuls b . Uniform convergence permits evaluation at the endpoint $s = 1$: the result is the Catalan moment. The finite representation of the character preserves its orientation; the depth index remains unlimited.

22.4 Composition with the constitutive response

Theorem 22.2 (Constitutive differential relation). *For $0 < s < 1$, with the orientation of (22.3),*

$$f(s) = \frac{1 + s + s^2}{s(1 + s)} \mathcal{D}^2 \mathcal{C}_2(s). \quad (22.8)$$

Proof. Substituting (22.5) into (22.7) and using $(1 + s)(1 + s^2) = 1 + s + s^2 + s^3$ gives the identity. All denominators are positive on the domain. The intertwiner (22.4) identifies the operator coefficient used before its evaluation. $\square$

Upon evaluating $s_{\pm} = q_{\pm}^{30}$, the same collection produces the two channel responses of (20.3). The Catalan limit, the trace defect, and the constitutive eigenvalues retain their distinct functions within this composition. Reversing the cycle with the markings u, v fixed changes the sign of k_4 and preserves the determinant $1 + s^2$. This identifies orientation information that the determinant alone does not distinguish.

The Pell development uses a particular evaluation of this collection, with $\rho = 2 - \sqrt{3}$ and $\rho^{-1} = 2 + \sqrt{3}$. Its proof is retained as a distinct unit of integration; the angular definition of $q_{\pm}$ remains independent of any eventual identification of $s_{\pm}$ with ρ .

Figure 12 represents the relation between the rational response and the oriented kernel before integration. The Pell section satisfies $\rho + \rho^{-1} = 4$, so that $k_4(\rho) = (\rho + \rho^{-1})^{-1} = 1/4$. The two constitutive values are obtained, by contrast, by evaluating f at the already constructed arguments $s_{\pm}$. The plot includes the continuous extensions of the two functions to $[0, 1]$; the domain of constitutive inversion remains the open interval $(0, 1)$.

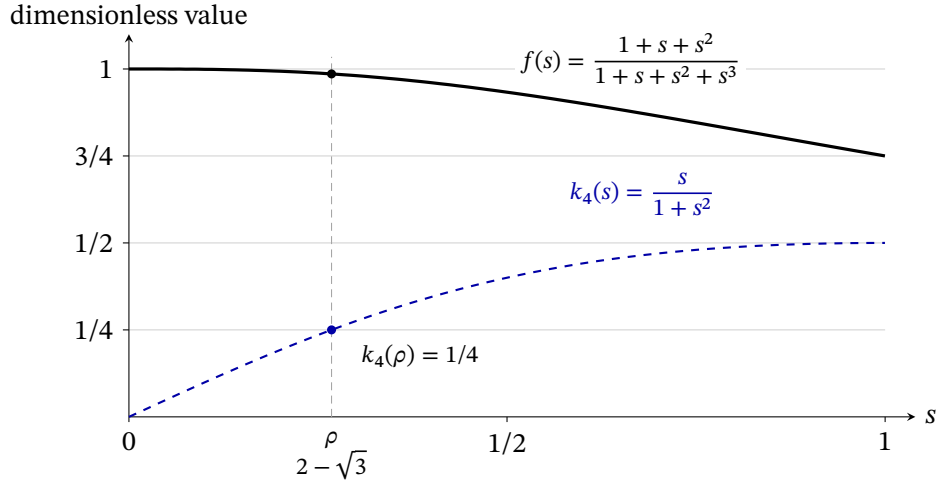


Figure 12: Constitutive response and kernel of the oriented moment. The solid curve is f and the dashed curve is $k_4 = \mathcal{D}^2 \mathcal{C}_2$; they are related by (22.8). The vertical line marks the Pell section, without identifying it with the constitutive arguments s_+ or s_- .

22.5 Return traces and moments of the constitutive quotient

The register update and the cycle response share the dodecaphasic transport and preserve different kinds of information. The signed event is obtained from the direction and orientation of each window of the APP-TRIT-TPK history. Its vector increment and balance (2.27) retain the interaction with the previous register. The construction below extends the spectral reading of the same cycle, once its incidence subspaces have been fixed. This order allows return, determinant and moments to be composed while preserving the history that precedes them.

Retain the operator C and projectors of (22.1), and define

$$E_{\text{cyc}} = P_3^{\text{cyc}} - P_4^{\text{cyc}}, \quad b_n = \text{Tr}(C^n E_{\text{cyc}}), \quad n \geq 0. \quad (22.9)$$

The symbol b_n distinguishes this spectral trace from the signed event a_m of a window. The trace uses the transport operator; the event also uses the trajectory actually traversed.

Proposition 22.3 (Trace sequence produced by incidence). *One has*

$$b_n = 3\mathbf{1}_{3|n} - 4\mathbf{1}_{4|n}, \quad (22.10)$$

$$(b_1, \dots, b_{12}) = (0, 0, 3, -4, 0, 3, 0, -4, 3, 0, 0, -1).$$

The least period is twelve and the sum over a period is zero. For $B_N = \sum_{n=1}^N b_n$,

$$B_N = 3[N/3] - 4[N/4], \quad |B_N| \leq 3. \quad (22.11)$$

Proof. On H_d , the operator C cyclically permutes the d coordinate classes. Its n th power has d fixed points if d divides n , and none otherwise. Since P_d^{cyc} commutes with C , $\text{Tr}(C^n P_d^{\text{cyc}})$ is the trace of this restriction. Subtracting the cases $d = 3, 4$ proves the formula. The least positive period is twelve: $b_n = -1$ occurs exactly at multiples of twelve. Over one period the contributions are $3 \cdot 4 - 4 \cdot 3 = 0$. The finite sum counts multiples and gives (22.11). Its values for $N = 0, \dots, 11$ are $(0, 0, 0, 3, -1, -1, 2, 2, -2, 1, 1, 1)$ and repeat after adding twelve. $\square$

Theorem 22.4 (Logarithmic reconstruction of the constitutive reader). *For $|s| < 1$, the quotient f of (22.2) satisfies*

$$-\log f(s) = \sum_{n \geq 1} \frac{b_n}{n} s^n, \quad (22.12)$$

$$-\frac{s f'(s)}{f(s)} = \frac{3s^3}{1-s^3} - \frac{4s^4}{1-s^4}. \quad (22.13)$$

The holomorphic branch is fixed by $\log f(0) = 0$. For $0 < s < 1$, it agrees with the real logarithm.

Proof. The polynomials $1 - s^3$ and $1 - s^4$ do not vanish in the open disc. The expansion $-\log(1 - w) = \sum_{k \geq 1} w^k/k$, obtained by integrating the geometric series from zero, converges uniformly on compact discs. Substitution of $w = s^3, s^4$ and subtraction give $-\log f(s) = \sum_{k \geq 1} s^{3k}/k - \sum_{k \geq 1} s^{4k}/k$. The coefficient of s^n is b_n/n . Local uniform convergence permits differentiation and summation of the two geometric series, proving (22.13). The condition at zero fixes the integration constant and the branch. $\square$

Thus $b_0/12 = -1/12$ records the normalized dimension defect, whereas the full sequence b_n reconstructs a function. These are readings of different scope on the same pair of subspaces. Upon substituting $s = q_{\pm}^{30}$, the arguments remain outputs of the preceding angular construction; the expansion does not select them.

Theorem 22.5 (Depth moments and Mellin representation). *The series*

$$M_b(z) = \sum_{n \geq 1} \frac{b_n}{n^z} \quad (22.14)$$

converges locally uniformly for $\operatorname{Re} z > 0$. Writing $\sigma = \operatorname{Re} z > 0$, its remainder satisfies

$$\left| M_b(z) - \sum_{n=1}^N \frac{b_n}{n^z} \right| \leq 3 \left(1 + \frac{|z|}{\sigma} \right) (N+1)^{-\sigma}. \quad (22.15)$$

In the domain of absolute convergence $\operatorname{Re} z > 1$,

$$M_b(z) = (3^{1-z} - 4^{1-z})\zeta(z), \quad \zeta(z) = \sum_{n \geq 1} n^{-z}. \quad (22.16)$$

For $\operatorname{Re} z > 0$ one also has

$$\Gamma(z)M_b(z) = \int_0^\infty t^{z-1} \left(\frac{3}{e^{3t}-1} - \frac{4}{e^{4t}-1} \right) dt, \quad (22.17)$$

where $\Gamma(z) = \int_0^\infty u^{z-1} e^{-u} du$.

Proof. Summation by parts and $|B_n| \leq 3$ give, after taking the upper limit,

$$\sum_{n > N} b_n n^{-z} = -B_N(N+1)^{-z} + \sum_{n > N} B_n(n^{-z} - (n+1)^{-z}).$$

The modulus of the difference of powers is bounded by $|z| \int_n^{n+1} x^{-\sigma-1} dx$. This proves (22.15), convergence and uniformity on compact subsets. For $\sigma > 1$, absolute convergence permits multiples of three and four to be grouped separately, giving (22.16).

The geometric series of the transport, evaluated at $s = e^{-t}$, produces

$$\sum_{n \geq 1} b_n e^{-nt} = \frac{3}{e^{3t}-1} - \frac{4}{e^{4t}-1} =: K_b(t).$$

For $\sigma > 1$, termwise integration is justified by $\sum |b_n| n^{-\sigma} < \infty$. Substitution of $u = nt$ gives $\Gamma(z)n^{-z}$ and proves (22.17) on that half-plane. At the origin the $1/t$ terms cancel: $K_b(t) = 1/2 + O(t)$. At infinity, $K_b(t) = O(e^{-3t})$. The integral is therefore holomorphic for $\operatorname{Re} z > 0$ by local uniform convergence. The series is also holomorphic there by the preceding bound; the identity theorem extends equality to that domain. $\square$

For $0 < \operatorname{Re} z \leq 1$, increasing depth fixes the convergent ordering of the series. Its realization as the absolute trace of the diagonal operator applies for $\operatorname{Re} z > 1$.

Corollary 22.6 (Evaluations compatible with the constitutive endpoint). *One has*

$$M_b(2) = \frac{\zeta(2)}{12}, \quad M_b(1) = \log(4/3) = -\log f(1), \quad f(1) = \frac{3}{4}. \quad (22.18)$$

Proof. The first equality follows from $3^{-1} - 4^{-1} = 1/12$. For the second, the exact partial sums are $H_{[N/3]} - H_{[N/4]}$, with $H_j = \sum_{k=1}^j 1/k$ and $H_0 = 0$. Integral comparison for $1/x$ between these two indices gives the limit $\log(4/3)$. Cancelling $1 - s$ before evaluating the determinantal quotient gives $f(1) = 3/4$. $\square$

The affine register retains events and their interaction with previous memory; the trace sequence retains the incidence of the two subspaces at every power of transport; the moment applies a depth weighting to that already-produced sequence. Its spectral periodicity coexists with an event history that continues after phase return. Replacing C by its inverse preserves b_n , since $d \mid n$ is invariant under $n \mapsto -n$; the oriented resolvent coefficient in (22.5), with fixed marks, changes sign. The joint reading preserves a distinction not recorded by the scalar determinant alone. The quadratic form and coordinate sum of the register retain their mathematical types; this composition assigns them neither energy units nor electric-charge units.

22.6 Residual composition and products of moments

The subsequent composition of readers also retains the operation performed on depth. In the realization $\mathcal{H} = \ell^2(\mathbb{N}_{>0})$, set $N\delta_n = n\delta_n$. Let a, c be two already-produced periodic residual tables and let $Q_a\delta_n = a(n)\delta_n$, $Q_c\delta_n = c(n)\delta_n$. After extension to a common period, the two readers act on the same index. For $\operatorname{Re} z > 1$, write $M_a(z) = \operatorname{Tr}(N^{-z}Q_a)$ and similarly for M_c .

Proposition 22.7 (Diagonal composition and tensor composition). *The following identities hold:*

$$\operatorname{Tr}(N^{-z}Q_aQ_c) = \sum_{n \geq 1} \frac{a(n)c(n)}{n^z}, \quad (22.19)$$

$$\begin{aligned} M_a(z)M_c(z) &= \operatorname{Tr}_{\mathcal{H} \otimes \mathcal{H}}((N \otimes N)^{-z}(Q_a \otimes Q_c)) \\ &= \sum_{k \geq 1} \frac{(a * c)(k)}{k^z}, \end{aligned} \quad (22.20)$$

$$(a * c)(k) = \sum_{d \mid k} a(d)c(k/d), \quad \operatorname{Re} z > 1. \quad (22.21)$$

Every operator appearing inside these traces is trace class.

Proof. The diagonal product acts on δ_n by $a(n)c(n)$, proving (22.19). On $\delta_m \otimes \delta_n$, the operator $N \otimes N$ has eigenvalue mn , and $Q_a \otimes Q_c$ acts by $a(m)c(n)$. If $\sigma = \operatorname{Re} z > 1$, the sum of the moduli of these spectral coefficients is bounded by $\|a\|_\infty \|c\|_\infty (\sum n^{-\sigma})^2 < \infty$. This bound establishes the trace-class condition and permits the double sum to be factorized or grouped by $k = mn$. Pairs with product k are exactly $(d, k/d)$ with $d \mid k$. The analogous bound with a single sum proves the diagonal case. $\square$

For the readers in this section, taking $a(n) = b_n$ and $c(n) = \langle v, C^n u \rangle$ gives, at $n = 3$, $a(3)c(3) = -3$, whereas $(a * c)(3) = 3$. This coefficient difference exhibits the information retained by each operation. Diagonal composition retains a common depth; tensor composition retains two depths before grouping by their product. The tensor realization does not ascribe statistical independence to the HMT histories that produced the readers. Regularized derivatives and normalization changes of other moments are subsequent operations distinct from the two compositions proved here.

22.7 Functional reconstruction of the oriented moment

The differential relation (22.8) admits an inverse preserving the complete collection of moments. Its domain is the constitutive function produced by the cycle sectors, with the orientation of (22.3). The incidences $90 = 3 \cdot 30$ and $120 = 4 \cdot 30$ determine that function before its evaluation in the channels $s_{\pm} = q_{\pm}^{30}$. The condition at the origin corresponds to the vanishing of the moment as its contractive argument tends to zero. This condition also determines the integration constants.

Theorem 22.8 (Integral reconstruction and uniqueness). *Let $f(s) = (1 - s^3)/(1 - s^4)$, for $0 < s < 1$, be the response of (22.2), and let $\mathcal{D} = s d/ds$. There is a unique function $F \in C^2((0, 1))$ satisfying*

$$\mathcal{D}^2 F(s) = \frac{s(1+s)}{1+s+s^2} f(s), \quad \lim_{s \downarrow 0} F(s) = 0. \quad (22.22)$$

It is given by

$$F(s) = \int_0^s \log \frac{s}{t} \frac{1+t}{1+t+t^2} f(t) dt = \int_0^s \frac{\log(s/t)}{1+t^2} dt = \mathcal{C}_2(s), \quad (22.23)$$

$$f(s) = \frac{1+s+s^2}{s(1+s)} \mathcal{D}^2 F(s). \quad (22.24)$$

Its continuous extension to $s = 1$ recovers $F(1) = G_{\text{Cat}}$.

Proof. The factorization $1+t+t^2+t^3 = (1+t)(1+t^2)$ gives

$$\frac{1+t}{1+t+t^2} f(t) = \frac{1}{1+t^2}.$$

The integrand of (22.23) is integrable at zero, since

$$0 \leq F(s) \leq \int_0^s \log(s/t) dt = s.$$

Hence $F(s) \rightarrow 0$. Differentiation on the open interval, first on a truncated interval and then by dominated convergence, gives

$$\mathcal{D}F(s) = \int_0^s \frac{dt}{1+t^2}, \quad \mathcal{D}^2 F(s) = \frac{s}{1+s^2} = \frac{s(1+s)}{1+s+s^2} f(s).$$

In particular, F is of class C^2 and satisfies the equation. If F_1, F_2 are two solutions, their difference $H = F_1 - F_2$ satisfies $\mathcal{D}^2 H = 0$. Setting $z = \log s$ gives $d^2 H(e^z)/dz^2 = 0$, so $H(s) = a \log s + b$. The existence of a zero limit as $s \downarrow 0$ first forces $a = 0$ and then $b = 0$. The condition on F therefore determines the solution and also the limit $\mathcal{D}F(s) \rightarrow 0$.

For $s < 1$, the geometric expansion of $(1+t^2)^{-1}$ and absolute integrability of its weighted terms allow termwise integration:

$$\int_0^s t^{2k} \log(s/t) dt = \frac{s^{2k+1}}{(2k+1)^2}, \quad F(s) = \sum_{k \geq 0} \frac{(-1)^k s^{2k+1}}{(2k+1)^2}.$$

This is precisely the collection (22.6). Absolute and uniform convergence of this last series on $[0, 1]$ justifies passage to the endpoint $s = 1$. For the integral, this passage is likewise justified by the integrable bound $\log(1/t)$, after extending the integrand by zero for $t > s$. Finally, solving the differential equation for f gives (22.24), with positive denominators on $(0, 1)$. $\square$

The limit of the moment and the limit of its second derivative are taken in their respective convergence classes. In particular,

$$\lim_{s \uparrow 1} \mathcal{D}^2 \mathcal{C}_2(s) = \frac{1}{2}$$

is the radial Abel limit of the differentiated series. Evaluation of $\mathcal{C}_2(1)$ directly uses its absolutely convergent series. This distinction preserves the same depth operator in the interior and in its boundary readout.

22.8 Compatibility of reconstruction with channels and speed

The theorem reconstructs a function from another previously constructed function. Cubic inversion (20.7), by contrast, acts on the evaluations $r_{\pm}$: it recovers the arguments $s_{\pm}$ within that fixed collection. Composition of the two operations preserves the quarter-turn orientation, the labels $P_{\pm}$, and the transport incidences. Thus $G_{\text{Cat}} = \mathcal{C}_2(1)$ is the endpoint readout of a functional object that also participates in the interior responses $r_{\pm} = f(s_{\pm})$.

For $\mathcal{V}_m = \mathcal{V} \otimes I_{9^m}$ and $\tau_m = \text{Tr} / (2 \cdot 9^m)$, the speed readout retains exactly the normalization of (20.31):

$$\exp[-2\tau_m(\log \mathcal{V}_m)] = \frac{1}{r_+ r_-} = (\det \mathcal{V})^{-1} = \hat{c}. \quad (22.25)$$

Indeed, each eigenvalue $r_{\pm}$ is repeated 9^m times, so $2\tau_m(\log \mathcal{V}_m) = \log r_+ + \log r_-$. The identity uses the determinant of the two-sheet realization and the normalized trace of its amplification. The previously fixed dimensional sections then give

$$c_{\text{int}} = c_{\text{vac}} = \hat{c} u_C, \quad \ell_0 = c_{\text{vac}} t_0,$$

with the same state, chart, and elementary duration as in (20.29). Functional reconstruction preserves this speed section when composed with action and length; spectral multiplicity is absorbed by the stated normalization.

23 Evaluation at the Pell scale and the constitutive Barbero–Immirzi functional

The relationship between the vacuum response and the Catalan moment allows two levels of transport to be composed: the weighted depth of the quarter-turn character and evaluation of the angular channels. The Pell scale distinguishes an argument of the moment collection; the vacuum channels, by contrast, receive the arguments determined by the angular pair. The Barbero–Immirzi functional is expressed on the latter through the same constitutive response and its inverse. The connection is thus established between defined operations, preserving the position of each evaluation within the APP–TRIT–TPK construction.

23.1 Oriented moment and evaluation at the Pell unit

Write

$$G_{\text{Cat}} := \mathcal{C}_2(1) = \sum_{k \geq 0} \frac{(-1)^k}{(2k+1)^2}, \quad \lambda_P = 2 + \sqrt{3}, \quad \rho_P = 2 - \sqrt{3} = \lambda_P^{-1}. \quad (23.1)$$

The constant G_{Cat} is the endpoint value of the moment constructed in (22.6). The pair λ_P, ρ_P corresponds to the expanding and contracting orientations of the Pell unit: $(2 + \sqrt{3})(2 - \sqrt{3}) = 1$. The norm identity $2^2 - 3 \cdot 1^2 = 1$ fixes reciprocity. Specialization of the moment at ρ_P follows construction of its oriented character.

Differential equation (22.7), with zero condition at the origin, gives

$$\mathcal{D}\mathcal{C}_2(s) = \arctan s, \quad \mathcal{C}_2(s) = \int_0^s \frac{\arctan t}{t} dt, \quad \mathcal{D} = s \frac{d}{ds}. \quad (23.2)$$

Indeed, integrating $\mathcal{D}^2 \mathcal{C}_2 = s/(1+s^2)$ with respect to ds/s produces $\arctan s$; a second integration produces the stated expression. The integrand extends continuously to $t = 0$ with value 1. The angular form of this integral allows exact determination of the evaluation at ρ_P .

Lemma 23.1 (Angular integral of the odd character). *For $0 < \theta \leq \pi/4$,*

$$-\int_0^\theta \log(\tan x) dx = \sum_{\substack{n \geq 1 \\ n \text{ odd}}} \frac{\sin(2n\theta)}{n^2}. \quad (23.3)$$

Proof. For $0 < a < 1$, the logarithm expansion inside the unit disk implies

$$\begin{aligned} L_a(x) &:= \log |1 - ae^{2ix}| - \log |1 + ae^{2ix}| \\ &= -2 \sum_{\substack{n \geq 1 \\ n \text{ odd}}} \frac{a^n \cos(2nx)}{n}. \end{aligned}$$

The series converges uniformly in x for every $a < 1$; termwise integration gives

$$-\int_0^\theta L_a(x) dx = \sum_{\substack{n \geq 1 \\ n \text{ odd}}} \frac{a^n \sin(2n\theta)}{n^2}.$$

As $a \uparrow 1$, the quotient of moduli converges to $|1 - e^{2ix}|/|1 + e^{2ix}| = \tan x$ for $0 < x \leq \theta$. The logarithm's singularity at the origin is integrable. To justify the limit, one may restrict to $a \geq 1/2$ and observe that

$$|1 - ae^{2ix}|^2 = (1 - a)^2 + 4a \sin^2 x \geq 2 \sin^2 x, \quad 1 \leq |1 + ae^{2ix}| \leq 2$$

on $0 \leq x \leq \pi/4$. Together with the upper bound $|1 - ae^{2ix}| \leq 2$, these inequalities give $|L_a(x)| \leq c + |\log x|$ with a constant independent of a . Dominated convergence permits passage to the limit in the integral. For the integrated series, domination by $\sum n^{-2}$ applies. The two limits prove (23.3). $\square$

Theorem 23.2 (Catalan–Pell evaluation). *The oriented moment satisfies*

$$\mathcal{C}_2(\rho_P) = \frac{2}{3}G_{\text{Cat}} - \frac{\pi}{12} \log \lambda_P, \quad (23.4)$$

$$\mathcal{D}\mathcal{C}_2(\rho_P) = \frac{\pi}{12}, \quad \mathcal{D}^2\mathcal{C}_2(\rho_P) = \frac{1}{4}. \quad (23.5)$$

Proof. The substitution $t = \tan x$ in (23.2), followed by integration by parts, gives

$$\mathcal{C}_2(\tan \theta) = \theta \log(\tan \theta) - \int_0^\theta \log(\tan x) dx. \quad (23.6)$$

The boundary term at zero vanishes because $x \log(\tan x) \rightarrow 0$. The tangent difference formula gives

$$\tan \frac{\pi}{12} = \tan\left(\frac{\pi}{4} - \frac{\pi}{6}\right) = \frac{\sqrt{3}-1}{\sqrt{3}+1} = 2 - \sqrt{3}.$$

It remains to evaluate the series in (23.3) at $\theta = \pi/12$. Let $\chi_{-4}(n) = (-1)^{(n-1)/2}$ for odd n . In the six odd classes modulo 12, one verifies

$$2 \sin \frac{n\pi}{6} = \chi_{-4}(n) + 3 \mathbf{1}_{3|n} \chi_{-4}(n/3). \quad (23.7)$$

The finite check is

$n \bmod 12$	1	3	5	7	9	11
$2 \sin(n\pi/6)$	1	2	1	-1	-2	-1
$\chi_{-4}(n)$	1	-1	1	-1	1	-1
$3 \mathbf{1}_{3 n} \chi_{-4}(n/3)$	0	3	0	0	-3	0

and both sides are periodic modulo 12 on odd indices. Absolute convergence permits separation of summands and the substitution $n = 3m$ in the second term:

$$\begin{aligned} \sum_{\substack{n \geq 1 \\ n \text{ odd}}} \frac{\sin(n\pi/6)}{n^2} &= \frac{1}{2} \sum_{n \text{ odd}} \frac{\chi_{-4}(n)}{n^2} + \frac{3}{2} \sum_{m \text{ odd}} \frac{\chi_{-4}(m)}{(3m)^2} \\ &= \left(\frac{1}{2} + \frac{1}{6}\right) G_{\text{Cat}} = \frac{2}{3} G_{\text{Cat}}. \end{aligned}$$

Sums without a lower bound in this expression run over positive integers. Substitution in (23.6), using $\log \rho_P = -\log \lambda_P$, yields (23.4). The first derivative follows from $\arctan \rho_P = \pi/12$; the second from

$$\frac{\rho_P}{1 + \rho_P^2} = \frac{1}{\rho_P + \rho_P^{-1}} = \frac{1}{4}.$$

□

The coefficient $2/3$ comes from the character decomposition in (23.7): restriction to multiples of 3 changes the depth weight by the factor 3^{-2} . Evaluation thus brings together the quarter-turn and dodecaphasic subdivision in an exact moment identity. The coefficients are fixed before any decimal comparison.

23.2 Composition of phase and weighted depth

The depth representation specifies what this evaluation preserves. On $\ell^2(\mathbb{N}_{>0})$, define

$$N\delta_n = n\delta_n, \quad U_4\delta_n = i^n\delta_n, \quad Q_4 = \frac{U_4 - U_4^*}{2i}.$$

For $0 < s < 1$, the operator $N^{-2}s^N Q_4$ is trace class, and

$$\mathcal{C}_2(s) = \text{Tr}(N^{-2}s^N Q_4). \quad (23.8)$$

This equality follows by evaluation on the basis δ_n ; the diagonal entries are $s^n \sin(n\pi/2)/n^2$. The interchange $U_4 \leftrightarrow U_4^*$ reverses Q_4 and with it the oriented moment. Index n and weight s^n remain fixed.

Composition is performed before taking the trace. For $a \in \mathbb{Z}/4\mathbb{Z}$, let $B(s, a) = s^N U_4^a$. Since both factors are diagonal in the same basis,

$$B(s, a)B(t, b) = B(st, a + b), \quad B(s, 1)^4 = s^{4N}. \quad (23.9)$$

Phase return preserves the accumulated contraction. In particular, $B(\rho_P, 1)^4 = \rho_P^{4N}$, whose eigenvalue at depth n is ρ_P^{4n} . This realization gives an operator meaning to the contractive evaluation; the scalar moment sum aggregates depths after preserving their index and orientation. The constitutive arguments $s_{\pm} = q_{\pm}^{30}$ remain determined by the angular pair of (20.8). The preceding formulas contain no equality between those arguments and ρ_P .

23.3 Dodecaphasic incidence and return functional

The Barbero–Immirzi contribution uses angular transport and hexad–octad incidence. Let $H \subset O$ be a hexad and an octad with the hexad’s marked pair preserved. The relevant flag sets are

$$\mathcal{F}_6 = H \times \binom{H}{2} \hookrightarrow \mathcal{F}_8 = O \times \binom{H}{2}, \quad |\mathcal{F}_6| = 90, \quad |\mathcal{F}_8| = 120. \quad (23.10)$$

The cardinalities are $6\binom{6}{2}$ and $8\binom{6}{2}$; the same incidence preserves the normalized defect $(90 - 120)/(24\binom{6}{2}) = -1/12$. This count identifies the two sectors used by return, while the calendar determines their relative multiplicities.

For $m \in \{90, 120\}$, the tritic return on a contractive channel is

$$S_m(q) = \sum_{j \geq 0} q^{m+3mj} = \frac{q^m}{1 - q^{3m}}, \quad 0 < q < 1. \quad (23.11)$$

Each term corresponds to the initial height m and j returns of length $3m$. The geometric series thus preserves both the initial height and return period. The dodecaphasic calendar $\mathbb{Z}/12\mathbb{Z}$ supplies twelve sectors and a unique oriented boundary incidence, denoted ∂_{ret} . In the free module of events, its fundamental chain is

$$c_{12}^+ = \sum_{j \in \mathbb{Z}/12\mathbb{Z}} [j, \mathcal{F}_6] - [\partial_{\text{ret}}, \mathcal{F}_8]. \quad (23.12)$$

The linear readout $\mathcal{L}_q$ assigns $S_{90}(q)$ to each generator $[j, \mathcal{F}_6]$ and $S_{120}(q)$ to the boundary generator. This yields

$$\mathcal{L}_q(c_{12}^+) = 12S_{90}(q) - S_{120}(q). \quad (23.13)$$

The weights 12 and -1 are the chain’s oriented multiplicities. The conjugate orientation satisfies $c_{12}^- = -c_{12}^+$, with the same incidence types.

Proposition 23.3 (Singular coefficient of the fundamental return). *The coefficient of $(1 - q)^{-1}$ in $\mathcal{L}_q(c_{12}^+)$ is $1/24$.*

Proof. For every integer $m > 0$, the factorization $1 - q^{3m} = (1 - q) \sum_{j=0}^{3m-1} q^j$ gives

$$\lim_{q \uparrow 1} (1 - q) S_m(q) = \frac{1}{3m}.$$

Applying this identity to the functional already determined in (23.13) gives

$$\lim_{q \uparrow 1} (1 - q) \mathcal{L}_q(c_{12}^+) = \frac{12}{270} - \frac{1}{360} = \frac{1}{24}. \quad (23.14)$$

In the complex coordinate $q - 1$, the residue is $-1/24$. $\square$

Singular evaluation verifies a consequence of incidence and tritic return. The proof order is fixed by (23.10)–(23.13); the singular coefficient is computed after the chain has been constructed.

23.4 Factorization through the constitutive operator

Retain the lifted angular chart of Section 19. To make the angular conversion explicit, write

$$A_{\text{deg}} = \frac{180x}{\pi}, \quad C_{\text{deg}}^* = \frac{180y}{\pi}, \quad q_{\pm} = e^{-x \mp y}, \quad s_{\pm} = q_{\pm}^{30}. \quad (23.15)$$

The orientation considered in (20.3) corresponds to $0 < y < x$; sheet interchange is expressed by $y \mapsto -y$ in the larger domain $x > |y|$. Real exponentials act on these lifts, which preserve the values of $x \pm y$ during composition.

The oriented Barbero–Immirzi functional is

$$\gamma = \frac{\pi A_{\text{deg}} C_{\text{deg}}^*}{180} + 12[S_{90}(q_-) - S_{90}(q_+)] - [S_{120}(q_-) - S_{120}(q_+)]. \quad (23.16)$$

The first component is the angular evaluation; the second is the difference of the fundamental-chain readout on the conjugate channels. The arguments are received from the previously constructed angular transport. The formula determines a dimensionless scalar on that starting structure.

Let $\psi : (3/4, 1) \rightarrow (0, 1)$ be the inverse of f constructed through (20.7). Define

$$\mathcal{H}(s) = \frac{12s^3}{1-s^9} - \frac{s^4}{1-s^{12}} - \frac{(\log s)^2}{20\pi}, \quad 0 < s < 1. \quad (23.17)$$

This function is real analytic on its domain. The exponents 3, 4, 9, 12 result from expressing heights 90, 120 and returns 270, 360 in units of 30; the logarithmic coefficient is fixed when the angular contribution is transported.

Theorem 23.4 (Oriented constitutive expression of Barbero–Immirzi). *In chart (23.15),*

$$\gamma = \mathcal{H}(s_-) - \mathcal{H}(s_+) = \mathcal{H}(\psi(r_-)) - \mathcal{H}(\psi(r_+)). \quad (23.18)$$

On the two-dimensional channel representation,

$$\gamma = -\text{Tr}[\mathcal{R} \mathcal{H}(\psi(\mathcal{V}))], \quad (23.19)$$

where Tr is the ordinary trace, with $\text{Tr } I = 2$.

Proof. Substitution of $s = q^{30}$ into (23.11) gives

$$S_{90}(q) = \frac{s^3}{1-s^9}, \quad S_{120}(q) = \frac{s^4}{1-s^{12}}.$$

Moreover, (23.15) implies

$$\log s_{\pm} = -30(x \pm y) = -\frac{\pi}{6}(A_{\text{deg}} \pm C_{\text{deg}}^*).$$

By the difference of squares,

$$\frac{(\log s_+)^2 - (\log s_-)^2}{20\pi} = \frac{900[(x+y)^2 - (x-y)^2]}{20\pi} \quad (23.20)$$

$$= \frac{180xy}{\pi} = \frac{\pi A_{\text{deg}} C_{\text{deg}}^*}{180}. \quad (23.21)$$

Adding these identities proves the first equality in (23.18). The second uses the inversion $\psi(f(s_{\pm})) = s_{\pm}$ proved in Section 20.

For the operator expression, the sheet projectors have rank one and $\mathcal{R}P_{\pm} = \pm P_{\pm}$. Spectral calculus gives

$$\mathcal{H}(\psi(\mathcal{V})) = \mathcal{H}(s_+)P_+ + \mathcal{H}(s_-)P_-.$$

Multiplying by $\mathcal{R}$ and taking the ordinary trace gives $\mathcal{H}(s_+) - \mathcal{H}(s_-)$. Its negative is γ . $\square$

The orientation mark $\mathcal{R}$ belongs to the functional. If that mark is held fixed and the channels of $\mathcal{V}$ are interchanged, the sign of γ changes. Evaluation of the spectrum without its labels preserves the unordered pair, whereas (23.19) preserves the oriented contribution. If the normalized trace $\tau_2 = \text{Tr}/2$ is used, the same formula reads $\gamma = -2\tau_2[\mathcal{R}\mathcal{H}(\psi(\mathcal{V}))]$.

By contrast, a simultaneous change of representation $(\mathcal{R}, \mathcal{V}) \mapsto (U\mathcal{R}U^{-1}, U\mathcal{V}U^{-1})$ preserves the functional: spectral calculus is transported by conjugation and the trace is invariant. Interchanging the response relative to a fixed orientation and jointly transporting response and orientation are therefore different operations.

This normalization is compatible with the amplifications of (20.16). For $\mathcal{V}_m = \mathcal{V} \otimes I_{9m}$ and $\mathcal{R}_m = \mathcal{R} \otimes I_{9m}$,

$$\gamma = -\frac{1}{9^m} \text{Tr} [\mathcal{R}_m \mathcal{H}(\psi(\mathcal{V}_m))]. \quad (23.22)$$

Indeed, functional calculus preserves the tensor product and the trace of I_{9m} is 9^m . Representation multiplicity is separated from the oriented evaluation.

23.5 Reconstruction from vacuum coordinates

Identities (20.6) express the same functional through normalized impedance and velocity:

$$\gamma = \mathcal{H}\left(\psi\left(\sqrt{\frac{\hat{Z}}{\hat{c}}}\right)\right) - \mathcal{H}\left(\psi\left(\frac{1}{\sqrt{\hat{Z}\hat{c}}}\right)\right). \quad (23.23)$$

The positive domain of this reconstruction is described by

$$1 < \hat{Z}\hat{c} < \frac{16}{9}, \quad \frac{9}{16} < \frac{\hat{Z}}{\hat{c}} < 1. \quad (23.24)$$

Indeed, these inequalities are equivalent to $3/4 < r_{\pm} < 1$ after substitution of the positive roots of (20.6). For orientation $y > 0$, one also retains $r_- < r_+$, equivalently $\hat{Z} < 1$. Domain points are used with the angular compatibility already established by the TPK; inversion is a subsequent reconstruction of its channels. The coordinates are the internal ones of (20.5): a change of units must be undone through (27.1) before ψ is applied.

Interchange $y \mapsto -y$ acts as $(\hat{Z}, \hat{c}) \mapsto (\hat{Z}^{-1}, \hat{c})$ and produces $\gamma \mapsto -\gamma$. In the balanced case $y = 0$, the channels coincide and (23.18) gives $\gamma = 0$. Normalized velocity preserves the product of the channels; its joint readout with impedance recovers both responses and the orientation required to evaluate the functional. In this balanced case $\mathcal{V} = rI$ has degenerate spectrum: the mark $\mathcal{R}$ is retained as representation data, although the scalar r no longer distinguishes its projectors. For simple spectrum, by contrast, $P_+ = (\mathcal{V} - r_-I)/(r_+ - r_-)$ and $P_- = I - P_+$.

The central result of this composition is factorization (23.18). The Catalan moment supplies the oriented coefficient determining f ; cubic inversion recovers each channel's argument; dodecaphasic incidence and the angular pair determine $\mathcal{H}$. The Pell scale distinguishes an exact evaluation of the collection, with the proof of (23.4). Each relation preserves its argument and operation: the endpoint constant, contractive moment and constitutive functional are linked evaluations of specified structures.

24 Constitutive normalization and spectral incidences of return

The two-sheet response and the return functional preserve two complementary kinds of information. The first comprises the product and ratio of the constitutive eigenvalues; the second comprises the heights, periods and oriented multiplicities of the incidences. This section assembles the operations that distinguish these readings. Its antecedent is the angular transport produced by APP-TRIT-TPK, with its projectors and lifted chart. Normalizations are applied after constructing that transport and preserving its orientation.

24.1 Exact separation of the common and oriented modes

Let $r_+, r_- > 0$ be the labelled eigenvalues of (20.3). Define

$$a = \sqrt{r_+ r_-}, \quad \eta = \frac{1}{2} \log \frac{r_+}{r_-}, \quad \tilde{\mathcal{V}} = a^{-1} \mathcal{V}. \quad (24.1)$$

In the chamber $x > y > 0$, $a \in (3/4, 1)$ and $\eta > 0$. The orientation marker $\mathcal{R}$ satisfies $\mathcal{R}P_{\pm} = \pm P_{\pm}$.

Proposition 24.1 (Positive factorization with preserved orientation). *The response admits the unique factorization*

$$\mathcal{V} = a \exp(\eta \mathcal{R}), \quad \det \tilde{\mathcal{V}} = 1. \quad (24.2)$$

Its four constitutive readings satisfy

$$\hat{\varepsilon} = a^2 e^{2\eta}, \quad \hat{\mu} = a^2 e^{-2\eta}, \quad \hat{Z} = e^{-2\eta}, \quad \hat{c} = a^{-2}. \quad (24.3)$$

With the orientation fixed, sheet interchange preserves a and sends η to $-\eta$.

Proof. Spectral calculus gives $\exp(\eta \mathcal{R}) = e^{\eta} P_+ + e^{-\eta} P_-$. The definitions imply $ae^{\eta} = r_+$ and $ae^{-\eta} = r_-$, thus recovering $\mathcal{V}$ exactly. The product of its eigenvalues fixes the unique positive root a ; their ratio fixes the unique real value of η . Substitution into (20.5) proves the four identities. Finally, interchanging the eigenvalues inverts their ratio and preserves their product. $\square$

Unimodular normalization preserves the entire oriented coordinate and separates the common coordinate. If a is also retained, the transformation is reversible. Publishing only $\tilde{\mathcal{V}}$ determines $\hat{Z}$, whereas $\hat{c}$ requires the archived common factor. This distinction explains the role of reciprocal sheets in a unit-product reader: such a reader realizes the unimodular component. The complete constitutive response additionally preserves its determinant.

In particular, historical readers of the form $\mu_r^{\pm} = e^{\pm 2u}$ and $\varepsilon_r^{\pm} = e^{\mp 2u}$ have product one on each sheet. Their impedance ratio is e^{4u} . The current reading in (24.3) has product $a^4 = \hat{c}^{-2}$. The historical parameter u and the coordinate η are identified only when the map between these realizations is specified; the preceding decomposition states exactly which additional factor restores the complete response.

24.2 Operational role of the Catalan moment

The reconstruction in Theorem 22.8 places Catalan's constant at the endpoint of a functional collection. The operation entering the interior response is

$$r_{\pm} = \frac{1 + s_{\pm} + s_{\pm}^2}{s_{\pm}(1 + s_{\pm})} \mathcal{D}^2 \mathcal{C}_2(s_{\pm}), \quad s_{\pm} = q_{\pm}^{30}, \quad \mathcal{D} = s \frac{d}{ds}. \quad (24.4)$$

The collection and its condition at the origin determine the moment; cubic inversion determines its two arguments from the responses. Two levels of reconstruction are thus retained: a function from a function, and arguments from evaluations of an already fixed function.

Composition with (24.1) obtains a and η from these same evaluations. Subsequent composition with (23.18) obtains the oriented Barbero–Immirzi functional. The endpoint value $\mathcal{C}_2(1)$, the responses

$\mathcal{D}^2\mathcal{C}_2(s_{\pm})$, and the functional $\mathcal{H}(s_-) - \mathcal{H}(s_+)$ therefore perform specific mathematical roles within the same chain. Reconstruction uses the complete collection and the uniqueness conditions already proved.

To specify which observation distinguishes the channels, write $u_{\pm} = \log r_{\pm}$. Variations within a common chart satisfy

$$\begin{pmatrix} d \log \hat{c} \\ d \log \hat{Z} \end{pmatrix} = \begin{pmatrix} -1 & -1 \\ -1 & 1 \end{pmatrix} \begin{pmatrix} du_+ \\ du_- \end{pmatrix}. \quad (24.5)$$

The determinant is -2 , so the joint reading recovers both variations. Each reading separately retains one linear combination. This identity expresses the complementary roles of speed and impedance, with orientation and dimensional chart fixed.

24.3 Three distinct readings of the 90/120 incidences

The flags in (23.10) determine the heights 90 and 120. The normalized defect of their cardinalities is $-1/12$. The chain in (23.12), in turn, determines the signed weights 12 and -1 . Each height has its return series $S_m(q) = q^m/(1 - q^{3m})$. Preserving these three operations allows an exact comparison of the two functionals

$$L_{\text{eq}}(q) = S_{90}(q) - S_{120}(q), \quad L_{\text{ret}}(q) = 12S_{90}(q) - S_{120}(q). \quad (24.6)$$

The first corresponds to the equal-weight antecedent; the second reads the current oriented chain.

Proposition 24.2 (Normalizations of the return pole). *As $q \uparrow 1$,*

$$\lim(1 - q)L_{\text{eq}}(q) = \frac{1}{1080}, \quad \lim(1 - q)L_{\text{ret}}(q) = \frac{1}{24}, \quad L_{\text{ret}} - L_{\text{eq}} = 11S_{90}. \quad (24.7)$$

Proof. Factoring $1 - q^{3m}$ proves $\lim(1 - q)S_m(q) = 1/(3m)$. By linearity, the two coefficients are respectively $1/270 - 1/360 = 1/1080$ and $12/270 - 1/360 = 1/24$. Subtracting the definitions proves the last identity. $\square$

The ratio 45 between these singular coefficients follows from the specified weights. The functions also retain their regular terms; the ratio of their poles does not identify their values throughout the interval. The cardinality defect, chain weight and singular coefficient are thereby assigned distinct operations and normalizations.

24.4 Fourier resolution of the dodecaphasic pattern

On the same cycle, the pattern associated with the oriented incidences is represented by

$$p_m = 12 \cos \frac{2\pi 3m}{12} - \cos \frac{2\pi 4m}{12}, \quad 0 \leq m < 12. \quad (24.8)$$

Frequencies 3 and 4 realize phase increments of 90° and 120° . The arguments of the trigonometric functions are expressed in radians. The weights come from the previously constructed chain; the harmonic representation is applied afterwards.

Proposition 24.3 (Spectral support and normalization). *For the unnormalized transform $\hat{p}_k = \sum_{m=0}^{11} p_m e^{-2\pi i k m / 12}$,*

$$\hat{p}_3 = \hat{p}_9 = 72, \quad \hat{p}_4 = \hat{p}_8 = -6, \quad \hat{p}_k = 0 \text{ at all other indices.} \quad (24.9)$$

The pattern has zero mean and $\sum_m |p_m|^2 = 870$.

Proof. Writing each cosine as the half-sum of two exponentials gives coefficients 6 at characters 3, 9 and $-1/2$ at 4, 8. Finite orthogonality, $\sum_{m=0}^{11} e^{2\pi i (j-k)m/12} = 12\delta_{jk}$, gives (24.9). The coefficient at index zero vanishes. The same orthogonality gives $\sum |p_m|^2 = (2 \cdot 72^2 + 2 \cdot 6^2)/12 = 870$. $\square$

For the transform divided by 12, the coefficients are 6 and $-1/2$; under unitary normalization, they are $12\sqrt{3}$ and $-\sqrt{3}$. Their magnitude ratio $12 : 1$ is preserved in both cases. The exact sequence

$$2(p_0, \dots, p_{11}) = (22, 1, -23, -2, 25, 1, -26, 1, 25, -2, -23, 1) \quad (24.10)$$

allows the transform to be checked with exact algebraic arithmetic.

The zero mean of the pattern comes from character cancellation on the cycle. The defect $-1/12$ comes from the normalized incidence count; $1/24$ comes from the pole of the return series. Keeping these domains specified explains what each reading preserves before composition with the angular pair and the Barbero–Immirzi functional.

25 Differential reciprocity and recovery of constitutive channels

The vacuum channels, the Barbero–Immirzi functional, and the dodecaphasic record belong to one chain of constructions. The following result assembles their inverses: normalized speed retains the common component of the response, while the oriented Barbero reading determines its distribution between sheets. The composition recovers the angular pair and, on the generated image with its dimensional section, the action coefficient. Retaining the regional coordinates also gives the periodic reading of K .

The preceding construction of APP–TRIT–TPK and its enriched state remains the antecedent. The parameters used below are the angular outputs already obtained; the analytic proofs describe their realization and recovery operations. The labelled projectors $P_{\pm}$, $\mathcal{R} = P_+ - P_-$, orientation, and angular lift remain explicit.

25.1 Spectral potential and constitutive reciprocity

On the cone $x > |y|$, let

$$T = e^{-xI-y\mathcal{R}}, \quad s_{\pm} = e^{-30(x\pm y)}, \quad f(s) = \frac{1-s^3}{1-s^4}, \quad V = (I - T^{90})(I - T^{120})^{-1} = r_+P_+ + r_-P_-,$$

$$r_{\pm} = f(s_{\pm}), \quad \hat{c} = (r_+r_-)^{-1}, \quad \hat{Z} = r_-/r_+, \quad c_{\ell} = \log \hat{c}, \quad z_{\ell} = \log \hat{Z}.$$

Invertibility of $I - T^{120}$ follows from $0 < s_{\pm} < 1$. The extended polynomial quotient is $f(s) = (1 + s + s^2)/(1 + s + s^2 + s^3)$; its endpoint values are 1 and 3/4.

Theorem 25.1 (Potential of the normalized response). *The series*

$$\mathcal{F}(x, y) = \sum_{\sigma=\pm 1} \sum_{n \geq 1} \left[\frac{e^{-120n(x+\sigma y)}}{120n^2} - \frac{e^{-90n(x+\sigma y)}}{90n^2} \right] \quad (25.1)$$

defines a strictly concave real-analytic potential, with

$$\nabla \mathcal{F} = (c_{\ell}, z_{\ell}), \quad \partial_y \log \hat{c} = \partial_x \log \hat{Z}.$$

Its normalization is $\mathcal{F} \rightarrow 0$ as $x - |y| \rightarrow \infty$.

Proof. On each compact subset of the cone there is $\epsilon > 0$ with $x \pm y \geq \epsilon$. Every fixed-order derivative of a summand is bounded by a power of n times $e^{-90n\epsilon}$; the sum converges uniformly. Termwise differentiation and $\sum_{n \geq 1} z^n/n = -\log(1 - z)$ give the stated components. For concavity, set

$$g(w) = \log f(e^{-w}), \quad u = 30(x + y), \quad v = 30(x - y), \quad a = g'(u), \quad b = g'(v).$$

$$g'(w) = \frac{3}{e^{3w} - 1} - \frac{4}{e^{4w} - 1} > 0, \quad D(c_{\ell}, z_{\ell}) = -30 \begin{pmatrix} a+b & a-b \\ a-b & a+b \end{pmatrix}.$$

The inequality follows because $t/(e^{tw} - 1)$ is strictly decreasing for $t > 0$: its derivative has numerator $e^{tw}(1 - tw) - 1 < 0$. The matrix eigenvalues are $-60a$ and $-60b$, both negative, and its determinant is $3600ab > 0$. Symmetry proves differential reciprocity. The exponential bound gives the normalizing limit. $\square$

The same spectral moment $\mathcal{D}_2(z) = \sum_{n \geq 1} z^n/n^2$ expresses (25.1) through real contractive evaluations. The oriented character already constructed satisfies $\mathcal{C}_2(s) = \text{Im } \mathcal{D}_2(is)$, $\mathcal{C}_2(1) = G_{\text{Cat}}$. Absolute convergence extends to $s = 1$; for differentiated series that cease to converge there, the radial Abel limit is retained. Real and oriented evaluations share the depth index and weighting while retaining their respective arguments. Under amplification by $T \otimes I_{9^m}$, the trace divided by 9^m preserves these readings, since every eigenvalue is repeated exactly 9^m times. This identity concerns the specified amplification.

Proposition 25.2 (Oriented sensitivity and convexity). *For fixed $x > 0$, $y \mapsto c_\ell(x, y)$ is even and strictly convex, with its unique minimum at zero.*

Proof.

$$g''(w) = -\frac{9}{4 \sinh^2(3w/2)} + \frac{16}{4 \sinh^2(2w)} < 0.$$

Indeed, $t/\sinh(tw/2)$ decreases: its derivative has the sign of $\sinh v - v \cosh v < 0$. Since $c_\ell = -g(u) - g(v)$, one obtains $\partial_y^2 c_\ell = -900[g''(u) + g''(v)] > 0$. Symmetry under $y \mapsto -y$ fixes the minimum. The local expansion is

$$c_\ell = -2g(30x) - 900g''(30x)y^2 + O(y^4), \quad z_\ell = -60g'(30x)y + O(y^3).$$

□

Speed retains channel separation at second order; impedance retains its orientation at first order. Evaluating the mean channel before applying the reader removes precisely those contributions.

25.2 Global inversion through speed and Barbero–Immirzi

The inverse $\psi = f^{-1} : (3/4, 1) \rightarrow (0, 1)$ is defined by the unique root of

$$rs^3 + (r-1)(1+s+s^2) = 0.$$

Uniqueness follows from

$$f'(s) = -\frac{s^2(s^2 + 2s + 3)}{(1+s+s^2+s^3)^2} < 0.$$

Retain the return functional already derived from hexad–octad incidence:

$$\mathcal{H}(s) = \frac{12s^3}{1-s^9} - \frac{s^4}{1-s^{12}} - \frac{(\log s)^2}{20\pi}, \quad F_B = \mathcal{H} \circ \psi.$$

Theorem 25.3 (Oriented constitutive recovery). *The map*

$$(r_+, r_-) \mapsto \left(\hat{c} = \frac{1}{r_+ r_-}, \quad \gamma = F_B(r_-) - F_B(r_+) \right)$$

is a real-analytic diffeomorphism from $(3/4, 1)^2$ onto $(1, 16/9) \times \mathbb{R}$. This statement describes the analytic domain of the reader; the actually generated image additionally retains the TPK compatibility conditions.

Proof.

$$\mathcal{H}'(s) = \frac{36s^2(1+2s^9)}{(1-s^9)^2} - \frac{4s^3(1+2s^{12})}{(1-s^{12})^2} - \frac{\log s}{10\pi s} > 0.$$

The ratio of the first positive term to the magnitude of the second is

$$\frac{9}{s} \frac{1+2s^9}{1+2s^{12}} \left(\frac{1-s^{12}}{1-s^9} \right)^2 > 9,$$

and the last summand is also positive. The endpoint asymptotics are $\mathcal{H}(s) \sim -(\log s)^2/(20\pi)$ at zero and $\mathcal{H}(s) \sim 5/[4(1-s)]$ at one. Hence $F'_B < 0$, with limits $+\infty$ at $3/4$ and $-\infty$ at 1 .

For fixed $c = \hat{c}$, set $z = \log \hat{Z}$. Then

$$r_+ = c^{-1/2} e^{-z/2}, \quad r_- = c^{-1/2} e^{z/2}, \quad |z| < a(c) = \min\{\log c, \log(16/(9c))\}.$$

The function $\mathcal{G}_c(z) = F_B(r_-) - F_B(r_+)$ satisfies

$$\mathcal{G}'_c(z) = \frac{1}{2}[r_- F'_B(r_-) + r_+ F'_B(r_+)] < 0.$$

At the upper endpoint, $r_- \rightarrow 1$, or $r_+ \rightarrow 3/4$, or both, and $\mathcal{G}_c \rightarrow -\infty$. At the lower endpoint it tends to $+\infty$. The intermediate value theorem and monotonicity give a unique solution for each γ . The nonzero derivative gives the analytic inverse by the implicit function theorem. The minimum in $a(c)$ only describes the boundary; the argument takes place in its open domain. □

25.3 Angular recovery, action, and periodic record

Once $r_{\pm}$ have been obtained, cubic inversion gives $s_{\pm}$. With the lift and orientation retained,

$$x = -\frac{1}{60} \log(s_+ s_-), \quad y = \frac{1}{60} \log \frac{s_-}{s_+}, \quad A_{\text{deg}} = -\frac{3}{\pi} \log(s_+ s_-), \quad C_{\text{deg}}^* = \frac{3}{\pi} \log \frac{s_-}{s_+}.$$

On the generated positive section $A_{\text{deg}} = 1000\alpha$, $C_{\text{deg}}^* = 2(\eta_{\text{ret}} + \alpha)$, one recovers

$$\alpha = A_{\text{deg}}/1000, \quad \eta_{\text{ret}} = C_{\text{deg}}^*/2 - \alpha, \quad \hbar_{\text{ret}} = s_0 \eta_{\text{ret}}, \quad s_0 := 10^{-34} \mathcal{U}_S.$$

The scale s_0 and the projectors are retained antecedents. Under conjugate orientation, the action label is transported as well. Recovery is an inverse on the corresponding image.

Proposition 25.4 (Composition with the twelve-block periodic reading). *The periodic reading prolongs the twelve blocks of K , while the regional coordinates retain their complete coinductive generation. With α_A denoting the previously defined arithmetic-closure coordinate,*

$$\kappa_{\text{per}} = \pi_{\text{HMT}} + e_{\text{HMT}} - \varphi_{\text{HMT}} - 4 - \alpha_A, \quad B = 1000, \quad N_K = (B^{12} - 1)\kappa_{\text{per}}.$$

On the image of the generated record, N_K is an integer and its twelve blocks, with fixed origin and retained leading zeros, are recovered exactly by

$$K_j = \left\lfloor \frac{N_K}{B^{12-j}} \right\rfloor - B \left\lfloor \frac{N_K}{B^{13-j}} \right\rfloor, \quad 1 \leq j \leq 12.$$

Proof. Successive Euclidean division of N_K in base B uniquely determines $K_j \in \{0, \dots, B-1\}$ and gives $N_K = \sum_{j=1}^{12} K_j B^{12-j}$. Periodic repetition of these blocks sums the geometric series $N_K \sum_{m \geq 1} B^{-12m} = N_K/(B^{12}-1)$. Both compositions are inverse. The finite reading N_K/B^{12} retains a different normalization from the periodic one. With approximate data, recovery requires a certified interval for N_K containing a unique integer. $\square$

This composition assembles constitutive restitution with recovery of the already constructed record. Exceptional and temporal selection determines K beforehand; the inverse explains which information its readers can restore. Its scope is the ordered twelve-coordinate record with the stated antecedents.

25.4 Stability bounds in the contractive domain

For $w > 0$, write $a(w) = g'(w)$. With $s = e^{-w}$,

$$a(w) = \frac{s^3(s^2 + 2s + 3)}{(1 + s + s^2)(1 + s + s^2 + s^3)}, \quad \frac{1}{2}e^{-3w} < a(w) < 3e^{-3w}.$$

The bounds follow by comparing the positive polynomials in the quotient; $g'' < 0$ proves its decrease. On a convex domain where $0 < m_0 \leq 30(x \pm y) \leq M_0 < \infty$, the map $\mathcal{C} = (c_\ell, z_\ell)$ satisfies, in the Euclidean norm,

$$60a(M_0)\|p - q\| \leq \|\mathcal{C}(p) - \mathcal{C}(q)\| \leq 60a(m_0)\|p - q\|.$$

For the upper bound, integrate the Jacobian along the segment. For the lower bound, take the scalar product with $p - q$, use negative definiteness of the Jacobian, and then apply Cauchy–Schwarz. The induced norm of the inverse differential is $1/[60 \min\{a(u), a(v)\}]$. In deep contraction it can grow exponentially even when the relative condition number remains close to one. Exact uniqueness and robustness to error are thus specified by distinct, compatible criteria.

26 Charge section and electrical quantization relations

The constitutive response establishes a relation between the two sheets and a relative propagation scale. The charge section is obtained by composing this response with the fine-structure constant and the action, both constructed in the preceding chapters. This order has a precise consequence: charge does not retrospectively determine the dodecaphase record. It is the record, together with the action and the vacuum response, that determines the charge section on the corresponding line.

26.1 Positive determination of the charge section

Consider the positive sections $\hbar_a$, with $a \in \{\text{pre}, \text{ret}\}$, and $h_a = 2\pi\hbar_a$. The impedance Z_{vac} has already been constructed through (20.18). The quotient h_a/Z_{vac} belongs to the quadratic charge line. This dimensional compatibility permits the following definition without yet choosing an SI chart.

Definition 26.1 (Positive charge section). The charge section corresponding to action orientation a is

$$e_a = \left(\frac{2\alpha h_a}{Z_{\text{vac}}} \right)^{1/2}. \quad (26.1)$$

The root is taken on the positive charge half-line. Its conjugate section is $-e_a$; the sign of the latter is not identified with the action return index.

Proposition 26.2 (Existence, uniqueness, and compatibility of the coupling). *The equation $Z_{\text{vac}} e_a^2 = 2\alpha h_a$ determines a unique positive section. In the constitutive realization, one also has*

$$\alpha = \frac{Z_{\text{vac}} e_a^2}{2h_a} = \frac{e_a^2}{4\pi\epsilon_{\text{vac}} \hbar_a c_{\text{vac}}}. \quad (26.2)$$

Proof. Positivity of α , h_a , and Z_{vac} guarantees a unique positive root. The constitutive identities give $\epsilon_{\text{vac}} c_{\text{vac}} = Z_{\text{vac}}^{-1}$. Substituting this equality and $h_a = 2\pi\hbar_a$ into the first expression of (26.2) yields the second. $\square$

The final expression is the customary form of electromagnetic coupling in a physical chart. Here it appears as a subsequent identity of the constructed sections. The proof does not recalculate α from a measured charge: it establishes the algebraic coherence of the preceding result with the section just defined. The physical identification additionally requires retaining the unit chart and coupling regime used in the comparison. An identity between quantities and an experimental comparison serve different functions in that recognition.

26.2 Resistance, conductance, flux, and frequency

The following four quantities use the same section e_a . Their dependencies can be fully resolved before any decimal is evaluated.

Theorem 26.3 (Composition of the electrical quanta). *The sections*

$$R_K = \frac{h_a}{e_a^2} = \frac{Z_{\text{vac}}}{2\alpha}, \quad G_0 = \frac{2e_a^2}{h_a} = \frac{4\alpha}{Z_{\text{vac}}}, \quad (26.3)$$

$$\Phi_{0,a} = \frac{h_a}{2e_a} = \sqrt{\frac{h_a Z_{\text{vac}}}{8\alpha}}, \quad K_{J,a} = \frac{2e_a}{h_a} = \Phi_{0,a}^{-1} \quad (26.4)$$

satisfy exactly

$$R_K G_0 = 2, \quad G_0 Z_{\text{vac}} = 4\alpha, \quad K_{J,a} \Phi_{0,a} = 1, \quad K_{J,a}^2 R_K = \frac{4}{h_a}. \quad (26.5)$$

Proof. Substituting $e_a^2 = 2\alpha h_a/Z_{\text{vac}}$ into the first two definitions cancels h_a and gives (26.3). Squaring the positive section $h_a/(2e_a)$ gives $h_a Z_{\text{vac}}/(8\alpha)$; positivity fixes the root in (26.4). Its inverse is $2e_a/h_a$. The first three products in (26.5) reduce, respectively, to 2, 4α , and 1. Finally, $(4e_a^2/h_a^2)(h_a/e_a^2) = 4/h_a$. $\square$

The physical realization recognizes in these sections the von Klitzing resistance, the conductance quantum, the flux quantum, and the Josephson constant. The factor 2 in the conductance belongs to the normalization of the quantity defined here; identifying it with a multiplicity of channels in a material system requires specifying that representation. The four equations do not constitute four independent generators: they share α , the action, and the vacuum response. Precisely this dependency produces the cross-identities and makes it possible to check an evaluation without altering the operations from which it arose.

26.3 Action return and transport of the sections

The quotient $R_{\text{act}} = h_{\text{pre}}/h_{\text{ret}}$, obtained before the angular pair, preserves the distinction between the two action sections. It does not change the sheet assignments of the constitutive response. These are two distinct operations: action return and channel interchange.

Proposition 26.4 (Covariance under return). *With common α and Z_{vac} , one has*

$$\frac{e_{\text{pre}}}{e_{\text{ret}}} = R_{\text{act}}^{1/2}, \quad R_K^{\text{pre}} = R_K^{\text{ret}}, \quad G_0^{\text{pre}} = G_0^{\text{ret}}, \quad (26.6)$$

$$\frac{\Phi_{0,\text{pre}}}{\Phi_{0,\text{ret}}} = R_{\text{act}}^{1/2}, \quad \frac{K_{J,\text{pre}}}{K_{J,\text{ret}}} = R_{\text{act}}^{-1/2}. \quad (26.7)$$

Proof. The first identity follows by dividing the two versions of (26.1) and taking the positive root. In (26.3), the action has already cancelled. The other two identities follow from the powers $h_a^{1/2}$ and $h_a^{-1/2}$ in (26.4). $\square$

Resistance and conductance preserve the constitutive quotient, whereas flux and frequency also retain the action orientation. This separation gives a structural reading of their dependencies: some quantities are invariants of the sheet quotient, while others additionally measure the action section used. The metrological comparison must specify which section it evaluates; the numerical proximity of the two actions does not authorize their identification.

27 Constitutive evaluation and dimensional realization

Evaluation now brings together the preceding constructions in their effective order. The regional expansions of π , φ , and e arise from nonadic refinement; the register K preserves its twelve positions. Its periodic readout determines the positional section of α . The action character and regional return act on it, followed by the angular pair and, finally, the constitutive response. This evaluation follows the selection of rules: none of its digits is used to choose a channel.

27.1 Evaluation of the internal chain

To reproduce the table, one uses the thousand-decimal regional expansions generated by the nonadic certificate and the integer N_K in (8.18). The readout used is

$$\alpha = \pi + e - \varphi - 4 - \frac{N_K}{1000^{12} - 1}.$$

The finite twelve-block section and the prolonged section have their definitions and exact difference in the register chapter. Here the latter is held fixed throughout evaluation. Readouts are not interchanged as precision increases, nor is the analytic operator of the second route replaced by a decimal coincidence with the first.

Object	Decimal evaluation (24 significant digits)
α	0.00729735256928380099728511 ...
H_5	1.05457181764615446962582 ...
η_{ret}	1.05457181691503906113369 ...
A_{deg}	7.29735256928380099728511 ...
C_{deg}^*	2.12373833896864572426195 ...
q_+	0.848377942576404374945737 ...
q_-	0.913660151150557285295519 ...
r_+	0.999999628548249363178695 ...
r_-	0.999724137189518364131517 ...
$\hat{\varepsilon}$	0.999999257096636702760442 ...
$\hat{\mu}$	0.999448350479326935089982 ...
$\hat{Z}$	0.999724508538937215455554 ...
$\hat{c}$	1.00027631048615752610639 ...
γ	0.274007672694459274747255 ...
G_{Cat}	0.915965594177219015054604 ...
ρ	0.267949192431122706472554 ...

The calculation uses 120 working digits; 24 significant digits are displayed to facilitate comparison between quantities of different orders. The 120 digits are an evaluation precision, not an experimental uncertainty or, by themselves, an interval error certificate. The package preserves the generated inputs, their hashes, the operations, and the residuals of the checked identities. The algebraic proofs in the main text establish these identities independently of this table.

The meaning of the final four constitutive coordinates is preserved under evaluation: $\hat{\varepsilon}$ and $\hat{\mu}$ are the quadratic responses per sheet, $\hat{Z}$ is their relative orientation, and $\hat{c}$ is the inverse of their product. The proximity of some digits to unity expresses the small defect of the 3–4 quotient on these channels. Its origin is seen exactly in

$$1 - f(s) = \frac{s^3}{1 + s + s^2 + s^3}.$$

The formula allows the response to be studied before a dimensional quantity is assigned to it: the sign of the defect, its cubic order, and its monotonicity are consequences of the same operator.

27.2 Unit charts and covariance

Let $\mathcal{L}_S$, $\mathcal{L}_Q$, and $\mathcal{L}_C$ be the action, charge, and velocity lines. A positive physical chart evaluates the sections u_S , u_Q , and u_C in action, charge, and velocity units. The realization (20.18) then produces an impedance, a velocity, a permeability, and a permittivity, with their respective dimensions. The coordinates in the preceding table remain dimensionless.

Proposition 27.1 (Covariance of the constitutive realization). *If the reference sections change by $u'_S = au_S$, $u'_Q = bu_Q$, $u'_C = du_C$, with $a, b, d > 0$, the coordinates representing the same quantities satisfy*

$$\hat{Z}' = a^{-1}b^2\hat{Z}, \quad \hat{c}' = d^{-1}\hat{c}, \quad \hat{\mu}' = a^{-1}db^2\hat{\mu}, \quad \hat{\varepsilon}' = adb^{-2}\hat{\varepsilon}.$$

The relations $\hat{\mu}'\hat{\varepsilon}' = (\hat{c}')^{-2}$ and $\hat{\mu}'/\hat{\varepsilon}' = (\hat{Z}')^2$ are preserved.

Proof. Equate each quantity in (20.18) to its expression in the new basis and solve for its coordinate. The factors in the product are d^2 , and those in the quotient are $a^{-2}b^4$, exactly those required by the transformed coordinates of c and Z . $\square$

This covariance distinguishes two acts: constructing a section and expressing that section in chosen units. The spectral theorem fixes the coordinates in the declared internal normalization. An SI comparison must also exhibit the evaluation of the three reference sections. Fitting them using the quantities to be reproduced would calibrate the chart, not independently verify those quantities.

To reconstruct the channels from coordinates expressed in the primed chart, undo that change before using the cubic root:

$$\hat{Z} = ab^{-2}\hat{Z}', \quad \hat{c} = d\hat{c}', \quad \hat{\mu} = ad^{-1}b^{-2}\hat{\mu}', \quad \hat{\varepsilon} = a^{-1}d^{-1}b^2\hat{\varepsilon}'. \quad (27.1)$$

For example, the inputs to the spectral inverse are $r_+ = (\hat{Z}\hat{c})^{-1/2}$ and $r_- = (\hat{Z}/\hat{c})^{1/2}$, calculated after (27.1). Directly inserting the primed coordinates into those expressions would change the operator normalization and could take them outside the interval $(3/4, 1)$. Dimensional covariance preserves physical quantities; cubic inversion reconstructs the transport in the internal chart in which it was defined.

27.3 Identity checks and scope of comparison

Reproducible comparison is organized by relations. First, $\hat{\mu}\hat{\varepsilon}\hat{c}^2 = 1$ and $\hat{\mu}/(\hat{\varepsilon}\hat{Z}^2) = 1$ check constitutive consistency. Second, cubic inversion recovers the original arguments and angular coordinates. Finally, evaluating Barbero through the angular functional and through functional calculus of the response agrees by the identity proved in the previous chapter. Interchanging sheets must reverse the oriented contribution and preserve the common mode; this check detects a loss of labeling that scalar products would not reveal.

Comparison of the reduced Planck constant with reference tables is preserved in the preceding action development. For the four vacuum quantities, the present table evaluates their internal coordinates. It is not presented as a table of SI matches: that additional assertion requires an independent numerical chart for u_S, u_Q, u_C . The proved constitutive result, including its inversion, remains complete on its domain and explicitly preserves the dimensional interface required to make that comparison.

28 Conclusions

The central result is a reversible constitutive determination on the contractive domain: the two oriented channels of a transport produce four related coordinates, and those coordinates recover the angular pair. Structure first fixes the response equation; evaluation determines the values while preserving the orientation needed to reconstruct their spectral antecedents. This two-way direction, constructive and inverse, gives precise content to the thesis of structure, equation and value.

The response preserves the angular pair

The incidences 90 and 120 determine a unique solution of the completion equation for the fixed transport. Its scalar function is a strictly decreasing bijection from $(0, 1)$ onto $(3/4, 1)$. Consequently, the cubic equation (20.7) has exactly one admissible root for each channel response. Inversion is not a numerical coincidence: it reconstructs the two contractions and, through their logarithms, the angular pair.

The four vacuum coordinates satisfy

$$\hat{\mu}\hat{\varepsilon} = \hat{c}^{-2}, \quad \hat{\mu}/\hat{\varepsilon} = \hat{Z}^2.$$

Within the fixed product, quotient and positivity rules, these identities determine the quadratic evaluations. Their common dependency also appears under sheet interchange: permittivity and permeability are permuted, impedance is inverted and speed is preserved. The symmetric product and oriented quotient thus transmit complementary information. Recovery preserves the sheet labels and internal normalization; the source state additionally preserves route, boundary and memory.

The factorization $\mathcal{V} = a \exp(\eta\mathcal{R})$ identifies these two kinds of information within the operator itself. The positive factor $a = \sqrt{\det \mathcal{V}}$ determines normalized speed; the unimodular component determines impedance. Preserving both reconstructs the response. This separation explains the relation between unit-product reciprocal sheets and the four vacuum readings: normalization separates the common mode, which remains necessary to recover the complete operator.

The inverse links responses evaluating different functions of the same channels. The ratio of their exponents reconstructs the anisotropy of the action ellipse and the relative LC impedance, as proved in Proposition 20.7. The action section and the frequency and impedance references complete their dimensional quantities. The identity $c_{\text{int}} = c_{\text{vac}}$ in turn establishes that the kinematic and constitutive readouts realize the same speed section in a common chart. The normalized trace preserves this readout under nonadic amplification.

Composition and reconstruction across sectors

The transported law $r \odot t = f(\psi(r)\psi(t))$ expresses transport composition in constitutive coordinates. Its identity is $3/4$ in the boundary extension, and $-\log \psi(r)/30$ converts composition into addition. Common projectors preserve the operator realisation and sheet labels. The velocity of a single state, in turn, retains the ordinary product of its two channels. The two operations are distinguished by their arguments and by the exact proof of their relationship.

Higgs–impedance reconstruction establishes another explicit connection. The gauge couplings recover the common contraction and the electromagnetic coupling; the Higgs self-coupling recovers the oriented coordinate. Within the tree-level realisation, these data determine the two channels and their vacuum responses. Domain inequalities and genealogical compatibility specify which triples belong to the realisation under consideration. Retaining marked projectors also allows reconstruction of the operator; the enriched history retains its additional provenance data.

The cycle relates response, moment and orientation

The dodecaphasic realization explains the provenance of the rational function through a determinant quotient in dimensions 3 and 4. The rank difference produces $-1/12$ as a normalized trace on this finite domain. The quarter-turn plane preserves other information: its operator satisfies $J^2 = -I$, and its resolvent coefficient retains the orientation sign. The two logarithmic integrations of that coefficient, with their initial conditions, determine the Catalan moment at unlimited depth.

The trace table b_n reconstructs $-\log f(s)$ and produces convergent moments for $\operatorname{Re} z > 0$, with an explicit remainder bound. Its value at depth one is $\log(4/3)$, consistent with the constitutive endpoint. The Mellin representation and the distinction between diagonal and tensorial products retain the operations preceding the scalar evaluation of each moment.

The integral reconstruction in Theorem 22.8 completes the functional correspondence in both directions:

$$\mathcal{C}_2(s) = \int_0^s \log \frac{s}{t} \frac{1+t}{1+t+t^2} f(t) dt, \quad f(s) = \frac{1+s+s^2}{s(1+s)} \mathcal{D}^2 \mathcal{C}_2(s).$$

The condition $\mathcal{C}_2(s) \rightarrow 0$ at the origin removes all additive and logarithmic freedom. The endpoint value G_{Cat} , the oriented kernel, and the constitutive response are thus placed in a single collection with explicit reconstruction operations. The arguments $s_{\pm}$ retain their angular determination; the normalized trace preserves the speed of the same state when linking it to action and length. This correspondence specifies the structural content preserved in passing from the collection to its values.

The Catalan–Pell identity specifies an algebraic evaluation of this collection:

$$\mathcal{C}_2(2 - \sqrt{3}) = \frac{2}{3} G_{\text{Cat}} - \frac{\pi}{12} \log(2 + \sqrt{3}).$$

The coefficient $2/3$ arises from the calendar's odd classes and the depth weight; the first two logarithmic derivatives at this section take the values $\pi/12$ and $1/4$. Reciprocity of the Pell unit links expansion and contraction, while the two constitutive arguments remain determined by the angular pair. This relation keeps the collection, its readouts and the sections of evaluation explicit.

The Barbero–Immirzi functional is subsequently expressed as an oriented difference of the same function of the reconstructed channels:

$$\gamma = \mathcal{H}(\psi(r_-)) - \mathcal{H}(\psi(r_+)).$$

Incidence and dodecaphasic return determine its terms; constitutive inversion transports them to the vacuum response. With the orientation mark fixed, channel interchange reverses the sign of γ ; joint conjugation preserves the functional. This distinction proves that its content exceeds an unlabelled spectrum and is preserved under normalized amplifications.

Section 25 assembles the spectral potential of constitutive response, global channel recovery through normalized speed and Barbero–Immirzi, and stability bounds. On the generated image, retaining labels and bases, the composition recovers the angular pair, action section, and periodic reading of the dodecaphasic record.

Harmonic resolution of the dodecaphasic pattern preserves exactly modes 3, 9, 4, 8, with unnormalized coefficients 72, 72, -6 , -6 . The magnitude ratio $12 : 1$ expresses the oriented multiplicities. The incidence defect $-1/12$ and singular coefficient $1/24$ remain assigned to their respective operations. Explicit comparison with the equal-weight reader, whose coefficient is $1/1080$, specifies how the oriented chain enters the normalization of return while preserving the complete functionals beyond their poles.

From arithmetic balance to electrical relations

The vacancy discrepancy provides a centred polarization whose magnitude remains below one unit and whose increment satisfies an exact temporal balance. Changes between floor and ceiling conventions are recorded by a boundary term. A charge section and an oriented duration realize that balance as centred charge and current, preserving its telescoping law.

The electronic prefactor likewise preserves its antecedent structure. Enumeration of the fibres in D_9^3 determines the tritic singular mode of value $1/2$; the complete spectrum of their pairing proves

$$\| [R_{2,3}, R_{1,3}] \| = \sqrt{3}/4.$$

Its quadratic defect $3/4$ coincides with the limit of the $90/120$ sector quotient. This equality relates results of specified operators, each with its own space and action: the prefactor is an arithmetic incompatibility norm, whereas the interior response depends on its contractive channels.

The positive charge section determined by $Z_{\text{vac}}e_a^2 = 2\alpha h_a$ combines coupling, action and impedance. The von Klitzing, conductance, flux and Josephson relations satisfy exactly

$$R_K G_0 = 2, \quad G_0 Z_{\text{vac}} = 4\alpha, \quad K_{J,a} \Phi_{0,a} = 1, \quad K_{J,a}^2 R_K = 4/h_a.$$

Action return preserves resistance and conductance; charge and flux transform with $R_{\text{act}}^{1/2}$ and Josephson with $R_{\text{act}}^{-1/2}$. These dependencies jointly explain which quantities retain the action and which preserve only its constitutive quotient.

Dimensional realization preserves all these identities when the action, charge and speed bases change. The resulting dimensionless values belong to the construction's internal chart; metrological recognition requires independent evaluation of the dimensional sections and the corresponding physical regime. The established mathematical scope is the positive, invertible response, its compositions and covariance on the stated domains.

Together, these results show how a single APP-TRIT-TPK genealogy preserves the relations required to pass from paths and arithmetic operations to channels, moments, constitutive response and electrical relations. The values thereby acquire an explained provenance and recoverable information: they express the operations producing them and reveal, within their representation, the structure retained in them.

A Dependencies, procedures and reproduction

The constructions used are developed within this article: the dodecaphasic register in Section 8, positional closure in Section 9, action in Section 14, and the angular pair in Section 14.3. The constitutive response in Section 20 composes these outputs while preserving the sheet and orientation labels.

A.1 Order of reproduction

Reproduction distinguishes the coordinate producer from the subsequent evaluators. The former enumerates admissible states and applies the rules of APP, TRIT and TPK. Its regional outputs are legitimate inputs to the subsequent operators of the same development. The vacuum evaluation program is not presented as a substitute for that producer.

The package reading order is: kernel definitions; regional enumeration and refinement; signed register and transformation of K ; positional closure of α ; action section and return; angular pair; contractive channels; constitutive operator; cyclic moments and the Barbero functional; dimensional sections and subsequent comparison.

Each stage preserves its domain. A regional-digit file carries the producer's fingerprint, not a metrological attribution. The twelve-component tuple of K has its source in the register chapter. The action coefficients, 90/120 incidence and $12 : -1$ multiplicities are read in their proofs preceding evaluation. The local program reproduces those compositions at the declared precision.

A.2 Three checks with different functions

Documentary preservation checks file identity and continuity of the copies. Mathematical proof establishes identities and limits in their domains. Numerical execution checks a particular evaluation, its stability and the residuals of the identities. None of these three checks replaces the others.

The accompanying package includes a source manifest, the LaTeX texts, the regional data used and the verification programs. The proofs of inversion, monotonicity, moments, transport and covariance are given in the body of the article. Documentary sources are cited for provenance; readers can check the arguments developed here without using a target numerical result as a generator.

The positional closure of α is proved in Proposition 9.3. Theorem 9.13 identifies its prefixes with those of the correlated analytic representation of the same state, through the explicit recurrence and compatible cylinders. Constitutive inversion in Section 20 uses the specified internal normalization: changes of units are undone before that inverse is applied. These conditions delimit the results used by the subsequent compositions.

B Exhaustive enumeration and finite determination of transport operators

This appendix develops two finite operations with different starting data. The first enumerates the initial conditions of the two cursors and constructs their emissions through the declared arithmetic recurrence. The second determines transport operators from the regional transition register. Separating their domains specifies what each proof produces: enumeration constructs the emission catalogue; matrix reconstruction determines the operators compatible with the register. The reconstruction theorem is not used as a proof of the prior production of that register.

The primitives of the first operation are the support I_9^2 , the four cardinal orientations, the tritic selector, the pre-displacement readout, orientation reversals and the emission rule. Their definitions are brought together below. Those of the second operation are the four transition matrices in (B.7). Their unique reconstruction is proved in Theorem B.2; the resulting calendar is given in (B.9). These determinations preserve the starting data specified for each operation.

B.1 Complete domain of initial conditions

Let $I_9 = \{0, \dots, 8\}$ and let

$$\mathcal{D} = \{N, E, S, O\}, \quad v_N = (-1, 0), \quad v_E = (0, 1), \quad v_S = (1, 0), \quad v_O = (0, -1).$$

An initial cursor condition is $s = (i, j, d) \in \mathcal{S} = I_9^2 \times \mathcal{D}$. The two cursors are independent in the initial domain

$$\Omega = \mathcal{S}_+ \times \mathcal{S}_\times, \quad |\mathcal{S}| = 9^2 \cdot 4 = 324, \quad |\Omega| = 324^2 = 104\,976. \quad (\text{B.1})$$

Index i corresponds to the positive APP label $i + 1$. For positive integers, positive reduction is $\rho_9(n) = 1 + [(n - 1)]_9$, where $[a]_b$ denotes the representative in $\{0, \dots, b - 1\}$. The residual evaluations of the two sheets are

$$\sigma(i, j) = \rho_9(i + j + 2), \quad \varpi(i, j) = \rho_9((i + 1)(j + 1)).$$

The finite multiplicative readout uses

$$\begin{array}{c|cccccccc} a & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ \hline \ell_9(a) & 0 & 1 & 0 & 2 & 5 & 0 & 4 & 3 & 0 \end{array}. \quad (\text{B.2})$$

On units, ℓ_9 is the discrete exponent in base 2. On 3, 6, 9, the value 0 constitutes the declared extension of the emission observable; membership in that sector remains in the trajectory register.

For $1 \leq t \leq 54$, define

$$r_t = 1 + [(t - 1)]_9, \quad \tau_t = \begin{cases} +1, & r_t \in \{1, 4, 7\}, \\ -1, & r_t \in \{2, 5, 8\}, \\ 0, & r_t \in \{3, 6, 9\}. \end{cases}$$

Each cursor preserves its position z_t immediately before step t and its orientation vector v_t . The additive cursor is active when $\tau_t = +1$; the multiplicative cursor, when $\tau_t = -1$. For each of them, with activity indicator a_t , the update is

$$z_{t+1} = [z_t + a_t v_t]_9, \quad v_{t+1} = \begin{cases} -v_t, & t \in \{27, 54\}, \\ v_t, & t \notin \{27, 54\}. \end{cases} \quad (\text{B.3})$$

The readout at step t is taken at z_t , before displacement. The neutral event preserves both positions. This convention fixes unambiguously the window endpoints and the effect of cardinal reversals.

In the windows $W_m = \{9m - 8, \dots, 9m\}$, $1 \leq m \leq 6$, let

$$\begin{aligned} S_m(s_+) &= \sum_{\substack{t \in W_m \\ \tau_t = +1}} \sigma(z_t^+), & E_m(s_\times) &= \sum_{\substack{t \in W_m \\ \tau_t = -1}} \ell_9(\varpi(z_t^\times)), \\ s_m &= [S_m]_{10}, & e_m &= [E_m]_{10}. \end{aligned} \quad (\text{B.4})$$

Each window contains three events of each type. The signatures s and e are the six components of these residues, not the integers S_m, E_m before reduction.

B.2 Census of signatures and emissions

The emission of a pair of initial conditions is determined by

$$U_m = 100s_m + 10[s_m + e_m]_{10} + [3s_m + 5e_m + 1]_{10}, \quad 1 \leq m \leq 6. \quad (\text{B.5})$$

The final term 1 is $[7 \cdot 3]_{10}$, corresponding to the three neutral events. This rule's coefficients belong to the declared recurrence. The real constants studied subsequently do not enter enumeration of its domain or its evaluation.

Proposition B.1 (Injective factorization of the finite catalogue). *Emission distinguishes every signature pair (s, e) . There are 18 additive and 26 multiplicative signatures; their pairs produce 468 emissions. The multiplicities of these emissions are 144, 432 and 1944, with 432, 18 and 18 emissions, respectively.*

Proof. The hundreds digit of U_m recovers s_m . If b_m is its tens digit, then $e_m = [b_m - s_m]_{10}$. The units digit verifies the third congruence of (B.5). Thus, the coupling is injective.

The census is obtained by traversing, in the full Cartesian order, $i = 0, \dots, 8$, $j = 0, \dots, 8$ and $d = N, E, S, O$, and applying the 54 steps of (B.3). The 324 additive conditions are distributed over the following 18 signatures, each with 18 preimages. In the tables, a sequence of six digits represents the six ordered components of a signature.

122564	122898	212645
212988	221456	221889
456221	465898	546988
564122	645212	654889
889221	889654	898122
898465	988212	988546

For the multiplicative cursor, the signatures and their multiplicities are

Signature	Multiplicity	Signature	Multiplicity
000000	108	555555	24
177393	8	555717	8
177555	8	555771	8
177771	8	555933	8
339555	8	717555	8
339771	8	717717	8
339933	8	717933	8
393177	8	771177	8
393393	8	771339	8
393555	8	771555	8
555177	8	933339	8
555339	8	933555	8
555393	8	933717	8

These tables result from the finite sums (B.4), with domain, readout order and reversals already specified. Their sums of multiplicities are $18 \cdot 18 = 324$ and $24 \cdot 8 + 24 + 108 = 324$. The Cartesian domain (B.1) permits every signature pair to be realized. Injectivity gives $18 \cdot 26 = 468$ emissions and the multiplicities

$$(18 \cdot 24, 18 \cdot 8) = (432, 144), \quad (18, 18 \cdot 24) = (18, 432), \quad (18, 18 \cdot 108) = (18, 1944),$$

where each pair indicates the number of emissions and multiplicity of their fibres. Finally,

$$432 \cdot 144 + 18 \cdot 432 + 18 \cdot 1944 = 104\,976.$$

Enumeration thus exhausts the complete domain of initial conditions. $\square$

The oriented ternary reduction is

$$\Psi(U) = ([-U_1]_3, \dots, [-U_6]_3).$$

Applied to the $18 \cdot 26$ emissions of the preceding tables, it produces the following fibre census. Here r counts distinct emissions, not initial conditions.

$$\begin{array}{c|cccccc} r & 1 & 2 & 3 & 4 & 5 & 6 \\ \hline \#\{w : \#\Psi^{-1}(w) = r\} & 132 & 66 & 18 & 3 & 6 & 18 \end{array}. \quad (\text{B.6})$$

The sum of the second row is 243, while its weighted sum with weights r is 468. The image of 243 words is therefore distinguished from the ambient space $\mathbb{F}_3^6$, containing 729 words. The accompanying certificate preserves all signatures with their preimages, the 468 emissions and their ternary reduction; the preceding formulas provide a complete procedure for reconstructing them.

B.3 Transition register and reconstruction of the lifts

This subsection fixes the regional register of the cited source. Let $b_0^\chi, \dots, b_4^\chi \in \mathbb{F}_3^6$, with $\chi \in \{C, P, A\}$, be its three sequences. The register organizes transitions $b_0 \rightarrow b_1 \rightarrow b_2$ in X_0, Y_0 and transitions $b_2 \rightarrow b_3 \rightarrow b_4$ in X_1, Y_1 , using row vectors and the same regional order:

$$\begin{aligned} X_0 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 2 & 0 & 1 & 1 & 0 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 2 & 1 & 2 & 0 & 0 \\ 1 & 1 & 2 & 2 & 0 & 2 \end{pmatrix}, & Y_0 &= \begin{pmatrix} 0 & 1 & 2 & 2 & 2 & 2 \\ 0 & 1 & 0 & 2 & 1 & 1 \\ 1 & 2 & 1 & 2 & 2 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 1 & 1 & 2 & 2 & 0 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \end{pmatrix}, \\ X_1 &= \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 1 \\ 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 2 & 1 & 0 & 2 & 0 \\ 0 & 1 & 0 & 2 & 1 & 0 \end{pmatrix}, & Y_1 &= \begin{pmatrix} 0 & 0 & 2 & 1 & 1 & 1 \\ 1 & 1 & 0 & 2 & 2 & 1 \\ 0 & 1 & 2 & 2 & 2 & 2 \\ 1 & 0 & 2 & 0 & 1 & 1 \\ 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 1 & 0 & 2 & 0 & 0 \end{pmatrix}. \end{aligned} \quad (\text{B.7})$$

The preceding production is expressed in its source through the composition of selection, transport and update:

$$x_{j+1}^\chi = \mathcal{U}_{9j+9} \circ \dots \circ \mathcal{U}_{9j+1}(x_j^\chi), \quad b_j^\chi = \Pi_6(x_j^\chi), \quad \mathcal{U}_t = \text{Upd}_t \circ \text{Tra}_t \circ \text{Sel}_t.$$

The following result starts from register (B.7). Its proof reconstructs the operators and verifies transition compatibility; it does not attribute coordinatewise production of states x_j^χ to matrix inversion.

Proposition B.2 (Uniqueness on the regional register). *The systems $X_a L_a = Y_a$, $a = 0, 1$, have a unique solution over $\mathbb{F}_3$. The two solutions are*

$$L_0 = \begin{pmatrix} 2 & 2 & 2 & 1 & 2 & 1 \\ 2 & 1 & 2 & 2 & 1 & 1 \\ 1 & 0 & 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 1 & 1 & 0 \\ 2 & 2 & 2 & 0 & 2 & 1 \\ 2 & 1 & 2 & 1 & 0 & 0 \end{pmatrix}, \quad L_1 = \begin{pmatrix} 2 & 1 & 2 & 0 & 2 & 2 \\ 1 & 2 & 1 & 1 & 2 & 0 \\ 1 & 2 & 1 & 1 & 1 & 2 \\ 0 & 1 & 0 & 0 & 2 & 1 \\ 2 & 0 & 2 & 1 & 0 & 1 \\ 0 & 2 & 2 & 2 & 1 & 1 \end{pmatrix}. \quad (\text{B.8})$$

Proof. Elimination in $\mathbb{F}_3$ gives $\det X_0 = 1$ and $\det X_1 = 2 = -1$. Each map $L \mapsto X_a L$ is therefore an automorphism of the 36-dimensional matrix space; under columnwise vectorization it is represented by $I_6 \otimes X_a$. One obtains $L_a = X_a^{-1} Y_a$. Multiplication of the matrices in (B.8) by the respective matrices X_a reproduces the two matrices Y_a of (B.7); existence and uniqueness are established on that register. $\square$

The three resulting sequences, explicitly separated into blocks, are

$$\begin{aligned} B^C &= 010211 \mid 012222 \mid 010211 \mid 002111 \mid 110221, \\ B^P &= 201101 \mid 121221 \mid 102011 \mid 012222 \mid 102011, \\ B^A &= 121200 \mid 112202 \mid 121020 \mid 010210 \mid 010200. \end{aligned}$$

Each block preserves its six positions. Concatenation is not identified with a base-ten integer.

B.4 Exhaustiveness of the calendar and stability of reconstruction

A binary calendar is $c = (c_0, c_1, c_2, c_3) \in \{0, 1\}^4$. Define the number of transitions incompatible with the register by

$$d(c) = \sum_{\chi \in \{C, P, A\}} \sum_{j=0}^3 \mathbf{1}\{b_j^\chi L_{c_j} \neq b_{j+1}^\chi\}. \quad (\text{B.9})$$

Multiplying the preceding sequences by (B.8) gives

$$d(c) = 3 d_H(c, (0, 0, 1, 1)),$$

where d_H is Hamming distance: at each of the four positions, the alternative operator fails to satisfy all three regional transitions. The complete census is

c	$d(c)$	c	$d(c)$	c	$d(c)$	c	$d(c)$
0000	6	0100	9	1000	9	1100	12
0001	3	0101	6	1001	6	1101	9
0010	3	0110	6	1010	6	1110	9
0011	0	0111	3	1011	3	1111	6

Consequently, 0011 is the unique calendar reproducing the three complete sequences.

Sensitivity to matrix entries likewise admits an exhaustive proof without approximations. A single-entry modification replaces L_a by $L_a + \lambda E_{ij}$, with $a \in \{0, 1\}$, $1 \leq i, j \leq 6$ and $\lambda \in \mathbb{F}_3 \setminus \{0\}$. There are $2 \cdot 6 \cdot 6 \cdot 2 = 144$ such modifications. Since X_a is invertible,

$$X_a(L_a + \lambda E_{ij}) - Y_a = \lambda X_a E_{ij} \neq 0.$$

Every modification destroys at least one transition of the corresponding register. The executable certificate also enumerates the 144 modifications and preserves, for each, the number of violated transitions.

The proved property is rigidity of the operators with respect to the fixed regional register. Reproducing a prior construction of the register itself requires the coordinatewise action of its selection, transport and update operators. That dependency is kept separate from the rigidity theorem, the finite emission catalogue and the subsequent analytic realizations.

B.5 Reproduction certificate and location of the data

The program `verificar_nucleo_finito.py` traverses the 324 conditions of each cursor and forms the product of the resulting classes. The documentary catalogue is loaded after this construction to compare emissions and their multiplicities. In the second part, the four transition matrices are extracted from the verbatim copy of the source; solutions are computed by exact elimination modulo 3 before comparison with the printed matrices. The receipt `NUCLEO_FINITO.json` preserves the domains, preimages, emissions, 16 calendars and 144 modifications.

Verification uses exact integers and residues. It receives no metrological values or real expansions of the constants. Its scope is that of the finite operations and starting data specified in this appendix. The proofs of compatibility at unbounded depth and physical identifications retain their respective domains and expository obligations.

C Finite transport realization with phase memory

The following developments bring together two complementary mechanisms: a finite transfer on APP positions, modes and phase, and a collection of kinematic and constitutive readouts on two angular channels. Each realization retains its state space, operations and parameters. The finite operator has the domain defined below; the enriched TPK developed in the main text also retains its own coordinates and prolongations. Reciprocal readouts are presented as auxiliary realizations with specific formulae and a declared domain.

C.1 Finite transfer on positions, modes and phase

Let $\text{dr}_9(n)$ be the representative in $\{1, \dots, 9\}$ of $n \bmod 9$, with 9 representing the zero residue. For positions $i, j \in \{0, \dots, 8\}$, consider the two tables

$$A_+(i, j) = \text{dr}_9(i + j + 2), \quad A_\times(i, j) = \text{dr}_9((i + 1)(j + 1)).$$

The arguments evaluated here are positive integers. Consequently, the alternative historical convention $\text{dr}_9(0) = 0$ does not enter these tables. The finite state space is

$$\Omega = (\mathbb{Z}/9\mathbb{Z})^2 \times \{+, \times\} \times \{N, E, S, O\} \times \mathbb{Z}/3\mathbb{Z}, \quad |\Omega| = 1944.$$

Write $x = (i, j, m, d, \tau)$, with O denoting the westward direction. The transition retains the following order: choose d' , move the position, read APP in the previous mode, reduce modulo 3, update the mode and advance the phase. With

$$\delta_N = (-1, 0), \quad \delta_E = (0, 1), \quad \delta_S = (1, 0), \quad \delta_O = (0, -1),$$

the operations are

$$\begin{aligned} (i', j') &= (i, j) + \delta_{d'} \pmod{9}, \\ a &= A_m(i', j') \pmod{3}, \\ m' &= \begin{cases} +, & a = 1, \\ \times, & a = 2, \\ m, & a = 0, \end{cases} & \tau' &= \tau + 1 \pmod{3}. \end{aligned}$$

Thus $F_{d'}(x) = (i', j', m', d', \tau')$. The turning and mode-switching cost is

$$c_\lambda(x, d') = \mathbf{1}_{d' \neq d} + \lambda \mathbf{1}_{m' \neq m}, \quad \beta > 0, \quad \lambda \geq 0,$$

and the operator acts on the state basis by

$$T_{\beta, \lambda} e_x = \sum_{d' \in \{N, E, S, O\}} e^{-\beta c_\lambda(x, d')} e_{F_{d'}(x)}. \quad (\text{C.1})$$

The parameters β, λ are declared parameters of this finite model. The reference table uses $(\beta, \lambda) = (1, 1)$; the case $\lambda = 0$ belongs to the sensitivity tests.

Proposition C.1 (Equivariance of the finite antecedent). *The action*

$$\Phi_{u, v, w}(i, j, m, d, \tau) = (i + 3u, j + 3v, m, d, \tau + w), \quad (u, v, w) \in (\mathbb{Z}/3\mathbb{Z})^3,$$

commutes with $T_{\beta, \lambda}$.

Proof. Since $\text{dr}_9(n) \equiv n \pmod{3}$, one has $A_+(i, j) \equiv i + j + 2$ and $A_\times(i, j) \equiv (i + 1)(j + 1) \pmod{3}$. Translation by multiples of three preserves both reduced readouts. It also commutes with the chosen displacement and preserves the previous mode, the subsequent mode and both directions. It therefore preserves each cost indicator. Finally, adding w to the phase commutes with adding one unit. Every transition and its weight are transported to the corresponding transition and weight; their sum in (C.1) retains the same equality. $\square$

Theorem C.2 (Complete character decomposition). *For $\omega = e^{2\pi i/3}$, the space $\mathbb{C}^\Omega$ decomposes unitarily into twenty-seven subspaces of dimension 72. Their blocks obey*

$$T(k_a, k_b, k_t) = \omega^{k_t} T(k_a, k_b, 0), \quad k_a, k_b, k_t \in \{0, 1, 2\}.$$

Proof. Write $i = 3a + r$, $j = 3b + s$, with $a, b, r, s \in \{0, 1, 2\}$. Every fine fibre $u = (r, s, m, d)$ contains 27 states (a, b, τ) . Define the isometry $U_k : \mathbb{C}^{72} \rightarrow \mathbb{C}^\Omega$ by

$$U_k e_u = \frac{1}{\sqrt{27}} \sum_{a,b,\tau=0}^2 \omega^{-(k_a a + k_b b + k_t \tau)} e_{(3a+r, 3b+s, m, d, \tau)}.$$

Character orthogonality gives $U_k^* U_\ell = \delta_{k\ell} I$ and $\sum_k U_k U_k^* = I$. If a transition crosses fine boundaries with coarse increments $\Delta a, \Delta b$, its coefficient in $U_k^* T U_k$ is

$$e^{-\beta c_\lambda \omega^{k_a \Delta a + k_b \Delta b + k_t}}.$$

This proves $T U_k = U_k T(k)$ and the operator decomposition. The temporal increment is always one, so the final factor is common to the whole block. Cubing removes that temporal factor, while the spatial characters retain their role. $\square$

This decomposition rests on equivariance and character orthogonality. Nonnegativity of the trivial block also guarantees a nonnegative real eigenvalue equal to the spectral radius. This property applies to the full space, retaining its transient states as well.

C.2 Spectral decomposition and parametric sensitivity

Let $\rho_1 \geq \rho_2$ be the two largest moduli, counted with multiplicity, and let $g = \rho_1 - \rho_2$. The following table contains the nine spatial blocks. The preceding theorem determines all twenty-seven temporal blocks: for every row and every $t = 0, 1, 2$, the dominant eigenvalue is $\omega^t \mu_1(k_a, k_b, 0)$, whereas ρ_1, ρ_2, g remain unchanged. This representation explicitly preserves all three phases. This first table fixes $\beta = \lambda = 1$.

k_a	k_b	$\mu_1(k_a, k_b, 0)$	ρ_1	ρ_2	g
0	0	1.730888896	1.730888896	1.083076285	0.647812612
0	1	1.459962951	1.459962951	1.116843785	0.343119167
0	2	1.459962951	1.459962951	1.116843785	0.343119167
1	0	1.459962951	1.459962951	1.116843785	0.343119167
1	1	-1.508515719	1.508515719	1.001353790	0.507161929
1	2	-1.514010372	1.514010372	0.971977167	0.542033205
2	0	1.459962951	1.459962951	1.116843785	0.343119167
2	1	-1.514010372	1.514010372	0.971977167	0.542033205
2	2	-1.508515719	1.508515719	1.001353790	0.507161929

The signs of the dominant eigenvalues and their phases are part of the output: the four sectors with $k_a, k_b \neq 0$ have a negative real leader for $k_t = 0$. Its modulus preserves the spectral rate, while its sign retains additional information that enters the composition of iterations.

The five complete sensitivity maps, with rows k_a and columns k_b , are as follows; all correspond to $k_t = 0$:

$$\begin{aligned}\rho(1, 1) &= \begin{pmatrix} 1.730888896 & 1.459962951 & 1.459962951 \\ 1.459962951 & 1.508515719 & 1.514010372 \\ 1.459962951 & 1.514010372 & 1.508515719 \end{pmatrix}, \\ \rho(1, 0) &= \begin{pmatrix} 2.103638324 & 1.685504327 & 1.685504327 \\ 1.685504327 & 1.853245966 & 1.853245966 \\ 1.685504327 & 1.853245966 & 1.853245966 \end{pmatrix}, \\ \rho(1, 2) &= \begin{pmatrix} 1.656915306 & 1.433869155 & 1.433869155 \\ 1.433869155 & 1.423885161 & 1.432116157 \\ 1.433869155 & 1.432116157 & 1.423885161 \end{pmatrix}, \\ \rho(0.5, 1) &= \begin{pmatrix} 2.480049445 & 2.124801655 & 2.124801655 \\ 2.124801655 & 2.281861809 & 2.286772914 \\ 2.124801655 & 2.286772914 & 2.281861809 \end{pmatrix}, \\ \rho(2, 1) &= \begin{pmatrix} 1.226049377 & 1.131390710 & 1.131390710 \\ 1.131390710 & 1.028041508 & 1.028201900 \\ 1.131390710 & 1.028201900 & 1.028041508 \end{pmatrix}.\end{aligned}$$

Here $\rho(\beta, \lambda)$ denotes the map of spectral radii, not the transfer operator. Interchanging i and j , together with the corresponding interchange of directions, preserves APP, transitions and cost; these maps are therefore symmetric. Varying the parameters changes the radii while preserving the character decomposition, which follows from symmetry of the operations. The tables display numerical approximations to finite spectra. Transition, equivariance and factorization identities follow from the preceding proofs; their numerical consequences are retained together with the complete data for all twenty-seven components.

C.3 Reciprocal readouts and sensitivity tests

Reproduction of the antecedent angular realization preserves the declared decimal numerical measures in degrees

$$a_h = 7.297352569283802, \quad c_h = 2.123738338968646.$$

The angles and the corresponding dimensionless coordinate are distinguished by

$$A_h = a_h^\circ, \quad C_h = c_h^\circ, \quad \alpha_h = a_h/1000.$$

These data permit a comparison with that antecedent. The constants in the current development retain their prior APP-TRIT-TPK generation and subsequent readout; historical reproduction does not replace that generation by the decimals written here. The angular conversion and the channels used are

$$x_h = \frac{\pi a_h}{180}, \quad y_h = \frac{\pi c_h}{180}, \quad q_{\pm} = e^{-(x_h \pm y_h)}, \quad 0 < q_+ < q_- < 1.$$

The first readout gives

$$\epsilon_{90} = q_-^{90} - q_+^{90}, \quad c_{\pm} = \frac{c_{\text{ref}}}{1 \pm \epsilon_{90}}.$$

Here c_{ref} only normalizes the kinematic test. Reproduction in the stated units uses the same historical reference value, $299792458 \text{ m s}^{-1}$.

Proposition C.3 (Harmonic mean and two-sheet product). *If $|\epsilon_{90}| < 1$, the harmonic mean of c_+ and c_- is c_{ref} . In the historical constitutive readout*

$$\mu_h^{\pm} = e^{\pm 2\alpha_h}, \quad \epsilon_h^{\pm} = e^{\mp 2\alpha_h},$$

one has $\mu_h^{\pm} \epsilon_h^{\pm} = 1$ and $Z_+/Z_- = e^{4\alpha_h}$.

Proof. The sum $c_+^{-1} + c_-^{-1} = 2/c_{\text{ref}}$ removes the oriented term. On each sheet, the two constitutive exponents sum to zero. Moreover, $Z_{\pm} = \sqrt{\mu_{\text{h}}^{\pm}/\varepsilon_{\text{h}}^{\pm}} = e^{\pm 2\alpha_{\text{h}}}$. The impedance ratio therefore retains the orientation removed by the constitutive product. $\square$

The second historical readout follows from convergent geometric series:

$$\begin{aligned} S_a(q) &= \frac{q^a}{1 - q^{3a}} = \sum_{j \geq 0} q^{a(1+3j)}, \\ \delta_{a,b}(q) &= S_a(q) - S_b(q), \\ R_{a,b} &= \frac{\delta_{a,b}(q_-) - \delta_{a,b}(q_+)}{q_-^a - q_+^a}. \end{aligned}$$

Convergence holds for $|q| < 1$; the denominator of the last quotient is positive for the channels and positive integers considered here. The historical auxiliary readouts are

$$\zeta_{12} = 1 - R_{90,120}^2, \quad \Delta_{4,\text{h}} = \frac{x_{\text{h}} - y_{\text{h}}}{810} \left(1 - \frac{7\pi}{12 \cdot 729} \right), \quad \xi_{\text{h}} = 1 - \frac{27}{160} \Delta_{4,\text{h}}.$$

The subscript h keeps this historical definition of torsion explicit, avoiding identification with a different current readout merely because it uses the same symbol.

Evaluation from the declared decimal strings gives

$$\begin{aligned} q_+ &= 0.848377942576404356 \dots, & q_- &= 0.913660151150557274 \dots, \\ \varepsilon_{90} &= 0.000295169439289827959 \dots, & R_{90,120} &= 0.933314535237713061 \dots, \\ \zeta_{12} &= 0.128923978314011665 \dots, & \xi_{\text{h}} &= 0.999981235497802379 \dots \end{aligned}$$

The complete table of discrete perturbations is

a	b	$R_{a,b}$
90	120	0.933314535238
90	119	0.927013603649
90	121	0.939071551912
89	120	0.939066204789
91	120	0.927019444194
30	40	0.569215041549
60	80	0.834174862952

Angular sensitivity distinguishes the two coordinates of the pair:

Variable	Relative change	$R_{90,120}$	Change in ppm
A_{h}	−0.001	0.933059250359	−273.525022305
A_{h}	+0.001	0.933568846564	+272.481908792
C_{h}	−0.001	0.933388163678	+78.889202964
C_{h}	+0.001	0.933240822366	−78.979667757

Evaluations used 80 and 120 digits of working precision, starting from the declared decimal strings. Evaluation precision preserves the finite specification of those angular data.

Finally, these readouts retain their place in the documentary genealogy. The current constitutive operator $V = (I - T^{90})(I - T^{120})^{-1}$ evaluates $(1 - q_{\pm}^{90})/(1 - q_{\pm}^{120})$ on its two sheets. The current oriented functional uses $12S_{90} - S_{120}$. The antecedent realization retains the readout $S_{90} - S_{120}$ and its normalization by $q_-^{90} - q_+^{90}$. Their relation is expressed by preserving each operation and its weights, rather than replacing them by a single number or the same symbol.

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